

Applied and Numerical Harmonic Analysis

$$\hat{f}(\gamma) = \int f(x) e^{-2\pi i x \gamma} dx$$

Justin Feuto
Bérenger Akon Kpata
Editors

Harmonic Analysis and Partial Differential Equations

Proceedings of the Workshop in Abidjan,
Côte d'Ivoire, May 22-26, 2023

 Birkhäuser

Applied and Numerical Harmonic Analysis

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*To the pioneers of harmonic analysis at the
Mathematics and Computer Science Training
and Research Unit (UFRMI) of Félix
Houphouët-Boigny University*

ANHA Series Preface

The *Applied and Numerical Harmonic Analysis* (ANHA) book series aims to provide the engineering, mathematical, and scientific communities with significant developments in harmonic analysis, ranging from abstract harmonic analysis to basic applications. The title of the series reflects the importance of applications and numerical implementation, but richness and relevance of applications and implementation depend fundamentally on the structure and depth of theoretical underpinnings. Thus, from our point of view, the interleaving of theory and applications and their creative symbiotic evolution is axiomatic.

Harmonic analysis is a wellspring of ideas and applicability that has flourished, developed, and deepened over time within many disciplines and by means of creative cross-fertilization with diverse areas. The intricate and fundamental relationship between harmonic analysis and fields such as signal processing, partial differential equations (PDEs), and image processing is reflected in our state-of-the-art ANHA series.

Our vision of modern harmonic analysis includes a broad array of mathematical areas, e.g., wavelet theory, Banach algebras, classical Fourier analysis, time-frequency analysis, deep learning, and fractal geometry, as well as the diverse topics that impinge on them.

For example, wavelet theory can be considered an appropriate tool to deal with some basic problems in digital signal processing, speech and image processing, geophysics, pattern recognition, biomedical engineering, and turbulence. These areas implement the latest technology from sampling methods on surfaces to fast algorithms and computer vision methods. The underlying mathematics of wavelet theory depends not only on classical Fourier analysis but also on ideas from abstract harmonic analysis, including von Neumann algebras and the affine group. This leads to a study of the Heisenberg group and its relationship to Gabor systems, and of the metaplectic group for a meaningful interaction of signal decomposition methods.

The unifying influence of wavelet theory in the aforementioned topics illustrates the justification for providing a means for centralizing and disseminating information from the broader, but still focused, area of harmonic analysis. This will be a key

role of ANHA. We intend to publish with the scope and interaction that such a host of issues demands.

Along with our commitment to publish mathematically significant works at the frontiers of harmonic analysis, we have a comparably strong commitment to publish major advances in the following applicable topics in which harmonic analysis plays a substantial role:

*Analytic Number theory * Antenna Theory * Artificial Intelligence * Biomedical
Signal Processing * Classical Fourier Analysis * Coding Theory *
Communications Theory * Compressed Sensing * Crystallography and
Quasi-Crystals * Data Mining * Data Science * Deep Learning * Digital Signal
Processing * Dimension Reduction and Classification * Fast Algorithms * Frame
Theory and Applications * Gabor Theory and Applications * Geophysics * Image
Processing * Machine Learning * Manifold Learning * Numerical Partial
Differential Equations * Neural Networks * Phaseless Reconstruction * Prediction
Theory * Quantum Information Theory * Radar Applications * Sampling Theory
(Uniform and Non-uniform) and Applications * Spectral Estimation * Speech
Processing * Statistical Signal Processing * Super-resolution * Time Series *
Time-Frequency and Time-Scale Analysis * Tomography * Turbulence *
Uncertainty Principles * Waveform Design * Wavelet Theory and Applications

The above point of view for the ANHA book series is inspired by the history of Fourier analysis itself, whose tentacles reach into so many fields.

In the last two centuries Fourier analysis has had a major impact on the development of mathematics, on the understanding of many engineering and scientific phenomena, and on the solution of some of the most important problems in mathematics and the sciences. Historically, Fourier series were developed in the analysis of some of the classical PDEs of mathematical physics; these series were used to solve such equations. In order to understand Fourier series and the kinds of solutions they could represent, some of the most basic notions of analysis were defined, e.g., the concept of “function.” Since the coefficients of Fourier series are integrals, it is no surprise that Riemann integrals were conceived to deal with uniqueness properties of trigonometric series. Cantor’s set theory was also developed because of such uniqueness questions.

A basic problem in Fourier analysis is to show how complicated phenomena, such as sound waves, can be described in terms of elementary harmonics. There are two aspects of this problem: first, to find, or even define properly, the harmonics or spectrum of a given phenomenon, e.g., the spectroscopy problem in optics; second, to determine which phenomena can be constructed from given classes of harmonics, as done, for example, by the mechanical synthesizers in tidal analysis.

Fourier analysis is also the natural setting for many other problems in engineering, mathematics, and the sciences. For example, Wiener’s Tauberian theorem in Fourier analysis not only characterizes the behavior of the prime numbers but is a fundamental tool for analyzing the ideal structures of Banach algebras. It also provides the proper notion of spectrum for phenomena such as white light. This latter process leads to the Fourier analysis associated with correlation functions in

filtering and prediction problems. These problems, in turn, deal naturally with Hardy spaces in complex analysis, as well as inspiring Wiener to consider communications engineering in terms of feedback and stability, his cybernetics. This latter theory develops concepts to understand complex systems such as learning and cognition and neural networks; it is arguably a precursor of deep learning and its spectacular interactions with data science and AI.

Nowadays, some of the theory of PDEs has given way to the study of Fourier integral operators. Problems in antenna theory are studied in terms of unimodular trigonometric polynomials. Applications of Fourier analysis abound in signal processing, whether with the fast Fourier transform (FFT), or filter design, or the adaptive modeling inherent in time-frequency-scale methods such as wavelet theory.

The coherent states of mathematical physics are translated and modulated Fourier transforms, and these are used, in conjunction with the uncertainty principle, for dealing with signal reconstruction in communications theory. We are back to the *raison d'être* of the ANHA series!

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Foreword

The introduction of Harmonic Analysis at the National University of Côte d'Ivoire called now "University Félix HOUPOUET-BOIGNY(UFHB)" since the Decree n° 2012-982 of October 10, 2012, dates back to the 1980s. At that time, we were interested in the theory of Representations of Lie Groups and Analysis on Riemannian Symmetric Spaces. Since then, Harmonic Analysis has developed a lot at the University Félix HOUPOUET-BOIGNY, in particular at Mathematics and Computer Science Training and Research Unit.

I was very happy when young colleagues led by Professor Justin Feuto (my former students in fact) came to ask me to sponsor scientific meetings around Harmonic Analysis. I did not hesitate to give them my full support. I congratulate this dynamic team which contributes a lot in developing research in Harmonic Analysis in Abidjan and which gives the opportunity to the UFRMI to receive eminent researchers in this field.

I personally participated in the opening of the workshop from May 22 to 26, 2023, and I was very interested by the quality of the guests and themes on the agenda. I wish this group a good luck by renewing my flawless support for them.

Abidjan, Côte d'Ivoire
November 21, 2023

Professor Saliou Toure

Preface

The Workshop in Harmonic Analysis and Applications has become a tradition in Mathematics and Computer Science Training and Research Unit of Félix Houphouët-Boigny University since the 2015 edition. It is organized by the Group of Harmonic Analysis and Applications of Abidjan (GAHAA) whose members are researchers and doctoral and master's students from all the universities of Côte d'Ivoire.

The recent Workshop took place at UFRMI from May 22 to 26, 2023. The members of the scientific committee were Aline Bonami, Kinvi Kangni, Kasso A. Okoudjou, and Augustin Touré. The local organizing committee included Justin Feuto, Bérenger A. Kpata, David Tuo, Clément Bahi, Martial A. Dakoury, Mouhamadou Dosso, Sylvain Zeze, Piè Coulibaly, and Modeste Essoh.

The objectives of these workshops are among other things the exploration of new trends in harmonic analysis, the creation of a space for exchange and presentation of research results, and the presentation to students of Master 2, research possibilities that are offered by this specialty and its applications.

The book is made up of articles presented at the workshop held in Abidjan from May 22 to 26, 2023, and contributions from the guest authors. All the chapters deal with harmonic analysis or partial differential equations or even of the two. The chapters are either a survey or an original research article. More specifically, there are two surveys in the book and nine original research articles.

The first group of contributions, that is to say the first five chapters, deals with classical harmonic analysis. This group begins with the contribution of Aline Bonami, Sandrine Grellier, and Benoît Sehba linked to the Stein theorem in the upper half-plan and the spaces of Bergman with weights. This chapter, which is partly a survey, relates to the complex harmonic analysis on the upper half-plan. This is also the case of the second chapter which is the contribution of Jean-Marcel Tanoh Dje and Justin Feuto and which gives Poisson and Cauchy representation theorems on Hardy-Orlicz spaces in the Upper half-plan. After these two chapters on the complex analysis, Nouffou Diarra gives a characterization of temperate distributions which are divergence of function in Lebesgue space with vector values. Pokou Nagacy, Bérenger Akon Kpata, and Justin Feuto give in the chapter before

last of this group, a control of the commutator of the maximal function in a Wiener amalgam type space in the context of the Dunkl analysis. This group ends with a survey. Entitled “The Ubiquitous Appearance of BUPUs”, this contribution of Hans G. Feichtinger gives a survey on the use of uniform partitions in harmonic analysis.

In the second group of chapters, the reader will have concepts of abstract harmonic analysis. This begins with a survey of Ali Baklouti and Junko Inoue, dealing with the Estimate of the L_p -Fourier transform norm for some Lie groups. The next contribution in this group comes from Pié Coulibaly, Kouakou Germain Brou, and Kinvi Kangni and it deals with the spherical function of type delta, on groupoids while Kouakou Germain Brou and Kinvi Kangni propose a chapter on the Convolution in Orlicz spaces on Hypergroups.

This book ends with a group of chapters dealing with partial differential equations. The first contribution in this group is that of Boureima Bella, Bruno Kabore, and Ibrahim Ly with a Variational Approach to the Cauchy Problem for Nonlinear Elliptic Systems, followed by a contribution of Armel Fabrice Evrard Yode and Kouacou Tanoh on stochastic partial differential equations. Both chapters propose a theoretical approach while the next from Ludovic Martaud and Christophe Berthon is a numerical resolution of partial differential equations and it is entitled “Entropy stable Godunov numerical schemes for the Euler equations.” The book ends with the contribution of Kolé Keita and Mahamar Dicko on a “Time Convergence of Bloch-Type Model with Pauli’s Master Equation and Wave-Matter Interaction.”

Abidjan, Côte d’Ivoire
November 30, 2023

Justin Feuto

Acknowledgments

We would like to mention that all our workshops are under the sponsorship of Professor Saliou Toure. He always knew how to convince us to continue organizing these meetings despite the difficulties encountered. We would like to warmly thank him for his wise advice and financial and moral support. We would like to emphasize here that the May 2023 session would never have taken place if he had not been personally involved. We also extend our sincere thanks to the Société Mathématiques de Côte d'Ivoire (SMCI) which always responds favorably to our requests. Thanks also to the president of Félix Houphouët-Boigny University and the director of the Mathematics and Computer Science Training and Research Unit who always facilitated the holding of our meetings. Aline Bonami has been with us since the start of this adventure. May she find here the expression of our gratitude. We thank all the teachers who gave mini-lessons during our workshops. They raised, through their participation, the level of these meetings. Many thanks to everyone who contributed by a chapter to the proceedings, especially those who agreed to contribute even though they did not attend the workshop.

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Chapter 1

Stein's Theorem in the Upper-Half Plane and Bergman Spaces with Weights



Aline Bonami, Sandrine Grellier, and Benoît Sehba

Abstract This is a companion paper to our previous one, *Avatars of Stein's Theorem in the complex setting*. In this previous paper, we gave a sufficient condition for an integrable function in the upper-half plane to have an integrable Bergman projection. Here we push forward methods and establish in particular a converse statement. This naturally leads us to study a family of weighted Bergman spaces for logarithmic weights $(1 + \ln_+(1/\Im m(z)) + \ln_+(|z|))^k$, which have the same kind of behavior, respectively, at the boundary and at infinity. We introduce their duals, which are logarithmic Bloch type spaces and interest ourselves in multipliers, pointwise products, and Hankel operators. In particular we characterize the symbols of bounded Hankel operators within these weighted Bergman spaces.

Classification 30C40, 30H20, 30H30

1.1 Introduction

This chapter revisits previous work of the authors, essentially [5] but also [4]. It may serve partly as a survey but also contains new information as well as some extensions of the results given there.

Let us first give the main results of these two papers.

In [4], we considered an analog for the whole space of Stein's Theorem ([10]), which says that, whenever f is an $L^1(\mathbb{R}^n)$ nonnegative function, which is supported in a ball B , its Hardy–Littlewood maximal function Mf is integrable on B if and

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only if $f \ln_+(f)$ is integrable. We proved the following, for which [1] was also a source of inspiration:

Theorem 1.1 *A function $f \in L^1(\mathbb{R}^n)$ is in $H^1(\mathbb{R}^n)$ when it satisfies the two conditions:*

$$\int_{\mathbb{R}^n} f dx = 0, \quad (1.1)$$

$$\int_{\mathbb{R}^n} |f|(\ln_+(|f|) + \ln_+(|x|)) dx < \infty. \quad (1.2)$$

Conversely, let $f \in L^1(\mathbb{R}^n)$ be a function which is nonnegative outside a compact set and such that

$$\int |f_-(x)| \ln_+(|f_-(x)|) dx < \infty.$$

If f belongs to $H^1(\mathbb{R}^n)$, then f satisfies Conditions (1.1) and (1.2).

Of course it is not possible for a nonnegative function f to be itself in $H^1(\mathbb{R}^n)$ since all functions in $H^1(\mathbb{R}^n)$ have integral 0. The first part of this statement may be found directly in [4], while the converse part looks more general. In fact, with the assumptions given above, we can write, assuming that f has integral 0, that

$$f = \left(f_+ - \left(\int f_+ dx \right) \chi_B \right) - \left(f_- - \left(\int f_- dx \right) \chi_B \right),$$

with B the unit ball centered at 0. The second term is in H^1 since it has integral 0, is in $L \log L$, and is compactly supported (just use the first statement of the theorem). The statement given in [4] expresses that the first one is in H^1 if and only if f_+ satisfies (1.2). This concludes the proof of this new formulation.

In [5] we consider an analogous situation, but in the upper-half-space

$$\mathbb{C}_+ := \{z \in \mathbb{C}, \Im m(z) > 0\}$$

endowed with the Lebesgue measure noted

$$dV(z) = dx dy \quad z = x + iy.$$

We denote by $\mathcal{H} := \mathcal{H}(\mathbb{C}_+)$ the space of holomorphic functions in $\mathbb{C}_+$. We are now interested in the Bergman projection, that is, the orthogonal projection from $L^2(\mathbb{C}_+)$ onto its subspace of holomorphic functions. More generally, we define

$$A^p(\mathbb{C}_+) := L^p(\mathbb{C}_+) \cap \mathcal{H}.$$

We denote by P the Bergman projection. Its kernel, called the Bergman kernel, is given by

$$K(z, \zeta) = \frac{1}{\pi(z - \bar{\zeta})^2}.$$

We call P^+ the operator with kernel $|K(z, \zeta)|$. The following statement is an improvement of the one of [5].

Theorem 1.2 *The Bergman projection of a function $f \in L^1(\mathbb{C}_+)$ (respectively, P^+f) is an integrable function if it satisfies the two conditions:*

$$\int_{\mathbb{C}_+} f dV(z) = 0, \quad (1.3)$$

$$\int_{\mathbb{C}_+} |f(z)| (\ln_+((\Im m z)^{-1}) + \ln_+(|z|)) dV(z) < \infty. \quad (1.4)$$

Conversely, let $f \in L^1(\mathbb{C}_+)$ be a nonnegative function outside a compact set of $\mathbb{C}_+$. When P^+f belongs to $L^1(\mathbb{C}_+)$, then f satisfies Conditions (1.3) and (1.4).

The converse part of the theorem is new. We postpone details and proofs to the next section. The first idea of the proof is the fact that, since f has integral 0, we may subtract $K(z, i)$ to the Bergman kernel $K(z, \zeta)$ and write

$$\begin{aligned} Pf(z) &= \int_{\mathbb{C}_+} K_{\text{mod}}(z, \zeta) f(\zeta) dV(\zeta), & K_{\text{mod}}(z, \zeta) &= K(z, \zeta) - K(z, i) \\ P^+f(z) &= \int_{\mathbb{C}_+} K_{\text{mod}}^+(z, \zeta) f(\zeta) dV(\zeta), & K_{\text{mod}}^+(z, \zeta) &= |K(z, \zeta)| - |K(z, i)|. \end{aligned}$$

These new kernels are integrable in z . Let us emphasize that this modification of the Bergman kernel is classical, see, for instance, [7, 9].

We will denote by P_{mod} and P_{mod}^+ the operators with kernels K_{mod} and K_{mod}^+ , respectively, considered as defined on the entire space $L^1(\mathbb{C}_+)$. Remark that while P and P^+ satisfy the inequality $|Pf| \leq P^+(|f|)$, there is no inequality between modified operators, and P_{mod}^+ is no more nonnegative.

One may be surprised to see a difference between the two statements, with $\ln_+(|f|)$ replaced by $\ln_+((\Im m z)^{-1})$ in the upper-half plane. Indeed, the Bergman projection is a singular integral, like Riesz transforms in $\mathbb{R}^n$. This is clarified by the following proposition, whose proof is given in [5].

Proposition 1.1 *Whenever $f \in L^1(\mathbb{C}_+)$ is such that*

$$\int_{\mathbb{C}_+} |f(z)|(1 + \ln_+(|f(z)|) + \ln_+(|z|))dV(z) < \infty, \quad (1.5)$$

then f satisfies Condition (1.4).

There exist functions that satisfy Condition (1.4) but not Condition (1.5): Indeed any L^1 function that is supported in a compact set of $\mathbb{C}_+$ satisfies (1.4) but does not in general satisfy (1.5). So one finds a strictly weaker condition in the Bergman situation. Moreover we do not need to appeal to a Musielak Orlicz space as in the case of $\mathbb{R}^n$ to have a critical condition for Pf to be in $L^1(\mathbb{C}_+)$.

In the second section we will give the proof of Theorem 1.2 and related statements. The third section is at the same time a survey and a generalization to weighted Bergman spaces $A_\Omega^1 := \mathcal{H} \cap L^1(\Omega dV)$. We will particularly be interested in weights ω^k , where

$$\omega(z) := 1 + \ln_+ \left(\frac{1}{\Im m(z)} \right) + \ln_+(|z|), \quad z \in \mathbb{C}_+. \quad (1.6)$$

Our starting point in [5] was the fact that, when looking at the product of a function in $A^1(\mathbb{C}_+)$ and a function in the Bloch class, that is, an element of its dual, one obtains a function in $A_{\omega^{-1}}^1$. So the same weight as in Theorem 1.2, but with a negative power, appeared there. Remark that one may find a better estimate, see [5] and also [3] for the case of the unit ball. Indeed, for $f \in A^1(\mathbb{C}_+)$ and $b \in \mathcal{B}$, the product $h = fb$ satisfies the inequality

$$\int_{\mathbb{C}_+} \frac{|h(z)|}{1 + \ln_+(|h(z)|) + \ln_+(|z|)}dV(z) < \infty, \quad (1.7)$$

which is stronger as it can be proved using atomic decomposition of both spaces. We could as well define spaces of functions f such that

$$\int_{\mathbb{C}_+} \frac{|h(z)|}{1 + (\ln_+(|h(z)|))^k + (\ln_+(|z|))^k}dV(z) < \infty.$$

But spaces $A_{\omega^k}^1$ have the advantage of being Banach spaces while having the same duals, so that we will limit ourselves to them.

We will consider products of functions, respectively, in $A_{\omega^k}^1$ and its dual, which is given by the Bloch type space $\mathcal{B}_{\omega^{-k}}(\mathbb{C}_+)$. Here $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$ is the space of all holomorphic functions f such that

$$\sup_{z \in \mathbb{C}_+} (\Im m z) \omega^k(z) |f'(z)| < \infty.$$

Only $\mathcal{B}_\omega(\mathbb{C}_+)$ was introduced in [5] under the name $\mathcal{B}_{\log}$. The behavior of these spaces depends on the value of k . Our first elementary observation is the fact that $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$ is an algebra for $k > 1$.

In each case we study Hankel operators and weak factorization. We establish in particular the following theorem which characterizes the symbols of bounded Hankel operators between the above weighted Bergman spaces.

Recall that the Hankel operator h_b , with $b \in \mathcal{B}$, is the operator defined on $A^2(\mathbb{C}_+)$ by $h_b(f) = P(\bar{b}f)$. Condition (1.3) for $\bar{b}\bar{f}$ induces to replace P by P_{mod} .¹

Theorem 1.3 *Let b be in $\mathcal{H}(\mathbb{C}_+)$. Suppose that $l \in \mathbb{R}$, and $k < 1$. Then the Hankel operator h_b extends to a bounded operator from $A^1_{\omega^l}(\mathbb{C}_+)$ to $A^1_{\omega^{-k}}(\mathbb{C}_+)$ if and only if $b \in \mathcal{B}_{\omega^{1-(l+k)}}(\mathbb{C}_+)$.*

In particular, from the above result, we have the following.

Corollary 1.1 *Let b be in $\mathcal{H}(\mathbb{C}_+)$. Suppose that $k < 1$. Then the Hankel operator h_b extends to a bounded operator from $A^1_{\omega^{1-k}}(\mathbb{C}_+)$ to $A^1_{\omega^{-k}}(\mathbb{C}_+)$ if and only if $b \in \mathcal{B}(\mathbb{C}_+)$.*

The following is also an interesting consequence as it provides another equivalent characterization of $\mathcal{B}_\omega(\mathbb{C}_+)$.

Corollary 1.2 *Let b be in $\mathcal{H}(\mathbb{C}_+)$. Suppose that $k < 1$. Then the Hankel operator h_b extends to a bounded operator from $A^1_{\omega^{-k}}(\mathbb{C}_+)$ to $A^1_{\omega^{-k}}(\mathbb{C}_+)$ if and only if $b \in \mathcal{B}_\omega(\mathbb{C}_+)$.*

We will give other characterizations of Hankel operators in Sect. 1.3.

In the following, we will use the notation $A \lesssim B$ (respectively, $A \gtrsim B$, $A \simeq B$) whenever there exists a uniform constant $C > 0$ such that $A \leq CB$ (respectively, $A \geq CB$, respectively, $A \lesssim B$ and $A \gtrsim B$).

We will use the notation $\log$ for the holomorphic logarithm and the notation $\ln$ for the real one.

1.2 Theorem 1.2: Proof and Related Statements

We refer to [5] for the proof of the sufficiency of Conditions (1.3) and (1.4). We start with a lemma, which completes Lemma 1.2 of this previous paper, and which will have as corollary that the theorem is sharp for P and P^+ . Here we recall the definition of the weight ω .

$$\omega(z) := 1 + \ln_+ \left(\frac{1}{\Im m(z)} \right) + \ln_+ (|z|), \quad z \in \mathbb{C}_+. \quad (1.8)$$

¹ When reading the prints, we realized that some clarifications of the proofs would be welcome. This will be done in a forthcoming paper.

Lemma 1.1 *We have the inequality, valid for $|\zeta - i| > 1$,*

$$I(\zeta) := \int_{\mathbb{C}_+} \left| |z - \bar{\zeta}|^{-2} - |z + i|^{-2} \right| dV(z) \simeq \omega(\zeta). \quad (1.9)$$

Proof The above bound has been proved in [5]. We come back to the proof in order to give the bounds below. We write $z = x + iy$ and $\zeta = \xi + i\lambda$. From the continuity of the integral as a function of ζ , one sees immediately that it is bounded below on any compact set of $\mathbb{C}_+$. So it is sufficient to prove the inequality for $\lambda < \varepsilon < 1$ and for $|\zeta| > M$. We have two cases to consider:

1. If ζ is such that $\lambda < 1/4$, we write that $I(\zeta)$ is bounded below by the integral over the domain $E_0 := \{z \in \mathbb{C}_+, |x - \xi| < y + \lambda, y < 1/4\}$. On this set, the inequality $|z - \bar{\zeta}|^2 \leq 2(y + \lambda)^2 < 1/2$ holds so that

$$\int_{E_0} |z - \bar{\zeta}|^{-2} dV(z) \geq \int_0^{1/4} \frac{dy}{y + \lambda} = \ln(1 + \frac{1}{4\lambda}),$$

while

$$\int_{E_0} |z + i|^{-2} dV \leq 1.$$

So for $\lambda < 1/4$ we have

$$I(\zeta) \gtrsim 1 + \ln_+(\lambda^{-1}).$$

2. If $|\zeta| > M$, we write that $I(\zeta)$ is bounded below by the integral on the domain $E_1 := \{z \in \mathbb{C}_+, |z + i| < \frac{1}{4}|\zeta - i|, |z| > 1\}$. The function to integrate is bounded below by $c|z + i|^{-2}$, so that

$$I(\zeta) \gtrsim \int_{E_1} |z|^{-2} dV(z) \gtrsim \ln(\frac{1}{4}|\zeta| - 2).$$

Here, we used the fact that the half disc $\{|z| < \frac{1}{4}|\zeta| - 2, y > 0\}$ is contained in E_1 . So finally, we find that $I(\zeta) \simeq 1 + \ln_+(|\zeta|)$ for $|\zeta| \geq M$ large enough. This completes the proof of the lemma.

Remark 1.1 The same is valid for the kernel $K_{\text{mod}}(z, \zeta)$ as well.

As a corollary, we find a family of functions f that are all nonnegative outside a fixed compact set and for which the L^1 norm of Pf is equivalent to the norm of f in the weighted L^1 space for the weight ω . This is given in the following proposition.

Proposition 1.2 *For each $\zeta = \xi + i\lambda \in \mathbb{C}_+$ such that $|\zeta - i| > 1$, let f_ζ be the function equal to $\frac{4}{\pi\lambda^2} \chi_{B(\zeta, \lambda/2)} - \frac{4}{\pi} \chi_{B(i, 1/2)}$. Then $\|f_\zeta\|_1 \leq 2$, while $\|Pf_\zeta\|_1 \simeq \omega(\zeta)$. Constants may be chosen independent of ζ .*

Proof It is sufficient to see that $Pf_\zeta(z) = \frac{1}{\pi} [(z - \bar{\zeta})^{-2} - (z + i)^{-2}]$ and to use Remark 1.1. This is a consequence of the mean value property of holomorphic functions.

This proposition proves that Condition (1.4) is optimal.

We have the same proposition with P^+ in place of P .

Proposition 1.3 *We have $\|f_\zeta\|_1 + \|P^+ f_\zeta\|_1 \simeq \omega(\zeta)$. Constants may be chosen independent of ζ .*

Proof We use again Lemma 1.1. It is sufficient to prove that $\frac{4}{\pi\lambda^2} P^+(\chi_{B(\zeta, \lambda/2)}) - |K(z, \zeta)|$ has an L^1 norm bounded by a uniform constant. But for $\zeta' \in B(\zeta, \lambda/2)$, the function $|K(\cdot, \zeta)| - |K(\cdot, \zeta')|$ has L^1 norm uniformly bounded, which allows to conclude. Indeed, this difference is bounded, up to a constant, by $\frac{\lambda}{|z - \bar{\zeta}|^3}$, which is uniformly in L^1 .

Next, we prove that the condition (1.3) is necessary for Pf to be integrable. Namely:

Proposition 1.4 *Let $f \in L^1(\mathbb{C}_+)$ such that Pf (resp., $P^+ f$) is in $L^1(\mathbb{C}_+)$. Then f has integral 0.*

Proof We prove it for Pf in L^1 . The same proof holds for $P^+ f$ in L^1 . We write that

$$z^2 Pf(z) = c \int_{\mathbb{C}_+} \frac{f(\zeta) dV(\zeta)}{(1 - \bar{\zeta}/z)^2}$$

and want to use Lebesgue Dominated Convergence Theorem. When $|x| < y$, as $|1 - \bar{\zeta}/z| > y/|z| > 1/2$, we get for $|z|$ tending to ∞ with $|x| < y$

$$z^2 Pf(z) \longrightarrow \int_{\mathbb{C}_+} f(\zeta) dV(\zeta).$$

Since $z \mapsto z^{-2}$ is not integrable over this domain, this forces the integral to be 0 if Pf is integrable.

Remark 1.2 As $z \mapsto \frac{\omega^k(z)}{z^2}$ is not integrable, the last proposition remains valid when we weaken the assumptions and only assume that Pf (resp., $P^+ f$) is integrable for the weighted measure $\omega^k dV$ for $k \geq -1$.

We are now ready to finish the proof of the necessity conditions in Theorem 1.2. So let $f \in L^1(\mathbb{C}_+)$ be a nonnegative function outside a compact set of $\mathbb{C}_+$ such that $P^+(f)$ belongs to $L^1(\mathbb{C}_+)$. We can assume without loss of generality that f is nonnegative for $\zeta \notin B(i, 1/2)$ and has norm in L^1 equal to 1. We already know that f has integral 0 so that

$$P^+ f(z) = \int (|K(z, \zeta)| - |K(z, i)|) f(\zeta) dV(\zeta).$$

Moreover, since $\int (|K(z, \zeta)| - |K(z, i)|) f(\zeta) dV(\zeta)$ is in L^1 as f_- has compact support, we can replace f by f_+ . So the question becomes: *Let $f \in L^1$ be a nonnegative function such that*

$$P_{\text{mod}}^+ f = \int (|K(\cdot, \zeta)| - |K(\cdot, i)|) f(\zeta) dV(\zeta) \in L^1.$$

Then prove that f satisfies (1.4). We also assume that $\|f\|_1 = 1$ for simplicity. The proof is given in the two next lemmas, whose assumptions are the ones that we have just given.

The first step mimics the proof of Lemma 1.1. We give its proof for completeness, even if the proof is already given in [5].

Lemma 1.2 *Under the assumptions that f is nonnegative and $P_{\text{mod}}^+ f$ is in L^1 , we have*

$$\int_{\mathbb{C}_+} |f(\zeta)| \ln_+(1/\Im m \zeta) dV(\zeta) < \infty.$$

Proof If we reduce the integral of $|P^+ f(z)|$ to the one over the set $y < 1$, then the integral coming from the second term is bounded since

$$\int_{\{x+iy \in \mathbb{C}_+; y < 1\}} \int_{\mathbb{C}_+} |K(z, i)| f(\zeta) dV(\zeta) dV(z) \lesssim \int_0^1 (y+1)^{-1} dy \leq \ln(2).$$

Finally by difference, we get that

$$\int_{\{x+iy \in \mathbb{C}_+; y < 1\}} \int_{\mathbb{C}_+} |K(z, \zeta)| f(\zeta) dV(\zeta) dV(z) < \infty.$$

We integrate first in z and find that

$$\int_{\mathbb{C}_+} f(\zeta) \ln(1 + 1/\Im m \zeta) dV(\zeta) < \infty.$$

Lemma 1.3 *Under the previous assumptions for f , we have*

$$\int_{\mathbb{C}_+} |f(\zeta)| \ln_+(|\zeta|) dV(\zeta) < \infty.$$

Proof We only consider the integral of $P_{\text{mod}}^+ f$ over the set

$$F := \{z = x + iy; |x| \leq y, y > 1\}.$$

We write for $z \in F$,

$$P_{\text{mod}}^+ f(z) := I_1(z) + I_2(z) + I_3(z),$$

where

$$I_j(z) := \int_{E_j} K_{\text{mod}}^+(z, \zeta) f(\zeta) dV(\zeta).$$

We will choose E_j , $j = 1, 2, 3$ so that $\int_F |I_j(z)| dV(z) \lesssim 1$ for $j = 2, 3$, while $K_{\text{mod}}^+(z, \zeta)$ is nonpositive on E_1 . More precisely, we take

$$\begin{aligned} E_1 &:= \left\{ |z + i| < \frac{1}{4} |\zeta - i| \right\} \\ E_2 &:= \left\{ \frac{1}{4} |\zeta - i| < |z + i| < 4 |\zeta - i| \right\} \\ E_3 &:= \{ |z + i| > 4 |\zeta - i| \}. \end{aligned}$$

Under the first condition, the second term of K_{mod}^+ is the largest one, which gives the nonpositivity. Under the second condition, since $|K_{\text{mod}}^+(z, \zeta)| \lesssim y^{-2}$, its integral in z on a domain of the form $R \leq |z + i| \leq CR$ is uniformly bounded. Finally, in the third domain, we use the fact that the kernel is given by a difference, which is bounded, up to a constant, by $|\zeta - i| \times |z + i|^{-3}$. Its integral in z is uniformly bounded. So these last two terms are bounded under the unique assumption that f is integrable. Finally, under the assumption that $P_{\text{mod}}^+ f$ is integrable, we know that

$$\int_F \int_{|z+i| < \frac{1}{4} |\zeta-i|} |z+i|^{-2} f(\zeta) dV(\zeta) dV(z) < \infty. \quad (1.10)$$

We integrate first in z to find that

$$\int_{\mathbb{C}_+} f(\zeta) \ln_+(|\zeta - i|) dV(\zeta) < \infty.$$

This finishes the proof.

Next, we consider weighted spaces with weighted measure $\omega^k dV$ where ω is defined by (1.6). We only give sufficient conditions for Pf to be in $L_{\omega^k}^1$ in the next proposition. We outline the fact that the result depends on the power k . Basically, when $k > -1$, the sufficient condition for Pf to be in $L_{\omega^k}^1$ may be written in terms

of the weight ω^{k+1} . For $k = -1$, the condition is given in terms of the $\ln \ln$ function. For $k < -1$ there is no further condition since the kernel is now uniformly integrable for the measure $\omega^k dV$.

Proposition 1.5 *Let f be an integrable function. Sufficient conditions for Pf (resp., P^+f) to be in $L^1_{\omega^k}(\mathbb{C}_+)$ are given in the following enumeration:*

- If $k > -1$, it is sufficient that f satisfies the two conditions:

$$\int_{\mathbb{C}_+} f dV(z) = 0, \quad (1.11)$$

$$\int_{\mathbb{C}_+} |f(z)| \omega^{k+1}(z) dV(z) < \infty. \quad (1.12)$$

- If $k = -1$, it is sufficient that f satisfies the condition (1.11) and the condition

$$\int_{\mathbb{C}_+} |f(z)| (1 + \ln_+(\ln_+(\Im m(z))^{-1})) + \ln_+(\ln_+(|z|)) dV(z) < \infty. \quad (1.13)$$

- If $k < -1$, Pf belongs to $L^1_{\omega^k}(\mathbb{C}_+)$ with no extra condition.

The proof is analogous to the sufficiency in Proposition 1.2. In the two first cases, we already know that (1.11) is a necessary condition, so that we assume they are satisfied and the kernel can be replaced by the modified kernel. One adapts easily Lemma 1.1. The main difference is the fact that the value of the function $\int_1^a y^{-1} (\ln y)^k dy$ replaces now the logarithm. More precisely, we use the elementary inequality

$$\int_2^t \frac{\ln(s)^k}{s} ds \leq \begin{cases} c_k & \text{if } k < -1, \\ \ln(\ln(t)) & \text{if } k = -1, \\ c_k (\ln(t))^{1+k} & \text{if } k > -1. \end{cases} \quad (1.14)$$

Details are given in [6] for $k = -1$. For $k < -1$, the kernel $K(\cdot, \zeta)$ belongs to $L^1_{\omega^k}$ uniformly in ζ hence $Pf \in L^1_{\omega^k}$ for any $f \in L^1(\mathbb{C}_+)$.

1.3 Bloch Functions, Multipliers, Duality

Let us recall that, for $k \in \mathbb{R}$, the weighted Bloch space $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$ is the space of holomorphic functions f such that

$$\|f\|_{\mathcal{B}_{\omega^k}} := |f(i)| + \sup_{z \in \mathbb{C}_+} (\Im m(z)) \omega^k(z) |f'(z)| < \infty. \quad (1.15)$$

We start by observing the following fact.

Lemma 1.4 *Let $k \in \mathbb{R}$. Then there is a constant $C_k > 0$ such that for any $f \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$ of norm 1,*

$$|f(z)| \leq \begin{cases} C_k & \text{if } k > 1, \\ C_{-1} \ln(\omega(z)) & \text{if } k = 1, \\ C_k (\omega(z))^{1-k} & \text{if } k < 1. \end{cases}$$

Remark that we have the equivalence

$$1 + \ln(\omega(z)) \simeq 1 + \ln_+(\ln_+(1/y)) + \ln_+(\ln_+(|z|))$$

and recognize the quantity that appears in (1.13).

Proof We write, for $|x| \leq y$,

$$|f(z) - f(i)| \leq |f(x + iy) - f(iy)| + |f(iy) - f(i)|$$

and, for $|x| \geq y$,

$$\begin{aligned} |f(z) - f(i)| &\leq |f(x + iy) - f(x + i|x|)| + |f(x + i|x|) - f(i|x|)| \\ &\quad + |f(i|x|) - f(iy)| + |f(iy) - f(i)|. \end{aligned}$$

We bound all differences by integrals of derivatives and use the inequalities (1.14) to conclude.

Critical examples are given by $z \mapsto 1$ when $k > 1$, by $z \mapsto \log \log(4i + z)$ when $k = 1$, and by $z \mapsto (\log(4i + z))^{1-k}$ when $k < 1$. For these last examples, we use the fact that $\log(4i + z)$ has a positive real part, so that its logarithm is well defined as a holomorphic function.

These estimates may be used to give two kinds of estimates: first on pointwise products of functions that are respectively in the Bergman space $A_{\omega^l}^1(\mathbb{C}_+)$ and in the weighted Bloch space $\mathcal{B}_{\omega^k}$, secondly for multipliers of the weighted Bloch spaces.

1.3.1 Pointwise Products

As a direct consequence, we have the following corollary.

Corollary 1.3 *Let $k < 1$, and $l \in \mathbb{R}$. The pointwise product of a function in the Bergman space $A_{\omega^l}^1(\mathbb{C}_+)$ and a function in the weighted Bloch space $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$ belongs to the space $A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$. Moreover, for $f \in A_{\omega^l}^1(\mathbb{C}_+)$ and $g \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$, one has*

$$\|fg\|_{A_{\omega^{l+k-1}}^1} \lesssim \|f\|_{A_{\omega^l}^1} \|g\|_{\mathcal{B}_{\omega^k}}.$$

Remark that multiplication by a function $b \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$, for $k > 1$, preserves all spaces $A_{\omega'}^1(\mathbb{C}_+)$ since b is bounded. One could also write a statement for multiplication by a function in $\mathcal{B}_{\omega}(\mathbb{C}_+)$. It would involve a logarithm.

We will see that the result given here is optimal.

1.3.2 Multipliers

We start by recalling that a holomorphic function b is a pointwise multiplier of the Bloch space $\mathcal{B}(\mathbb{C}_+)$ if and only if b is bounded and belongs to $\mathcal{B}_{\omega}(\mathbb{C}_+)$ (see [5]). The fact that multipliers satisfy a logarithmic condition at infinity was remarked by Nakai [8] in the case of BMO .

The next proposition extends this result to all $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$, $k > -1$ and gives other characterizations of multipliers for other values of k .

Proposition 1.6 *The space of pointwise multipliers of the weighted Bloch space $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$ consists of:*

- (a) *Bounded functions that are in the class $\mathcal{B}_{\omega}(\mathbb{C}_+)$, when $k < 1$*
- (b) *Functions in $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$, when $k > 1$*
- (c) *Bounded functions that satisfy the condition*

$$\sup_{z \in \mathbb{C}_+} (\Im m(z)) \omega(z) (1 + \ln(\omega(z))) |f'(z)| < \infty \quad (1.16)$$

when $k = 1$

Proof *Proof of the sufficient condition.*

Let $f \in \mathcal{B}_{\omega^k}$. We write $(fb)' = f'b + b'f$. If we assume that b is bounded, we immediately conclude in each case that the first term satisfies the same inequality as f' . Then, in each case we conclude for the second term because of the condition on the derivative of b and the estimate on f given in Lemma 1.4. So, the conditions on b are sufficient.

Proof of the Necessary Condition

In all cases, when we test on the function $f = 1$, we see that the condition $b \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$ is necessary. This is sufficient to conclude for Case (b). In other cases it is also sufficient to conclude that $b(z)$ is bounded as well as $y|b'(z)|$ when $\omega(z) \simeq 1$. So we only need to concentrate on Cases (a) and (c) and points for which $\omega(z) \simeq \ln_+(|z|)$ or $\omega(z) \simeq \ln_+(1/y)$.

Let us explain how we will proceed for the proof. Assuming that b is a multiplier of $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$, with $k < 1$, we first prove that b is necessarily a bounded function. From Lemma 1.4, we know that there is a constant $C_k > 0$ such that for any $f \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$,

$$|f(z)b(z)| \leq C_k \|f\|_{\mathcal{B}_{\omega^k}} \omega(z)^{1-k}. \quad (1.17)$$

We need to find test functions $f \in \mathcal{B}_{\omega^k}$ (which may depend on the point z) with norm 1 such that $|f(z)| \geq c_k \omega(z)^{1-k}$. This will imply that $|b(z)| \leq c_k^{-1} C_k$. Once we know that b is bounded, we have $|yb'(z)||f(z)| \leq C\omega(z)^{-k}$ by definition of a multiplier. So it is sufficient to find test functions $f \in \mathcal{B}_{\omega^k}$ such that $|f(z)| \simeq \omega(z)^{-k+1}$. The construction of such test functions is given by the following lemma.

Lemma 1.5 *We consider the function*

$$(z, w) \mapsto \theta_w(z) := 1 - \log(z - \bar{w}) + \ln|i + w| + 2 \log(i + z).$$

It has the following properties:

- (i) $\Re e(\theta_w) > 1 - \ln(2) + \ln(|i + z|)$, so that, in particular, its logarithm and its powers are well defined.
- (ii)

$$|\theta_w(z)| \simeq 1 + \ln_+(|z|) + \ln_+(1/|z - \bar{w}|)$$

with constants that do not depend on $w \in \mathbb{C}_+$.

- (iii) *For all $k < 1$, the function θ_w^{-k+1} belongs to $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$ and*

$$\|\theta_w^{-k+1}\|_{\mathcal{B}_{\omega^k}} \simeq 1$$

with constants that do not depend on w .

- (iv) *The function $\log(\theta_w)$ belongs to $\mathcal{B}_{\omega}(\mathbb{C}_+)$ and*

$$\|\log(\theta_w)\|_{\mathcal{B}_{\omega}} \simeq 1$$

with constants that do not depend on w .

End of the Proof of the Necessary Condition Let us assume this lemma and finish the proof for Case (a). We first prove the boundedness when $\omega(z) \simeq \ln_+(|z|)$. It is sufficient to take as a test function θ_w with $w = i$. Indeed, with this choice, we know from (i) that $|\theta_w| \simeq \omega(z)$ and we conclude. Next, assuming that $\omega(z) \simeq \ln_+(1/y)$, $z = x + iy$, we choose $w = z$, so that $|z - \bar{w}| = 2y$. Again $|\theta_w| \simeq \omega(z)$ and we conclude in the same way. Once we have the boundedness of b , we conclude that b is in $\mathcal{B}_{\omega}$ by testing in each case on the same functions.

The proof of Case (c) is identical, but by taking $\log(\theta_w)$ as test functions, with the same values of the parameter w . We do not give the details again.

Proof (of Lemma 1.5) Assertion (i) was proved in [5]. Let us prove (ii). The bound below is straightforward. The bound above is straightforward when $|z - \bar{w}| \leq 1$. For $|z - \bar{w}| \geq 1$, the bound is a consequence of the additivity of the logarithm $t \mapsto \ln(t)$ for $t > 1$, so that

$$\ln|i + w| \lesssim (1 + \ln(|i + z|) + \ln_+(|z - \bar{w}|)).$$

Let us now prove (iii), that is,

$$\frac{y}{|z+i|} \left(\frac{|\theta_w(z)|}{\omega(z)} \right)^{-k} + \frac{y}{|z-\bar{w}|} \left(\frac{|\theta_w(z)|}{\omega(z)} \right)^{-k} \lesssim 1, \forall z = x + iy \in \mathbb{C}_+.$$

For $k \leq 0$ we conclude directly since, from (ii), we know that $|\theta_w(z)| \lesssim \omega(z)$. We go on with the proof for $k > 0$. The inequality

$$\frac{y}{|z+i|} \lesssim \left(\frac{|\theta_w(z)|}{\omega(z)} \right)^k$$

is straightforward because of (ii) when $\omega(z) \simeq 1$ or $\omega(z) \simeq \ln_+(|z|)$: The right-hand side is bounded below by a constant. So it is sufficient to consider the case when $\omega(z) \simeq \ln_+(1/y)$. Then we remark that

$$y (\ln_+(1/y))^k \lesssim 1,$$

while both $|z+i|$ and $|\theta_w(z)|$ are bounded below. This allows to conclude for the term in $\frac{y}{|z+i|}$.

We then prove that

$$\frac{y}{|z-\bar{w}|} \lesssim \left(\frac{|\theta_w(z)|}{\omega(z)} \right)^k.$$

The proof is analogous except when $\omega(z) \simeq \ln_+(1/y)$. In this case we remark that

$$y (\ln_+(1/y))^k \leq y (\ln(e+1/y))^k \leq |z-\bar{w}| \ln(e+1/|z-\bar{w}|)^k$$

by monotonicity of the function $t \mapsto t (\ln(e+1/t))^k$. The right-hand side is bounded by $|z-\bar{w}| |\theta_w(z)|^k$, which allows us to conclude the proof of (iii).

The proof of (iv) follows the same lines.

1.3.3 Reproducing Formula for $A_{\omega^k}^1$ -Functions

We first recall some notations. For Ω a weight, the space $L_{\Omega}^p(\mathbb{C}_+)$ is the L^p space for the measure ΩdV and $A_{\Omega}^p(\mathbb{C}_+) := L_{\Omega}^p(\mathbb{C}_+) \cap \mathcal{H}(\mathbb{C}_+)$ the corresponding Bergman space. When $\Omega(z) = (\Im m(z))^{\alpha}$, we recover the classical (weighted) Bergman space denoted by $A_{\alpha}^p(\mathbb{C}_+)$. This space is nontrivial only if $\alpha > -1$. Finally, for $\alpha > -1$, the Bergman projection P_{α} is the orthogonal projection from $L_{\alpha}^2(\mathbb{C}_+)$ onto its closed subspace $A_{\alpha}^2(\mathbb{C}_+)$. It is defined by

$$P_{\alpha} f(z) = \int_{\mathbb{C}_+} K_{\alpha}(z, w) f(w) dV_{\alpha}(w),$$

where

$$K_\alpha(z, w) = \frac{c_\alpha}{(z - \bar{w})^{2+\alpha}}$$

is the Bergman kernel, $dV_\alpha(w) = (\Im m(w))^\alpha dV(w)$, and c_α is a constant that depends only on α .

The aim of this section is to prove the following proposition.

Proposition 1.7 *For $k \in \mathbb{R}$, the Bergman projection P_α , $\alpha > 0$, reproduces the functions in $A^1_{\omega^k}(\mathbb{C}_+)$.*

We need preliminary results.

Let us start with the following estimates that generalize the ones from [5], and deal with this kind of kernels. The proof follows the lines of [5].

Lemma 1.6 *Let $\alpha > 0$, $\beta > -1$, and $k, s \geq 0$. Let*

$$\begin{aligned} \Omega(z) = & \left(\ln^{\varepsilon_1}(e + |z|) + \ln^{\varepsilon_2} \left(e + \frac{1}{\Im m z} \right) \right)^{\pm k} \\ & \left[\ln \left(e + \ln^{\varepsilon_3}(e + |z|) + \ln^{\varepsilon_4} \left(e + \frac{1}{\Im m z} \right) \right) \right]^{\pm s}, \end{aligned}$$

$\varepsilon_j \in \{0, 1\}$, $j = 1, 2, 3, 4$. Then there is a constant $C = C_{\alpha, \beta} > 0$ such that for any $z_0 \in \mathbb{C}_+$,

$$\int_{\mathbb{C}_+} \frac{\Omega(z) dV_\beta(z)}{|z - \bar{z}_0|^{2+\alpha+\beta}} \leq C (\Im m(z_0))^{-\alpha} \Omega(z_0). \quad (1.18)$$

We will also need the following proposition.

Proposition 1.8 *Let $k > 0$. Then the set $A^1(\mathbb{C}_+)$ is dense in $A^1_{\omega^{-k}}(\mathbb{C}_+)$.*

We refer to [5] for the proof, which is inspired in particular by [2, 11]. It is easily adapted from the statement given there when $k = 1$. As in this reference, one first consider the weighted Bergman space $A^1_{\rho^k}(\mathbb{C}_+)$, with $\rho := (1 + \ln(e + 1/y))^{-1}$, $k > 0$. We leave details of verification to the reader.

Proof (of Proposition 1.7) For $k > 0$, this follows from the continuous embedding $A^1_{\omega^k}(\mathbb{C}_+) \hookrightarrow A^1(\mathbb{C}_+)$ and that P_α reproduces functions of $A^1(\mathbb{C}_+)$.

To obtain that P_α also reproduces functions in $A^1_{\omega^{-k}}(\mathbb{C}_+)$, $k > 0$, we first observe that P_α maps $L^1_{\rho^k}(\mathbb{C}_+)$ boundedly into $A^1_{\rho^k}(\mathbb{C}_+)$. The conclusion for this case then follows from the latter and Proposition 1.8.

1.3.4 Duality

Let us use the notation $L^1_{\omega^{-k},\alpha}(\mathbb{C}_+) = L^1(\mathbb{C}_+, \omega^{-k}(z)dV_\alpha(z))$.

Proposition 1.9 *Let $\alpha > 0$, and $\beta > -1$. Then the Bergman projector $P_{\alpha+\beta}$ is bounded from $L^1_{\omega^{-k},\beta}(\mathbb{C}_+)$ to $L^1_{\omega^{-k},\alpha}(\mathbb{C}_+)$.*

Proof This is a direct consequence of Fubini's Theorem and Lemma 1.6.

Let us define the following weighted space.

$$L^\infty_{\omega^k,\alpha}(\mathbb{C}_+) := \{f \text{ measurable} : \sup_{z \in \mathbb{C}_+} (\Im m(z))^\alpha \omega^k(z) |f(z)| < \infty\}.$$

Then $L^\infty_{\omega^k,\alpha}(\mathbb{C}_+)$ is a Banach space with norm

$$\|f\|_{\infty,\omega^k,\alpha} := \sup_{z \in \mathbb{C}_+} (\Im m(z))^\alpha \omega^k(z) |f(z)|.$$

The following is elementary.

Proposition 1.10 *Let $\alpha > 0$, and $\beta > -1$. Then the Bergman projection $P_{\alpha+\beta}$ maps $L^\infty_{\omega^k,\alpha}(\mathbb{C}_+)$ boundedly into itself.*

We have the following duality result.

Theorem 1.4 *The dual space $\left(A^1_{\omega^{-k}}(\mathbb{C}_+)\right)^*$ of $A^1_{\omega^{-k}}(\mathbb{C}_+)$ identifies with $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$ modulo constants under the pairing*

$$\langle f, g \rangle_* = \int_{\mathbb{C}_+} f(z) \overline{g'(z)} (\Im m(z)) dV(z), \quad (1.19)$$

$f \in A^1_{\omega^{-k}}(\mathbb{C}_+)$, and $g \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$.

Proof Let $g \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$. Then that (1.19) defines a bounded linear operator is direct from the definition of the spaces involved.

Conversely, assume that Λ is an element of $\left(A^1_{\omega^{-k}}(\mathbb{C}_+)\right)^*$. Then Λ can be extended as an element $\tilde{\Lambda}$ of $\left(L^1_{\omega^{-k}}(\mathbb{C}_+)\right)^*$ with the same operator norm. Then classical arguments give that there is an element $h \in L^\infty(\mathbb{C}_+)$ such that for any $f \in L^1_{\omega^{-k}}(\mathbb{C}_+)$,

$$\tilde{\Lambda}(f) = \int_{\mathbb{C}_+} f(z) \overline{h(z)} \omega^{-k}(z) dV(z).$$

In particular, we have that for any $f \in A^1_{\omega^{-k}}(\mathbb{C}_+)$,

$$\Lambda(f) = \int_{\mathbb{C}_+} f(z) \overline{h(z)} \omega^{-k}(z) dV(z).$$

By Corollary 1.7, P_1 reproduces functions in $A^1_{\omega^{-k}}(\mathbb{C}_+)$. Hence, for $f \in A^1_{\omega^{-k}}(\mathbb{C}_+)$,

$$\begin{aligned} \Lambda(f) &= \int_{\mathbb{C}_+} f(z) \overline{h(z)} \omega^{-k}(z) dV(z) \\ &= \int_{\mathbb{C}_+} P_1(f)(z) \overline{h(z)} \omega^{-k}(z) dV(z) \\ &= \int_{\mathbb{C}_+} f(z) \overline{P_1 g(z)} (\Im m(z)) dV(z), \end{aligned}$$

where $g(z) = \omega^{-k}(z) (\Im m(z))^{-1} h(z)$. Here, we applied Fubini's Theorem to obtain the last equality. We note that $g \in L^{\infty}_{\omega^k, 1}(\mathbb{C}_+)$ and that using Proposition 1.10, one obtains that P_1 is bounded on $L^{\infty}_{\omega^k, 1}(\mathbb{C}_+)$. It follows that if we define G to be a solution of $G'(z) = P_1 g(z)$, then $G \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$. Hence

$$\Lambda(f) = \int_{\mathbb{C}_+} f(z) \overline{G'(z)} (\Im m(z)) dV(z).$$

That is, Λ is given by (1.19). The proof is complete.

Let us derive the following duality whose proof is essentially the same as above. This result will be useful in the next section.

Proposition 1.11 *The dual space $\left(A^1_{\omega^{-k}}(\mathbb{C}_+)\right)^*$ of $A^1_{\omega^{-k}}(\mathbb{C}_+)$ identifies with $P_1 \left(L^{\infty}_{\omega^k, 1}(\mathbb{C}_+)\right)$ under the pairing*

$$\langle f, g \rangle_1 = \int_{\mathbb{C}_+} f(z) \overline{g(z)} (\Im m(z)) dV(z), \quad (1.20)$$

$f \in A^1_{\omega^{-k}}(\mathbb{C}_+)$, and $g \in P_1 \left(L^{\infty}_{\omega^k, 1}(\mathbb{C}_+)\right)$.

1.4 Products of Functions in $A_{\omega^l}^1$ and $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$ and Hankel Operators

1.4.1 Weak Factorization and Boundedness of Hankel Operators

We discuss in this part the relation between the boundedness of Hankel operators and the weak factorization of the above weighted Bergman space. For this, we introduce some definitions.

We have seen that the product of a function f in $A_{\omega^l}^1(\mathbb{C}_+)$ and a function g in $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$ belongs to $A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$, $k < 1$ (see Corollary 1.3). The question of weak factorization is a kind of converse of the latter. It consists in knowing whether a function in $A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$ can be weakly factorized, that is, may be written as a sum $\sum f_j g_j$ with $f_j \in A_{\omega^l}^1(\mathbb{C}_+)$ and $g_j \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$, with an adapted normalization. In [5], in the case $l = k = 0$, this question was answered affirmatively, with the help of an atomic decomposition. We will generalize this result to weighted Bergman spaces but in a simpler way that does not use atomic decomposition of the weighted Bergman spaces. More precisely, this will be derived from the boundedness of Hankel operators. Indeed, we prove below the equivalence of both questions.

We denote by $A_{\omega^{l+k-1},w}^1(\mathbb{C}_+)$ the space of all functions of the form $f = \sum f_j g_j$ such that $\sum_j \|f_j\|_{A_{\omega^l}^1} \|g_j\|_{\mathcal{B}_{\omega^k}} < \infty$. We denote by $\|f\|_{A_{\omega^{l+k-1},w}^1}$, the infimum for all such writings of f of the sums $\sum_j \|f_j\|_{A_{\omega^l}^1} \|g_j\|_{\mathcal{B}_{\omega^k}}$. This makes $A_{\omega^{l+k-1},w}^1(\mathbb{C}_+)$ a Banach space. By Corollary 1.3, $A_{\omega^{l+k-1},w}^1(\mathbb{C}_+)$ is continuously embedded in $A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$.

We are interested by the converse statement in which case we say there is a weak factorization of $A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$ or that $A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$ weakly factorizes. The weak factorization will follow once one has proved that the injection of $A_{\omega^{l+k-1},w}^1(\mathbb{C}_+)$ into $A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$ is onto. Furthermore, by the open mapping theorem, we will obtain the existence of some universal constant $C > 0$ such that, for any $f \in A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$

$$\|f\|_{A_{\omega^{l+k-1},w}^1} \leq C \|f\|_{A_{\omega^{l+k-1}}^1}.$$

We recall that for b in $L^\infty(\mathbb{C}_+)$, the Hankel operator h_b is the operator primarily defined for f in $A^2(\mathbb{C}_+)$, by $h_b(f) := P(b\bar{f})$. Observing that for any $g \in A^2(\mathbb{C}_+)$,

$$\langle h_b(f), g \rangle = \langle b, fg \rangle,$$

the above definition of Hankel operators on $A^2(\mathbb{C}_+)$ can then be extended to symbol in $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$ as follows:

$$\langle h_b(f), g \rangle = \langle b, fg \rangle_{A_{\omega^{-k}}^1, \mathcal{B}_{\omega^k}}.$$

Here $\langle \cdot, \cdot \rangle_{A_{\omega^{-k}}^1, \mathcal{B}_{\omega^k}}$ stands for the duality bracket between $A_{\omega^{-k}}^1(\mathbb{C}_+)$ and $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$, given by (1.19). Remark that it coincides with the scalar product on $L^2(\mathbb{C}_+)$ when well defined.

The following gives the equivalence between the boundedness of the Hankel operators and the above weak factorization.

Theorem 1.5 *Let l, k be real, $k < 1$. Then the following are equivalent:*

- (a) *The space $A_{\omega^{l+k-1}}^1$ weakly factorizes.*
- (b) *The Hankel operator h_b extends into a bounded operator from $A_{\omega^l}^1(\mathbb{C}_+)$ into $A_{\omega^{-k}}^1(\mathbb{C}_+)$ if and only if $b \in \left(A_{\omega^{l+k-1}}^1(\mathbb{C}_+)\right)^* = \mathcal{B}_{\omega^{1-(l+k)}}(\mathbb{C}_+)$.*

Proof Let us assume that (a) holds. That $b \in \mathcal{B}_{\omega^{1-(l+k)}}(\mathbb{C}_+)$ implies the boundedness of h_b from $A_{\omega^l}^1(\mathbb{C}_+)$ into $A_{\omega^{-k}}^1(\mathbb{C}_+)$ is harmless, and we refer to the next section for details. Now assume that h_b is well defined and bounded from $A_{\omega^l}^1(\mathbb{C}_+)$ into $A_{\omega^{-k}}^1(\mathbb{C}_+)$. Let us prove that necessarily, b belongs to $\mathcal{B}_{\omega^{1-(l+k)}}(\mathbb{C}_+)$. By Theorem 1.4, it is sufficient to prove that, for any $f \in A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$, we have

$$|\langle b, f \rangle_{A_{\omega^{l+k-1}}^1, \mathcal{B}_{\omega^{1-(l+k)}}}| = \left| \int_{\mathbb{C}_+} b'(z) \overline{f(z)} (\Im m z) dV(z) \right| \leq C \|f\|_{A_{\omega^{l+k-1}}^1} \quad (1.21)$$

for some uniform constant. From the hypothesis, we have that any $f \in A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$ can be represented as

$$f = \sum_j f_j g_j,$$

with $\sum \|f_j\|_{A_{\omega^l}^1} \|g_j\|_{\mathcal{B}_{\omega^k}} \leq C \|f\|_{A_{\omega^{l+k-1}}^1}$. We obtain

$$\langle b, f \rangle_{A_{\omega^{l+k-1}}^1, \mathcal{B}_{\omega^{1-(l+k)}}} = \sum_j \langle b', f_j g_j \rangle_1$$

where

$$\langle b', f \rangle_1 := \int_{\mathbb{C}_+} b'(z) \overline{f(z)} \Im m(z) dV(z).$$

The boundedness of h_b from $A_{\omega^l}^1(\mathbb{C}_+)$ into $A_{\omega^{-k}}^1(\mathbb{C}_+)$ allows to say that

$$|\langle b', f_j g_j \rangle_1| = |\langle h_b(f_j), g_j \rangle| \leq C \|f_j\|_{A_{\omega^l}^1} \|g_j\|_{\mathcal{B}_{\omega^k}}$$

for some uniform constant C . Inequality (1.21) follows at once.

Now assume that (b) holds. Let us prove (a). For this, it is enough to show that the injection of $A_{\omega^{l+k-1}, w}^1(\mathbb{C}_+)$ into $A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$ is onto. Let b an element of the dual of $A_{\omega^{l+k-1}, w}^1(\mathbb{C}_+)$ where the duality pairing is $\langle \cdot, \cdot \rangle_1$. Then we have in particular that b is such that for some constant $C > 0$, for any $f \in A_{\omega^l}^1(\mathbb{C}_+)$ and any $g \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$,

$$|\langle b, fg \rangle_1| = \left| \int_{\mathbb{C}_+} f(z) g(z) \overline{b'(z)} (\Im m z) dV(z) \right| \leq C \|f\|_{A_{\omega^l}^1} \|g\|_{\mathcal{B}_{\omega^k}}.$$

This implies that the Hankel operator h_b extends into a bounded operator from $A_{\omega^l}^1(\mathbb{C}_+)$ into $A_{\omega^{-k}}^1(\mathbb{C}_+)$. Hence, b belongs $\mathcal{B}_{\omega^{1-(l+k)}}(\mathbb{C}_+)$, which is the dual space of $A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$ under the pairing $\langle \cdot, \cdot \rangle_1$. This concludes the proof.

Let us consider the following function:

$$f(z) = \frac{(\Im m(w)) \omega^{1-(l+k)}(w)}{(z - \overline{w})^3}. \quad (1.22)$$

We observe that using Lemma 1.6, one obtains $\|f\|_{L^1(\omega^{l+k-1})} \simeq 1$. For this weighted kernel, we actually have a strong factorization as shown below. This result will be useful in proving the necessary condition in the boundedness of Hankel operators.

Proposition 1.12 *Let f be an atom given by (1.22). There exist $g \in A_{\omega^l}^1(\mathbb{C}_+)$ and $\theta \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$, $k < 1$ such that*

$$f = g \times \theta$$

with

$$\|g\|_{A_{\omega^l}^1} \times \|\theta\|_{\mathcal{B}_{\omega^k}} \lesssim 1.$$

Proof We choose $\theta = \theta_w^{-k+1}$ where the function θ_w is defined in Lemma 1.5. In this case, the function g is given as follows:

$$g : z \mapsto \frac{(\Im m(w)) \omega^{1-(l+k)}(w)}{(z - \overline{w})^3 \theta(z)},$$

which is a well-defined holomorphic function in the upper-half plane. It remains to prove that the inequality $\|g\|_{A_{\omega^l}^1} \lesssim 1$ holds. When $\omega(w) \simeq 1$, then $\theta \simeq 1$, and there

is nothing to prove. We will now use the two notations $\omega_1(z) = \ln\left(e + \frac{1}{\Im m z}\right)$ and $\omega_2(z) = \ln(e + |z|)$. Lemma 1.6 is still valid with the weight ω replaced by weights ω_1 and ω_2 . We distinguish two cases:

- (i) Case $v < 1/2$ and $\ln(v^{-1}) \simeq \omega(w) \simeq \omega_1(w)$. We proceed as in [5]. We cover the upper-half plane by the union of E'_j s, with $E_0 := Q_w$ the Carleson square centered at w , and $E_j := Q_{w_j} \setminus Q_{w_{j-1}}$, $j \geq 1$. Here $w = u + iv$ and $w_j = u + i2^j v$.

We recall that for $z \in E_j$, we have $|z - \bar{w}| \leq 2\Im m(w_j)$. Hence using Lemma 1.5, we obtain

$$\begin{aligned} \Re e(\theta(z)) &\gtrsim \left(1 - \ln 2 + \ln_+ \left(\frac{1}{|z - \bar{w}|}\right)\right)^{-k+1} \\ &\gtrsim \left(\ln\left(e + (\Im m(w_j))^{-1}\right)\right)^{-k+1} = \omega_1^{-k+1}(w_j). \end{aligned} \quad (1.23)$$

Using Lemma 1.6, it allows to get

$$\int |g(z)|\omega^l(z)dV(z) \lesssim \sum_j \frac{\omega_1^{1-(l+k)}(w)}{\omega_1^{1-(l+k)}(w_j)} 2^{-2j}.$$

To conclude, in the case $1 - (l + k) > 0$, we use the fact that $\omega_1(w_j) \geq \omega_1(w) - j \ln 2$, and in the case $1 - (l + k) \leq 0$, the fact that $\omega_1(w) \geq \omega_1(w_j)$.

- (ii) Case $\omega(w) \simeq \ln(e + |w|) = \omega_2(w)$. As $|\theta(z)| \gtrsim (\ln(e + |z|))^{-k+1} = \omega_2^{-k+1}(z)$, the estimate follows as in the previous case.

1.4.2 Hankel Operators

It is a well-known and easy-to-verify fact that a necessary and sufficient condition on b to get a bounded operator on $A^2(\mathbb{C}_+)$ is b in the Bloch class. It was proved in [5] that the Hankel operator h_b is a bounded operator on $A^1(\mathbb{C}_+)$ if and only if the symbol b belongs to $\mathcal{B}_\omega(\mathbb{C}_+)$. The next theorem generalizes this result.

Theorem 1.6 *Let b be in $\mathcal{H}(\mathbb{C}_+)$. Let $l \in \mathbb{R}$, and $k < 1$. Then the Hankel operator h_b extends to a bounded operator from $A_{\omega^l}^1(\mathbb{C}_+)$ to $A_{\omega^{-k}}^1(\mathbb{C}_+)$ if and only if $b \in \mathcal{B}_{\omega^{1-(l+k)}}(\mathbb{C}_+)$. Moreover,*

$$\|h_b\| \simeq \|b\|_{\mathcal{B}_{\omega^{1-(l+k)}}}.$$

Proof Recall that if f and g are in $A^2(\mathbb{C}_+)$, and b is a Bloch function, then

$$\langle h_b(f), g \rangle = \langle b, fg \rangle_{A^1, \mathcal{B}}.$$

This can be extended for $b \in \mathcal{B}_{\omega^{1-(l+k)}}(\mathbb{C}_+)$, $f \in A_{\omega^l}^1(\mathbb{C}_+)$, and $g \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$ since as $fg \in A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$ the following quantity is well defined

$$\int_{\mathbb{C}_+} f(z)g(z)\overline{b'(z)}(\Im z)dV(z).$$

Moreover, the corresponding operator is bounded from $A_{\omega^l}^1(\mathbb{C}_+)$ into $A_{\omega^{-k}}^1(\mathbb{C}_+)$ since

$$\left| \int_{\mathbb{C}_+} f(z)g(z)\overline{b'(z)}\Im z dV(z) \right| \leq \|b\|_{\mathcal{B}_{\omega^{1-(l+k)}}} \|f\|_{A_{\omega^l}^1} \|g\|_{\mathcal{B}_{\omega^k}}.$$

Conversely, assume that h_b is well defined and bounded from $A_{\omega^l}^1(\mathbb{C}_+)$ into $A_{\omega^{-k}}^1(\mathbb{C}_+)$. This means in particular that the symbol b is such that for any $f \in A_{\omega^l}^1(\mathbb{C}_+)$ and any $g \in \mathcal{B}_{\omega^k}(\mathbb{C}_+)$,

$$|\langle b', fg \rangle_1| = |\langle h_b(f), g \rangle| \leq \|h_b\| \|g\|_{A_{\omega^l}^1} \|f\|_{\mathcal{B}_{\omega^k}}. \quad (1.24)$$

Let us prove that necessarily, b belongs to $\mathcal{B}_{\omega^{1-(l+k)}}(\mathbb{C}_+)$.

Taking $g = 1$ in (1.24), we obtain that b' belongs to the dual of $A_{\omega^l}^1(\mathbb{C}_+)$ under the duality pairing $\langle \cdot, \cdot \rangle_1$. Thus by Proposition 1.11, $b' = P_1(\tilde{b})$, where $\tilde{b} \in L_{\omega^{-l},1}^\infty(\mathbb{C}_+)$. Using the factorization in Proposition 1.12 and our hypothesis, we obtain that

$$\begin{aligned} (\Im w)\omega^{1-(l+k)}(w)|b'(w)| &= \left| \int_{\mathbb{C}_+} \tilde{b}(z) \frac{(\Im w)\omega^{1-(l+k)}(w)}{(w-\bar{z})^3} (\Im z)dV(z) \right| \\ &= \left| \int_{\mathbb{C}_+} \tilde{b}(z)\overline{g(z)}\overline{\theta(z)}(\Im z)dV(z) \right| \\ &= |\langle \tilde{b}, g\theta \rangle_1| \\ &= |\langle P_1(\tilde{b}), g\theta \rangle_1| \\ &= |\langle h_b(g), \theta \rangle_1| \\ &\lesssim \|h_b\|. \end{aligned}$$

The proof is complete. $\square$

As a consequence of the above theorem, following Theorem 1.5, we have the following weak factorization of weighted Bergman spaces considered in this note.

Theorem 1.7 *Let $l \in \mathbb{R}$, and $k < 1$. The product of a function in $A_{\omega^l}^1(\mathbb{C}_+)$ and a function in $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$ is in $A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$. Conversely, a weak factorization result holds: If f is a holomorphic function in $A_{\omega^{l+k-1}}^1(\mathbb{C}_+)$, then*

$$f = \sum g_j h_j,$$

with g_j 's and h_j 's holomorphic and such that

$$\|f\|_{L^1_{\omega^l+\kappa-1}} \simeq \sum_j \|g_j\|_{A^1_{\omega^l}} \|h_j\|_{\mathcal{B}_{\omega^k}}.$$

The following result is a consequence of the duality result Theorem 1.4 and the fact that for $k > 1$, functions in $\mathcal{B}_{\omega^k}(\mathbb{C}_+)$ are bounded.

Theorem 1.8 *Let b be in $\mathcal{H}(\mathbb{C}_+)$. Let $l \in \mathbb{R}$, and $k > 1$. Then the Hankel operator h_b extends to a bounded operator from $A^1_{\omega^l}(\mathbb{C}_+)$ to $A^1_{\omega^{-k}}(\mathbb{C}_+)$ if and only if $b \in \mathcal{B}_{\omega^{-l}}(\mathbb{C}_+)$. Moreover,*

$$\|h_b\| \simeq \|b\|_{\mathcal{B}_{\omega^{-l}}}.$$

It remains to consider the case $k = 1$. This case requires to introduce some other weighted Bergman spaces. More precisely, for l a real number, we denote by $A^1_{\omega^l(\ln \omega)^{-1}}(\mathbb{C}_+)$ the space of all holomorphic functions f such that

$$\|f\|_{A^1_{\omega^l(\ln \omega)^{-1}}} := \int_{\mathbb{C}_+} |f(z)| \omega^l(z) (\ln(e + \omega(z)))^{-1} dV(z) < \infty.$$

We note that $A^1_{\omega^l(\ln \omega)^{-1}}(\mathbb{C}_+)$ embeds continuously into $A^1_{\omega^{l-1}}(\mathbb{C}_+)$. Consequently, the Bergman projection P_1 is also bounded on $A^1_{\omega^l(\ln \omega)^{-1}}(\mathbb{C}_+)$.

The weighted Bloch space $\mathcal{B}_{\omega^k \ln \omega}(\mathbb{C}_+)$ is the space of holomorphic functions f such that

$$\|f\|_{\mathcal{B}_{\omega^k \ln \omega}} := |f(i)| + \sup_{z \in \mathbb{C}_+} (\Im m(z)) \omega^k(z) (\ln(e + \omega(z))) |f'(z)| < \infty. \quad (1.25)$$

Proceeding as in Theorem 1.4, we obtain the following.

Proposition 1.13 *Let k be real. Then the dual space $\left(A^1_{\omega^{-k}(\ln \omega)^{-1}}(\mathbb{C}_+)\right)^*$ of the Bergman space $A^1_{\omega^{-k}(\ln \omega)^{-1}}(\mathbb{C}_+)$ identifies with $\mathcal{B}_{\omega^k \ln \omega}(\mathbb{C}_+)$ modulo constants under the integral pairing $\langle \cdot, \cdot \rangle$.*

For w fixed in $\mathbb{C}_+$, we introduce the following function:

$$f : z \mapsto \frac{(\Im m w) \omega^{-k}(w) \ln(e + \omega(w))}{(z - \bar{w})^3}. \quad (1.26)$$

Then f is a well-defined holomorphic function in $\mathbb{C}_+$, and we have, using Lemma 1.6, that $\|f\|_{A^1_{\omega^k(\ln \omega)^{-1}}} \lesssim 1$. We have the following.

Proposition 1.14 *Let f be an atom given by (1.26). There exist $g \in A_{\omega^k}^1(\mathbb{C}_+)$ and $\theta \in \mathcal{B}_\omega(\mathbb{C}_+)$ such that*

$$f = g \times \theta$$

with

$$\|g\|_{A_{\omega^k}^1} \times \|\theta\|_{\mathcal{B}_\omega} \lesssim 1.$$

Proof Take $\theta = \log(\theta_w)$ where θ_w is given in Lemma 1.5. One then only has to show that the function

$$g : z \mapsto \frac{(\Im w)\omega^{-k}(w) \ln(e + \omega(w))}{(z - \overline{w})^3 \theta(z)}$$

is uniformly in $A_{\omega^k}^1(\mathbb{C}_+)$. This follows as in Proposition 1.12.

Reasoning as in Theorem 1.6, with the help of Proposition 1.4 and Proposition 1.14, we get:

Theorem 1.9 *Let b be in $\mathcal{H}(\mathbb{C}_+)$. Let $l \in \mathbb{R}$. Then the Hankel operator h_b extends to a bounded operator from $A_{\omega^l}^1(\mathbb{C}_+)$ to $A_{\omega^{-l}}^1(\mathbb{C}_+)$ if and only if $b \in \mathcal{B}_{\omega^{-l} \ln \omega}(\mathbb{C}_+)$. Moreover,*

$$\|h_b\| \simeq \|b\|_{\mathcal{B}_{\omega^{-l} \ln \omega}}.$$

The following weak factorization result then follows as above.

Theorem 1.10 *Let $k \in \mathbb{R}$. The product of a function in $A_{\omega^k}^1(\mathbb{C}_+)$ and a function in $\mathcal{B}_\omega(\mathbb{C}_+)$ is in $A_{\omega^k(\ln \omega)^{-1}}^1(\mathbb{C}_+)$. Conversely, a weak factorization result holds: If f is a holomorphic function in $A_{\omega^k(\ln \omega)^{-1}}^1(\mathbb{C}_+)$, then*

$$f = \sum g_j h_j,$$

with g_j 's and h_j 's holomorphic and such that

$$\|f\|_{L_{\omega^k(\ln \omega)^{-1}}^1} \simeq \sum_j \|g_j\|_{A_{\omega^k}^1} \|h_j\|_{\mathcal{B}_\omega}.$$

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Chapter 2

Representation Theorems in Hardy–Orlicz Spaces of the Upper Half Plane



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Abstract In this chapter, we give Poisson and Cauchy representation theorems in Hardy–Orlicz spaces on the upper half plane. We use these theorems for the construction of dual spaces of certain Hardy–Orlicz spaces and also for the characterization of some classical operators in Orlicz spaces.

2.1 Introduction

Representation theorem is very important tool when working in Hardy spaces. Regarding Hardy space on the upper half plane, there are two versions of this theorem depending on whether the space is defined as a subspace of harmonic spaces or holomorphic spaces.

Recall that the upper half plane is the subset of the complex plane $\mathbb{C}$ defined by

$$\mathbb{C}_+ := \{z = x + iy \in \mathbb{C} : y > 0\}.$$

We denote by $h^p(\mathbb{C}_+)$ the space of all harmonic functions F on $\mathbb{C}_+$ which satisfy

$$\|F\|_{h^p} := \sup_{y>0} \|F(\cdot + iy)\|_{L^p} < \infty,$$

where $\|\cdot\|_{L^p}$ stands for the usual Lebesgue norm on $\mathbb{R}$. For $p \geq 1$, $(h^p(\mathbb{C}_+), \|\cdot\|_{h^p})$ is a Banach space.

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The Poisson representation theorem given in [13, Theorem 11.6] can be reformulated as follows.

Let $1 < p < \infty$ and F be a harmonic function on $\mathbb{C}_+$. The following assertions are equivalent:

- $F \in h^p(\mathbb{C}_+)$.
- There exists a unique $f \in L^p(\mathbb{R})$ such that

$$F(x + iy) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x - t)^2 + y^2} f(t) dt, \quad \forall x + iy \in \mathbb{C}_+. \quad (2.1)$$

Moreover,

$$\|F\|_{h^p} = \|f\|_{L^p}.$$

The above result coupled with the fact that the Poisson integral (2.1) is an isometric isomorphism of $h^p(\mathbb{C}_+)$ onto $L^p(\mathbb{R})$ allows us to identify the Hardy space $h^p(\mathbb{C}_+)$ with the Lebesgue space $L^p(\mathbb{R})$.

Let $\mathcal{M}(\mathbb{R})$ stand for the set of bounded Radon measures on $\mathbb{R}$. Equipped with the norm

$$\mu \longmapsto \|\mu\|_{\mathcal{M}} := |\mu|(\mathbb{R}),$$

$\mathcal{M}(\mathbb{R})$ is a Banach space, where $|\mu|$ is the total variation of μ .

For a harmonic function F on $\mathbb{C}_+$, the following assertions are equivalent:

- (i) $F \in h^1(\mathbb{C}_+)$.
- (ii) There exists a unique $\mu \in \mathcal{M}(\mathbb{R})$ such that

$$F(x + iy) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x - t)^2 + y^2} d\mu(t), \quad \forall x + iy \in \mathbb{C}_+.$$

Moreover,

$$\|F\|_{h^1} = \|\mu\|_{\mathcal{M}}.$$

As for the case $p > 1$, we can identify $h^1(\mathbb{C}_+)$ with $\mathcal{M}(\mathbb{R})$. We now consider the spaces $H^p(\mathbb{C}_+)$ consisting of all analytic functions F on $\mathbb{C}_+$ satisfying

$$\|F\|_{H^p} := \sup_{y>0} \|F(\cdot + iy)\|_{L^p} < \infty$$

and put

$$H^p(\mathbb{R}) = \left\{ f \in L^p(\mathbb{R}) : \int_{\mathbb{R}} \frac{f(t)}{t - \bar{z}} dt = 0, \quad \forall z \in \mathbb{C}_+ \right\}.$$

For $p \geq 1$, $(H^p(\mathbb{R}), \|\cdot\|_{L^p})$ (resp. $(H^p(\mathbb{C}_+), \|\cdot\|_{H^p})$) is a closed subspace of $(L^p(\mathbb{R}), \|\cdot\|_{L^p})$ (resp. $(h^p(\mathbb{C}_+), \|\cdot\|_{h^p})$).

Cauchy’s representation theorem in Hardy spaces (see, for example, [10, 13]) can be reformulated as follows: Let $1 \leq p < \infty$ and F be an analytic function on $\mathbb{C}_+$. The following assertions are equivalent:

- (i) $F \in H^p(\mathbb{C}_+)$.
- (ii) There exists a unique $f \in H^p(\mathbb{R})$ such that

$$F(z) = \frac{1}{2\pi i} \int_{\mathbb{R}} \frac{f(t)}{t - z} dt, \quad \forall z \in \mathbb{C}_+. \quad (2.2)$$

Moreover,

$$\|F\|_{H^p} = \|f\|_{L^p}.$$

The Cauchy integral (2.2) is an isometric isomorphism of the Hardy space $H^p(\mathbb{C}_+)$ onto the space $H^p(\mathbb{R})$. Thus, the Hardy space $H^p(\mathbb{C}_+)$ is identified with a closed subspace of $H^p(\mathbb{R})$.

Note that the use of the Poisson representation theorem (resp. Cauchy) has made it possible to simplify the solution of several problems related to Hardy spaces. Let us quote among others, the control of certain classical operators in Hardy spaces (see, for example, [2, 9, 10, 12, 13]) and the characterization of dual of Hardy’s space (see [10]).

In view of what precedes, it seemed useful to extend these results to Orlicz’s space which generalizes Lebesgue’s space. This is one fold of this chapter which is organized as follows:

In the next section, we state our main results. Section 2.3 is devoted to some useful properties and prerequisites on Hardy–Orlicz spaces. In Sect. 2.4, we give the proof of the results stated in Sect. 2.2.

We propose the following abbreviation $A \lesssim B$ for the inequalities $A \leq CB$, where C is a positive constant independent of the main parameters. If $A \lesssim B$ and $B \lesssim A$, then we write $A \approx B$. In all what follows, the letter C will be used for nonnegative constants independent of the relevant variables that may change from one occurrence to another. Constants with subscript, such as C_s , may also change in different occurrences but depend on the parameters mentioned in it.

2.2 Statement of Main Results

In this paper, a continuous and nondecreasing function Φ from $\mathbb{R}_+$ onto itself is called a growth function. Observe that if Φ is a growth function, then $\Phi(0) = 0$ and $\lim_{t \rightarrow +\infty} \Phi(t) = +\infty$. If Φ is convex and $\Phi(t) > 0$ for all $t > 0$, then Φ is a homeomorphism of $\mathbb{R}_+$ onto $\mathbb{R}_+$.

Let Φ be a growth function:

- The Orlicz space $L^\Phi(\mathbb{R})$ is the set of all equivalent classes (in the usual sense) of measurable functions $f : \mathbb{R} \rightarrow \mathbb{C}$ which satisfy

$$\|f\|_{L^\Phi}^{lux} := \inf \left\{ \lambda > 0 : \int_{\mathbb{R}} \Phi \left(\frac{|f(x)|}{\lambda} \right) dx \leq 1 \right\} < \infty.$$

- The Hardy–Orlicz space $h^\Phi(\mathbb{C}_+)$ is the set of harmonic functions on $\mathbb{C}_+$ which satisfy

$$\|F\|_{h^\Phi}^{lux} := \sup_{y>0} \|F(\cdot + y)\|_{L^\Phi}^{lux} < \infty.$$

If Φ is convex (resp. convex and $\Phi(t) > 0$ for all $t > 0$), then $(L^\Phi(\mathbb{R}), \|\cdot\|_{L^\Phi}^{lux})$ (resp. $(h^\Phi(\mathbb{C}_+), \|\cdot\|_{h^\Phi}^{lux})$) is a Banach space. The space L^Φ (resp. h^Φ) generalizes the Lebesgue space L^p (resp. Hardy space h^p) for $0 < p < \infty$.

Let Φ be a convex growth function. We say that Φ is an N -function if the following assertions are satisfied:

- (i) $\Phi(t) = 0$ if and only if $t = 0$.
- (ii) $\lim_{t \rightarrow 0} \frac{\Phi(t)}{t} = 0$ and $\lim_{t \rightarrow \infty} \frac{\Phi(t)}{t} = +\infty$.

The map $t \mapsto \exp(t) - t - 1$ is an example of N -function.

Theorem 2.1 *Let Φ be an N -function and F a harmonic function on $\mathbb{C}_+$. The following assertions are equivalent:*

- (i) $F \in h^\Phi(\mathbb{C}_+)$.
- (ii) There exists a unique $f \in L^\Phi(\mathbb{R})$ such that

$$F(x + iy) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x-t)^2 + y^2} f(t) dt, \quad \forall x + iy \in \mathbb{C}_+. \quad (2.3)$$

Moreover, we have

$$\|F\|_{h^\Phi}^{lux} = \lim_{y \rightarrow 0} \|F(\cdot + iy)\|_{L^\Phi}^{lux} = \|f\|_{L^\Phi}^{lux}.$$

Notice that if $\Phi(t) = t^p$ with $1 < p < \infty$, then Theorem 2.1 corresponds to [13, Theorem 11.6]. The Poisson integral (2.3) is an isometric isomorphism of $h^\Phi(\mathbb{C}_+)$ onto $L^\Phi(\mathbb{R})$ whenever Φ is an N -function.

Theorem 2.2 *Let Φ be a convex growth function such that $\Phi(t) > 0$ for all $t > 0$. For $F \in h^\Phi(\mathbb{C}_+)$, there exists a unique Borel measure μ such that*

$$\int_{\mathbb{R}} \frac{d|\mu|(t)}{1+t^2} < \infty$$

and

$$F(x + iy) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x - t)^2 + y^2} d\mu(t), \quad \forall x + iy \in \mathbb{C}_+.$$

Moreover,

$$\lim_{y \rightarrow 0} \int_{\mathbb{R}} \varphi(x) F(x + iy) dx = \int_{\mathbb{R}} \varphi(x) d\mu(x), \quad \forall \varphi \in C_c(\mathbb{R}), \quad (2.4)$$

where $C_c(\mathbb{R})$ is the set of continuous functions with compact support on $\mathbb{R}$.

We say that a growth function Φ :

- Satisfies the Δ_2 -condition (or $\Phi \in \Delta_2$) if there exists a constant $K > 1$ such that

$$\Phi(2t) \leq K \Phi(t), \quad \forall t > 0 \quad (2.5)$$

- Is of upper type $q > 0$ if there exists a constant $C_q > 0$ such that

$$\Phi(st) \leq C_q t^q \Phi(s), \quad \forall s > 0, \forall t \geq 1 \quad (2.6)$$

For $q \geq 1$, we denote by $\mathcal{U}^q$ the set of all growth functions of upper type q such that the function $t \mapsto \frac{\Phi(t)}{t}$ is nondecreasing on $\mathbb{R}_+^* = \mathbb{R}_+ \setminus \{0\}$, and we put $\mathcal{U} := \bigcup_{q \geq 1} \mathcal{U}^q$. It is easy to see that the function $t \mapsto \frac{t^q}{\ln(e+t)}$ belongs to $\mathcal{U}^q$ for $q \geq 1$. Growth functions in $\mathcal{U}$ satisfy the Δ_2 -condition. Moreover, any element in $\mathcal{U}$ is a homeomorphism of $\mathbb{R}_+$ onto $\mathbb{R}_+$.

We will assume in the sequel that any element of $\mathcal{U}$ is convex and belongs to $\mathcal{C}^1(\mathbb{R}_+)$ (see for example [6–8, 18]).

Let Φ be a growth function. We now consider $H^\Phi(\mathbb{R})$ the set of functions $f \in L^\Phi(\mathbb{R})$ such that

$$\int_{\mathbb{R}} \frac{f(t)}{t - \bar{z}} dt = 0, \quad \forall z \in \mathbb{C}_+,$$

and $H^\Phi(\mathbb{C}_+)$ the subspace of $h^\Phi(\mathbb{C}_+)$ consisting of analytic functions on $\mathbb{C}_+$.

If Φ is convex and $\Phi(t) > 0$ for all $t > 0$ (resp. $\Phi \in \mathcal{U}$), then $(H^\Phi(\mathbb{C}_+), \|\cdot\|_{H^\Phi}^{lux})$ (resp. $(H^\Phi(\mathbb{R}), \|\cdot\|_{L^\Phi}^{lux})$) is a closed subspace of $(h^\Phi(\mathbb{C}_+), \|\cdot\|_{h^\Phi}^{lux})$ (resp. $(L^\Phi(\mathbb{R}), \|\cdot\|_{L^\Phi}^{lux})$), where $\|\cdot\|_{H^\Phi}^{lux}$ is the restriction of $\|\cdot\|_{h^\Phi}^{lux}$ on $H^\Phi(\mathbb{C}_+)$.

Theorem 2.3 *Let $\Phi \in \mathcal{U}$ and F be an analytic function on $\mathbb{C}_+$. The following assertions are equivalent:*

- (i) $F \in H^\Phi(\mathbb{C}_+)$.

(ii) There exists $f \in H^\Phi(\mathbb{R})$ such that

$$F(x + iy) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x - t)^2 + y^2} f(t) dt, \quad \forall x + iy \in \mathbb{C}_+.$$

If so, f is unique and we also have for all $z = x + iy \in \mathbb{C}_+$

$$F(z) = \frac{i}{\pi} \int_{\mathbb{R}} \frac{x - t}{(x - t)^2 + y^2} f(t) dt = \frac{1}{2\pi i} \int_{\mathbb{R}} \frac{f(t)}{t - z} dt.$$

Moreover,

$$\lim_{y \rightarrow 0} \|F(\cdot + iy) - f\|_{L^\Phi}^{lux} = 0$$

and

$$\|F\|_{H^\Phi}^{lux} = \lim_{y \rightarrow 0} \|F(\cdot + iy)\|_{L^\Phi}^{lux} = \|f\|_{L^\Phi}^{lux}.$$

This theorem corresponds to [13, Theorem 13.2] whenever $\Phi(t) = t^p$ with $1 \leq p < \infty$. If $\Phi \in \mathcal{U}$, then the Cauchy integral is an isometric isomorphism between $H^\Phi(\mathbb{C}_+)$ and $H^\Phi(\mathbb{R})$.

We presented in the above results representation theorems in Hardy–Orlicz spaces. In the rest of the paragraph, we give some applications of these theorems in certain theories related to Hardy spaces.

Let Φ be a convex growth function. The complementary function of Φ is the function Ψ defined by

$$\Psi(s) = \sup_{t \geq 0} \{st - \Phi(t)\}, \quad \forall s \geq 0.$$

In general, Ψ is not a growth function; but if Φ is an N -function, then Ψ is too and the complementary of Ψ is Φ (see for example, [3, 11, 15, 16]).

Let Φ be an N -function. We say that Φ satisfies ∇_2 -condition (or $\Phi \in \nabla_2$) if both Φ and its complementary function satisfy Δ_2 -condition. For $\alpha > 1$ and $\beta > 0$, the function $t \mapsto t^\alpha [\ln(1 + t)]^\beta$ is an example of an N -function satisfying the ∇_2 -condition.

In the sequel, we will call complementary pair of N -functions the pair (Φ, Ψ) of N -functions in which Φ is the complementary function of Ψ . We will also assume that the N -functions are of class $\mathcal{C}^1(\mathbb{R}_+)$.

The Hilbert transform $\mathcal{H}(f)$ and the Cauchy integral $S(f)$ of a measurable function f on $\mathbb{R}$ are, respectively, defined by

$$\mathcal{H}(f)(x) := \lim_{\varepsilon \rightarrow 0} \frac{1}{\pi} \int_{|x-t| > \varepsilon} \frac{f(t)}{x - t} dt, \quad \forall x \in \mathbb{R}$$

and

$$S(f)(z) = \frac{1}{i\pi} \int_{\mathbb{R}} \frac{f(t)}{t-z} dt, \quad \forall z \in \mathbb{C}_+.$$

Theorem 2.4 *Let Φ be an N -function and $f \in L^\Phi(\mathbb{R})$ with real values. If $\Phi \in \nabla_2$, then the following equivalent assertions are satisfied:*

- (i) $S(f) \in H^\Phi(\mathbb{C}_+)$.
- (ii) $(f + i\mathcal{H}(f)) \in H^\Phi(\mathbb{R})$.

Moreover, there is a constant $C := C_\Phi > 0$ such that

$$\|f\|_{L^\Phi}^{lux} \leq \|S(f)\|_{H^\Phi}^{lux} \leq C \|f\|_{L^\Phi}^{lux}. \quad (2.7)$$

We also defined the Hilbert maximal function $\tilde{\mathcal{H}}(f)$ of a measurable function f on $\mathbb{R}$ by

$$\tilde{\mathcal{H}}(f)(x) := \sup_{\varepsilon > 0} \left| \frac{1}{\pi} \int_{|x-t| > \varepsilon} \frac{f(t)}{x-t} dt \right|, \quad \forall x \in \mathbb{R}.$$

Let $\alpha > 0$ and F be a harmonic function on $\mathbb{C}_+$. The radial maximal function $\mathcal{M}_{rad}(F)$ and the non-tangential maximal function $\mathcal{M}_{ntg}^\alpha(F)$ of F are respectively defined by

$$\mathcal{M}_{rad}(F)(t) := \sup_{y>0} |F(t+iy)|, \quad \forall t \in \mathbb{R}$$

and

$$\mathcal{M}_{ntg}^\alpha(F)(t) := \sup_{z \in \Gamma_\alpha(t)} |F(z)|, \quad \forall t \in \mathbb{R},$$

where $\Gamma_\alpha(t) := \{x+iy \in \mathbb{C}_+ : |x-t| < \alpha y\}$.

Theorem 2.5 *Let Φ be an N -function such that $\Phi \in \Delta_2$. The following assertions are equivalent:*

- (i) $\Phi \in \nabla_2$.
- (ii) There exists a constant $C_1 > 0$ such that for all $F \in h^\Phi(\mathbb{C}_+)$,

$$\|F\|_{h^\Phi}^{lux} \leq \|\mathcal{M}_{rad}(F)\|_{L^\Phi}^{lux} \leq \|\mathcal{M}_{ntg}^\alpha(F)\|_{L^\Phi}^{lux} \leq C \|F\|_{h^\Phi}^{lux}. \quad (2.8)$$

- (iii) There exists a constant $C_2 > 0$ such that for all $f \in L^\Phi(\mathbb{R})$,

$$\|f\|_{L^\Phi}^{lux} \leq \|\mathcal{H}(f)\|_{L^\Phi}^{lux} \leq \|\tilde{\mathcal{H}}(f)\|_{L^\Phi}^{lux} \leq C_2 \|f\|_{L^\Phi}^{lux}. \quad (2.9)$$

Let (Φ, Ψ) be a complementary pair of N -functions. If both Φ and Ψ satisfy Δ_2 -condition, then the map

$$F \mapsto \|F\|_{h^\Phi}^0 := \sup \left\{ \lim_{y \rightarrow 0} \int_{\mathbb{R}} |F(x + iy)G(x + iy)| dx : G \in h^\Psi(\mathbb{C}_+) \right. \\ \left. \text{with } \|G\|_{h^\Psi}^{lux} \leq 1 \right\}$$

is a norm on $h^\Phi(\mathbb{C}_+)$. Moreover,

$$\|F\|_{h^\Phi}^{lux} \leq \|F\|_{h^\Phi}^0 \leq 2\|F\|_{h^\Phi}^{lux}, \quad \forall F \in h^\Phi(\mathbb{C}_+). \quad (2.10)$$

Theorem 2.6 *Let (Φ, Ψ) be a complementary pair of N -functions. If $\Phi \in \nabla_2$, then the following assertions are satisfied:*

- (i) *The topological dual $(h^\Phi(\mathbb{C}_+))^*$ of $h^\Phi(\mathbb{C}_+)$ is isometrically isomorphic to $h^\Psi(\mathbb{C}_+)$, in the sense that for $T \in (h^\Phi(\mathbb{C}_+))^*$, there is a unique $G \in h^\Psi(\mathbb{C}_+)$ such that*

$$T(F) = \lim_{y \rightarrow 0} \int_{\mathbb{R}} F(x + iy)G(x + iy)dx, \quad \forall F \in h^\Phi(\mathbb{C}_+)$$

and

$$\|T\|_{(h^\Phi)^*} = \|G\|_{h^\Psi}^0.$$

- (ii) *The topological dual $(H^\Phi(\mathbb{C}_+))^*$ of $H^\Phi(\mathbb{C}_+)$ is isomorphic to $H^\Psi(\mathbb{C}_+)$, in the sense that for all $T \in (H^\Phi(\mathbb{C}_+))^*$, there is a unique $G \in H^\Psi(\mathbb{C}_+)$ such that*

$$T(F) = \lim_{y \rightarrow 0} \int_{\mathbb{R}} F(x + iy)\overline{G(x + iy)}dx, \quad \forall F \in H^\Phi(\mathbb{C}_+).$$

2.3 Some Definitions and Useful Properties

We present in this section some useful results needed in our presentation.

2.3.1 Some Properties of Orlicz Space on $\mathbb{R}$

Let $\Phi \in \mathcal{C}^1(\mathbb{R}_+)$ be a growth function. The lower and upper indices of Φ are respectively, defined by

$$a_\Phi := \inf_{t > 0} \frac{t\Phi'(t)}{\Phi(t)} \quad \text{and} \quad b_\Phi := \sup_{t > 0} \frac{t\Phi'(t)}{\Phi(t)}.$$

The following assertions are satisfied:

- (i) If Φ is convex, then (see [7, Page 171])

$$\Phi \in \Delta_2 \Leftrightarrow \Phi \in \mathcal{U} \Leftrightarrow 1 \leq a_\Phi \leq b_\Phi < \infty.$$

- (ii) If Φ is an N -function, then $\Phi \in \nabla_2$ if and only if $1 \leq a_{\Phi h} \leq b_{\Phi h} < \infty$ (see [5, Lemma 2.1]). Moreover, if $\Phi \in \mathcal{U}$, then $\Phi \in \nabla_2$ if and only if there is a constant $C := C_\Phi > 0$ such that for all $t > 0$,

$$\int_0^t \frac{\Phi(s)}{s^2} ds \leq C \frac{\Phi(t)}{t} \quad (2.11)$$

(see [7, Lemma 3.1]).

- (iii) If $0 < a_\Phi \leq b_\Phi < \infty$, then the function $t \mapsto \frac{\Phi(t)}{t^{a_\Phi}}$ is increasing on $\mathbb{R}_+^*$, while the function $t \mapsto \frac{\Phi(t)}{t^{b_\Phi}}$ is decreasing on $\mathbb{R}_+^*$ (see [18, Lemma 2.1]).

Let $p > 0$ and Φ be a growth function. We say that Φ is of lower type p if there exists a constant $C_p > 0$ such that

$$\Phi(st) \leq C_p t^p \Phi(s), \quad \forall s > 0, \forall 0 < t \leq 1. \quad (2.12)$$

For $0 < p \leq 1$, we denote by $\mathcal{L}^p$ the set of all growth functions of lower type p such that the function $t \mapsto \frac{\Phi(t)}{t}$ is nonincreasing on $\mathbb{R}_+^*$, and we put $\mathcal{L} := \bigcup_{0 < p \leq 1} \mathcal{L}^p$. If Φ is a one-to-one growth function, then $\Phi \in \mathcal{U}^q$ if and only if $\Phi^{-1} \in \mathcal{L}^{1/q}$ (see [19, Proposition 2.1]).

Let Φ be a growth function. The Morse–Transue space $M^\Phi(\mathbb{R})$ is the set of all equivalent classes (in the usual sense) of measurable functions $f : \mathbb{R} \rightarrow \mathbb{C}$ such that

$$\int_{\mathbb{R}} \Phi \left(\frac{|f(x)|}{\lambda} \right) dx < \infty, \quad \text{for all } \lambda > 0.$$

Most of the results below can be found in the following references [3, 11, 15].

Let Φ be a convex growth function. The following assertions are valid:

- (i) $(M^\Phi(\mathbb{R}), \|\cdot\|_{L^\Phi}^{lux})$ is a closed subspace of $(L^\Phi(\mathbb{R}), \|\cdot\|_{L^\Phi}^{lux})$.
- (ii) If $\Phi(t) > 0$, for all $t > 0$ then $M^\Phi(\mathbb{R})$ is separable.
- (iii) If $\Phi \in \Delta_2$, then $M^\Phi(\mathbb{R}) = L^\Phi(\mathbb{R})$.

The following result can be found in [1].

Lemma 2.1 *Let $y > 0$ and $\alpha \in \mathbb{R}$. The integral*

$$J_\alpha(y) = \int_{\mathbb{R}} \frac{dx}{|x + iy|^\alpha}$$

converges if and only if $\alpha > 1$. In this case,

$$J_\alpha(y) = B\left(\frac{1}{2}, \frac{\alpha-1}{2}\right) y^{1-\alpha},$$

where B is the function Bêta defined by

$$B(m, n) = \int_0^\infty \frac{u^{m-1}}{(1+u)^{m+n}} du, \quad \forall m, n > 0.$$

For $y > 0$, the Poisson kernel P_y and its conjugate Q_y are the functions defined respectively by

$$P_y(x) = \frac{1}{\pi} \frac{y}{x^2 + y^2}, \quad \forall x \in \mathbb{R},$$

and

$$Q_y(x) = \frac{1}{\pi} \frac{x}{x^2 + y^2}, \quad \forall x \in \mathbb{R}.$$

The following result is an immediate consequence of Lemma 2.1. Therefore, the proof is omitted.

Proposition 2.1 *Let $y > 0$ and Φ be a convex growth function. The following assertions are satisfied:*

- (i) $P_y \in M^\Phi(\mathbb{R})$.
- (ii) If Φ is $\mathcal{C}^1(\mathbb{R}_+)$ and $1 < a_\Phi \leq b_\Phi < \infty$ then $Q_y \in L^\Phi(\mathbb{R})$.

Let Φ be a growth function. We have the following inclusions:

- (a) If Φ is convex, then

$$C_c(\mathbb{R}) \subseteq M^\Phi(\mathbb{R}) \subseteq L^\Phi(\mathbb{R}) \subseteq L^1\left(\frac{dt}{1+t^2}\right).$$

- (b) If $\Phi \in \mathcal{C}^1(\mathbb{R}_+)$ and $1 \leq a_\Phi \leq b_\Phi < \infty$, then

$$\begin{aligned} L^{a_\Phi}(\mathbb{R}) \cap L^{b_\Phi}(\mathbb{R}) &\subseteq L^\Phi(\mathbb{R}) \subseteq L^{a_\Phi}(\mathbb{R}) + L^{b_\Phi}(\mathbb{R}) \\ &\subseteq L^1\left(\frac{dt}{1+|t|}\right) \cap L^1\left(\frac{dt}{1+t^2}\right). \end{aligned}$$

We have the following density result.

Proposition 2.2 *Let Φ be a convex growth function. If $\Phi \in \Delta_2$, then $C_c(\mathbb{R})$ is dense in $L^\Phi(\mathbb{R})$.*

Proof Let $f \in L^\Phi(\mathbb{R})$ and $\varepsilon > 0$. Since the set of simple functions on $\mathbb{R}$ is dense in $L^\Phi(\mathbb{R})$ (see [15, Corollary 5]), there exists $g = \sum_{i=1}^n \alpha_i \chi_{A_i}$ with $|\{x \in \mathbb{R} : g(x) \neq 0\}| < \infty$, where α_i 's are real numbers and the A_i 's are measurable subsets of $\mathbb{R}$, such that $\|f - g\|_{L^\Phi}^{lux} \leq \frac{\varepsilon}{2}$. There also exists $h \in C_c(\mathbb{R})$ such that

$$\|h\|_\infty \leq \|g\|_\infty \quad \text{and} \quad |\{x \in \mathbb{R} : g(x) \neq h(x)\}| < \frac{1}{\Phi\left(\frac{4\|g\|_\infty}{\varepsilon}\right)},$$

according to Lusin's theorem (see [17, page 52]). It comes that

$$\begin{aligned} \int_{\mathbb{R}} \Phi\left(\frac{|h(x) - g(x)|}{\varepsilon/2}\right) dx &= \int_{\{x \in \mathbb{R} : g(x) \neq h(x)\}} \Phi\left(\frac{|h(x) - g(x)|}{\varepsilon/2}\right) dx \\ &\leq \Phi\left(\frac{2\|g - h\|_\infty}{\varepsilon}\right) |\{x \in \mathbb{R} : g(x) \neq h(x)\}| \\ &\leq \Phi\left(\frac{4\|g\|_\infty}{\varepsilon}\right) |\{x \in \mathbb{R} : g(x) \neq h(x)\}| \leq 1. \end{aligned}$$

Therefore $\|h - g\|_{L^\Phi}^{lux} \leq \frac{\varepsilon}{2}$. It follows that

$$\|f - h\|_{L^\Phi}^{lux} \leq \|f - g\|_{L^\Phi}^{lux} + \|g - h\|_{L^\Phi}^{lux} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

Let $\{\varphi_t\}_{t>0}$ be a family of integrable functions on $\mathbb{R}$. We say that $\{\varphi_t\}_{t>0}$ is an approximate identity on $\mathbb{R}$ if the following properties are satisfied:

- (a) For all $t > 0$, for all $u \in \mathbb{R}$, $\varphi_t(u) \geq 0$ and $\int_{\mathbb{R}} \varphi_t(u) du = 1$.
- (b) For each fixed $\delta > 0$,

$$\lim_{t \rightarrow 0} \int_{|u| > \delta} \varphi_t(u) du = 0 \tag{2.13}$$

and

$$\lim_{t \rightarrow 0} \sup_{|u| > \delta} \varphi_t(u) = 0. \tag{2.14}$$

The family $\{P_y\}_{y>0}$ is a classic example of an approximate identity on $\mathbb{R}$.

Theorem 2.7 *Let Φ be a convex growth function. Let $\{\varphi_t\}_{t>0}$ be an approximate identity on $\mathbb{R}$ and $f \in L^\Phi(\mathbb{R})$. The following assertions are satisfied:*

(i) For all $t > 0$, $(\varphi_t \star f) \in L^\Phi(\mathbb{R})$ and

$$\|(\varphi_t \star f)\|_{L^\Phi}^{lux} \leq \|f\|_{L^\Phi}^{lux}.$$

(ii) Moreover, if $\Phi \in \Delta_2$, then we have

$$\lim_{t \rightarrow 0} \|(\varphi_t \star f) - f\|_{L^\Phi}^{lux} = 0.$$

Proof The first assertion follows from Jensen's inequality and Fubini's theorem. As the second is concerned, we assume that $f \not\equiv 0$ since otherwise there is nothing to prove.

Let $0 < \varepsilon < 1$. There exists $g \in C_c(\mathbb{R})$ such that $\|f - g\|_{L^\Phi}^{lux} \leq \frac{\varepsilon}{3}$, thanks to Proposition 2.2. For $u \in \mathbb{R}$, we define the functions f_u and g_u by

$$f_u(x) = f(x - u) \quad \text{and} \quad g_u(x) = g(x - u), \quad \forall x \in \mathbb{R}.$$

It is immediate that

$$\|f_u - g_u\|_{L^\Phi}^{lux} = \|f - g\|_{L^\Phi}^{lux} \quad \text{and} \quad |\{x \in \mathbb{R} : g(x - u) - g(x) \neq 0\}| \leq 2|K|,$$

where K is the support of g . Since g is uniformly continuous on $\mathbb{R}$, there exists $\alpha_\varepsilon > 0$ such that for all $u \in \mathbb{R}$ with $|u| \leq \alpha_\varepsilon$,

$$|g(x - u) - g(x)| \leq \frac{\varepsilon}{3} \Phi^{-1}\left(\frac{1}{2|K|}\right), \quad \forall x \in \mathbb{R}.$$

It follows that

$$\begin{aligned} \int_{\mathbb{R}} \Phi\left(\frac{|g_u(x) - g(x)|}{\varepsilon/3}\right) dx &= \int_{\{x \in \mathbb{R} : g(x - u) - g(x) \neq 0\}} \Phi\left(\frac{|g(x - u) - g(x)|}{\varepsilon/3}\right) dx \\ &\leq \Phi\left(\Phi^{-1}\left(\frac{1}{2|K|}\right)\right) |\{x \in \mathbb{R} : g(x - u) \\ &\quad - g(x) \neq 0\}| \leq 1. \end{aligned}$$

We deduce that $\|g_u - g\|_{L^\Phi}^{lux} \leq \frac{\varepsilon}{3}$. Therefore

$$\|f_u - f\|_{L^\Phi}^{lux} \leq \|f_u - g_u\|_{L^\Phi}^{lux} + \|g_u - g\|_{L^\Phi}^{lux} + \|g - f\|_{L^\Phi}^{lux} \leq \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

For $t > 0$, put

$$I := \int_{|u| \leq \alpha_\varepsilon} \int_{\mathbb{R}} \varphi_t(u) \Phi\left(\frac{|f_u(x) - f(x)|}{2\varepsilon}\right) dudx$$

and

$$II := \int_{|u| > \alpha_\varepsilon} \int_{\mathbb{R}} \varphi_t(u) \Phi \left(\frac{|f_u(x) - f(x)|}{2\varepsilon} \right) du dx.$$

By Fubini's theorem we have

$$I \leq \frac{1}{2} \int_{|u| \leq \alpha_\varepsilon} \varphi_t(u) \left(\int_{\mathbb{R}} \Phi \left(\frac{|f_u(x) - f(x)|}{\|f_u - f\|_{L^\Phi}^{ux}} \right) dx \right) du \leq \frac{1}{2} \int_{|u| \leq \alpha_\varepsilon} \varphi_t(u) du \leq \frac{1}{2}.$$

From Relation (2.13), we have that there exists $t_\varepsilon > 0$ such that for $0 < t < t_\varepsilon$,

$$\int_{|u| > \alpha_\varepsilon} \varphi_t(u) du \leq \frac{1}{2 \int_{\mathbb{R}} \Phi \left(\frac{|f(x)|}{\varepsilon} \right) dx}, \quad \forall 0 < t < t_\varepsilon.$$

Hence for $0 < t < t_\varepsilon$, we have

$$\begin{aligned} II &\leq \int_{|u| > \alpha_\varepsilon} \varphi_t(u) \left(\int_{\mathbb{R}} \Phi \left(\frac{1}{2} \frac{|f_u(x)|}{\varepsilon} + \frac{1}{2} \frac{|f(x)|}{\varepsilon} \right) dx \right) du \\ &\leq \int_{|u| > \alpha_\varepsilon} \varphi_t(u) \left(\frac{1}{2} \int_{\mathbb{R}} \Phi \left(\frac{|f_u(x)|}{\varepsilon} \right) dx + \frac{1}{2} \int_{\mathbb{R}} \Phi \left(\frac{|f(x)|}{\varepsilon} \right) dx \right) du \\ &\leq \int_{|u| > \alpha_\varepsilon} \varphi_t(u) \left(\int_{\mathbb{R}} \Phi \left(\frac{|f(x)|}{\varepsilon} \right) dx \right) du < \frac{1}{2} \end{aligned}$$

It follows from Jensen's inequality and estimates I and II that

$$\begin{aligned} \int_{\mathbb{R}} \Phi \left(\frac{|(\varphi_t \star f)(x) - f(x)|}{2\varepsilon} \right) dx &\leq \int_{\mathbb{R}} \int_{\mathbb{R}} \varphi_t(u) \Phi \left(\frac{|f_u(x) - f(x)|}{2\varepsilon} \right) du dx \\ &= I + II \leq \frac{1}{2} + \frac{1}{2} = 1, \end{aligned}$$

which ends the proof of Assertion (ii). $\square$

If $\Phi(t) = t^p$ with $1 \leq p < \infty$, then Theorem 2.7 corresponds to [13, Theorem 10.11].

2.3.2 Some Properties of Hardy–Orlicz Space on $\mathbb{C}_+$

Let f be a measurable function on $\mathbb{R}$. The Poisson integral U_f of f and its conjugate V_f are the functions defined respectively by

$$U_f(x + iy) := \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x - t)^2 + y^2} f(t) dt, \quad \forall x + iy \in \mathbb{C}_+$$

and

$$V_f(x + iy) := \frac{1}{\pi} \int_{\mathbb{R}} \frac{x - t}{(x - t)^2 + y^2} f(t) dt, \quad \forall x + iy \in \mathbb{C}_+,$$

when the integrals exist. If $f \in L^1\left(\frac{dt}{1+t^2}\right)$, then U_f is a harmonic function on $\mathbb{C}_+$ and

$$\lim_{y \rightarrow 0} U_f(x + iy) = f(x),$$

for almost all $x \in \mathbb{R}$. Moreover, if $f \in L^1\left(\frac{dt}{1+|t|}\right) \cap L^1\left(\frac{dt}{1+t^2}\right)$, then V_f is the harmonic conjugate of U_f and

$$\lim_{y \rightarrow 0} V_f(x + iy) = \mathcal{H}(f)(x),$$

for almost all $x \in \mathbb{R}$ (see [13]).

The following consequence of Theorem 2.7 is obvious.

Corollary 2.1 *Let Φ be a convex growth function. If $\Phi \in \Delta_2$, then for all $f \in L^\Phi(\mathbb{R})$, we have*

$$\lim_{y \rightarrow 0} \|U_f(\cdot + iy) - f\|_{L^\Phi}^{lux} = 0.$$

Theorem 2.8 *Let Φ be an N -function and $f \in L^\Phi(\mathbb{R})$. If $\Phi \in \nabla_2$, then $V_f \in h^\Phi(\mathbb{C}_+)$, $\mathcal{H}(f) \in L^\Phi(\mathbb{R})$, and there is a constant $C := C_\Phi > 0$ such that*

$$\|V_f\|_{h^\Phi}^{lux} = \|\mathcal{H}(f)\|_{L^\Phi}^{lux} \leq C \|f\|_{L^\Phi}^{lux}. \quad (2.15)$$

Moreover,

$$V_f(x + iy) = U_{\mathcal{H}(f)}(x + iy), \quad \forall x + iy \in \mathbb{C}_+$$

and

$$\lim_{y \rightarrow 0} \|V_f(\cdot + iy) - \mathcal{H}(f)\|_{L^\Phi}^{lux} = 0.$$

For the proof of this result, we need the following lemma.

Lemma 2.2 *Let Φ be a convex growth function such that $\Phi(t) > 0$ for all $t > 0$ and $f \in L^1\left(\frac{dt}{1+t^2}\right)$. The following assertions are equivalent:*

- (i) $f \in L^\Phi(\mathbb{R})$.
- (ii) $U_f \in h^\Phi(\mathbb{C}_+)$.

Moreover,

$$\|U_f\|_{h^\Phi}^{lux} = \|f\|_{L^\Phi}^{lux}.$$

Proof Let f be a not null element of $L^\Phi(\mathbb{R})$. By definition, we have

$$U_f(x + iy) = (P_y \star f)(x), \quad \forall x + iy \in \mathbb{C}_+.$$

Since $\{P_y\}_{y>0}$ is an approximate identity on $\mathbb{R}$, we have

$$\sup_{y>0} \|(P_y \star f)\|_{L^\Phi}^{lux} \leq \|f\|_{L^\Phi}^{lux},$$

thanks to point (i) of Theorem 2.7. It follows that

$$\|U_f\|_{h^\Phi}^{lux} = \sup_{y>0} \|(P_y \star f)\|_{L^\Phi}^{lux} \leq \|f\|_{L^\Phi}^{lux}, \quad (2.16)$$

which allows us to say that $U_f \in h^\Phi(\mathbb{C}_+)$ since U_f is harmonic on $\mathbb{C}_+$.

Let us prove the converse. Since

$$\lim_{y \rightarrow 0} U_f(x + iy) = f(x),$$

for almost all $x \in \mathbb{R}$, we have

$$\begin{aligned} \int_{\mathbb{R}} \Phi \left(\frac{|f(x)|}{\|U_f\|_{h^\Phi}^{lux}} \right) dx &\leq \liminf_{y \rightarrow 0} \int_{\mathbb{R}} \Phi \left(\frac{|U_f(x + iy)|}{\|U_f\|_{h^\Phi}^{lux}} \right) dx \\ &\leq \sup_{y>0} \int_{\mathbb{R}} \Phi \left(\frac{|U_f(x + iy)|}{\|U_f(\cdot + iy)\|_{L^\Phi}^{lux}} \right) dx \leq 1, \end{aligned}$$

thanks to Fatou's lemma. It follows that $f \in L^\Phi(\mathbb{R})$ and

$$\|f\|_{L^\Phi}^{lux} \leq \|U_f\|_{h^\Phi}^{lux}. \quad (2.17)$$

We deduce that

$$\|U_f\|_{h^\Phi}^{lux} = \|f\|_{L^\Phi}^{lux},$$

thanks to Relations (2.16) and (2.17). □

It follows from Lemma 2.2 that the map $f \mapsto U_f$ is a linear isometry of $L^\Phi(\mathbb{R})$ into $h^\Phi(\mathbb{C}_+)$.

Proof of Theorem 2.8 We assume $f \neq 0$ because there is nothing to show otherwise. Since $\mathcal{H}$ is bounded on $L^\Phi(\mathbb{R})$, there exists a constant $C > 0$ independent of f such that

$$\|\mathcal{H}(f)\|_{L^\Phi}^{lux} \leq C \|f\|_{L^\Phi}^{lux}$$

(see [21, Theorem 3.1.4]). It follows that

$$\|U_{\mathcal{H}(f)}\|_{h^\Phi}^{lux} = \|\mathcal{H}(f)\|_{L^\Phi}^{lux}$$

and

$$\lim_{y \rightarrow 0} \|U_{\mathcal{H}(f)}(\cdot + iy) - \mathcal{H}(f)\|_{L^\Phi}^{lux} = 0,$$

thanks to Lemma 2.2 and Corollary 2.1. To have $V_f \in h^\Phi(\mathbb{C}_+)$ we need only to prove that

$$V_f(x + iy) = U_{\mathcal{H}(f)}(x + iy), \quad \forall x + iy \in \mathbb{C}_+.$$

Since both Φ and its complementary function Ψ satisfy ∇_2 -condition, we have $1 < a_\Phi \leq b_\Phi < \infty$ and $1 < a_\Psi \leq b_\Psi < \infty$. In view of Proposition 2.2, we have that $L^{a_\Phi}(\mathbb{R}) \cap L^{b_\Phi}(\mathbb{R})$ is a dense subset of $L^\Phi(\mathbb{R})$. Therefore there exists a sequence $\{f_n\}_n$ in $L^{a_\Phi}(\mathbb{R}) \cap L^{b_\Phi}(\mathbb{R})$ such that

$$\lim_{n \rightarrow \infty} \|f_n - f\|_{L^\Phi}^{lux} = 0.$$

For $n \in \mathbb{N}$, we have $\mathcal{H}(f_n) \in L^{a_\Phi}(\mathbb{R}) \cap L^{b_\Phi}(\mathbb{R})$ and $V_{f_n}(x + iy) = U_{\mathcal{H}(f_n)}(x + iy)$, for $x + iy \in \mathbb{C}_+$ (see [13, Theorem 14.7]). It follows that

$$\begin{aligned} \left| V_f(x + iy) - U_{\mathcal{H}(f)}(x + iy) \right| &= |(Q_y \star (f - f_n))(x) + (P_y \star (\mathcal{H}(f_n - f)))(x)| \\ &\lesssim \|Q_y\|_{L^\Psi}^{lux} \|f_n - f\|_{L^\Phi}^{lux} \\ &\quad + \|P_y\|_{L^\Psi}^{lux} \|\mathcal{H}(f_n - f)\|_{L^\Phi}^{lux} \\ &\lesssim \|Q_y\|_{L^\Psi}^{lux} \|f_n - f\|_{L^\Phi}^{lux} + \|P_y\|_{L^\Psi}^{lux} \|f_n \\ &\quad - f\|_{L^\Phi}^{lux}, \quad x + iy \in \mathbb{C}_+, \end{aligned}$$

where the second inequality comes from Hölder's inequality in Orlicz spaces. Letting n tend to infinity yields the required result. $\square$

Proposition 2.3 *Let Φ be a convex growth function such that $\Phi(t) > 0$ for all $t > 0$. For $F \in h^\Phi(\mathbb{C}_+)$, we have*

$$|F(x + iy)| \leq \Phi^{-1}\left(\frac{2}{\pi y}\right) \|F\|_{h^\Phi}^{lux}, \quad \forall x + iy \in \mathbb{C}_+. \quad (2.18)$$

Proof Let $0 \neq F \in h^\Phi(\mathbb{C}_+)$. For $z_0 = x_0 + iy_0 \in \mathbb{C}_+$ and $0 < r < y_0$, we have

$$\begin{aligned} \Phi\left(\frac{|F(z_0)|}{\|F\|_{h^\Phi}^{lux}}\right) &\leq \Phi\left(\frac{1}{\pi r^2} \int \int_{\mathcal{D}(z_0, r)} \frac{|F(u+iv)|}{\|F\|_{h^\Phi}^{lux}} dudv\right) \\ &\leq \frac{1}{\pi r^2} \int \int_{\mathcal{D}(z_0, r)} \Phi\left(\frac{|F(u+iv)|}{\|F\|_{h^\Phi}^{lux}}\right) dudv \\ &\leq \frac{1}{\pi r^2} \int_0^{2r} \int_{\mathbb{R}} \Phi\left(\frac{|F(u+iv)|}{\|F\|_{h^\Phi}^{lux}}\right) dudv \leq \frac{1}{\pi r^2} \int_0^{2r} dv = \frac{2}{\pi r}, \end{aligned}$$

where the second inequality follows from Jensen's inequality. We deduce that for $r < y_0$, $\Phi\left(\frac{|F(z_0)|}{\|F\|_{h^\Phi}^{lux}}\right) \leq \frac{2}{\pi r}$. $\square$

Lemma 2.3 *Let Φ be a convex growth function such that $\Phi(t) > 0$ for all $t > 0$ and $F \in h^\Phi(\mathbb{C}_+)$. For $\beta > 0$, consider F_β the function defined by*

$$F_\beta(z) = F(z + i\beta), \quad \forall z \in \mathbb{C}_+.$$

Then F_β is continuous and bounded on $\overline{\mathbb{C}_+} := \mathbb{C}_+ \cup \mathbb{R}$ and harmonic on $\mathbb{C}_+$. Moreover,

$$F(z + i\beta) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x-t)^2 + y^2} F(t + i\beta) dt, \quad \forall z = x + iy \in \mathbb{C}_+.$$

Proof By construction F_β is continuous on $\overline{\mathbb{C}_+}$ and harmonic on $\mathbb{C}_+$. For $x + iy \in \overline{\mathbb{C}_+}$, we have

$$\begin{aligned} |F_\beta(x + iy)| &= |F(x + i(y + \beta))| \leq \Phi^{-1}\left(\frac{2}{\pi(y + \beta)}\right) \|F\|_{h^\Phi}^{lux} \\ &\leq \Phi^{-1}\left(\frac{2}{\pi\beta}\right) \|F\|_{h^\Phi}^{lux}. \end{aligned}$$

We deduce that F_β is bounded on $\overline{\mathbb{C}_+}$, according to Proposition 2.3. It follows that

$$F_\beta(x + iy) = \int_{\mathbb{R}} P_y(x - t) F_\beta(t) dt = \int_{\mathbb{R}} P_y(x - t) F(t + i\beta) dt, \quad \forall x + iy \in \mathbb{C}_+$$

(see [10, Lemma 3.4]). $\square$

From the above lemma and Jensen inequalities, we have the following result.

Corollary 2.2 *Let Φ be a convex growth function such that $\Phi(t) > 0$ for all $t > 0$. For $F \in h^\Phi(\mathbb{C}_+)$, the function $y \mapsto \|F(\cdot + iy)\|_{L^\Phi}^{lux}$ is decreasing on $\mathbb{R}_+^*$. Moreover,*

$$\|F\|_{h^\Phi}^{lux} = \sup_{y>0} \|F(\cdot + iy)\|_{L^\Phi}^{lux} = \lim_{y \rightarrow 0} \|F(\cdot + iy)\|_{L^\Phi}^{lux}.$$

We recall that the complex unit disk and the unit circle are respectively defined by

$$\mathbb{D} := \{z \in \mathbb{C} : |z| < 1\} \quad \text{and} \quad \mathbb{T} := \{z \in \mathbb{C} : |z| = 1\}.$$

Let $0 < p < \infty$. The Hardy space on $\mathbb{D}$, $h^p(\mathbb{D})$ is the set of harmonic functions F on $\mathbb{D}$ which satisfy

$$\|F\|_{h^p(\mathbb{D})} := \sup_{0 \leq r < 1} \left(\frac{1}{2\pi} \int_0^{2\pi} |F(re^{i\theta})|^p d\theta \right)^{1/p} < \infty.$$

For $1 \leq p < \infty$, $(h^p(\mathbb{D}), \|\cdot\|_{h^p(\mathbb{D})})$ is a Banach space.

The Cayley transform is the map φ defined by

$$\varphi(\omega) = i \frac{1 - \omega}{1 + \omega}, \quad \forall \omega \in \mathbb{D} \cup \mathbb{T} \setminus \{-1\}. \quad (2.19)$$

It is well known that the restriction of φ to $\mathbb{D}$ (resp. $\mathbb{T} \setminus \{-1\}$) is an analytic function on $\mathbb{D}$ with values in $\mathbb{C}_+$ (resp. a homeomorphism from $\mathbb{T} \setminus \{-1\}$ onto $\mathbb{R}$).

Theorem 2.9 *Let Φ be a convex growth function such that $\Phi(t) > 0$ for all $t > 0$. For $F \in h^\Phi(\mathbb{C}_+)$, the function G defined by*

$$G(\omega) = F\left(i \frac{1 - \omega}{1 + \omega}\right), \quad \forall \omega \in \mathbb{D},$$

is in $h^1(\mathbb{D})$. Moreover,

$$\|G\|_{h^1(\mathbb{D})} \leq \Phi^{-1}(1) \|F\|_{h^\Phi(\mathbb{C}_+)}^{lux}.$$

Proof Let $0 \neq F \in h^\Phi(\mathbb{C}_+)$. Consider a decreasing sequence $\{\beta_n\}_{n \in \mathbb{N}}$ of positive reals such that $\lim_{n \rightarrow \infty} \beta_n = 0$. For $n \in \mathbb{N}$, let F_n be the function defined by

$$F_n(z) = F(z + i\beta_n), \quad \forall z \in \mathbb{C}_+.$$

We have

$$\Phi\left(\frac{|F(z + i\beta_n)|}{\|F\|_{h^\Phi}^{lux}}\right) \leq \int_{\mathbb{R}} P_y(x - t) \Phi\left(\frac{|F(t + i\beta_n)|}{\|F\|_{h^\Phi}^{lux}}\right) dt, \quad \forall z = x + iy \in \mathbb{C}_+, \quad (2.20)$$

thanks to Lemma 2.3 and Jensen's inequality. Consider f_n the function defined by

$$f_n(t) = \Phi \left(\frac{|F(t + i\beta_n)|}{\|F\|_{h\Phi}^{lux}} \right), \quad \forall t \in \mathbb{R}.$$

By construction, the sequence $\{f_n\}_{n \in \mathbb{N}}$ is bounded in $L^1(\mathbb{R})$. Indeed,

$$\int_{\mathbb{R}} |f_n(t)| dt = \int_{\mathbb{R}} \Phi \left(\frac{|F(t + i\beta_n)|}{\|F\|_{h\Phi}^{lux}} \right) dt \leq \sup_{y>0} \int_{\mathbb{R}} \Phi \left(\frac{|F(t + iy)|}{\|F\|_{h\Phi}^{lux}} \right) dt \leq 1.$$

So there is a subsequence of $\{f_n\}_{n \in \mathbb{N}}$, say $\{f_{n_k}\}_{k \in \mathbb{N}}$, and λ a positive Borel measure on $\mathbb{R}$ such that $\{f_{n_k}\}_{k \in \mathbb{N}}$ converges to λ in the weak-* topology and

$$\lambda(\mathbb{R}) \leq \liminf_{k \rightarrow \infty} \|f_{n_k}\|_{L^1} \leq 1.$$

Since $P_y \in C_0(\mathbb{R})$ (the set of functions f continuous on $\mathbb{R}$ such that $\lim_{|t| \rightarrow \infty} f(t) = 0$), by letting k tend to infinity in the Relation (2.20), we deduce that

$$\Phi \left(\frac{|F(x + iy)|}{\|F\|_{h\Phi}^{lux}} \right) \leq \int_{\mathbb{R}} P_y(x - t) d\lambda(t), \quad \forall x + iy \in \mathbb{C}_+.$$

Put for $A \in \mathcal{B}(\mathbb{T} \setminus \{-1\})$,

$$\nu(A) = \lambda(\varphi(A)) \quad \text{and} \quad \nu(\{-1\}) = 0,$$

where $\mathcal{B}(\mathbb{T} \setminus \{-1\})$ is the tribu of borelians on $\mathbb{T} \setminus \{-1\}$. Since λ is a positive measure on $\mathbb{R}$ and φ a homeomorphism from $\mathbb{T} \setminus \{-1\}$ onto $\mathbb{R}$, we deduce ν is a positive measure on $\mathbb{T}$. Moreover,

$$\nu(\mathbb{T}) = \int_{\mathbb{T} \setminus \{-1\}} d\nu(e^{i\theta}) = \int_{\mathbb{R}} d\lambda(t) = \lambda(\mathbb{R}) \leq 1.$$

Fix $0 \leq r < 1$. For $\omega = re^{iu} \in \mathbb{D}$ and $z = x + iy \in \mathbb{C}_+$ such that $z = i \frac{1-\omega}{1+\omega}$, we have

$$\begin{aligned} \Phi \left(\frac{|G(re^{iu})|}{\|F\|_{h\Phi}^{lux}} \right) &= \Phi \left(\frac{|F \circ \varphi(re^{iu})|}{\|F\|_{h\Phi}^{lux}} \right) = \Phi \left(\frac{|F(x + iy)|}{\|F\|_{h\Phi}^{lux}} \right) \\ &\leq \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x - t)^2 + y^2} d\lambda(t) \leq \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x - t)^2 + y^2} (1 + t^2) d\lambda(t) \\ &= \frac{1}{\pi} \int_{\mathbb{T}} \frac{1 - r^2}{1 - 2r \cos(u - \theta) + r^2} d\nu(e^{i\theta}). \end{aligned}$$

It follows that

$$\begin{aligned}
\Phi\left(\frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{|G(re^{iu})|}{\|F\|_{h\Phi}^{lux}}du\right) &\leq \frac{1}{2\pi}\int_{-\pi}^{\pi}\Phi\left(\frac{|G(re^{iu})|}{\|F\|_{h\Phi}^{lux}}\right)du \\
&\leq \frac{1}{2\pi}\int_{-\pi}^{\pi}\int_{\mathbb{T}}\frac{1-r^2}{1-2r\cos(u-\theta)+r^2}dv(e^{i\theta})du \\
&= \int_{\mathbb{T}}\left(\frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{1-r^2}{1-2r\cos(u-\theta)+r^2}du\right)dv(e^{i\theta}) \\
&= \int_{\mathbb{T}}dv(e^{i\theta}) = v(\mathbb{T}) \leq 1.
\end{aligned}$$

□

Corollary 2.3 *Let Φ be a convex growth function such that $\Phi(t) > 0$ for all $t > 0$. For all $F \in H^\Phi(\mathbb{C}_+)$, the function G defined by*

$$G(\omega) = F\left(i\frac{1-\omega}{1+\omega}\right), \quad \forall \omega \in \mathbb{D},$$

is in $H^1(\mathbb{D})$. Moreover,

$$\|G\|_{H^1(\mathbb{D})} \leq \Phi^{-1}(1)\|F\|_{H^\Phi(\mathbb{C}_+)}^{lux},$$

where $H^1(\mathbb{D})$ is the subspace of $h^1(\mathbb{D})$ consisting of analytic functions on $\mathbb{D}$.

2.3.3 Control of Some Maximal Operators

Let E be a measurable set of $\mathbb{R}$. We denote by

$$|E| := \int_E dx.$$

For $\beta \in \{0; 1/3\}$, we call interval β -dyadic any interval I of $\mathbb{R}$ of the form

$$2^{-j}([0, 1) + k + (-1)^j\beta),$$

where $k, j \in \mathbb{Z}$. We put $\mathcal{D}_j^\beta := \cup_{k \in \mathbb{Z}}[2^{-j}([0, 1) + k + (-1)^j\beta)]$, for all $j \in \mathbb{Z}$ and $\mathcal{D}^\beta := \bigcup_{j \in \mathbb{Z}} \mathcal{D}_j^\beta$. We have the following properties (see for example [4, 14, 20]):

- For all $I, J \in \mathcal{D}^\beta$, we have $I \cap J \in \{\emptyset; I; J\}$.

- For each fixed $j \in \mathbb{Z}$, if $I \in \mathcal{D}_j^\beta$, then there exists a unique $J \in \mathcal{D}_{j-1}^\beta$ such that $I \subset J$.
- For each fixed $j \in \mathbb{Z}$, if $I \in \mathcal{D}_j^\beta$, then there exists $I_1, I_2 \in \mathcal{D}_{j+1}^\beta$ such that $I = I_1 \cup I_2$ and $I_1 \cap I_2 = \emptyset$.

We can find the following result in [6].

Lemma 2.4 *Let I be an interval. There exist $\beta \in \{0, 1/3\}$ and $J \in \mathcal{D}^\beta$ such that $I \subset J$ and $|J| \leq 6|I|$.*

Let f be a measurable function on $\mathbb{R}$. We define the maximal Hardy–Littlewood function, $\mathcal{M}_{HL}(f)$ of f by

$$\mathcal{M}_{HL}(f)(x) := \sup_{I \subset \mathbb{R}} \frac{\chi_I(x)}{|I|} \int_I |f(t)| dt, \quad \forall x \in \mathbb{R},$$

where the supremum is taken over all intervals of $\mathbb{R}$. Similarly, for $\beta \in \{0, 1/3\}$, we define its dyadic version $\mathcal{M}_{HL}^{\mathcal{D}^\beta}(f)$, where the supremum is taken this time on the intervals contained in $\mathcal{D}^\beta$. We have

$$\mathcal{M}_{HL}(f) \leq 6 \sum_{\beta \in \{0, 1/3\}} \mathcal{M}_{HL}^{\mathcal{D}^\beta}(f). \quad (2.21)$$

For $\beta \in \{0, 1/3\}$, $\lambda > 0$, and f be a measurable function on $\mathbb{R}$, consider $\mathcal{D}_\lambda(f)$ (resp. $\mathcal{D}_\lambda^*(f)$) the set of intervals I , β -dyadic (resp. maximal for the inclusion) satisfying the relation

$$\frac{1}{|I|} \int_I |f(y)| dy > \lambda.$$

If $\mathcal{D}_\lambda^*(f) \neq \emptyset$, then its elements are pairwise disjoint. Moreover, each element of $\mathcal{D}_\lambda(f)$ is contained in an element of $\mathcal{D}_\lambda^*(f)$. If $f \in L_{loc}^1(\mathbb{R})$ and $\frac{1}{|I|} \int_I |f(y)| dy \rightarrow 0$ when $|I| \rightarrow \infty$, then $\mathcal{D}_\lambda^*(f) \neq \emptyset$ whenever $\mathcal{D}_\lambda(f) \neq \emptyset$. Moreover,

$$\left\{ x \in \mathbb{R} : \mathcal{M}_{HL}^{\mathcal{D}^\beta}(f)(x) > \lambda \right\} = \bigcup_{I \in \mathcal{D}_\lambda^*(f)} I$$

and

$$\lambda < \frac{1}{|I|} \int_I |f(y)| dy \leq 2\lambda, \quad \forall I \in \mathcal{D}_\lambda^*(f).$$

We have the following inequality

$$\frac{1}{\lambda} \int_{\{x \in \mathbb{R} : |f(x)| > \lambda\}} |f(x)| dx \leq \left| \left\{ x \in \mathbb{R} : \mathcal{M}_{HL}(f)(x) > \frac{\lambda}{12} \right\} \right|. \quad (2.22)$$

Let $\alpha \geq 0$ and f a measurable function on $\mathbb{R}$. If $f \in L^1\left(\frac{dt}{1+t^2}\right)$, then we have

$$\begin{aligned} \frac{1}{2\pi} \mathcal{M}_{HL}(f)(x) &\leq \mathcal{M}_{rad}(U_f)(x) \leq \mathcal{M}_{ntg}^\alpha(U_f)(x) \\ &\leq \left(1 + \frac{2\alpha}{\pi}\right) \mathcal{M}_{HL}(f)(x), \quad \forall x \in \mathbb{R}. \end{aligned} \quad (2.23)$$

Moreover, if $f \in L^1\left(\frac{dt}{1+|t|}\right) \cap L^1\left(\frac{dt}{1+t^2}\right)$, then we have also

$$\tilde{\mathcal{H}}(f)(x) \leq (1 + 1/\pi) \mathcal{M}_{HL}(f)(x) + \mathcal{M}_{rad}(V_f)(x), \quad \forall x \in \mathbb{R} \quad (2.24)$$

(see [13]).

Theorem 2.10 *Let $\alpha \geq 0$ and Φ an N -function such that $\Phi \in \Delta_2$. The following assertions are equivalent:*

- (i) $\Phi \in \nabla_2$.
- (ii) There exists a constant $C_1 > 0$ such that for all $f \in L^\Phi(\mathbb{R})$,

$$\|\mathcal{M}_{HL}(f)\|_{L^\Phi}^{lux} \leq C_1 \|f\|_{L^\Phi}^{lux}. \quad (2.25)$$

- (iii) There exists a constant $C_2 > 0$ such that for all $f \in L^\Phi(\mathbb{R})$,

$$\|f\|_{L^\Phi}^{lux} \leq \|\mathcal{M}_{rad}(U_f)\|_{L^\Phi}^{lux} \leq \|\mathcal{M}_{ntg}^\alpha(U_f)\|_{L^\Phi}^{lux} \leq C_2 \|f\|_{L^\Phi}^{lux}. \quad (2.26)$$

- (iv) There exists a constant $C_3 > 0$ such that for all $f \in L^\Phi(\mathbb{R})$,

$$\|\tilde{\mathcal{H}}(f)\|_{L^\Phi}^{lux} \leq C_3 \|f\|_{L^\Phi}^{lux}. \quad (2.27)$$

For the proof of Theorem 2.10, we need the following results.

Lemma 2.5 *Let $\Phi_1, \Phi_2 \in \mathcal{U}$. The following assertions are equivalent:*

- (i) There exists a constant $C_1 > 0$ such that for all $t > 0$,

$$\int_0^t \frac{\Phi_2(s)}{s^2} ds \leq C_1 \frac{\Phi_1(t)}{t}. \quad (2.28)$$

- (ii) There exists a constant $C_2 > 0$ such that for all $f \in L^{\Phi_1}(\mathbb{R})$,

$$\|\mathcal{M}_{HL}(f)\|_{L^{\Phi_2}}^{lux} \leq C_2 \|f\|_{L^{\Phi_1}}^{lux}. \quad (2.29)$$

(iii) There exists a constant $C_3 > 0$ such that for all $f \in L^{\Phi_1}(\mathbb{R})$

$$\|f\|_{L^{\Phi_2}}^{lux} \leq \|\mathcal{M}_{rad}(Uf)\|_{L^{\Phi_2}}^{lux} \leq \|\mathcal{M}_{ntg}^{\alpha}(Uf)\|_{L^{\Phi_2}}^{lux} \leq C_3 \|f\|_{L^{\Phi_1}}^{lux}. \quad (2.30)$$

Proof The proof of the equivalence between points (ii) and (iii) follows from the Relation (2.23) and Lemma 2.2. For (i) implies (ii) uses the same arguments as the proof of the [7, Proposition 3.6]. It remains for us to prove the converse. Suppose (i) is not true. There exists $(t_k)_{k \geq 1}$, a sequence of positive real numbers such that

$$\int_0^{2^k t_k} \frac{\Phi_2(s)}{s^2} ds \geq \frac{2^k \Phi_1(2^k t_k)}{t_k}, \quad \forall k \geq 1.$$

For $k \geq 1$, consider the function f_k defined by

$$f_k(x) := 2^k t_k \chi_{I_k}(x), \quad \forall x \in \mathbb{R},$$

where $I_k := \{x \in \mathbb{R} : a_k \leq x < a_{k+1}\}$ with $a_k = \sum_{j=0}^{k-1} \frac{1}{2^j \Phi_1(2^j t_j)}$. Since inequality (2.22) is satisfied, we have

$$\begin{aligned} \int_{\mathbb{R}} \Phi_2(\mathcal{M}_{HL}(12f_k)(x)) dx &= \int_0^\infty \Phi_2'(\lambda) \left| \left\{ x \in \mathbb{R} : \mathcal{M}_{HL}(f_k)(x) > \frac{\lambda}{12} \right\} \right| d\lambda \\ &\gtrsim \int_0^\infty \Phi_2'(\lambda) \left(\frac{1}{\lambda} \int_{\{y \in \mathbb{R} : |f_k(y)| > \lambda\}} |f_k(x)| dx \right) d\lambda \\ &\gtrsim \int_{I_k} 2^k t_k \left(\int_0^{2^k t_k} \frac{\Phi_2'(\lambda)}{\lambda} d\lambda \right) dx \\ &\gtrsim 2^k t_k |I_k| \times \frac{2^k \Phi_1(2^k t_k)}{t_k}. \end{aligned}$$

Consider the function f defined by

$$f(x) = \sum_{k=1}^{\infty} 12f_k(x), \quad \forall x \in \cup_{k \geq 1} I_k \quad \text{and} \quad f(x) = 0, \quad \forall x \in \mathbb{R} \setminus \cup_{k \geq 1} I_k.$$

Since the $\{I_k\}_k$ are pairwise disjoint and

$$|I_k| = a_{k+1} - a_k = \frac{1}{2^k \Phi_1(2^k t_k)},$$

we deduce that $f \in L^{\Phi_1}(\mathbb{R})$. Indeed,

$$\begin{aligned} \int_{\mathbb{R}} \Phi_1(|f(x)|) dx &\lesssim \sum_{k=1}^{\infty} \int_{I_k} \Phi_1(2^k t_k) \chi_{I_k}(x) dx \\ &= \sum_{k=1}^{\infty} \Phi_1(2^k t_k) |I_k| = \sum_{k=1}^{\infty} \frac{1}{2^k} < \infty. \end{aligned}$$

It follows that

$$\int_{\mathbb{R}} \Phi_2(\mathcal{M}_{HL}(f)(x)) dx \gtrsim \int_{\mathbb{R}} \Phi_2(\mathcal{M}_{HL}(12f_k)(x)) dx \gtrsim 2^k, \quad \forall k \geq 1.$$

As 2^k tends to infinity when $(k \rightarrow \infty)$, we deduce that $\mathcal{M}_{HL}(f) \notin L^{\Phi_2}(\mathbb{R})$. $\square$

Proposition 2.4 *Let Φ be an N -function such that $\Phi \in \Delta_2$. Then the following assertions are equivalent:*

- (i) $\Phi \in \nabla_2$.
- (ii) $\mathcal{M}_{HL} : L^{\Phi}(\mathbb{R}) \rightarrow L^{\Phi}(\mathbb{R})$ is bounded.
- (iii) $\mathcal{M}_{HL}^{\mathcal{D}^{\beta}} : L^{\Phi}(\mathbb{R}) \rightarrow L^{\Phi}(\mathbb{R})$ is bounded, for all $\beta \in \{0; 1/3\}$.
- (iv) There exists a constant $C > 0$ such that for all $f \in L^{\Phi}(\mathbb{R})$

$$\|f\|_{L^{\Phi}}^{lux} \leq \|\mathcal{M}_{rad}(U_f)\|_{L^{\Phi}}^{lux} \leq \|\mathcal{M}_{ntg}^{\alpha}(U_f)\|_{L^{\Phi}}^{lux} \leq C \|f\|_{L^{\Phi}}^{lux}. \quad (2.31)$$

Proof The proof follows from Relation (2.11) and Lemma 2.5. $\square$

Proof of Theorem 2.10 The equivalence (i) $\Leftrightarrow$ (ii) follows from Proposition 2.4. The equivalence between the points (ii) and (iii) follows from the Relation (2.23) and from Lemma 2.5 (for $\Phi_1 = \Phi_2$). For (i) implies (ii), the proof follows from Relation (2.24) and from Theorem 2.8. It remains for us to prove (iv) implies (i). Suppose that $\Phi \notin \nabla_2$. There exist $\beta \in \{0; 1/3\}$ and $f \in L^{\Phi}(\mathbb{R})$ such that $\mathcal{M}_{HL}^{\mathcal{D}^{\beta}}(f) \notin L^{\Phi}(\mathbb{R})$, according to Proposition 2.4. For $\lambda > 0$, there exists a family $\{I_j\}_{j \in \mathbb{N}}$ of pairwise disjoint β -dyadic intervals such that

$$\left\{x \in \mathbb{R} : \mathcal{M}_{HL}^{\mathcal{D}^{\beta}}(f)(x) > \lambda\right\} = \bigcup_{j \in \mathbb{N}} I_j$$

and

$$\lambda < \frac{1}{|I_j|} \int_{I_j} |f(y)| dy \leq 2\lambda, \quad \forall j \in \mathbb{N}.$$

For $j \in \mathbb{N}$, consider the interval $\tilde{I}_j$ defined by

$$\tilde{I}_j = I_j \cup I'_j,$$

where I'_j is the interval such that $I_j \cap I'_j = \emptyset$ and $|I_j| = |I'_j|$. For $x \in I'_j$, we have

$$\begin{aligned} \frac{1}{2\pi} \lambda &< \frac{1}{2\pi} \frac{1}{|I_j|} \int_{I_j} |f(t)| dt = \frac{1}{\pi} \frac{1}{|\tilde{I}_j|} \int_{\mathbb{R}} |f(t) \chi_{I_j}(t)| dt \\ &\leq \sup_{\varepsilon > 0} \left| \frac{1}{\pi} \int_{|x-t| > \varepsilon} \frac{|f(t) \chi_{I_j}(t)|}{x-t} dt \right|. \end{aligned}$$

We deduce that

$$I'_j \subset \left\{ x \in \mathbb{R} : \tilde{\mathcal{H}}(|f \chi_{I_j}|)(x) > \frac{\lambda}{2\pi} \right\}.$$

Since the inequality (2.9) is satisfied, we have

$$\begin{aligned} \int_{\mathbb{R}} \Phi \left(\mathcal{M}_{HL}^{\mathcal{D}^\beta}(f)(x) \right) dx &= \int_0^\infty \Phi'(\lambda) \left| \left\{ x \in \mathbb{R} : \mathcal{M}_{HL}^{\mathcal{D}^\beta} f(x) > \lambda \right\} \right| d\lambda \\ &= \int_0^\infty \Phi'(\lambda) \left| \bigcup_{j \in \mathbb{N}} I_j \right| d\lambda = \int_0^\infty \Phi'(\lambda) \sum_{j \in \mathbb{N}} |I_j| d\lambda \\ &= \sum_{j \in \mathbb{N}} \int_0^\infty \Phi'(\lambda) |I'_j| d\lambda \\ &\lesssim \sum_{j \in \mathbb{N}} \int_0^\infty \Phi'(\lambda) \left| \left\{ x \in \mathbb{R} : \tilde{\mathcal{H}}(f \chi_{I_j})(x) > \frac{\lambda}{2\pi} \right\} \right| d\lambda \\ &\lesssim \sum_{j \in \mathbb{N}} \int_{\mathbb{R}} \Phi \left(\tilde{\mathcal{H}}(|f \chi_{I_j}|)(x) \right) dx \\ &\lesssim \sum_{j \in \mathbb{N}} \int_{\mathbb{R}} \Phi(|f(x) \chi_{I_j}(x)|) dx \\ &= \sum_{j \in \mathbb{N}} \int_{I_j} \Phi(|f(x)|) dx = \int_{\bigcup_{j \in \mathbb{N}} I_j} \Phi(|f(x)|) dx < \infty, \end{aligned}$$

which is absurd because $\mathcal{M}_{HL}^{\mathcal{D}^\beta}(f) \notin L^\Phi(\mathbb{R})$. Therefore, $\Phi \in \nabla_2$. $\square$

2.4 Proofs of Main Results

2.4.1 Proof of Theorems 2.1 and 2.2

Proof of Theorem 2.1 The implication (ii) $\Rightarrow$ (i) follows from Lemma 2.2. Let us prove (i) implies (ii). Suppose that $F \in h^\Phi(\mathbb{C}_+)$. Since the map $f \mapsto U_f$ is injective on $L^\Phi(\mathbb{R})$, uniqueness is trivial in (ii). We now prove the existence.

For $z_0 = x_0 + iy_0 \in \mathbb{C}_+$, consider the sequence $(y_n)_n$ such that $0 < y_n < y_0$ and decreasing to 0, and put for $z \in \mathbb{C}_+$

$$F_n(z) = F(z + iy_n).$$

By construction, F_n is continuous and bounded on $\overline{\mathbb{C}_+}$ and harmonic on $\mathbb{C}_+$, according to Lemma 2.3. Moreover, we have

$$F(z_0) = \lim_{n \rightarrow \infty} F(z_0 + iy_n) = \lim_{n \rightarrow \infty} \int_{\mathbb{R}} P_{y_0}(x_0 - t) F(t + iy_n) dt. \quad (2.32)$$

For $n \in \mathbb{N}$, consider T_n the map defined by

$$T_n(g) = \int_{\mathbb{R}} F(t + iy_n) g(t) dt, \quad \forall g \in M^\Psi(\mathbb{R}),$$

where Ψ is the complementary function of Φ . For $g \in M^\Psi(\mathbb{R})$, we have

$$|T_n(g)| \leq \int_{\mathbb{R}} |F(t + iy_n) g(t)| dt \leq 2 \|F(\cdot + y_n)\|_{L^\Phi}^{lux} \|g\|_{L^\Psi}^{lux} \leq 2 \|F\|_{h^\Phi}^{lux} \|g\|_{L^\Psi}^{lux},$$

thanks to Hölder's inequality in Orlicz spaces. It follows that the sequence $\{T_n\}_n$ is bounded in $(M^\Psi(\mathbb{R}))^*$. Since $M^\Psi(\mathbb{R})$ is a separable Banach space, there exist $T \in (M^\Psi(\mathbb{R}))^*$ and a sub-sequence $\{T_{n_k}\}_k$ of the sequence $\{T_n\}_n$ such that $\{T_{n_k}\}_k$ converges to T for the topology $*$ -weak. There is a unique function $f \in L^\Phi(\mathbb{R})$ such that

$$T(g) = \int_{\mathbb{R}} f(x) g(x) dx, \quad \forall g \in M^\Psi(\mathbb{R}),$$

since $(M^\Psi(\mathbb{R}))^* = L^\Phi(\mathbb{R})$. As P_{y_0} belongs to $M^\Psi(\mathbb{R})$, we deduce that

$$\lim_{k \rightarrow \infty} \int_{\mathbb{R}} P_{y_0}(x_0 - t) F(t + iy_{n_k}) dt = \int_{\mathbb{R}} P_{y_0}(x_0 - t) f(t) dt. \quad (2.33)$$

It follows that

$$F(z_0) = \int_{\mathbb{R}} P_{y_0}(x_0 - t) f(t) dt,$$

thanks to Relations (2.32) and (2.33). Moreover,

$$\|F\|_{h^\Phi}^{lux} = \lim_{y \rightarrow 0} \|F(\cdot + iy)\|_{L^\Phi}^{lux} = \|f\|_{L^\Phi}^{lux},$$

according to Lemma 2.2 and Corollary 2.2. $\square$

Proof of Theorem 2.2 Let $F \in h^\Phi(\mathbb{C}_+)$, and for $\omega \in \mathbb{D}$, put

$$G(\omega) = F\left(i \frac{1-\omega}{1+\omega}\right).$$

Since $G \in h^1(\mathbb{D})$, there exists a unique finite Borel measure ν on $\mathbb{T}$ such that for all $0 \leq r < 1$

$$G(re^{iu}) = \int_{\mathbb{T}} \frac{1-r^2}{1-2r \cos(\theta-u) + r^2} d\nu(e^{i\theta}).$$

For $A \in \mathcal{B}(\mathbb{R})$ and $t \in \mathbb{R}$, put respectively

$$\lambda(A) = \nu\left(\varphi^{-1}(A)\right)$$

and

$$d\mu(t) = \pi(1+t^2)d, \lambda(t),$$

where φ is the Cayley transform defined in Relation (2.19) and $\mathcal{B}(\mathbb{R})$ is the Borel algebra on $\mathbb{R}$. By construction, λ and μ are measures on $\mathbb{R}$. Moreover,

$$\frac{1}{\pi} \int_{\mathbb{R}} \frac{d|\mu|(t)}{1+t^2} = |\lambda|(\mathbb{R}) = |\nu|\left(\varphi^{-1}(\mathbb{R})\right) = |\nu|(\mathbb{T} \setminus \{-1\}) < \infty.$$

For $z = x + iy \in \mathbb{C}_+$ and $\omega = re^{iu} \in \mathbb{D}$ such that $z = i \frac{1-\omega}{1+\omega}$, we have

$$\begin{aligned} F(x + iy) &= G(re^{iu}) = \int_{\mathbb{T}} \frac{1-r^2}{1-2r \cos(\theta-u) + r^2} d\nu(e^{i\theta}) \\ &= \frac{1-r^2}{1+2r \cos u + r^2} \nu(\{-1\}) + \int_{\mathbb{T} \setminus \{-1\}} \frac{1-r^2}{1-2r \cos(\theta-u) + r^2} d\nu(e^{i\theta}) \\ &= \nu(\{-1\})y + \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x-t)^2 + y^2} d\mu(t). \end{aligned}$$

We deduce that

$$F(x + iy) = \alpha y + \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x-t)^2 + y^2} d\mu(t), \quad \forall x + iy \in \mathbb{C}_+, \quad (2.34)$$

where $\alpha = \nu(\{-1\}) \in \mathbb{R}$. Consider the sequence $(y_n)_{n \in \mathbb{N}}$ of positive real numbers such that $y_n > 0$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} y_n = +\infty$. For $n \in \mathbb{N}$, we have

$$|\alpha| \leq \frac{|F(iy_n)|}{y_n} + \frac{1}{\pi} \int_{\mathbb{R}} \frac{1}{t^2 + y_n^2} d|\mu|(t),$$

thanks to Relation (2.34). We deduce that

$$|F(iy_n)| \leq \Phi^{-1} \left(\frac{2}{\pi y_n} \right) \|F\|_{h^\Phi}^{lux},$$

according to Proposition 2.3. It follows that

$$\lim_{n \rightarrow \infty} \frac{|F(iy_n)|}{y_n} = 0. \quad (2.35)$$

For $t \in \mathbb{R}$,

$$f_n(t) = \frac{1}{t^2 + y_n^2}.$$

We have

$$\lim_{n \rightarrow \infty} f_n(t) = 0,$$

for almost all $t \in \mathbb{R}$. For $t \in \mathbb{R}$, we have

$$f_n(t) = \frac{1 + t^2}{t^2 + y_n^2} \times \frac{1}{1 + t^2},$$

and it follows that

$$|f_n(t)| \leq \frac{1}{1 + t^2}, \quad \forall y_n \geq 1.$$

Since $\int_{\mathbb{R}} \frac{d|\mu|(t)}{1+t^2} < \infty$, we deduce that

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} |f_n(t)| d\lambda(t) = 0, \quad (2.36)$$

thanks to dominated convergence theorem. From Relations (2.35) and (2.36), we have $\alpha = 0$. Therefore,

$$F(x + iy) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x - t)^2 + y^2} d\mu(t), \quad \forall x + iy \in \mathbb{C}_+.$$

Relation (2.4) follows from [13, Theorem 11.9]. □

2.4.2 Proof of Theorems 2.3 and 2.4

Lemma 2.6 *Let Φ be a convex growth function such that $\Phi(t) > 0$ for all $t > 0$ and F an analytic function on $\mathbb{C}_+$. The following assertions are equivalent:*

- (i) $F \in H^\Phi(\mathbb{C}_+)$.
- (ii) *There exists a unique $f \in L^\Phi(\mathbb{R})$ such that $\log |f| \in L^1\left(\frac{dt}{1+t^2}\right)$ and*

$$F(x + iy) = U_f(x + iy), \quad \forall x + iy \in \mathbb{C}_+.$$

Moreover,

$$\|F\|_{H^\Phi}^{lux} = \lim_{y \rightarrow 0} \|F(\cdot + iy)\|_{L^\Phi}^{lux} = \|f\|_{L^\Phi}^{lux}.$$

Proof The implication (ii) $\Rightarrow$ (i) is obvious. For the converse, for $\omega \in \mathbb{D}$, put

$$G(\omega) = F\left(i \frac{1-\omega}{1+\omega}\right).$$

Since $G \in H^1(\mathbb{D})$, there exists a unique function $g \in L^1(\mathbb{T})$ such that $\log |g| \in L^1(\mathbb{T})$ and for all $0 \leq r < 1$,

$$G(\omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1-r^2}{1-2r \cos(u-\theta) + r^2} g(e^{i\theta}) d\theta$$

(see [13, Theorem 5.11]).

For $z = x + iy \in \mathbb{C}_+$ and $\omega = re^{iu} \in \mathbb{D}$ such that $z = i \frac{1-\omega}{1+\omega}$, we have

$$\begin{aligned} F(x + iy) &= G(re^{iu}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1-r^2}{1-2r \cos(u-\theta) + r^2} g(e^{i\theta}) d\theta \\ &= \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x-t)^2 + y^2} g \circ \varphi^{-1}(t) dt. \end{aligned}$$

We deduce that

$$F(x + iy) = U_{g \circ \varphi^{-1}}(x + iy), \quad \forall x + iy \in \mathbb{C}_+.$$

Since g and $\log |g|$ belong to $L^1(\mathbb{T})$, we deduce that $g \circ \varphi^{-1}$ and $\log |g \circ \varphi^{-1}|$ belong to $L^1\left(\frac{dt}{1+t^2}\right)$. It follows that $g \circ \varphi^{-1} \in L^\Phi(\mathbb{R})$ and

$$\|F\|_{H^\Phi}^{lux} = \lim_{y \rightarrow 0} \|F(\cdot + iy)\|_{L^\Phi}^{lux} = \|g \circ \varphi^{-1}\|_{L^\Phi}^{lux},$$

according to Lemma 2.2 and Corollary 2.2. □

Lemma 2.7 Let $\beta > 0$ and $\Phi \in \mathcal{U}$. If $F \in H^\Phi(\mathbb{C}_+)$, then we have

$$\int_{\mathbb{R}} \frac{F(t + i\beta)}{t - \bar{z}} dt = 0, \quad \forall z \in \mathbb{C}_+.$$

Proof Let us prove that the integral

$$\int_{\mathbb{R}} \frac{F(t + i\beta)}{t - \bar{z}} dt, \quad \forall z \in \mathbb{C}_+, \quad (2.37)$$

is well defined. For $z \in \mathbb{C}_+$, consider $\varphi_{\bar{z}}$ the function defined by

$$\varphi_{\bar{z}}(t) = \frac{1}{t - \bar{z}}, \quad \forall t \in \mathbb{R}.$$

Let us show that $\varphi_{\bar{z}} \in L^\Psi(\mathbb{R})$, where Ψ is the complementary function of Φ . Since $L^\Phi(\mathbb{R})$ is contained in $L^{a_\Phi}(\mathbb{R}) + L^{b_\Phi}(\mathbb{R})$, with $1 \leq a_\Phi \leq b_\Phi < \infty$, for $f \in L^\Phi(\mathbb{R})$, there exist $f_1 \in L^{a_\Phi}(\mathbb{R})$ and $f_2 \in L^{b_\Phi}(\mathbb{R})$ such that $f = f_1 + f_2$. We have

$$\begin{aligned} \int_{\mathbb{R}} |\varphi_{\bar{z}} f(t)| dt &\leq \int_{\mathbb{R}} |\varphi_{\bar{z}} f_1(t)| dt + \int_{\mathbb{R}} |\varphi_{\bar{z}} f_2(t)| dt \\ &\leq \|\varphi_{\bar{z}}\|_{q_1} \|f_1\|_{L^{a_\Phi}} + \|\varphi_{\bar{z}}\|_{q_2} \|f_2\|_{L^{b_\Phi}} < \infty, \end{aligned}$$

where $\frac{1}{a_\Phi} + \frac{1}{q_1} = 1$ and $\frac{1}{b_\Phi} + \frac{1}{q_2} = 1$. We deduce that

$$\int_{\mathbb{R}} |\varphi_{\bar{z}} f(t)| dt < \infty, \quad \forall f \in L^\Phi(\mathbb{R}).$$

It follows that $\varphi_{\bar{z}} \in L^\Psi(\mathbb{R})$, according to the inverse Hölder inequality in Orlicz spaces (see [15, Proposition 1.1]). Therefore, the integral (2.37) is well defined. Indeed,

$$\begin{aligned} \left| \int_{\mathbb{R}} \frac{F(t + i\beta)}{t - \bar{z}} dt \right| &\leq \int_{\mathbb{R}} |F(t + i\beta) \varphi_{\bar{z}}(t)| dt \leq 2 \|\varphi_{\bar{z}}\|_{L^\Psi}^{lux} \|F(\cdot + i\beta)\|_{L^\Phi}^{lux} \\ &\leq 2 \|\varphi_{\bar{z}}\|_{L^\Psi}^{lux} \|F\|_{H^\Phi}^{lux} < \infty. \end{aligned}$$

Let $(O, \vec{u}, \vec{v})$ be the canonical orthonormal frame of the complex plane. For $R > \max\{1, \beta + |z|\}$, consider $\Gamma_{R,\beta}$ the curve defined by

$$\Gamma_{R,\beta} := \{\omega \in \mathbb{C}_+ : |\omega - i\beta| \leq R \text{ and } \text{Im}(\omega) \geq \beta\}.$$

Since $i\beta + \bar{z}$ is not in the interior of $\Gamma_{R,\beta}$, we deduce that

$$\int_{\Gamma_{R,\beta}} \frac{F(\zeta)}{\zeta - i\beta - \bar{z}} d\zeta = 0,$$

according to Cauchy's integral formula. Consider the triangle AOB rectangular at O and contained in the frame $(O, \vec{u}, \vec{v})$ such that the distances $AO = \beta$ and $AB = R$, and put $\theta_R = \text{mes}(\overline{OBA})$, the measure of the angle OBA . We have

$$\int_{\theta_R}^{\pi-\theta_R} \Phi^{-1}\left(\frac{2}{\pi R \sin \theta}\right) d\theta \lesssim \int_{\theta_R}^{\pi/2} \Phi^{-1}\left(\frac{1}{R \sin \theta}\right) d\theta \lesssim \int_{\theta_R}^{\pi/2} \Phi^{-1}\left(\frac{1}{R\theta}\right) d\theta.$$

Let $q \geq 1$ such that $\Phi \in \mathcal{U}^q$. It follows that

$$\int_{\theta_R}^{\pi/2} \Phi^{-1}\left(\frac{1}{R\theta}\right) d\theta \lesssim \int_{\theta_R}^{\pi/2} \Phi^{-1}\left(\frac{2}{R\pi}\right) \times \frac{1}{\theta} d\theta \lesssim \left(\frac{2}{R\pi}\right)^{1/q} \times \ln\left(\frac{\pi}{2\theta_R}\right),$$

since Φ^{-1} is of lower type $1/q$ and the function $t \mapsto \frac{\Phi^{-1}(t)}{t}$ is nonincreasing on $\mathbb{R}_+^*$. We deduce that

$$\int_{\theta_R}^{\pi-\theta_R} \Phi^{-1}\left(\frac{2}{\pi R \sin \theta}\right) d\theta \lesssim \frac{\ln\left(\frac{\pi R}{2\beta}\right)}{R^{1/q}},$$

as $\theta_R \geq \sin \theta_R = \frac{\beta}{R}$. By Proposition 2.3, we have

$$\begin{aligned} \left| \int_{-R \cos \theta_R}^{R \cos \theta_R} \frac{F(t + i\beta)}{t - \bar{z}} dt \right| &= \left| \int_{\pi-\theta_R}^{\theta_R} \frac{F(Re^{i\theta})}{Re^{i\theta} - i\beta - \bar{z}} i Re^{i\theta} d\theta \right| \\ &= \left| - \int_{\theta_R}^{\pi-\theta_R} \frac{F(R \cos \theta + i R \sin \theta)}{Re^{i\theta} - i\beta - \bar{z}} i Re^{i\theta} d\theta \right| \\ &\lesssim \int_{\theta_R}^{\pi-\theta_R} \left| \frac{F(R \cos \theta + i R \sin \theta)}{Re^{i\theta} - i\beta - \bar{z}} i Re^{i\theta} \right| d\theta \\ &\lesssim \frac{R}{R - |i\beta + \bar{z}|} \|F\|_{H^\Phi}^{lux} \int_{\theta_R}^{\pi-\theta_R} \Phi^{-1}\left(\frac{2}{\pi R \sin \theta}\right) d\theta. \end{aligned}$$

We deduce that

$$\left| \int_{-R \cos \theta_R}^{R \cos \theta_R} \frac{F(t + i\beta)}{t - \bar{z}} dt \right| \lesssim \|F\|_{H^\Phi}^{lux} \times \frac{R}{R - |i\beta + \bar{z}|} \times \frac{\ln\left(\frac{\pi R}{2\beta}\right)}{R^{1/q}} \longrightarrow 0,$$

when $R \longrightarrow \infty$. □

Proof of Theorem 2.3 The implication (ii) $\Rightarrow$ (i) is obvious. For the converse, suppose that $F \in H^\Phi(\mathbb{C}_+)$. There exists a unique function $f \in L^\Phi(\mathbb{R})$ such that

$$F(x + iy) = U_f(x + iy), \quad \forall x + iy \in \mathbb{C}_+$$

and

$$\|F\|_{H^\Phi}^{lux} = \lim_{y \rightarrow 0} \|F(\cdot + iy)\|_{L^\Phi}^{lux} = \|f\|_{L^\Phi}^{lux},$$

according to Lemma 2.6. Moreover,

$$\lim_{y \rightarrow 0} \|F(\cdot + iy) - f\|_{L^\Phi}^{lux} = 0,$$

according to Corollary 2.1. For $z \in \mathbb{C}_+$, consider $\varphi_{\bar{z}}$ the function defined above by

$$\varphi_{\bar{z}}(t) = \frac{1}{t - \bar{z}}, \quad \forall t \in \mathbb{R}.$$

Since $\varphi_{\bar{z}} \in L^\Psi(\mathbb{R})$ (see the proof of Lemma 2.7), where Ψ is the complementary function of Φ , we have for $\beta > 0$

$$\begin{aligned} \left| \int_{\mathbb{R}} \frac{f(t)}{t - \bar{z}} dt - \int_{\mathbb{R}} \frac{F(t + i\beta)}{t - \bar{z}} dt \right| &\leq \int_{\mathbb{R}} |\varphi_{\bar{z}}(t)| |F(t + i\beta) - f(t)| dt \\ &\leq 2 \|\varphi_{\bar{z}}\|_{L^\Psi}^{lux} \|F(\cdot + i\beta) - f\|_{L^\Phi}^{lux}, \end{aligned}$$

thanks to Hölder's inequality in Orlicz spaces. We deduce that

$$\int_{\mathbb{R}} \frac{f(t)}{t - \bar{z}} dt = \lim_{\beta \rightarrow 0} \int_{\mathbb{R}} \frac{F(t + i\beta)}{t - \bar{z}} dt.$$

It follows that $f \in H^\Phi(\mathbb{R})$ since

$$\int_{\mathbb{R}} \frac{f(t)}{t - \bar{z}} dt = 0, \quad \forall z \in \mathbb{C}_+,$$

according to Lemma 2.7. For $z = x + iy \in \mathbb{C}_+$, we have

$$\frac{y}{(x - t)^2 + y^2} = \frac{i/2}{t - \bar{z}} - \frac{i/2}{t - z}$$

and

$$\frac{x - t}{(x - t)^2 + y^2} = \frac{-1/2}{t - \bar{z}} + \frac{-1/2}{t - z}.$$

We deduce that

$$F(z) = \frac{1}{2\pi i} \int_{\mathbb{R}} \frac{f(t)}{t - z} dt = \frac{i}{\pi} \int_{\mathbb{R}} \frac{x - t}{(x - t)^2 + y^2} f(t) dt.$$

□

Proof of Theorem 2.4 For $z = x + iy \in \mathbb{C}_+$, we have

$$\frac{1}{i\pi} \frac{1}{t - z} = \frac{1}{\pi} \frac{y}{(x - t)^2 + y^2} + \frac{i}{\pi} \frac{x - t}{(x - t)^2 + y^2}, \quad \forall t \in \mathbb{R}.$$

It follows that

$$S(f)(x + iy) = U_f(x + iy) + iV_f(x + iy) = U_f(x + iy) + iU_{\mathcal{H}(f)}(x + iy)$$

and

$$\lim_{y \rightarrow 0} S(f)(x + iy) = f(x) + i\mathcal{H}(f)(x),$$

for almost all $x \in \mathbb{R}$, according to Theorem 2.8. Since U_f is a real-valued harmonic function on $\mathbb{C}_+$, V_f its harmonic conjugate, and $\mathbb{C}_+$ a simply connected open set, we deduce that $S(f)$ is an analytic function on $\mathbb{C}_+$. It follows that $S(f) \in H^\Phi(\mathbb{C}_+)$ if and only if $f + i\mathcal{H}(f) \in H^\Phi(\mathbb{R})$, according to Theorem 2.3. Therefore,

$$\|f\|_{L^\Phi}^{lux} = \|U_f\|_{h^\Phi}^{lux} \leq \|S(f)\|_{H^\Phi}^{lux} \leq \|U_f\|_{h^\Phi}^{lux} + \|V_f\|_{h^\Phi}^{lux} \lesssim \|f\|_{L^\Phi}^{lux},$$

thanks to Lemma 2.2 and Theorem 2.8. □

2.4.3 Proof of Theorem 2.5

Proposition 2.5 *Let Φ be an N -function and $f \in L^\Phi(\mathbb{R})$. If $\Phi \in \nabla_2$, then the following assertions are satisfied:*

(i) *For almost all $x \in \mathbb{R}$,*

$$\mathcal{H}(\mathcal{H}(f))(x) = -f(x).$$

(ii) *There exist c_Φ and C_Φ constants depending only on Φ such that*

$$c_\Phi \|f\|_{L^\Phi}^{lux} \leq \|\mathcal{H}(f)\|_{L^\Phi}^{lux} \leq C_\Phi \|f\|_{L^\Phi}^{lux}. \quad (2.38)$$

Proof Without loss of generality, we assume that f is real-valued. Since $S(f) \in H^\Phi(\mathbb{C}_+)$ and

$$\lim_{y \rightarrow 0} S(f)(x + iy) = f(x) + i\mathcal{H}(f)(x), \quad (2.39)$$

for almost all $x \in \mathbb{R}$, according to Theorem 2.4, it follows that

$$S(f)(z) = \frac{1}{2\pi i} \int_{\mathbb{R}} \frac{f(t) + i\mathcal{H}(f)(t)}{t - z} dt, \quad \forall z \in \mathbb{C}_+,$$

according to Theorem 2.3. For $z = x + iy \in \mathbb{C}_+$, we have

$$2S(f)(z) = \frac{1}{\pi i} \int_{\mathbb{R}} \frac{f(t) + i\mathcal{H}(f)(t)}{t - z} dt = S(f)(z) + \frac{1}{\pi} \int_{\mathbb{R}} \frac{\mathcal{H}(f)(t)}{t - z} dt.$$

We deduce that

$$S(f)(z) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{\mathcal{H}(f)(t)}{t - z} dt = -V_{\mathcal{H}(f)}(x + iy) + iU_{\mathcal{H}(f)}(x + iy)$$

and

$$\lim_{y \rightarrow 0} S(f)(x + iy) = -\mathcal{H}(\mathcal{H}(f))(x) + i\mathcal{H}(f)(x), \quad (2.40)$$

for almost all $x \in \mathbb{R}$. It follows that

$$\lim_{y \rightarrow 0} S(f)(x + iy) = f(x) + i\mathcal{H}(f)(x) = -\mathcal{H}(\mathcal{H}(f))(x) + i\mathcal{H}(f)(x),$$

for almost all $x \in \mathbb{R}$, thanks to Relations (2.39) and (2.40). By identification, we get

$$\mathcal{H}(\mathcal{H}(f))(x) = -f(x),$$

for almost all $x \in \mathbb{R}$. Since $\mathcal{H}$ is bounded on $L^\Phi(\mathbb{R})$, we have

$$\|f\|_{L^\Phi}^{lux} = \|\mathcal{H}(\mathcal{H}(f))\|_{L^\Phi}^{lux} \lesssim \|\mathcal{H}(f)\|_{L^\Phi}^{lux} \lesssim \|f\|_{L^\Phi}^{lux}.$$

□

The proof of Theorem 2.5 follows from Theorems 2.1 and 2.10 and Proposition 2.5. Therefore, it will not be written.

2.4.4 Proof of Theorem 2.6

Proposition 2.6 *Let (Φ, Ψ) be a complementary pair of N -functions. If both Φ and Ψ satisfy to Δ_2 -condition, then for $(f, g) \in L^\Phi(\mathbb{R}) \times L^\Psi(\mathbb{R})$, we have*

$$\lim_{y \rightarrow 0} \int_{\mathbb{R}} U_f(x + iy) U_g(x + iy) dx = \int_{\mathbb{R}} f(x) g(x) dx,$$

and

$$\lim_{y \rightarrow 0} \int_{\mathbb{R}} |U_f(x + iy)U_g(x + iy)| dx = \int_{\mathbb{R}} |f(x)g(x)| dx.$$

Proof The proof follows from Corollary 2.1 and Hölder's inequality in Orlicz spaces. $\square$

Let (Φ, Ψ) be a complementary pair of N -functions. The function

$$f \mapsto \|f\|_{L^\Phi}^0 := \sup \left\{ \int_{\mathbb{R}} |f(x)g(x)| dx : g \in L^\Psi(\mathbb{R}) \text{ with } \|g\|_{L^\Psi}^{lux} \leq 1 \right\}$$

is a norm on $L^\Phi(\mathbb{R})$. Moreover,

$$\|f\|_{L^\Phi}^{lux} \leq \|f\|_{L^\Phi}^0 \leq 2\|f\|_{L^\Phi}^{lux}, \quad \forall f \in L^\Phi(\mathbb{R}). \quad (2.41)$$

Proposition 2.7 *Let Φ be an N -function and $f \in L^\Phi(\mathbb{R})$. If $\Phi \in \nabla_2$, then we have*

$$\lim_{y \rightarrow 0} \|U_f(\cdot + iy) - f\|_{L^\Phi}^0 = 0$$

and

$$\lim_{y \rightarrow 0} \|U_f(\cdot + iy)\|_{L^\Phi}^0 = \|f\|_{L^\Phi}^0.$$

Proof The proof follows from Corollary 2.1 and Relation (2.41). $\square$

Proposition 2.8 *Let Φ be an N -function such that $\Phi \in \nabla_2$. The map $F \mapsto \|F\|_{h^\Phi}^0$ is a norm on $h^\Phi(\mathbb{C}_+)$. Moreover,*

$$\|F\|_{h^\Phi}^{lux} \leq \|F\|_{h^\Phi}^0 \leq 2\|F\|_{h^\Phi}^{lux}, \quad \forall F \in h^\Phi(\mathbb{C}_+). \quad (2.42)$$

Proof It follows from the Hölder inequality in Orlicz spaces that the map $F \mapsto \|F\|_{h^\Phi}^0$ is well defined on $h^\Phi(\mathbb{C}_+)$. The homogeneity and the triangle inequality are obvious. It remains to prove that for $F \in h^\Phi(\mathbb{C}_+)$, if $\|F\|_{h^\Phi}^0 = 0$, then $F \equiv 0$. Let $F \in h^\Phi(\mathbb{C}_+)$. Suppose that $F \not\equiv 0$. There exists a unique function $f \in L^\Phi(\mathbb{R})$ such that

$$F(x + iy) = U_f(x + iy), \quad \forall x + iy \in \mathbb{C}_+$$

and $\|F\|_{h^\Phi}^{lux} = \|f\|_{L^\Phi}^{lux}$, according to Theorem 2.1. Put

$$A := \{x \in \mathbb{R} : |f(x)| > 0\}.$$

Since $|A| \neq 0$, we can find a measurable subset K of $\mathbb{R}$ such that $K \subset A$ and $0 < |K| < \infty$. Consider the function g defined by

$$g(x) = \Psi^{-1} \left(\frac{1}{|K|} \right) \chi_K(x), \quad \forall x \in \mathbb{R},$$

where Ψ is the complementary function of Φ . By construction, $g \in L^\Psi(\mathbb{R})$ and $\|g\|_{L^\Psi}^{lux} \leq 1$. Indeed,

$$\int_{\mathbb{R}} \Psi(|g(x)|) dx = \int_{\mathbb{R}} \Psi \left(\Psi^{-1} \left(\frac{1}{|K|} \right) \chi_K(x) \right) dx = \frac{1}{|K|} \int_K \Psi(k) dx = 1.$$

It follows that $U_g \in h^\Psi(\mathbb{C}_+)$ and $\|U_g\|_{h^\Psi}^{lux} = \|g\|_{L^\Psi}^{lux}$. We deduce that

$$\|F\|_{h^\Phi}^0 \geq \int_{\mathbb{R}} \left| f(x) \Psi^{-1} \left(\frac{1}{|K|} \right) \chi_K(x) \right| dx = \Psi^{-1} \left(\frac{1}{|K|} \right) \int_K |f(x)| dx > 0,$$

thanks to Proposition 2.6. Therefore, $\|F\|_{h^\Phi}^0 \neq 0$.

Let us now prove Relation (2.42). For $F \in h^\Phi(\mathbb{C}_+)$, there exists a unique function $f \in L^\Phi(\mathbb{R})$ such that

$$F(x + iy) = U_f(x + iy), \quad \forall x + iy \in \mathbb{C}_+$$

and $\|F\|_{h^\Phi}^{lux} = \|f\|_{L^\Phi}^{lux}$. It will suffice to show that

$$\|F\|_{h^\Phi}^0 = \|f\|_{L^\Phi}^0, \quad (2.43)$$

in view of Relation (2.41). Let $G \in h^\Psi(\mathbb{C}_+)$ such that $\|G\|_{h^\Psi}^{lux} \leq 1$. There exists a unique function $g \in L^\Psi(\mathbb{R})$ such that

$$G(x + iy) = U_g(x + iy), \quad \forall x + iy \in \mathbb{C}_+$$

and $\|G\|_{h^\Psi}^{lux} = \|g\|_{L^\Psi}^{lux}$. We deduce that

$$\lim_{y \rightarrow 0} \int_{\mathbb{R}} |F(x + iy)G(x + iy)| dx = \int_{\mathbb{R}} |f(x)g(x)| dx \leq \|f\|_{L^\Phi}^0,$$

according to Proposition 2.6. It follows that

$$\|F\|_{h^\Phi}^0 \leq \|f\|_{L^\Phi}^0. \quad (2.44)$$

Let $g \in L^\Psi(\mathbb{R})$ such that $\|g\|_{L^\Psi}^{lux} \leq 1$. Since $U_g \in h^\Psi(\mathbb{C}_+)$ and $\|U_g\|_{h^\Psi}^{lux} = \|g\|_{L^\Psi}^{lux}$, and as

$$\int_{\mathbb{R}} |f(x)g(x)|dx = \lim_{y \rightarrow 0} \int_{\mathbb{R}} |U_f(x+iy)U_g(x+iy)|dx \leq \|F\|_{h^\Phi}^0,$$

we deduce that

$$\|f\|_{L^\Phi}^0 \leq \|F\|_{h^\Phi}^0. \quad (2.45)$$

The conclusion follows from Relations (2.44) and (2.45). $\square$

Lemma 2.8 *Let Φ be an N -function such that $\Phi \in \nabla_2$. For all $F \in h^\Phi(\mathbb{C}_+)$ there exists a unique $f \in L^\Phi(\mathbb{R})$ such that*

$$F(x+iy) = U_f(x+iy), \quad \forall x+iy \in \mathbb{C}_+.$$

Moreover,

$$\lim_{y \rightarrow 0} \|F(\cdot + iy) - f\|_{L^\Phi}^0 = 0$$

and

$$\|F\|_{h^\Phi}^0 = \lim_{y \rightarrow 0} \|U_f(\cdot + iy)\|_{L^\Phi}^0 = \|f\|_{L^\Phi}^0.$$

Proof The proof follows from Theorem 2.1, Proposition 2.7 and Relation (2.43). $\square$

Corollary 2.4 *Let (Φ, Ψ) be a complementary pair of N -functions such that both Φ and Ψ satisfy to Δ_2 -condition. For $G \in h^\Psi(\mathbb{C}_+)$, the map T_G defined by*

$$T_G(F) = \lim_{y \rightarrow 0} \int_{\mathbb{R}} F(x+iy)G(x+iy)dx, \quad \forall F \in h^\Phi(\mathbb{C}_+),$$

is a continuous linear form on $h^\Phi(\mathbb{C}_+)$. Moreover,

$$\|T_G\|_{(h^\Phi)^*} = \|G\|_{h^\Psi}^0.$$

Proof The proof follows from Lemma 2.8. $\square$

Lemma 2.9 *Let (Φ, Ψ) be a complementary pair of N -functions such that both Φ and Ψ satisfy to Δ_2 -condition. For $f \in L^\Phi(\mathbb{R})$ and $g \in L^\Psi(\mathbb{R})$, we have*

$$\int_{\mathbb{R}} \mathcal{H}(f)(x)\mathcal{H}(g)(x)dx = \int_{\mathbb{R}} f(x)g(x)dx \quad (2.46)$$

and

$$\int_{\mathbb{R}} \mathcal{H}(f)(x)g(x)dx = - \int_{\mathbb{R}} f(x)\mathcal{H}(g)(x)dx. \quad (2.47)$$

Proof Let $f \in L^\Phi(\mathbb{R})$ and $g \in L^\Psi(\mathbb{R})$. We can assume without loss of generality that f and g are real-valued. Since $S(f) \in H^\Phi(\mathbb{C}_+)$ and $S(g) \in H^\Psi(\mathbb{C}_+)$, we deduce that the product $S(f)S(g)$ belongs to $H^1(\mathbb{C}_+)$, according to Hölder's inequality in Orlicz spaces. It follows that

$$p(x) := \lim_{y \rightarrow 0} S(f)(x+iy)S(g)(x+iy) = (f(x) + i\mathcal{H}(f)(x))(g(x) + i\mathcal{H}(g)(x)),$$

for almost any $x \in \mathbb{R}$ and

$$\int_{\mathbb{R}} p(x)dx = 0$$

(see [13, corollary 13.7]). Since

$$p = (fg - \mathcal{H}(f)\mathcal{H}(g)) + i(\mathcal{H}(f)g + \mathcal{H}(g)f),$$

the equalities (2.46) and (2.47) follow. $\square$

Proof of Theorem 2.6

(i) For $G \in h^\Phi(\mathbb{C}_+)$, consider T_G the map defined by

$$T_G(F) = \lim_{y \rightarrow 0} \int_{\mathbb{R}} F(x+iy)G(x+iy)dx, \quad \forall F \in h^\Phi(\mathbb{C}_+).$$

The map $G \mapsto T_G$ is continuous, linear, and isometric from $h^\Phi(\mathbb{C}_+)$ to $(h^\Phi(\mathbb{C}_+))^*$, according to Corollary 2.4. We still have to prove that $G \mapsto T_G$ is surjective. Let $T \in (h^\Phi(\mathbb{C}_+))^*$ and put for $f \in L^\Phi(\mathbb{R})$,

$$T_1(f) = T(U_f).$$

By construction, $T_1 \in (L^\Phi(\mathbb{R}))^*$. Indeed,

$$|T_1(f)| = |T(U_f)| \leq \|T\|_{(h^\Phi)^*} \|U_f\|_{h^\Phi}^{lux} \leq \|T\|_{(h^\Phi)^*} \|f\|_{L^\Phi}^{lux}.$$

So there is a unique function $g \in L^\Psi(\mathbb{R})$ such that

$$T_1(f) = \int_{\mathbb{R}} f(x)g(x)dx, \quad \forall f \in L^\Phi(\mathbb{R}).$$

For $F \in h^\Phi(\mathbb{C}_+)$, there exists a unique function $f \in L^\Phi(\mathbb{R})$ such that

$$F(x + iy) = U_f(x + iy), \quad \forall x + iy \in \mathbb{C}_+.$$

Since $G := U_g \in h^\Psi(\mathbb{C}_+)$, it follows that

$$\int_{\mathbb{R}} f(x)g(x)dx = \lim_{y \rightarrow 0} \int_{\mathbb{R}} F(x + iy)G(x + iy)dx,$$

according to Proposition 2.6. We deduce that

$$T(F) = \lim_{y \rightarrow 0} \int_{\mathbb{R}} F(x + iy)G(x + iy)dx, \quad \forall F \in h^\Phi(\mathbb{C}_+).$$

Therefore, $G \mapsto T_G$ is surjective from $h^\Phi(\mathbb{C}_+)$ to $(h^\Phi(\mathbb{C}_+))^*$.

(ii) For $G \in H^\Phi(\mathbb{C}_+)$, consider T_G the map defined by

$$T_G(F) = \lim_{y \rightarrow 0} \int_{\mathbb{R}} F(x + iy)\overline{G(x + iy)}dx, \quad \forall F \in H^\Phi(\mathbb{C}_+).$$

By construction, the anti-linearity, the injectivity, and the continuity of the map $G \mapsto T_G$ on $H^\Phi(\mathbb{C}_+)$ are obvious. We still have to prove that $G \mapsto T_G$ is surjective. For $L \in (H^\Phi(\mathbb{C}_+))^*$, consider $L_{\mathbb{R}}$, the map defined by

$$L_{\mathbb{R}}(F) = \operatorname{Re}(L(F)), \quad \forall F \in H^\Phi(\mathbb{C}_+).$$

By construction, $L_{\mathbb{R}}$ is a real-valued continuous linear form on $H^\Phi(\mathbb{C}_+)$. Moreover,

$$L(F) = L_{\mathbb{R}}(F) - iL_{\mathbb{R}}(iF), \quad \forall F \in H^\Phi(\mathbb{C}_+).$$

Consider L_1 the map defined by

$$L_1(f) = L_{\mathbb{R}}\left(U_f + iU_{\mathcal{H}(f)}\right), \quad \forall f \in L_{\mathbb{R}}^\Phi,$$

where $L_{\mathbb{R}}^\Phi$ is the subset of $L^\Phi(\mathbb{R})$ consisting of real-valued functions. By construction, L_1 is a continuous linear form on $L_{\mathbb{R}}^\Phi$. Indeed,

$$\begin{aligned} |L_1(f)| &= \left| L_{\mathbb{R}}\left(U_f + iU_{\mathcal{H}(f)}\right) \right| \lesssim \|L\|_{(H^\Phi)^*} \|U_f + iU_{\mathcal{H}(f)}\|_{H^\Phi}^{lux} \\ &\lesssim \|L\|_{(H^\Phi)^*} \|f\|_{L^\Phi}^{lux}, \quad \forall f \in L_{\mathbb{R}}^\Phi. \end{aligned}$$

Thus there exists $\widetilde{L}_1 \in (L^\Phi(\mathbb{R}))^*$ such that

$$\widetilde{L}_1(f) = L_1(f), \quad \forall f \in L^\Phi_{\mathbb{R}},$$

thanks to Hahn's Banach theorem. There thus exists a unique $g \in L^\Psi(\mathbb{R})$ with real values such that

$$\widetilde{L}_1(f) = \int_{\mathbb{R}} f(x)g(x)dx, \quad \forall f \in L^\Phi_{\mathbb{R}}.$$

We deduce that

$$L_{\mathbb{R}}(U_f + iU_{\mathcal{H}(f)}) = \int_{\mathbb{R}} f(x)g(x)dx, \quad \forall f \in L^\Phi_{\mathbb{R}}. \quad (2.48)$$

For $f \in L^\Phi_{\mathbb{R}}$, we have

$$\begin{aligned} L_{\mathbb{R}}(i(U_f + iU_{\mathcal{H}(f)})) &= L_{\mathbb{R}}(U_{\mathcal{H}(-f)} + iU_{\mathcal{H}(\mathcal{H}(-f))}) \\ &= \int_{\mathbb{R}} \mathcal{H}(-f)(x)g(x)dx, \end{aligned}$$

since $\mathcal{H}(\mathcal{H}(f)) = -f$, almost everywhere on $\mathbb{R}$. We deduce that

$$L_{\mathbb{R}}(i(U_f + iU_{\mathcal{H}(f)})) = - \int_{\mathbb{R}} \mathcal{H}(f)(x)g(x)dx, \quad \forall f \in L^\Phi_{\mathbb{R}}. \quad (2.49)$$

For $F \in H^\Phi(\mathbb{C}_+)$, there is a unique function $f \in H^\Phi(\mathbb{R})$ such that

$$\begin{aligned} F(x + iy) &= \frac{1}{2\pi i} \int_{\mathbb{R}} \frac{f(t)}{t - z} dt = \frac{1}{2} (U_f(x + iy) + iU_{\mathcal{H}(f)}(x + iy)), \\ \forall z &= x + iy \in \mathbb{C}_+, \end{aligned}$$

according to Theorem 2.3. Let $f = f_1 + if_2$, where f_1 and f_2 are real-valued functions, and we have

$$\begin{aligned} L(F) &= \frac{1}{2} (L_{\mathbb{R}}(U_{f_1 - \mathcal{H}(f_2)} + iU_{\mathcal{H}(f_1) - \mathcal{H}(\mathcal{H}(f_2))}) \\ &\quad - iL_{\mathbb{R}}(i(U_{f_1 - \mathcal{H}(f_2)} + iU_{\mathcal{H}(f_1) - \mathcal{H}(\mathcal{H}(f_2))}))) \\ &= \frac{1}{2} \left(\int_{\mathbb{R}} (f_1(x) - \mathcal{H}(f_2)(x)) g(x) dx \right) \end{aligned}$$

$$\begin{aligned}
& + i \int_{\mathbb{R}} (\mathcal{H}(f_1)(x) - \mathcal{H}(\mathcal{H}(f_2))(x)) g(x) dx \Big) \\
& = \frac{1}{2} \left(\int_{\mathbb{R}} f(x) g(x) dx + i \int_{\mathbb{R}} \mathcal{H}(f)(x) g(x) dx \right),
\end{aligned}$$

thanks to Relations (2.48) and (2.49). We deduce that

$$L(F) = \frac{1}{4} \int_{\mathbb{R}} (f(x) + i\mathcal{H}(f)(x)) (g(x) - i\mathcal{H}(g)(x)) dx,$$

according to Lemma 2.9. Taking $G = U_{g/4} + iU_{\mathcal{H}(g/4)} \in H^\Psi(\mathbb{C}_+)$, it follows that

$$L(F) = \lim_{y \rightarrow 0} \int_{\mathbb{R}} F(x + iy) \overline{G(x + iy)} dx.$$

□

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Chapter 3

Characterizations of L^p -Vector Fields

Divergence Distributions



Nouffou Diarra

Abstract We characterize the set $\mathcal{D}iv_p(\mathbb{R}^d)$ of all tempered distributions μ on $\mathbb{R}^d$ for which the equation $\mathcal{D}iv u = \mu$ has a solution in $L^p(\mathbb{R}^d, \mathbb{R}^d)$ when $d \geq 2$. Our first characterization of $\mathcal{D}iv_p(\mathbb{R}^d)$ is linked to the dual of the Riesz potential space $\mathcal{R}^{1,p'}(\mathbb{R}^d) := \left\{ I_1 f : f \in L^{p'}(\mathbb{R}^d) \right\}$, where $\frac{d}{d-1} < p < \infty$, $p' = \frac{p}{p-1}$, and I_1 denotes the Riesz potential operator of order 1. We also show that when $\frac{d}{d-1} < p \leq \infty$, $\mathcal{D}iv_p(\mathbb{R}^d)$ coincides with the dual of the homogeneous Sobolev space $\dot{W}^{1,p'}(\mathbb{R}^d)$ defined as the completion, with respect to the Dirichlet norm $f \mapsto \|\nabla f\|_{p'}$, of the set $C_c^\infty(\mathbb{R}^d)$ of all compactly supported and infinitely differentiable functions on $\mathbb{R}^d$. In order to establish the abovementioned characterizations of $\mathcal{D}iv_p(\mathbb{R}^d)$, we provide a direct proof of the fact that, for $1 < p < d$, $\mathcal{R}^{1,p}(\mathbb{R}^d)$ coincides with $\dot{W}^{1,p}(\mathbb{R}^d)$, with equivalent norms, and consequently, the Schwartz space $\mathcal{S}(\mathbb{R}^d)$ of rapidly decreasing smooth functions is dense in $\mathcal{R}^{1,p}(\mathbb{R}^d)$.

3.1 Introduction

Let d be a fixed positive integer. The space $\mathbb{R}^d$ is endowed with its usual scalar product $(x, y) \mapsto x \cdot y = \sum_{j=1}^d x_j y_j$, Euclidean norm $x \mapsto |x| = \sqrt{x \cdot x}$, and Lebesgue measure $E \mapsto |E|$. For $1 \leq p \leq \infty$, we denote by $\|\cdot\|_p$ the usual norm on the classical Lebesgue space L^p on $\mathbb{R}^d$ and by p' the conjugate exponent of p : $\frac{1}{p} + \frac{1}{p'} = 1$ with the convention $\frac{1}{\infty} = 0$. L_{loc}^p is the space of equivalence classes (modulo equality almost everywhere) of measurable complex functions f on $\mathbb{R}^d$ such that, for every compact subset K of $\mathbb{R}^d$, $f \chi_K$ belongs to L^p . The partial derivative (in the

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distributional sense) of a function f with respect to the j -th coordinate is denoted by $\partial_j f$ or $\frac{\partial f}{\partial x_j}$ for $j \in \{1, 2, \dots, d\}$.

Let μ be an element of the space $\mathcal{S}'$ of tempered distributions on $\mathbb{R}^d$ and X a linear subspace of L^1_{loc} . The solvability in $X^d := X \times \dots \times X$ (d times) of the equation

$$\text{div } u := \sum_{j=1}^d \partial_j u_j = \mu \quad (3.1)$$

has been studied by several authors (see [1, 3, 4, 10, 11, 14]).

Our aim in this chapter is to investigate the solvability of Eq. (3.1) when X is a given Lebesgue space. In order to shed some light on this problem, let us suppose for a while that μ belongs to the Schwartz space $\mathcal{S}$ of rapidly decreasing smooth functions and X is included in $\mathcal{S}$. It is known that the Fourier transform $\mathcal{F}$, defined by

$$\mathcal{F}(\varphi)(x) =: \widehat{\varphi}(x) = \int_{\mathbb{R}^d} \varphi(y) e^{-i2\pi y \cdot x} dy, \quad x \in \mathbb{R}^d, \quad \varphi \in \mathcal{S}, \quad (3.2)$$

satisfies

$$\mathcal{F}(\partial_j \varphi)(x) = i2\pi x_j \mathcal{F}(\varphi)(x), \quad x \in \mathbb{R}^d, \quad \varphi \in \mathcal{S}, \quad j \in \{1, 2, \dots, d\}. \quad (3.3)$$

Moreover, $\mathcal{F}$ is an isomorphism of $\mathcal{S}$ into itself. Therefore Eq. (3.1) is equivalent to

$$\sum_{j=1}^d i2\pi x_j \mathcal{F}(u_j)(x) = \mathcal{F}(\mu)(x), \quad x \in \mathbb{R}^d. \quad (3.4)$$

It is easy to see that we obtain a solution $u = (u_j)_{1 \leq j \leq d}$ of (3.4) by setting

$$\mathcal{F}(u_j)(x) = \left(-\frac{ix_j}{|x|} \right) \frac{1}{2\pi |x|} \mathcal{F}(\mu)(x), \quad x \in \mathbb{R}^d, \quad j \in \{1, 2, \dots, d\}. \quad (3.5)$$

Notice that (3.5) may be rephrased as follows:

$$u_j = R_j (I_1 \mu), \quad j \in \{1, 2, \dots, d\}, \quad (3.6)$$

where the Riesz transforms R_j ($1 \leq j \leq d$) and the Riesz potential operator I_1 are defined by

$$R_j \varphi(x) = \frac{\Gamma\left(\frac{d+1}{2}\right)}{\pi^{\frac{d+1}{2}}} \lim_{\epsilon \rightarrow 0^+} \int_{|x-y| \geq \epsilon} \frac{x_j - y_j}{|x-y|^{d+1}} \varphi(y) dy, \quad x \in \mathbb{R}^d, \quad \varphi \in \mathcal{S} \quad (3.7)$$

and

$$I_1\varphi(x) = \frac{\Gamma(\frac{d-1}{2})}{2\pi^{\frac{d}{2}}\Gamma(\frac{1}{2})} \int_{\mathbb{R}^d} |x-y|^{1-d} \varphi(y) dy, \quad x \in \mathbb{R}^d, \quad \varphi \in \mathcal{S}, \quad (3.8)$$

where Γ stands for the classical gamma function.

It is well known that if $1 \leq p < \infty$, then $\mathcal{S}$ is a dense subset of L^p , and the Riesz transforms R_j ($1 \leq j \leq d$) are bounded operators on this space. So the above discussion suggests the following claim which has actually been proved by Phuc and Torres (see [10, Theorem 3.2]).

Theorem 3.1 *Assume that $d \geq 2$, $\frac{d}{d-1} < p < \infty$, and μ is a nonnegative Radon measure on $\mathbb{R}^d$. Then the following assertions are equivalent:*

- (i) *Equation (3.1) has a solution in $(L^p)^d$.*
- (ii) *The Riesz potential function $I_1\mu$ defined by*

$$I_1\mu(x) = \frac{\Gamma(\frac{d-1}{2})}{2\pi^{\frac{d}{2}}\Gamma(\frac{1}{2})} \int_{\mathbb{R}^d} |x-y|^{1-d} d\mu(y), \quad x \in \mathbb{R}^d \quad (3.9)$$

belongs to L^p .

In the case where $p = \infty$, Phuc and Torres have obtained the following result (see [11, Lemma 4.1]).

Theorem 3.2 *For any element μ of $\mathcal{S}'$, the following assertions are equivalent:*

- (i) *Equation (3.1) has a solution in $(L^\infty)^d$.*
- (ii) *μ belongs to the dual space of the homogeneous Sobolev space $\dot{W}^{1,1}$ on $\mathbb{R}^d$.*

We recall that, for $1 \leq p \leq \infty$, the homogeneous Sobolev space $\dot{W}^{1,p}$ on $\mathbb{R}^d$ is the completion, with respect to the seminorm

$$f \mapsto \|f\|_{\dot{W}^{1,p}} := \left\| \left(\sum_{j=1}^d |\partial_j f|^2 \right)^{\frac{1}{2}} \right\|_p, \quad (3.10)$$

of the space C_c^∞ of all compactly supported and infinitely differentiable complex functions on $\mathbb{R}^d$.

Notice that, in Theorem 3.2, it is not assumed that μ is a nonnegative Radon measure on $\mathbb{R}^d$, contrary to Theorem 3.1.

In the present chapter, we shall prove two characterizations of the set

$$\mathcal{Div}_p = \left\{ \mu \in \mathcal{S}' / \text{Equation (3.1) has a solution in } (L^p)^d \right\}$$

when $d \geq 2$ and $\frac{d}{d-1} < p \leq \infty$. We note that our first characterization generalizes Theorem 3.1 and the second improves Theorem 3.2. In their proofs we shall make use of the fact that, for $1 < p < d$, $\dot{W}^{1,p}$ is the same as the Riesz potential space $\mathcal{R}^{1,p} = \{I_1 f : f \in L^p\}$. This equality follows from Corollary 1 and Theorem 2 in Section 7.1.2 of [8]. However we shall provide a direct proof of it. By taking advantage of the identity between the abovementioned two spaces, we show that the Schwartz space $\mathcal{S}$ is dense in the Riesz potential space $\mathcal{R}^{1,p}$ when $d \geq 2$ and $1 < p < d$.

The remainder of this chapter is organized as follows. Section 3.2 contains basic results about the classical operators which will be used in the sequel and the abovementioned proof of the identity between $\dot{W}^{1,p}$ and $\mathcal{R}^{1,p}$ when $1 < p < d$. Section 3.3 deals with characterizations of the set Div_p .

3.2 Preliminary Results

Let us recall that the Fourier transform on $\mathcal{S}$ has an extension $T \mapsto \mathcal{F}(T) = \widehat{T}$ on $\mathcal{S}'$, defined as follows:

$$\langle \mathcal{F}(T), \varphi \rangle = \langle T, \mathcal{F}(\varphi) \rangle, \quad T \in \mathcal{S}', \quad \varphi \in \mathcal{S}.$$

It is well known that this extension, called the Fourier transform on $\mathcal{S}'$, is an isomorphism of $\mathcal{S}'$ into itself. It is easy to see that the function $x \mapsto k(x) = |x|^{1-d}$ represents an element of $\mathcal{S}'$ in the sense that $\varphi \mapsto \langle k, \varphi \rangle = \int_{\mathbb{R}^d} k(x)\varphi(x)dx$ is a bounded linear form on $\mathcal{S}$. Moreover, it has the following basic properties.

Proposition 3.1 ([13])

$$(1) \quad \widehat{k}(x) = (2\pi|x|)^{-1} \quad (3.11)$$

in the sense that, for any $f \in \mathcal{S}$

$$\int_{\mathbb{R}^d} k(x)\widehat{f}(x)dx = \int_{\mathbb{R}^d} (2\pi|x|)^{-1}f(x)dx.$$

(2) For any element f of $\mathcal{S}$

$$\widehat{I_1 f}(x) = (2\pi|x|)^{-1}\widehat{f}(x) \quad (3.12)$$

in the sense that for any $g \in \mathcal{S}$

$$\int_{\mathbb{R}^d} I_1 f(x)\widehat{g}(x)dx = \int_{\mathbb{R}^d} (2\pi|x|)^{-1}\widehat{f}(x)g(x)dx.$$

(3) There is a finite Radon measure μ on $\mathbb{R}^d$ such that its Fourier transform $\widehat{\mu}$ satisfies

$$\widehat{\mu}(x) = \frac{2\pi|x|}{(1 + 4\pi^2|x|^2)^{\frac{1}{2}}} , \quad x \in \mathbb{R}^d. \quad (3.13)$$

It is well known that the Riesz potential operator I_1 may be extended to Lebesgue spaces, as stated below.

Proposition 3.2 ([13]) Suppose that $1 < p < d$ and $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{d}$:

(a) For any f in L^p , the integral

$$I_1 f(x) = \frac{\Gamma(\frac{d-1}{2})}{2\pi^{\frac{d}{2}}\Gamma(\frac{1}{2})} \int_{\mathbb{R}^d} |x-y|^{1-d} f(y) dy \quad (3.14)$$

converges absolutely for almost every x in $\mathbb{R}^d$.

(b) There is a real number A_p such that

$$\|I_1 f\|_{p^*} \leq A_p \|f\|_p , \quad f \in L^p. \quad (3.15)$$

Remark 3.1 Suppose that $1 < p < d$ and $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{d}$. We notice that $1 < p < p^* < \infty$ and $1 < (p^*)' < d$. Therefore, from (3.15) and Fubini's theorem it follows that

$$\int_{\mathbb{R}^d} g(x) I_1 f(x) dx = \int_{\mathbb{R}^d} I_1 g(x) f(x) dx , \quad (f, g) \in L^p \times L^{(p^*)'}. \quad (3.16)$$

It is worth noticing the following result.

Proposition 3.3 For any element φ of $\mathcal{S}$, there is an infinitely differentiable function f on $\mathbb{R}^d$ which belongs to $\bigcap_{1 \leq p \leq \infty} L^p$, satisfying $I_1 f = \varphi$.

Proof Let φ be an element of $\mathcal{S}$. It is clear that $x \mapsto \widehat{\varphi}(x) (1 + 4\pi^2|x|^2)^{\frac{1}{2}}$ belongs to $\mathcal{S}$. Therefore, there is an element h of $\mathcal{S}$ such that

$$\widehat{h}(x) = \widehat{\varphi}(x) (1 + 4\pi^2|x|^2)^{\frac{1}{2}} , \quad x \in \mathbb{R}^d.$$

Let $f = h * \mu$, where μ is a finite Radon measure satisfying (3.13). Classical properties of the integral show that f is infinitely differentiable and belongs to

$\bigcap_{1 \leq p \leq \infty} L^p$. Furthermore, according to (3.12), for any element x of $\mathbb{R}^d \setminus \{0\}$

$$\widehat{\varphi}(x) = (2\pi |x|)^{-1} \widehat{f}(x) = \widehat{I_1 f}(x).$$

This shows that $I_1 f = \varphi$. □

Remark 3.1 and Proposition 3.3 yield what follows.

Proposition 3.4 *Let $1 < p < d$. If f_1 and f_2 are two elements of L^p satisfying $I_1 f_1 = I_1 f_2$, then $f_1 = f_2$.*

Proof Let f_1 and f_2 be two elements of L^p satisfying $I_1 f_1 = I_1 f_2$.

If φ is an element of $\mathcal{S}$, then, by Proposition 3.3, there is an infinitely differentiable function f of $\bigcap_{1 \leq p \leq \infty} L^p$ such that $\varphi = I_1 f$ and therefore by (3.16)

we have

$$\int_{\mathbb{R}^d} f_1(x) \varphi(x) dx = \int_{\mathbb{R}^d} f_1(x) I_1 f(x) dx = \int_{\mathbb{R}^d} I_1 f_1(x) f(x) dx.$$

By the same argument, we obtain

$$\int_{\mathbb{R}^d} f_2(x) \varphi(x) dx = \int_{\mathbb{R}^d} I_1 f_2(x) f(x) dx.$$

Thus

$$\int_{\mathbb{R}^d} [f_1(x) - f_2(x)] \varphi(x) dx = 0, \quad \varphi \in \mathcal{S}.$$

Furthermore, since $\mathcal{S}$ is dense in $L^{p'}$, we have $f_1 = f_2$. □

The above result shows that the following definition is consistent.

Definition 3.1 Let $1 < p < d$. The Riesz potential space $\mathcal{R}^{1,p}$ on $\mathbb{R}^d$ is defined by

$$\mathcal{R}^{1,p} = \{ I_1 f : f \in L^p \},$$

and we set

$$\|I_1 f\|_{\mathcal{R}^{1,p}} = \|f\|_p. \quad (3.17)$$

It is not difficult to see that $g \mapsto \|g\|_{\mathcal{R}^{1,p}}$ is a norm on the Riesz potential space $\mathcal{R}^{1,p}$, and when equipped with this norm, $\mathcal{R}^{1,p}$ is a Banach space. Moreover, (3.15) implies that $\mathcal{R}^{1,p}$ is a subset of the Lebesgue space L^{p^*} , where $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{d}$. We refer to [7, 8, 12] for a historical background of Riesz potential spaces.

We point out that it is indirectly proved in [8] that, for $1 < p < d$, the Riesz potential space $\mathcal{R}^{1,p}$ coincides with the homogeneous Sobolev space $\dot{W}^{1,p}$ (see Sect. 3.1 for its definition). Furthermore, some recent descriptions of homogeneous Sobolev spaces can be found in [2, 9] and the references therein. For instance, it is well known (thanks to the celebrated Sobolev inequality) that, for $1 \leq p < d$, $\dot{W}^{1,p}$ may be identified with the functional set

$$\left\{ f \in L^{p^*} / \nabla f \in (L^p)^d \right\}, \quad \frac{1}{p^*} = \frac{1}{p} - \frac{1}{d}, \quad (3.18)$$

where $\nabla f = (\partial_j f)_{1 \leq j \leq d}$ is the gradient of f . Moreover, when endowed with (3.10), $\dot{W}^{1,p}$ becomes a Banach space.

In the sequel we shall use the following well-known properties of Riesz transforms (see [5, 6, 13]).

Proposition 3.5

(a) *It holds true that:*

$$I_1 \left[\sum_{j=1}^d R_j (\partial_j \varphi) \right] = \varphi, \quad \varphi \in \mathcal{S}. \quad (3.19)$$

(b) *Let j be an element of $\{1, 2, \dots, d\}$ and f be in L^p with $1 \leq p < \infty$. We set:*

$$R_j^\epsilon f(x) = \frac{\Gamma\left(\frac{d+1}{2}\right)}{\pi^{\frac{d+1}{2}}} \int_{|x-y| \geq \epsilon} \frac{x_j - y_j}{|x - y|^{d+1}} f(y) dy, \quad x \in \mathbb{R}^d, \quad \epsilon > 0 \quad (3.20)$$

and

$$R_j^* f(x) = \sup_{\epsilon > 0} \left| R_j^\epsilon f(x) \right|, \quad x \in \mathbb{R}^d. \quad (3.21)$$

- *For almost every x in $\mathbb{R}^d$, there is a complex number $R_j f(x)$ such that*

$$\lim_{\epsilon \rightarrow 0} R_j^\epsilon f(x) = R_j f(x). \quad (3.22)$$

- *If $1 < p < \infty$, then there is a real number B_p , not depending on f and such that*

$$\|R_j f\|_p \leq \|R_j^* f\|_p \leq B_p \|f\|_p. \quad (3.23)$$

Moreover

$$\lim_{\epsilon \rightarrow 0} \|R_j f - R_j^\epsilon f\|_p = 0 \quad (3.24)$$

and in $\mathcal{S}'$,

$$R_j f = -\partial_j(I_1 f) = -I_1(\partial_j f). \quad (3.25)$$

We now give a direct and simple proof of the identity between $\dot{W}^{1,p}$ and $\mathcal{R}^{1,p}$, using essentially Propositions 3.2 and 3.5.

Proposition 3.6 *Assume that $d \geq 2$ and $1 < p < d$. Then the Riesz potential space $\mathcal{R}^{1,p}$ is included in the homogeneous Sobolev space $\dot{W}^{1,p}$, and there exists a real constant $C_{d,p}$ such that*

$$\|h\|_{\dot{W}^{1,p}} \leq C_{d,p} \|h\|_{\mathcal{R}^{1,p}}, \quad h \in \mathcal{R}^{1,p}. \quad (3.26)$$

Proof Let h be an element of $\mathcal{R}^{1,p}$. Then there exists f in L^p such that $h = I_1 f$.

Let us consider two elements ρ and ω of C_c^∞ satisfying

$$\begin{cases} \text{supp } \rho \subset \overline{B(0,1)}, \quad \rho \geq 0, \quad \text{and} \quad \int_{\mathbb{R}^d} \rho(x) dx = 1 \\ \chi_{\overline{B(0,1)}} \leq \omega \leq \chi_{\overline{B(0,2)}} \end{cases}$$

where, for any real number $r > 0$, $B(0, r) = \{x \in \mathbb{R}^d : |x| < r\}$ and $\overline{B(0, r)}$ denotes its closure. We set, for any integer $n \geq 1$,

$$\begin{cases} \rho_n(x) = n^d \rho(nx) \quad \text{and} \quad \omega_n(x) = \omega(n^{-1}x), & x \in \mathbb{R}^d \\ h_n = (h \omega_n) * \rho_n. \end{cases}$$

We notice that, for any integer $n \geq 1$, h_n belongs to C_c^∞ and therefore to $\mathcal{S}$:

(1) Let us consider an element j of $\{1, 2, \dots, d\}$ and an integer $n \geq 1$:

(a) We have, by (3.25),

$$\frac{\partial h_n}{\partial x_j} = \left[\frac{\partial h}{\partial x_j} \omega_n + h \frac{\partial \omega_n}{\partial x_j} \right] * \rho_n = -[(R_j f) \omega_n] * \rho_n + \left[h \frac{\partial \omega_n}{\partial x_j} \right] * \rho_n. \quad (3.27)$$

Notice that

$$\left| \left[h \frac{\partial \omega_n}{\partial x_j} \right] * \rho_n \right| \leq \frac{1}{n} \left\| \frac{\partial \omega}{\partial x_j} \right\|_\infty (|h| \chi_{C_n}) * \rho_n \quad \text{with } C_n = \overline{B(0, 2n)} \setminus \overline{B(0, n)}$$

and therefore

$$\begin{aligned} \left\| \left[h \frac{\partial \omega_n}{\partial x_j} \right] * \rho_n \right\|_p &\leq \frac{1}{n} \left\| \frac{\partial \omega}{\partial x_j} \right\|_\infty \| |h| \chi_{C_n} \|_p \|\rho_n\|_1 \\ &= \frac{1}{n} \left\| \frac{\partial \omega}{\partial x_j} \right\|_\infty \| |h| \chi_{C_n} \|_p. \end{aligned}$$

Setting $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{d}$, we have

$$\| h \chi_{C_n} \|_p \leq \| h \chi_{C_n} \|_{p^*} |C_n|^{\frac{1}{d}} = 2 |B(0, 1)|^{\frac{1}{d}} n \| |h| \chi_{C_n} \|_{p^*}.$$

Consequently,

$$\left\| \left[h \frac{\partial \omega_n}{\partial x_j} \right] * \rho_n \right\|_p \leq 2 |B(0, 1)|^{\frac{1}{d}} \left\| \frac{\partial \omega}{\partial x_j} \right\|_\infty \| |h| \chi_{C_n} \|_{p^*}, \quad (3.28)$$

and so by inequality (3.15), we obtain

$$\begin{aligned} \left\| \left[h \frac{\partial \omega_n}{\partial x_j} \right] * \rho_n \right\|_p &\leq 2 |B(0, 1)|^{\frac{1}{d}} \left\| \frac{\partial \omega}{\partial x_j} \right\|_\infty \|h\|_{p^*} \\ &\leq 2A_p |B(0, 1)|^{\frac{1}{d}} \left\| \frac{\partial \omega}{\partial x_j} \right\|_\infty \|f\|_p. \end{aligned} \quad (3.29)$$

Furthermore,

$$|[-(R_j f) \omega_n] * \rho_n| \leq |R_j f| * \rho_n$$

and so

$$\|[-(R_j f) \omega_n] * \rho_n\|_p \leq \|R_j f\|_p \|\rho_n\|_1 = \|R_j f\|_p.$$

Consequently, by (3.23), we get

$$\|[-(R_j f) \omega_n] * \rho_n\|_p \leq B_p \|f\|_p. \quad (3.30)$$

From (3.27), (3.29), and (3.30), we deduce that

$$\left\| \frac{\partial h_n}{\partial x_j} \right\|_p \leq \left[2A_p |B(0, 1)|^{\frac{1}{d}} \left\| \frac{\partial \omega}{\partial x_j} \right\|_\infty + B_p \right] \|f\|_p.$$

(b) By (3.25) and (3.27), we have

$$\begin{aligned} \frac{\partial h}{\partial x_j} - \frac{\partial h_n}{\partial x_j} &= -R_j f + [(R_j f) \omega_n] * \rho_n - \left[h \frac{\partial \omega_n}{\partial x_j} \right] * \rho_n \\ &= -R_j f + (R_j f) * \rho_n - (R_j f) * \rho_n + [(R_j f) \omega_n] * \rho_n - \left[h \frac{\partial \omega_n}{\partial x_j} \right] * \rho_n, \end{aligned}$$

and so

$$\begin{aligned} &\left\| \frac{\partial h}{\partial x_j} - \frac{\partial h_n}{\partial x_j} \right\|_p \\ &\leq \|R_j f - (R_j f) * \rho_n\|_p + \|[(R_j f) - (R_j f) \omega_n] * \rho_n\|_p \\ &\quad + \left\| \left[h \frac{\partial \omega_n}{\partial x_j} \right] * \rho_n \right\|_p \\ &\leq \|R_j f - (R_j f) * \rho_n\|_p + \|R_j f - (R_j f) \omega_n\|_p \|\rho_n\|_1 \\ &\quad + 2 |B(0, 1)|^{\frac{1}{d}} \left\| \frac{\partial \omega}{\partial x_j} \right\|_\infty \| |h| \chi_{C_n} \|_{p^*} \quad (\text{by (3.28)}) \\ &\leq \|R_j f - (R_j f) * \rho_n\|_p + \|(R_j f) \chi_{\mathbb{R}^d \setminus B(0, n)}\|_p \\ &\quad + 2 |B(0, 1)|^{\frac{1}{d}} \left\| \frac{\partial \omega}{\partial x_j} \right\|_\infty \|h \chi_{\mathbb{R}^d \setminus B(0, n)}\|_{p^*}. \end{aligned}$$

(2) Since $(R_j f, h)$ belongs to $L^p \times L^{p^*}$ and $1 < p < p^* < \infty$, we have

$$\begin{cases} \lim_{n \rightarrow \infty} \|R_j f - (R_j f) * \rho_n\|_p = 0 \\ \lim_{n \rightarrow \infty} \|(R_j f) \chi_{\mathbb{R}^d \setminus B(0, n)}\|_p = 0 = \lim_{n \rightarrow \infty} \|h \chi_{\mathbb{R}^d \setminus B(0, n)}\|_{p^*}, \end{cases}$$

and so, from the results obtained in Point 1), we get

$$\begin{cases} \lim_{n \rightarrow \infty} \left\| \frac{\partial h}{\partial x_j} - \frac{\partial h_n}{\partial x_j} \right\|_p = 0 \\ \left\| \frac{\partial h}{\partial x_j} \right\|_p \leq \left[2A_p |B(0, 1)|^{\frac{1}{d}} \left\| \frac{\partial \omega}{\partial x_j} \right\|_\infty + B_p \right] \|f\|_p. \end{cases}$$

Therefore, since h belongs to L^{p^*} , it is in $\dot{W}^{1,p}$ (see (3.18)) and satisfies

$$\|h\|_{\dot{W}^{1,p}} \leq C_{d,p} \|h\|_{\mathcal{R}^{1,p}},$$

where

$$C_{d,p} = d \left[2A_p |B(0, 1)|^{\frac{1}{d}} \left\| \frac{\partial \omega}{\partial x_j} \right\|_{\infty} + B_p \right].$$

This completes the proof. $\square$

Proposition 3.7 *Assume that $d \geq 2$ and $1 < p < d$. Then the homogeneous Sobolev space $\dot{W}^{1,p}$ is included in the Riesz potential space $\mathcal{R}^{1,p}$, and there exists a real constant D_p such that*

$$\|f\|_{\mathcal{R}^{1,p}} \leq D_p \|f\|_{\dot{W}^{1,p}}, \quad f \in \dot{W}^{1,p}. \quad (3.31)$$

Proof Let f be an element of $\dot{W}^{1,p}$. By (3.23), the function $g = \sum_{j=1}^d R_j \left(\frac{\partial f}{\partial x_j} \right)$ belongs to L^p . Furthermore, there exists a sequence $(\varphi_n)_{n \geq 1}$ of C_c^∞ which converges to f in $\dot{W}^{1,p}$. For any integer $n \geq 1$, we have, by (3.19),

$$\varphi_n = I_1 \left[\sum_{j=1}^d R_j \left(\frac{\partial \varphi_n}{\partial x_j} \right) \right],$$

and therefore, by setting $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{d}$,

$$\begin{aligned} \|f - I_1 g\|_{p^*} &\leq \|f - \varphi_n\|_{p^*} + \left\| I_1 \left[\sum_{j=1}^d R_j \left(\frac{\partial \varphi_n}{\partial x_j} \right) \right] - I_1 \left[\sum_{j=1}^d R_j \left(\frac{\partial f}{\partial x_j} \right) \right] \right\|_{p^*} \\ &\leq \|f - \varphi_n\|_{p^*} + A_p \left\| \sum_{j=1}^d R_j \left(\frac{\partial \varphi_n}{\partial x_j} - \frac{\partial f}{\partial x_j} \right) \right\|_p \quad (\text{by (3.15)}) \\ &\leq \|f - \varphi_n\|_{p^*} + A_p B_p \sum_{j=1}^d \left\| \frac{\partial}{\partial x_j} (\varphi_n - f) \right\|_p, \end{aligned}$$

where the last inequality follows from Minkowsky's inequality and (3.23). Therefore, by letting n go to infinity in the above inequality, we get $f = I_1 g$. Thus f belongs to $\mathcal{R}^{1,p}$, and we have, by (3.23),

$$\|f\|_{\mathcal{R}^{1,p}} = \|g\|_p = \left\| \sum_{j=1}^d R_j \left(\frac{\partial f}{\partial x_j} \right) \right\|_p \leq B_p \sum_{j=1}^d \left\| \frac{\partial f}{\partial x_j} \right\|_p \leq D_p \|f\|_{\dot{W}^{1,p}}.$$

This ends the proof. $\square$

From the above two propositions, we deduce the announced result as an immediate corollary.

Corollary 3.1 *Assume that $d \geq 2$ and $1 < p < d$. Then the Riesz potential space $\mathcal{R}^{1,p}$ and the homogeneous Sobolev space $\dot{W}^{1,p}$ coincide, with equivalent norms.*

The following result will play a key role in the proof of our characterizations. It has been proved by Samko in [12] with very elaborated arguments. But it follows easily from Corollary 3.1.

Theorem 3.3 *Assume that $d \geq 2$ and $1 < p < d$. Then C_c^∞ (and therefore $\mathcal{S}$) is dense in the Riesz potential space $\mathcal{R}^{1,p}$.*

Proof It is well known that C_c^∞ is included in $\mathcal{S}$, and by Proposition 3.3, $\mathcal{S}$ is a subspace of $\mathcal{R}^{1,p}$. Thus the claim is a consequence of Corollary 3.1 and the denseness of C_c^∞ in the homogeneous Sobolev space $\dot{W}^{1,p}$. $\square$

3.3 Characterizations of the Set $\mathcal{D}iv_p$

We shall take advantage of the results obtained in Sect. 3.2, to establish characterizations of the set $\mathcal{D}iv_p$. We start with the following result.

Proposition 3.8 *Assume that $d \geq 2$, $\frac{d}{d-1} < p < \infty$ and μ is a distribution on $\mathbb{R}^d$. Then the following assertions hold:*

- (1) *μ belongs to the dual space $(\mathcal{R}^{1,p'})^*$ of the Riesz potential space $\mathcal{R}^{1,p'}$ if and only if there exists a unique element of L^p which we denote by $I_1\mu$ and satisfying*

$$\langle \mu, I_1 f \rangle = \int_{\mathbb{R}^d} I_1 \mu(x) f(x) dx \quad (:= \langle I_1 \mu, f \rangle), \quad f \in L^{p'}. \quad (3.32)$$

- (2) *$\mu \mapsto I_1 \mu$ is an isometric isomorphism of $(\mathcal{R}^{1,p'})^*$ onto L^p .*

Proof

- (1- a) Suppose that μ is in $(\mathcal{R}^{1,p'})^*$ and set

$$\langle v, f \rangle = \langle \mu, I_1 f \rangle, \quad f \in L^{p'}.$$

It is clear that v is a linear form on $L^{p'}$ such that

$$|\langle v, f \rangle| \leq \|\mu\| \|I_1 f\|_{\mathcal{R}^{1,p'}} = \|\mu\| \|f\|_{p'}, \quad f \in L^{p'}.$$

Therefore ν belongs to the dual space $(L^{p'})^*$ of $L^{p'}$. Moreover, $1 < p' < d$, and so, there is a unique element of L^p , which we will denote by $I_1\mu$ and satisfying

$$\langle \mu, I_1 f \rangle = \langle \nu, f \rangle = \int_{\mathbb{R}^d} I_1 \mu(x) f(x) dx, \quad f \in L^{p'}.$$

Moreover, we have $\|I_1 \mu\|_p = \|\mu\|$.

(b) Let h be any element of L^p and set

$$\langle \mu, I_1 f \rangle = \int_{\mathbb{R}^d} h(x) f(x) dx, \quad f \in L^{p'}.$$

Clearly, μ is a linear form on $\mathcal{R}^{1,p'}$. In addition

$$|\langle \mu, I_1 f \rangle| \leq \|h\|_p \|f\|_{p'} = \|h\|_p \|I_1 f\|_{\mathcal{R}^{1,p'}}, \quad f \in L^{p'}.$$

Consequently, μ belongs to $(\mathcal{R}^{1,p'})^*$, $I_1 \mu = h$, and $\|\mu\| \leq \|h\|_p$.

(2) From Point (1), it follows that $\mu \mapsto I_1 \mu$ is an isometry of $(\mathcal{R}^{1,p'})^*$ onto L^p whose linearity is easy to check. □

Remark 3.2 Assume that $d \geq 2$, $\frac{d}{d-1} < p < \infty$, and $g \in L^{r'}$ with $\frac{1}{r} = \frac{1}{p'} - \frac{1}{d}$.

We notice that $1 < p' < d$, and therefore, by (3.15), the map T_g defined by

$$I_1 f \mapsto \langle T_g, I_1 f \rangle = \int_{\mathbb{R}^d} g(x) I_1 f(x) dx$$

belongs to $(\mathcal{R}^{1,p'})^*$. In addition, from (3.16) we get

$$\langle T_g, I_1 f \rangle = \int_{\mathbb{R}^d} g(x) I_1 f(x) dx = \int_{\mathbb{R}^d} I_1 g(x) f(x) dx \quad f \in L^{p'}.$$

Hence $I_1 g = I_1(T_g)$, and this justifies our notation in Proposition 3.8.

We will use the following antisymmetric property of Riesz transforms, of which we provide a simple proof.

Proposition 3.9 Assume that $1 < p < \infty$. Then for any element j of $\{1, 2, \dots, d\}$, we have

$$\int_{\mathbb{R}^d} R_j f(x) g(x) dx = - \int_{\mathbb{R}^d} f(x) R_j g(x) dx, \quad (f, g) \in L^p \times L^{p'}. \quad (3.33)$$

Proof Let (f, g) be any element of $L^p \times L^{p'}$. From (3.21) and (3.23), we get, for any $\epsilon > 0$,

$$\left| (R_j^\epsilon f) g \right| \leq \left| (R_j^* f) g \right| \in L^1 \quad \text{and} \quad \left| f (R_j^\epsilon g) \right| \leq \left| f (R_j^* g) \right| \in L^1$$

and therefore by Fubini's theorem,

$$\begin{aligned} \int_{\mathbb{R}^d} \int_{|x-y| \geq \epsilon} \frac{x_j - y_j}{|x - y|^{d+1}} f(y) g(x) dy dx = \\ \int_{\mathbb{R}^d} f(y) \int_{|x-y| \geq \epsilon} \left[-g(x) \frac{y_j - x_j}{|x - y|^{d+1}} \right] dx dy, \end{aligned}$$

that is,

$$\int_{\mathbb{R}^d} R_j^\epsilon f(x) g(x) dx = - \int_{\mathbb{R}^d} f(y) R_j^\epsilon g(y) dy.$$

Furthermore, by (3.22), we have, for almost every x and y in $\mathbb{R}^d$,

$$\lim_{\epsilon \rightarrow 0} R_j^\epsilon f(x) g(x) = R_j f(x) g(x) \quad \text{and} \quad \lim_{\epsilon \rightarrow 0} f(y) R_j^\epsilon g(y) = f(y) R_j g(y),$$

and therefore, by the dominated convergence theorem, we obtain

$$\int_{\mathbb{R}^d} R_j f(x) g(x) dx = - \int_{\mathbb{R}^d} f(x) R_j g(x) dx.$$

This ends the proof. □

We also have the following result.

Proposition 3.10 Assume that $d \geq 2$, $\frac{d}{d-1} < p < \infty$, and μ is a distribution belonging to $(\mathcal{R}^{1,p'})^*$. Then there exists an element $F = (F_j)_{1 \leq j \leq d}$ of $(L^p)^d$ satisfying $\text{div } F = \mu$.

Proof By Proposition 3.8, there exists a unique element $I_1 \mu$ of L^p such that

$$\langle \mu, I_1 f \rangle = \int_{\mathbb{R}^d} I_1 \mu(x) f(x) dx, \quad f \in L^{p'}. \quad (3.34)$$

Let us set $F = (F_j)_{1 \leq j \leq d}$ with $F_j = R_j (I_1 \mu)$ for j in $\{1, 2, \dots, d\}$. By (3.23)

$$F_j \in L^p \quad \text{and} \quad \|F_j\|_p \leq B_p \|I_1 \mu\|_p, \quad j \in \{1, 2, \dots, d\}.$$

Let φ be an element of $\mathcal{S}$. We have

$$\langle \operatorname{div} F, \varphi \rangle = - \sum_{j=1}^d \int_{\mathbb{R}^d} F_j(x) \frac{\partial \varphi}{\partial x_j}(x) dx = - \sum_{j=1}^d \int_{\mathbb{R}^d} R_j(I_1 \mu)(x) \frac{\partial \varphi}{\partial x_j}(x) dx$$

and therefore

$$\begin{aligned} \langle \operatorname{div} F, \varphi \rangle &= \sum_{j=1}^d \int_{\mathbb{R}^d} I_1 \mu(x) R_j \left(\frac{\partial \varphi}{\partial x_j} \right)(x) dx && \text{(by (3.33))} \\ &= \int_{\mathbb{R}^d} I_1 \mu(x) \left[\sum_{j=1}^d R_j \left(\frac{\partial \varphi}{\partial x_j} \right)(x) \right] dx \\ &= \left\langle \mu, I_1 \left[\sum_{j=1}^d R_j \left(\frac{\partial \varphi}{\partial x_j} \right) \right] \right\rangle && \text{(by (3.34))} \\ &= \langle \mu, \varphi \rangle && \text{(by (3.19)).} \end{aligned}$$

Thus $\operatorname{div} F = \mu$. □

Proposition 3.11 Assume that $d \geq 2$, $\frac{d}{d-1} < p < \infty$, and $F = (F_j)_{1 \leq j \leq d}$ is an element of $(L^p)^d$. Then the tempered distribution $\operatorname{div} F$ belongs to $(\mathcal{R}^{1,p'})^*$.

Proof

(a) Let us consider an element f of $L^{p'}$. By (3.25) we have

$$\frac{\partial}{\partial x_j}(I_1 f) = -R_j f \in L^{p'}, \quad j \in \{1, 2, \dots, d\}.$$

Consequently,

$$- \sum_{j=1}^d \int_{\mathbb{R}^d} F_j(x) \frac{\partial}{\partial x_j}(I_1 f)(x) dx$$

is well defined in $\mathbb{C}$ and satisfies

$$\begin{aligned} \left| - \sum_{j=1}^d \int_{\mathbb{R}^d} F_j(x) \frac{\partial}{\partial x_j}(I_1 f)(x) dx \right| &\leq \sum_{j=1}^d \|F_j\|_p \left\| \frac{\partial}{\partial x_j}(I_1 f) \right\|_{p'} \\ &= \sum_{j=1}^d \|F_j\|_p \|R_j f\|_{p'} \end{aligned}$$

$$\begin{aligned}
&\leq \left[B_{p'} \sum_{j=1}^d \|F_j\|_p \right] \|f\|_{p'} \quad (\text{by (3.23)}) \\
&\leq C \|I_1 f\|_{\mathcal{R}^{1,p'}}
\end{aligned}$$

with $C = B_{p'} \sum_{j=1}^d \|F_j\|_p$.

From what precedes, it is easy to see that the map μ defined by

$$I_1 f \mapsto \langle \mu, I_1 f \rangle = - \sum_{j=1}^d \int_{\mathbb{R}^d} F_j(x) \frac{\partial}{\partial x_j} (I_1 f)(x) dx$$

belongs to $(\mathcal{R}^{1,p'})^*$.

- (b) Let g be an element of $\mathcal{S}$. By Proposition 3.3, we have $g = I_1 f$ for some f in $L^{p'}$. Therefore, by Point a), we deduce that

$$\begin{aligned}
\langle \operatorname{div} F, g \rangle &= - \sum_{j=1}^d \int_{\mathbb{R}^d} F_j(x) \frac{\partial}{\partial x_j} (g)(x) dx \\
&= - \sum_{j=1}^d \int_{\mathbb{R}^d} F_j(x) \frac{\partial}{\partial x_j} (I_1 f)(x) dx \\
&= \langle \mu, I_1 f \rangle.
\end{aligned}$$

Hence, since $\mathcal{S}$ is dense in $\mathcal{R}^{1,p'}$ (see Theorem 3.3), $\operatorname{div} F$ is the restriction to $\mathcal{S}$ of the element μ of $(\mathcal{R}^{1,p'})^*$ and so $\operatorname{div} F$ is a tempered distribution belonging to $(\mathcal{R}^{1,p'})^*$. $\square$

The conjunction of Propositions 3.8, 3.10, and 3.11 leads to the following result which is our first characterization of the set Div_p .

Theorem 3.4 Assume that $d \geq 2$, $\frac{d}{d-1} < p < \infty$, and μ is a tempered distribution on $\mathbb{R}^d$. Then the following assertions are equivalent:

- (i) μ belongs to the set Div_p .
- (ii) μ belongs to the dual space $(\mathcal{R}^{1,p'})^*$ of the Riesz potential space $\mathcal{R}^{1,p'}$.
- (iii) There exists a unique element $I_1 \mu$ of L^p satisfying

$$\langle \mu, I_1 f \rangle = \int_{\mathbb{R}^d} I_1 \mu(x) f(x) dx, \quad f \in L^{p'}.$$

Remark 3.3 If we assume that μ is a nonnegative Radon measure on $\mathbb{R}^d$, then Theorem 3.1 is nothing else than the equivalence between the assertions (i) and (iii) of Theorem 3.4. This shows that Theorem 3.4 is a generalization of Theorem 3.1.

Notice that the characterization of the set Div_p in Theorem 3.4 does not take into account the case $p = \infty$. However, since $\dot{W}^{1,p'} = \mathcal{R}^{1,p'}$ when $\frac{d}{d-1} < p < \infty$ (by Corollary 3.1), it clearly follows from Theorem 3.2 and Theorem 3.4 that the following characterization of Div_p , using homogeneous Sobolev spaces, holds true.

Theorem 3.5 Assume that $d \geq 2$, $\frac{d}{d-1} < p \leq \infty$, and μ is a tempered distribution on $\mathbb{R}^d$. Then the following assertions are equivalent:

- (i) μ belongs to the set Div_p .
- (ii) μ belongs to the dual space $\left(\dot{W}^{1,p'}\right)^*$ of the homogeneous Sobolev space $\dot{W}^{1,p'}$.

Remark 3.4 Let us notice that the characterizations of Div_p , in both Theorem 3.2 and Theorem 3.5, use homogeneous Sobolev spaces. But Theorem 3.2 is only proved for $p = \infty$, while Theorem 3.5 is proved for $p \in \left(\frac{d}{d-1}, \infty\right]$. This shows that Theorem 3.5 is an improvement of Theorem 3.2.

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Chapter 4

Boundedness of Some Commutators in Dunkl–Fofana Spaces



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Abstract In the framework of rational Dunkl analysis on the real line, we give conditions for the boundedness of the Dunkl-type fractional maximal commutators, the commutators of Dunkl-type fractional maximal operators, and the commutators of Dunkl-type fractional integral operators in Dunkl–Fofana spaces.

4.1 Introduction—Main Results

The boundedness properties of commutators of certain operators arising in harmonic analysis have been widely studied (see [11, 15–17], and the references therein). Also, the commutators are closely related to the regularity properties of the solutions of certain partial differential equations (see, for example, [1, 2, 5, 6, 8], and [23]).

Let b be a given locally integrable function on $\mathbb{R}^n$. The commutator generated by a Calderon–Zygmund operator T is denoted by $[b, T]$ and is defined for smooth functions f by

$$[b, T]f = bT(f) - T(bf).$$

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The maximal commutator C_b is defined as

$$C_b(f)(x) = \sup_{Q \ni x} \frac{1}{|Q|} \int_Q |b(x) - b(y)| |f(y)| dy, \quad x \in \mathbb{R}^n,$$

where the supremum is taken over all cubes Q containing x and $|Q|$ denotes the Lebesgue measure of Q .

It is well known (see [4]) that for a Calderon–Zygmund singular integral operator T with smooth kernel, the operator $[b, T]$ is bounded on $L^p(\mathbb{R}^n)$, $1 < p < \infty$, when b is a BMO function. In [3], Chanillo proved that commutators characterize Riesz potentials on the function space $BMO(\mathbb{R}^n)$. Gogatishvili et al. showed in [12] that $b \in BMO(\mathbb{R}^n)$ if and only if the maximal commutator C_b is bounded in some Morrey spaces. By interpolation techniques, Milman and Schonbek established in [20] a commutator result. As an application, they obtained the L^p -boundedness of the commutators of the classical Hardy–Littlewood maximal operator when b belongs to $BMO(\mathbb{R}^n)$ and $b \geq 0$.

Several results concerning the boundedness of commutators, which were established in classical Fourier analysis, have been extended within the framework of rational Dunkl analysis on the real line. In [18], Mammadov and Hasanli proved the boundedness of the Dunkl-type maximal commutator on Dunkl-type Morrey spaces when b belongs to a Dunkl-type BMO space. Next, they obtained in [19] a necessary and sufficient condition for the boundedness of the commutator of Dunkl-type fractional integral operators in Dunkl-type modified Morrey spaces. Recently, Guliyev et al. [14] studied the characterization of fractional maximal operator and its commutators on Orlicz spaces in the Dunkl setting.

In this chapter we give, in the setting of rational Dunkl analysis on the real line, conditions for the boundedness of the Dunkl-type fractional maximal commutators, the commutators of Dunkl-type fractional maximal operators, and the commutators of Dunkl-type fractional integral operators in Dunkl–Fofana spaces. Recall that on the real line, the Dunkl operators are differential-difference operators associated with the reflection group $\mathbb{Z}_2$ on $\mathbb{R}$.

Before stating our main results, let us start with some notations and basic definitions. Let $k > -\frac{1}{2}$ be a fixed number and μ be the weighted Lebesgue measure on $\mathbb{R}$, given by

$$d\mu(x) = \left(2^{k+1} \Gamma(k+1)\right)^{-1} |x|^{2k+1} dx.$$

By $L^0(\mu)$, we denote the complex vector space of equivalence classes (modulo equality μ -almost everywhere) of complex-valued functions μ -measurable on $\mathbb{R}$. For $1 \leq p \leq \infty$, we denote by $L^p(\mu)$ the Lebesgue space associated with the measure μ . For $f \in L^p(\mu)$, $\|f\|_p$ stands for the classical norm of f . The class of locally integrable functions with respect to μ is denoted by $L^1_{loc}(\mu)$. For any subset

A of $\mathbb{R}$, χ_A denotes the characteristic function of A . For $x \in \mathbb{R}$ and for $r > 0$, we set

$$B(x, r) = \{y \in \mathbb{R} : \max\{0, |x| - r\} < |y| < |x| + r\}$$

if $x \neq 0$ and

$$B_r = B(0, r) = (-r, r).$$

Let $1 \leq q, p \leq \infty$, and $r > 0$. For $f \in L^0(\mu)$ we define

$${}_r \|f\|_{q,p} = \begin{cases} \left\| \left[\int_{\mathbb{R}} (\tau_y |f|^q) \chi_{B_r}(x) d\mu(x) \right]^{\frac{1}{q}} \right\|_p & \text{if } q < \infty, \\ \left\| \|f \chi_{B(y,r)}\|_{\infty} \right\|_p & \text{if } q = \infty, \end{cases} \quad (4.1)$$

with the $L^p(\mu)$ -norm taken with respect to the variable y . Here and in the sequel, τ_y ($y \in \mathbb{R}$) stands for the Dunkl translation operator (see Sect. 4.2 for more details about this operator).

For $1 \leq q \leq \alpha \leq p \leq \infty$, the Dunkl–Fofana space $(L^q, L^p)^\alpha(\mu)$ is defined as the subspace of $L^0(\mu)$ such that $\|f\|_{q,p,\alpha} < \infty$, where

$$\|f\|_{q,p,\alpha} = \sup_{r>0} (\mu(B_r))^{\frac{1}{\alpha} - \frac{1}{q} - \frac{1}{p}} {}_r \|f\|_{q,p}.$$

It is proved in [21] that endowed with the norm $\|\cdot\|_{q,p,\alpha}$, $(L^q, L^p)^\alpha(\mu)$ is a complex Banach space.

Let $f \in L^0(\mu)$ and $0 \leq \beta < 2k+2$. The Dunkl-type fractional maximal operator M_β is defined by

$$M_\beta f(x) = \sup_{r>0} (\mu(B(0, r)))^{\frac{\beta}{2k+2} - 1} \int_{B(0,r)} \tau_x |f|(y) d\mu(y), \quad x \in \mathbb{R}.$$

In the case $\beta = 0$, the fractional maximal operator reduces to the Hardy–Littlewood maximal operator M associated with the Dunkl operator. For a given measurable function b , the commutator of Dunkl-type fractional maximal operator is denoted by $[b, M_\beta]$ and is formally defined by

$$[b, M_\beta]f = bM_\beta(f) - M_\beta(bf).$$

The Dunkl-type fractional maximal commutator $M_{b,\beta}$ is defined by

$$M_{b,\beta} f(x) = \sup_{r>0} (\mu(B_r))^{-1 + \frac{\beta}{2k+2}} \int_{B_r} \tau_x (|b(x) - b(y)| |f(y)|) d\mu(y), \quad x \in \mathbb{R}.$$

In the case $\beta = 0$, the fractional maximal commutator reduces to the Dunkl-type maximal commutator denoted by M_b .

The Dunkl-type fractional integral operator I_β , $0 < \beta < 2k + 2$, is defined as

$$I_\beta f(x) = \int_{\mathbb{R}} \tau_x f(z) |z|^{\beta-(2k+2)} d\mu(z), \quad x \in \mathbb{R}.$$

The commutator of Dunkl-type fractional integral operator is denoted by $[b, I_\beta]$ and is defined by

$$[b, I_\beta]f(x) = \int_{\mathbb{R}} |y|^{\beta-2k-2} \tau_x((b(x) - b(y))f(y)) d\mu(y), \quad x \in \mathbb{R}.$$

The Dunkl-type BMO space denoted by $BMO_k(\mathbb{R})$ is the class of functions f belonging to $L^1_{loc}(\mu)$ and satisfying $\|f\|_{BMO_k} < \infty$, where

$$\|f\|_{BMO_k} = \sup_{r>0, x \in \mathbb{R}} (\mu(B_r))^{-1} \int_{B_r} |\tau_x f(y) - f_{B_r}(x)| d\mu(y)$$

and $f_{B_r}(x) = (\mu(B_r))^{-1} \int_{B_r} \tau_x f(y) d\mu(y)$.

We shall prove the following:

Theorem 4.1 *Let $1 < q \leq \alpha \leq p < \infty$. Then the Dunkl-type maximal commutator M_b is bounded on $(L^q, L^p)^\alpha(\mu)$ if and only if $b \in BMO_k(\mathbb{R})$.*

As a consequence of Theorem 4.1, we obtain the following boundedness result for the commutator of Dunkl-type maximal operator.

Corollary 4.1 *Let $1 < q \leq \alpha \leq p < \infty$, $b \in BMO_k(\mathbb{R})$, and $b \geq 0$. Then*

$$\|[b, M]f\|_{q,p,\alpha} \leq C \|b\|_{BMO_k} \|f\|_{q,p,\alpha}, \quad f \in (L^q, L^p)^\alpha(\mu).$$

For the commutator of the Dunkl-type fractional integral operator, we have the following boundedness property.

Theorem 4.2 *Let $1 < q \leq \alpha \leq p < \infty$, $0 < \beta < \frac{2k+2}{\alpha}$, and $b \in BMO_k(\mathbb{R})$. Put*

$$\frac{1}{\alpha^*} = \frac{1}{\alpha} - \frac{\beta}{2k+2}, \quad \frac{1}{\bar{p}} = \frac{1}{p} \left(1 - \frac{\alpha\beta}{2k+2}\right) \text{ and } \frac{1}{\bar{q}} = \frac{1}{q} \left(1 - \frac{\alpha\beta}{2k+2}\right).$$

Then the commutator $[b, I_\beta]$ is bounded from $(L^q, L^p)^\alpha(\mu)$ to $(L^{\bar{q}}, L^{\bar{p}})^{\alpha^}(\mu)$.*

The result below gives a necessary and sufficient condition for the boundedness of the Dunkl-type fractional maximal commutator in Dunkl–Fofana spaces.

Theorem 4.3 *Let $1 < q \leq \alpha \leq p < \infty$ and $0 < \beta < \frac{2k+2}{\alpha}$. Put*

$$\frac{1}{\alpha^*} = \frac{1}{\alpha} - \frac{\beta}{2k+2}, \quad \frac{1}{\bar{p}} = \frac{1}{p} \left(1 - \frac{\alpha\beta}{2k+2}\right) \text{ and } \frac{1}{\bar{q}} = \frac{1}{q} \left(1 - \frac{\alpha\beta}{2k+2}\right).$$

Then the Dunkl-type fractional maximal commutator $M_{b,\beta}$ is bounded from $(L^q, L^p)^\alpha(\mu)$ to $(L^{\tilde{q}}, L^{\tilde{p}})^{\alpha^*}(\mu)$ if and only if $b \in BMO_k(\mathbb{R})$.

As a consequence of Theorem 4.3 we have the following result.

Corollary 4.2 *Let $1 < q \leq \alpha \leq p < \infty$, $0 < \beta < \frac{2k+2}{\alpha}$, $b \in BMO_k(\mathbb{R})$, and $b \geq 0$. Put*

$$\frac{1}{\alpha^*} = \frac{1}{\alpha} - \frac{\beta}{2k+2}, \quad \frac{1}{\tilde{p}} = \frac{1}{p} \left(1 - \frac{\alpha\beta}{2k+2}\right) \text{ and } \frac{1}{\tilde{q}} = \frac{1}{q} \left(1 - \frac{\alpha\beta}{2k+2}\right).$$

Then

$$\|[b, M_\beta]f\|_{\tilde{q}, \tilde{p}, \alpha^*} \leq C \|b\|_{BMO_k} \|f\|_{q, p, \alpha}, \quad f \in (L^q, L^p)^\alpha(\mu).$$

Since Dunkl–Fofana spaces are related to Dunkl–Lebesgue spaces and Dunkl-type Morrey spaces as described below, we have the following observations.

Remark 4.1 It is proved in [21] that, for $\alpha \in \{p, q\}$, $(L^q, L^p)^\alpha(\mu) = L^\alpha(\mu)$. Therefore, on the real line, Corollary 4.1 generalizes Theorem 4.6 in [20], while Theorem 4.2 and Corollary 4.2 generalize Theorem 4.7 in [20].

Furthermore, it is easy to see that for $1 \leq q \leq \alpha < \infty$, the space $(L^q, L^\infty)^\alpha(\mu)$ is the Dunkl-type Morrey space $L_{q, (2k+2)(1-\frac{q}{\alpha}), k}$ defined in [18]. Therefore, Theorem 9 established in [18] can be viewed as the analogue of Theorem 4.1 in the limiting case $p = \infty$ and $1 < q \leq \alpha < \infty$.

The letter C will be used for positive constants not depending on the relevant variables, and these constants may change from one occurrence to another.

The remainder of this note is organized as follows: Sect. 4.2 contains the prerequisite for Dunkl analysis on the real line. In Sect. 4.3 we recall some properties of the Dunkl–Fofana spaces. Section 4.4 is devoted to the proofs of Theorem 4.1 and Corollary 4.1. In Sect. 4.5, we give the proofs of Theorem 4.2, Theorem 4.3, and Corollary 4.2.

4.2 Prerequisite on Dunkl Analysis on the Real Line

The Dunkl operator associated with the reflection group $\mathbb{Z}_2$ on $\mathbb{R}$ is defined by

$$\Lambda_k f(x) = \frac{df}{dx}(x) + \frac{2k+1}{x} \left(\frac{f(x) - f(-x)}{2} \right).$$

For $\lambda \in \mathbb{C}$, the Dunkl kernel denoted by $\mathfrak{E}_k(\lambda)$ (see [9]) is the only solution of the initial value problem

$$\Lambda_k f(x) = \lambda f(x), \quad f(0) = 1, \quad x \in \mathbb{R}.$$

It is given by the formula

$$\mathfrak{E}_k(\lambda x) = j_k(i\lambda x) + \frac{\lambda x}{2(k+1)} j_{k+1}(i\lambda x), \quad x \in \mathbb{R},$$

where

$$j_k(z) = 2^k \Gamma(k+1) \frac{J_k(z)}{z^k} = \Gamma(k+1) \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{n! 2^{2n} \Gamma(n+k+1)}, \quad z \in \mathbb{C},$$

is the normalized Bessel function of the first kind and of order k . Notice that $\Lambda_{-\frac{1}{2}} = \frac{d}{dx}$ and $\mathfrak{E}_{-\frac{1}{2}}(\lambda x) = e^{\lambda x}$. It is also well known (see [24]) that $|\mathfrak{E}_k(ix)| \leq 1$ for every $x \in \mathbb{R}$.

The Dunkl kernel $\mathfrak{E}_k$ gives rise to an integral transform on $\mathbb{R}$ denoted $\mathcal{F}_k$ and called the Dunkl transform (see [7]). For $f \in L^1(\mu)$

$$\mathcal{F}_k f(\lambda) = \int_{\mathbb{R}} \mathfrak{E}_k(-i\lambda x) f(x) d\mu(x), \quad \lambda \in \mathbb{R}.$$

We have the following properties of the Dunkl transform given in [7] (see also [10]).

Proposition 4.1 *1. Let $f \in L^1(\mu)$. If $\mathcal{F}_k(f)$ is in $L^1(\mu)$, then we have the following inversion formula:*

$$f(x) = c \int_{\mathbb{R}} \mathfrak{E}_k(ixy) \mathcal{F}_k(f)(y) d\mu(y).$$

2. The Dunkl transform has a unique extension to an isometric isomorphism on $L^2(\mu)$.

Let $x, y, z \in \mathbb{R}$. We put

$$W_k(x, y, z) = [1 - \sigma_{x,y,z} + \sigma_{z,x,y} + \sigma_{z,y,x}] \Delta_k(|x|, |y|, |z|),$$

where

$$\sigma_{x,y,z} = \begin{cases} \frac{x^2 + y^2 - z^2}{2xy} & \text{if } x, y \in \mathbb{R} \setminus \{0\}, \\ 0 & \text{otherwise} \end{cases}$$

and Δ_k is the Bessel kernel given by

$$\Delta_k(|x|, |y|, |z|) = \begin{cases} b_k \frac{[(|x|+|y|)^2 - z^2](z^2 - (|x|-|y|)^2)^{k-\frac{1}{2}}}{|xyz|^{2k}} & \text{if } |z| \in \mathcal{A}_{x,y}, \\ 0 & \text{otherwise,} \end{cases}$$

with $b_k = 2^{1-k} \frac{(\Gamma(k+1))^2}{\sqrt{\pi} \Gamma(k+\frac{1}{2})}$ and $\mathcal{A}_{x,y} = [|x| - |y|, |x| + |y|]$.

Remark 4.2 ([24])

$$W_k(x, y, z) = W_k(y, x, z) = W_k(-x, z, y)$$

$$W_k(x, y, z) = W_k(-z, y, -x) = W_k(-x, -y, -z)$$

and

$$\int_{\mathbb{R}} |W_k(x, y, z)| d\mu_k(z) \leq 4.$$

In the sequel, we consider the signed measure $\nu_{x,y}$ on $\mathbb{R}$ given by

$$d\nu_{x,y}(z) = \begin{cases} W_k(x, y, z) d\mu_k(z) & \text{if } x, y \in \mathbb{R} \setminus \{0\}, \\ d\delta_x(z) & \text{if } y = 0, \\ d\delta_y(z) & \text{if } x = 0, \end{cases}$$

with $\text{supp}(\nu_{x,y}) = (-\mathcal{A}_{x,y}) \cup \mathcal{A}_{x,y} \quad \forall (x, y) \in \mathbb{R} \setminus \{0\} \times \mathbb{R} \setminus \{0\}$.

Definition 4.1 For $x, y \in \mathbb{R}$ and f a continuous function on $\mathbb{R}$, we put

$$\tau_x f(y) = \int_{\mathbb{R}} f(z) d\nu_{x,y}(z).$$

The operators $\tau_x, x \in \mathbb{R}$, are called Dunkl translation operators on $\mathbb{R}$.

Let f and g be two continuous functions on $\mathbb{R}$ with compact support. We define the generalized convolution $*_k$ of f and g by

$$f *_k g(x) = \int_{\mathbb{R}} \tau_x f(-y) g(y) d\mu(y).$$

The generalized convolution $*_k$ is associative and commutative (see [24]). We also have the below result.

Proposition 4.2 (See Soltani [25])

1. For all $x \in \mathbb{R}$, the operator τ_x extends to $L^p(\mu)$, $p \geq 1$, and

$$\|\tau_x f\|_p \leq 4 \|f\|_p \tag{4.2}$$

for all $f \in L^p(\mu)$.

2. Assume that $p, q, r \in [1, \infty]$ and satisfy $\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r}$. Then the generalized convolution defined on $C_c \times C_c$ extends to a continuous map from $L^p(\mu) \times L^q(\mu)$ to $L^r(\mu)$, and we have

$$\|f *_k g\|_r \leq 4 \|f\|_p \|g\|_q.$$

It is also proved in [13] that if $f \in L^1(\mu)$ and $g \in L^p(\mu)$, $1 \leq p < \infty$, then

$$\tau_x(f *_{\mathbf{k}} g) = \tau_x f *_{\mathbf{k}} g = f *_{\mathbf{k}} \tau_x g, x \in \mathbb{R}.$$

For any $f \in L^1_{loc}(\mu)$, the following analogue of the Lebesgue differentiation theorem holds (see [25]):

$$\lim_{r \rightarrow 0} \frac{1}{\mu(B_r)} \int_{B_r} |\tau_x f(y) - f(x)| d\mu(y) = 0 \quad \text{for a.e. } x \in \mathbb{R} \quad (4.3)$$

and

$$\lim_{r \rightarrow 0} \frac{1}{\mu(B_r)} \int_{B_r} \tau_x f(y) d\mu(y) = f(x) \quad \text{for a.e. } x \in \mathbb{R}. \quad (4.4)$$

We end this section by the following well-known equality:

$$\mu(B_r) = d_k r^{2k+2}, \quad r > 0,$$

where $d_k = (2^{k+1}(k+1)\Gamma(k+1))^{-1}$.

4.3 Some Properties of the Dunkl–Fofana Spaces

For $1 \leq q \leq \infty$, we will denote by q' its conjugate exponent: $\frac{1}{q} + \frac{1}{q'} = 1$, with the usual convention that $\frac{1}{\infty} = 0$. The results of this section are proved in [21] and [22].

Hölder's inequality remains valid in Dunkl–Fofana spaces as stated below.

Proposition 4.3 *Let (q_1, p_1, α_1) and (q_2, p_2, α_2) be two elements of $[1, +\infty]^3$ such that $q_1 \leq \alpha_1 \leq p_1$ and $q_2 \leq \alpha_2 \leq p_2$.*

If

$$\frac{1}{q_1} + \frac{1}{q_2} = \frac{1}{q} \leq 1, \quad \frac{1}{p_1} + \frac{1}{p_2} = \frac{1}{p} \leq 1 \quad \text{and} \quad \frac{1}{\alpha_1} + \frac{1}{\alpha_2} = \frac{1}{\alpha},$$

then for all f and g belonging to $L^0(\mu)$, we have

$$\|fg\|_{q,p,\alpha} \leq \|f\|_{q_1,p_1,\alpha_1} \|g\|_{q_2,p_2,\alpha_2}.$$

Proposition 4.4 *Let $1 \leq q \leq \alpha \leq p \leq \infty$. Then we have*

$$\|f\|_{q,p,\alpha} \leq 4^{\frac{1}{q}} \|f\|_{\alpha}.$$

The family of spaces $(L^q, L^p)^\alpha(\mu)$ is increasing with respect to the exponent p , and if $p_1 \leq p_2$, there exists a constant $C > 0$ such that

$$\|f\|_{q,p_2,\alpha} \leq C \|f\|_{q,p_1,\alpha}. \quad (4.5)$$

The following result states the boundedness property of M (the Hardy–Littlewood maximal operator associated with the Dunkl operator) on Dunkl–Fofana spaces.

Proposition 4.5 *Let $1 < q \leq \alpha \leq p \leq \infty$. There exists a constant $C > 0$ such that*

$$\|Mf\|_{q,p,\alpha} \leq C \|f\|_{q,p,\alpha}, \quad f \in (L^q, L^p)^\alpha(\mu).$$

For the Dunkl-type fractional integral operator we have the following:

Proposition 4.6 *Let $1 < q \leq \alpha \leq p < \infty$ and $0 < \beta < \frac{2k+2}{\alpha}$. Put*

$$\frac{1}{\alpha^*} = \frac{1}{\alpha} - \frac{\beta}{2k+2}, \quad \frac{1}{\bar{p}} = \frac{1}{p} \left(1 - \frac{\alpha\beta}{2k+2}\right) \text{ and } \frac{1}{\bar{q}} = \frac{1}{q} \left(1 - \frac{\alpha\beta}{2k+2}\right).$$

Then

$$\|I_\beta f\|_{\bar{q},\bar{p},\alpha^*} \leq C \|f\|_{q,p,\alpha}^{1-\frac{\alpha\beta}{2k+2}} \|f\|_{q,\infty,\alpha}^{\frac{\alpha\beta}{2k+2}}, \quad f \in (L^q, L^p)^\alpha(\mu)$$

and

$$\|I_\beta f\|_{\bar{q},\bar{p},\alpha^*} \leq C \|f\|_{q,p,\alpha}, \quad f \in (L^q, L^p)^\alpha(\mu).$$

As a consequence of Proposition 4.6, we obtain the corresponding boundedness results for the Dunkl-type fractional maximal operator M_β .

Corollary 4.3 *Let $1 < q \leq \alpha \leq p < \infty$ and $0 < \beta < \frac{2k+2}{\alpha}$. Put*

$$\frac{1}{\alpha^*} = \frac{1}{\alpha} - \frac{\beta}{2k+2}, \quad \frac{1}{\bar{p}} = \frac{1}{p} \left(1 - \frac{\alpha\beta}{2k+2}\right) \text{ and } \frac{1}{\bar{q}} = \frac{1}{q} \left(1 - \frac{\alpha\beta}{2k+2}\right).$$

Then

$$\|M_\beta f\|_{\bar{q},\bar{p},\alpha^*} \leq C \|f\|_{q,p,\alpha}, \quad f \in (L^q, L^p)^\alpha(\mu).$$

4.4 Proofs of Theorem 4.1 and Corollary 4.1

The following result (see [18]) will be useful in the proof of Theorem 4.1.

Lemma 4.1 *Let $1 < s < \infty$ and $b \in BMO_k(\mathbb{R})$. Then, there exists a constant $C > 0$ such that for all $f \in L^1_{loc}(\mu)$ and for all $x \in \mathbb{R}$,*

$$M(M_b f)(x) \leq C \|b\|_{BMO_k} \left((M(Mf)^s)^{\frac{1}{s}}(x) + M(M|f|^s)^{\frac{1}{s}}(x) \right).$$

Proof (Proof of Theorem 4.1) Let $1 < q \leq \alpha \leq p < \infty$.

(i) Assume that $b \in BMO_k(\mathbb{R})$ and $f \in (L^q, L^p)^\alpha(\mu)$. Put $s = \frac{1+q}{2}$. It follows from (4.4) and Lemma 4.1 that

$$M_b f(x) \leq C \|b\|_{BMO_k} \left((M(Mf)^s)^{\frac{1}{s}}(x) + M(M|f|^s)^{\frac{1}{s}}(x) \right), \quad \text{for a.e. } x \in \mathbb{R}.$$

By taking the $(L^q, L^p)^\alpha(\mu)$ -norm of the both sides of the above estimation, we get

$$\|M_b f\|_{q,p,\alpha} \leq C \|b\|_{BMO_k} \left(\left\| (M(Mf)^s)^{\frac{1}{s}} \right\|_{q,p,\alpha} + \left\| M(M|f|^s)^{\frac{1}{s}} \right\|_{q,p,\alpha} \right).$$

According to Proposition 4.5, we have on the one hand

$$\begin{aligned} \left\| (M(Mf)^s)^{\frac{1}{s}} \right\|_{q,p,\alpha} &= \left\| (M(Mf)^s)^{\frac{1}{s}} \right\|_{\frac{q}{s}, \frac{p}{s}, \frac{\alpha}{s}} \\ &\leq C \left\| (Mf)^s \right\|_{\frac{q}{s}, \frac{p}{s}, \frac{\alpha}{s}}^{\frac{1}{s}}. \end{aligned}$$

It follows that

$$\left\| (M(Mf)^s)^{\frac{1}{s}} \right\|_{q,p,\alpha} \leq C \|f\|_{q,p,\alpha}. \quad (4.6)$$

On the other hand, applying again Proposition 4.5, we have

$$\begin{aligned} \left\| M(M|f|^s)^{\frac{1}{s}} \right\|_{q,p,\alpha} &\leq C \left\| (M|f|^s)^{\frac{1}{s}} \right\|_{q,p,\alpha} \\ &\leq C \left\| M|f|^s \right\|_{\frac{q}{s}, \frac{p}{s}, \frac{\alpha}{s}}^{\frac{1}{s}}. \end{aligned}$$

It follows that

$$\left\| M(M|f|^s)^{\frac{1}{s}} \right\|_{q,p,\alpha} \leq C \|f\|_{q,p,\alpha}. \quad (4.7)$$

From (4.6) and (4.7) we deduce that

$$\|M_b f\|_{q,p,\alpha} \leq C \|b\|_{BMO_k} \|f\|_{q,p,\alpha}.$$

(ii) Conversely, assume that the commutator M_b is bounded on $(L^q, L^p)^\alpha(\mu)$.

Let $t > 0$. Since $\chi_{B_t} \in L^\alpha(\mu)$, it follows from Proposition 4.4 that $\chi_{B_t} \in (L^q, L^p)^\alpha(\mu)$. Therefore, there exists a constant $C > 0$ such that

$$\|M_b \chi_{B_t}\|_{q,p,\alpha} \leq C \|\chi_{B_t}\|_{q,p,\alpha}.$$

Also, we have (see [18])

$$(\mu(B_t))^{-1} \int_{B_t} |\tau_x b(y) - b_{B_t}| d\mu(y) \leq (\mu(B_t))^{-1} \int_{B_t} M_b \chi_{B_t}(y) d\mu(y).$$

Applying Proposition 4.3 and Proposition 4.4, we have

$$\begin{aligned} (\mu(B_t))^{-1} \int_{B_t} M_b \chi_{B_t}(y) d\mu(y) &\leq (\mu(B_t))^{-1} \|M_b \chi_{B_t}\|_{q,p,\alpha} \|\chi_{B_t}\|_{q',p',\alpha'} \\ &\leq C (\mu(B_t))^{-1} \|\chi_{B_t}\|_{q,p,\alpha} \|\chi_{B_t}\|_{q',p',\alpha'} \\ &\leq C (\mu(B_t))^{-1} 4^{\frac{1}{q}} \|\chi_{B_t}\|_\alpha 4^{\frac{1}{q'}} \|\chi_{B_t}\|_{\alpha'} \\ &\leq C (\mu(B_t))^{-1 + \frac{1}{\alpha} + \frac{1}{\alpha'}}. \end{aligned}$$

Thus, there exists a constant $C > 0$ such that

$$(\mu(B_t))^{-1} \int_{B_t} |\tau_x b(y) - b_{B_t}| d\mu(y) \leq C.$$

By taking in the left-hand side the supremum over all $t > 0$ and $x \in \mathbb{R}$, we obtain $\|b\|_{BMO_k} < \infty$. This ends the proof. $\square$

Proof (Proof of Corollary 4.1) Let $1 < q \leq \alpha \leq p < \infty$, $b \in BMO_k(\mathbb{R})$ and $b \geq 0$. Let $r > 0$, $f \in (L^q, L^p)^\alpha(\mu)$ and $x \in \mathbb{R}$.

Put

$$A = (\mu(B_r))^{-1} \int_{B_r} \tau_x (|b(y)| |f(y)|) d\mu(y)$$

and

$$B = b(x) (\mu(B_r))^{-1} \int_{B_r} \tau_x |f(y)| d\mu(y).$$

We have

$$\begin{aligned} &|A - B| \\ &= \left| (\mu(B_r))^{-1} \int_{B_r} \tau_x (|b(y)| |f(y)|) d\mu(y) - b(x) (\mu(B_r))^{-1} \int_{B_r} \tau_x |f(y)| d\mu(y) \right| \end{aligned}$$

$$\begin{aligned}
&\leq (\mu(B_r))^{-1} \int_{\mathbb{R}} ||b(y)f(y)| - |b(x)f(y)|| \tau_{-x} \chi_{B_r}(y) d\mu(y) \\
&\leq (\mu(B_r))^{-1} \int_{\mathbb{R}} |b(y)f(y) - b(x)f(y)| \tau_{-x} \chi_{B_r}(y) d\mu(y) \\
&\leq (\mu(B_r))^{-1} \int_{B_r} \tau_x |(b(y) - b(x))f(y)| d\mu(y) \leq M_b f(x).
\end{aligned}$$

The following implications hold.

$$\begin{aligned}
A - B \leq M_b f(x) &\implies A \leq M_b f(x) + B \\
&\implies M(bf)(x) \leq M_b f(x) + b(x)M(f)(x).
\end{aligned}$$

Hence,

$$M(bf)(x) - b(x)M(f)(x) \leq M_b f(x). \quad (4.8)$$

Also,

$$\begin{aligned}
-M_b f(x) \leq A - B &\implies B \leq A + M_b f(x) \\
&\implies b(x)M(f)(x) \leq M(bf)(x) + M_b f(x).
\end{aligned}$$

Hence,

$$b(x)M(f)(x) - M(bf)(x) \leq M_b f(x). \quad (4.9)$$

Thus, from (4.8) and (4.9), we obtain

$$|[b, M]f| \leq M_b f. \quad (4.10)$$

We end the proof by taking the norm $\|\cdot\|_{q,p,\alpha}$ on the both sides of (4.10) and by applying Theorem 4.1. $\square$

4.5 Proofs of Theorems 4.2, 4.3, and Corollary 4.2

For a given measurable function b , let us introduce the commutator $|b, I_\beta|$, $0 < \beta < 2k + 2$, defined by

$$|b, I_\beta|f(x) = \int_{\mathbb{R}} |y|^{\beta-2k-2} \tau_x |(b(x) - b(y))f(y)| d\mu(y), \quad x \in \mathbb{R}.$$

The following result is a necessary and sufficient condition for the boundedness of $|b, I_\beta|$ in Dunkl–Fofana spaces. It is the main tool of this section.

Theorem 4.4 *Let $1 < q \leq \alpha \leq p < \infty$ and $0 < \beta < \frac{2k+2}{\alpha}$. Put*

$$\frac{1}{\alpha^*} = \frac{1}{\alpha} - \frac{\beta}{2k+2}, \quad \frac{1}{\bar{p}} = \frac{1}{p} \left(1 - \frac{\alpha\beta}{2k+2}\right) \text{ and } \frac{1}{\bar{q}} = \frac{1}{q} \left(1 - \frac{\alpha\beta}{2k+2}\right).$$

Then the commutator $|b, I_\beta|$ is bounded from $(L^q, L^p)^\alpha(\mu)$ to $(L^{\bar{q}}, L^{\bar{p}})^{\alpha^}(\mu)$ if and only if $b \in BMO_k(\mathbb{R})$.*

Proof (i) Assume that $b \in BMO_k(\mathbb{R})$ and $f \in (L^q, L^p)^\alpha(\mu) \setminus \{0\}$ since the case $f = 0$ is obvious.

Let $r > 0$ and $x \in \mathbb{R}$.

$$\begin{aligned} |b, I_\beta|f(x) &= \int_{B(0,r)} |y|^{\beta-2k-2} \tau_x(b(x) - b(y)) f(y) d\mu(y) \\ &\quad + \int_{B(0,r)^c} |y|^{\beta-2k-2} \tau_x(b(x) - b(y)) f(y) d\mu(y) = F_1(x, r) + F_2(x, r). \end{aligned}$$

By proceeding as in the proof of Theorem 3.1 in [19] and by using Hölder's inequality (with $s = \frac{1+q}{2}$), we get the following estimation for $F_1(x, r)$.

$$F_1(x, r) \leq Cr^\beta \|b\|_{BMO_k} (M(|f|^s)^{\frac{1}{s}})(x). \quad (4.11)$$

We now estimate $F_2(x, r)$. By proceeding again as in the proof of Theorem 3.1 in [19] and by using Hölder's inequality, we have

$$\begin{aligned} F_2(x, r) &\leq \|b\|_{BMO_k} \sum_{i=0}^{\infty} (2^i r)^{\beta-2k-2} \mu(B(0, 2^{i+1}r))^{\frac{1}{q}} \mu(B(0, 2^{i+1}r))^{-\frac{1}{\alpha} + \frac{1}{q}} \\ &\quad \times \mu(B(0, 2^{i+1}r))^{\frac{1}{\alpha} - \frac{1}{q}} \left\| \left(\int_{B(0, 2^{i+1}r)} \tau_{(\cdot)} |f(y)|^q d\mu(y) \right)^{\frac{1}{q}} \right\|_{\infty} \\ &\leq \|b\|_{BMO_k} \|f\|_{q, \infty, \alpha} \sum_{i=0}^{\infty} (2^i r)^{\beta-2k-2} \mu(B(0, 2^{i+1}r))^{-\frac{1}{\alpha} + 1} \\ &\leq C \|b\|_{BMO_k} \|f\|_{q, \infty, \alpha} \sum_{i=0}^{\infty} (2^i r)^{\beta-2k-2} \mu(B(0, 2^i r))^{-\frac{1}{\alpha} + 1} \end{aligned}$$

because μ is doubling.

Since $\mu(B(0, 2^i r))^{1-\frac{1}{\alpha}} = d_k^{1-\frac{1}{\alpha}} (2^i r)^{(2k+2)(1-\frac{1}{\alpha})}$, we have

$$\sum_{i=0}^{\infty} (2^i r)^{\beta-2k-2} \mu(B(0, 2^i r))^{1-\frac{1}{\alpha}} = d_k^{1-\frac{1}{\alpha}} r^{\beta-\frac{2k+2}{\alpha}} \sum_{i=0}^{\infty} (2^i)^{\beta-\frac{2k+2}{\alpha}}$$

and $\sum_{i=0}^{\infty} (2^i)^{\beta-\frac{2k+2}{\alpha}} < \infty$ because $\beta - \frac{2k+2}{\alpha} < 0$.

Hence,

$$F_2(x, r) \leq C \|b\|_{BMO_k} \|f\|_{q, \infty, \alpha} r^{\beta-\frac{2k+2}{\alpha}}. \quad (4.12)$$

It follows from (4.11) and (4.12) that

$$|b, I_{\beta}|f(x) \leq C \|b\|_{BMO_k} (r^{\beta} (M(|f|^s))^{\frac{1}{s}}(x) + \|f\|_{q, \infty, \alpha} r^{\beta-\frac{2k+2}{\alpha}})$$

for all $r > 0$. Taking

$$r = \left(\frac{\|f\|_{q, \infty, \alpha}}{(M(|f|^s))^{\frac{1}{s}}(x)} \right)^{\frac{\alpha}{2k+2}},$$

we get

$$\begin{aligned} r \|b, I_{\beta}|f\|_{\bar{q}, \bar{p}} &\leq C \|b\|_{BMO_k} \|f\|_{q, \infty, \alpha}^{\frac{\alpha\beta}{2k+2}} r \|(M(|f|^s))^{\frac{1}{s}}(1-\frac{\alpha\beta}{2k+2})\|_{\bar{q}, \bar{p}} \\ &\leq C \|b\|_{BMO_k} \|f\|_{q, \infty, \alpha}^{\frac{\alpha\beta}{2k+2}} r \|M(|f|^s)\|_{\frac{\bar{q}}{s}, \frac{\bar{p}}{s}(1-\frac{\alpha\beta}{2k+2})}^{\frac{1}{s}(1-\frac{\alpha\beta}{2k+2})} \\ &\leq C \|b\|_{BMO_k} \|f\|_{q, \infty, \alpha}^{\frac{\alpha\beta}{2k+2}} r \|M(|f|^s)\|_{\frac{q}{s}, \frac{p}{s}}^{\frac{1}{s}(1-\frac{\alpha\beta}{2k+2})}. \end{aligned}$$

Multiplying the both sides of the previous inequality by $\mu(B(0, r))^{\frac{1}{\alpha^*}-\frac{1}{q}-\frac{1}{p}}$, we obtain

$$\begin{aligned} &\mu(B(0, r))^{\frac{1}{\alpha^*}-\frac{1}{q}-\frac{1}{p}} r \|b, I_{\beta}|f\|_{\bar{q}, \bar{p}} \\ &\leq C \|b\|_{BMO_k} \|f\|_{q, \infty, \alpha}^{\frac{\alpha\beta}{2k+2}} \mu(B(0, r))^{\frac{1}{\alpha^*}-\frac{1}{q}-\frac{1}{p}} r \|M(|f|^s)\|_{\frac{q}{s}, \frac{p}{s}}^{\frac{1}{s}(1-\frac{\alpha\beta}{2k+2})} \\ &\leq C \|b\|_{BMO_k} \|f\|_{q, \infty, \alpha}^{\frac{\alpha\beta}{2k+2}} (\mu(B(0, r))^{\frac{1}{\alpha}-\frac{1}{q}-\frac{1}{p}})^{(1-\frac{\alpha\beta}{2k+2})} r \|M(|f|^s)\|_{\frac{q}{s}, \frac{p}{s}}^{\frac{1}{s}(1-\frac{\alpha\beta}{2k+2})} \\ &\leq C \|b\|_{BMO_k} \|f\|_{q, \infty, \alpha}^{\frac{\alpha\beta}{2k+2}} \left(\mu(B(0, r))^{\frac{s}{\alpha}-\frac{s}{q}-\frac{s}{p}} r \|M(|f|^s)\|_{\frac{q}{s}, \frac{p}{s}} \right)^{\frac{1}{s}(1-\frac{\alpha\beta}{2k+2})}. \end{aligned}$$

Hence,

$$\| |b, I_\beta| f \|_{\bar{q}, \bar{p}, \alpha^*} \leq C \|b\|_{BMO_k} \|f\|_{q, \infty, \alpha}^{\frac{\alpha\beta}{2k+2}} \|M(|f|^s)\|_{\frac{q}{s}, \frac{p}{s}, \frac{\alpha}{s}}^{\frac{1}{s}(1-\frac{\alpha\beta}{2k+2})}.$$

By applying Proposition 4.5 and (4.5), we get

$$\| |b, I_\beta| f \|_{\bar{q}, \bar{p}, \alpha^*} \leq C \|b\|_{BMO_k} \|f\|_{q, p, \alpha}.$$

(ii) Conversely, assume that the commutator $|b, I_\beta|$ is bounded from $(L^q, L^p)^\alpha(\mu)$ to $(L^{\bar{q}}, L^{\bar{p}})^{\alpha^*}(\mu)$. Let $t > 0$. Since $\chi_{B_t} \in L^\alpha(\mu)$, it follows from Proposition 4.4 that $\chi_{B_t} \in (L^q, L^p)^\alpha(\mu)$. Therefore, there exists a constant $C > 0$ such that

$$\| |b, I_\beta| \chi_{B_t} \|_{\bar{q}, \bar{p}, \alpha^*} \leq C \| \chi_{B_t} \|_{q, p, \alpha}.$$

Also, we have (see [19])

$$\begin{aligned} & (\mu(B_t))^{-1} \int_{B_t} |\tau_x b(y) - b_{B_t}| d\mu(y) \\ & \leq d_k^{-1+\frac{\beta}{2k+2}} (\mu(B_t))^{-1-\frac{\beta}{2k+2}} \int_{B_t} |b, I_\beta| \chi_{B_t}(y) d\mu(y). \end{aligned}$$

Applying Proposition 4.3 and Proposition 4.4, we have

$$\begin{aligned} & (\mu(B_t))^{-1-\frac{\beta}{2k+2}} \int_{B_t} |b, I_\beta| \chi_{B_t}(y) d\mu(y) \\ & \leq (\mu(B_t))^{-1-\frac{\beta}{2k+2}} \| |b, I_\beta| \chi_{B_t} \|_{\bar{q}, \bar{p}, \alpha^*} \| \chi_{B_t} \|_{\bar{q}', \bar{p}', \alpha^{*'}} \\ & \leq (\mu(B_t))^{-1-\frac{\beta}{2k+2}} \| \chi_{B_t} \|_{q, p, \alpha} \| \chi_{B_t} \|_{\bar{q}', \bar{p}', \alpha^{*'}} \\ & \leq (\mu(B_t))^{-1-\frac{\beta}{2k+2}} 4^{\frac{1}{q}} \| \chi_{B_t} \|_{\alpha} 4^{\frac{1}{q'}} \| \chi_{B_t} \|_{\alpha^{*'}} \\ & \leq 4^{\frac{1}{q}(2-\frac{\alpha\beta}{2k+2})} (\mu(B_t))^{-1-\frac{\beta}{2k+2}+\frac{1}{\alpha}+\frac{1}{\alpha^{*'}}} = 4^{\frac{1}{q}(2-\frac{\alpha\beta}{2k+2})}. \end{aligned}$$

Thus, there exists a constant $C > 0$ such that

$$(\mu(B_t))^{-1} \int_{B_t} |\tau_x b(y) - b_{B_t}| d\mu(y) \leq C.$$

By taking in the left-hand side the supremum over all $t > 0$ and $x \in \mathbb{R}$, we obtain $\|b\|_{BMO_k} < \infty$. This ends the proof. $\square$

Theorem 4.2 is a direct consequence of Theorem 4.4.

Proof (Proof of Theorem 4.2) Let $1 < q \leq \alpha \leq p < \infty$, $0 < \beta < \frac{2k+2}{\alpha}$, and $f \in (L^q, L^p)^\alpha(\mu)$. Let $x \in \mathbb{R}$. We have

$$\begin{aligned} |[b, I_\beta]f(x)| &= \left| \int_{\mathbb{R}} \tau_{-x}|y|^{\beta-2k-2}((b(x) - b(y))f(y))d\mu(y) \right| \\ &\leq \int_{\mathbb{R}} \tau_{-x}|y|^{\beta-2k-2}|(b(x) - b(y))f(y)|d\mu(y) \\ &\leq |b, I_\beta|f(x). \end{aligned}$$

So, the desired result follows by applying Theorem 4.4. $\square$

Theorem 4.3 is another consequence of Theorem 4.4.

Proof (Proof of Theorem 4.3) Let $1 < q \leq \alpha \leq p < \infty$ and $0 < \beta < \frac{2k+2}{\alpha}$. Put

$$\frac{1}{\alpha^*} = \frac{1}{\alpha} - \frac{\beta}{2k+2}, \quad \frac{1}{\bar{p}} = \frac{1}{p} \left(1 - \frac{\alpha\beta}{2k+2}\right) \text{ and } \frac{1}{\bar{q}} = \frac{1}{q} \left(1 - \frac{\alpha\beta}{2k+2}\right).$$

(i) Assume that $b \in BM O_k(\mathbb{R})$ and $f \in (L^q, L^p)^\alpha(\mu) \setminus \{0\}$ since the case $f = 0$ is obvious. Let $x \in \mathbb{R}$ and $r > 0$. We have

$$\begin{aligned} &\frac{1}{\mu(B(0, r))^{1-\frac{\beta}{2k+2}}} \int_{B(0, r)} \tau_x(|b(x) - b(z)| |f(z)|)d\mu(z) \\ &= \int_{B(0, r)} \tau_x(|b(x) - b(z)| |f(z)|) \frac{1}{\mu(B(0, r))^{1-\frac{\beta}{2k+2}}} d\mu(z) \\ &= \int_{B(0, r)} \tau_x(|b(x) - b(z)| |f(z)|) \frac{1}{(d_k r^{2k+2})^{1-\frac{\beta}{2k+2}}} d\mu(z) \\ &= d_k^{\frac{\beta}{2k+2}-1} \int_{B(0, r)} \tau_x(|b(x) - b(z)| |f(z)|) \frac{1}{r^{2k+2-\beta}} d\mu(z) \\ &\leq d_k^{\frac{\beta}{2k+2}-1} \int_{B(0, r)} \tau_x(|b(x) - b(z)| |f(z)|) |z|^{\beta-2k-2} d\mu(z) \\ &\leq d_k^{\frac{\beta}{2k+2}-1} |b, I_\beta|f(x). \end{aligned}$$

Hence,

$$M_{b, \beta} f(x) \leq d_k^{\frac{\beta}{2k+2}-1} |b, I_\beta|f(x).$$

So, by using Theorem 4.4, we obtain the desired result.

(ii) Conversely, assume that the commutator $M_{b, \beta}$ is bounded from $(L^q, L^p)^\alpha(\mu)$ to $(L^{\bar{q}}, L^{\bar{p}})^{\alpha^*}(\mu)$. Let $t > 0$. Since $\chi_{B_t} \in L^\alpha(\mu)$, it follows from Proposition 4.4

that $\chi_{B_t} \in (L^q, L^p)^\alpha(\mu)$. Therefore, there exists a constant $C > 0$ such that

$$\|M_{b,\beta} \chi_{B_t}\|_{\vec{q},\vec{p},\alpha^*} \leq C \|\chi_{B_t}\|_{q,p,\alpha}.$$

Proceeding as in the proof of Theorem 9 in [18], we have

$$\begin{aligned} & (\mu(B_t))^{-1} \int_{B_t} |\tau_x b(y) - b_{B_t}| d\mu(y) \\ & \leq d_k^{-1+\frac{\beta}{2k+2}} (\mu(B_t))^{-1-\frac{\beta}{2k+2}} \int_{B_t} M_{b,\beta} \chi_{B_t}(y) d\mu(y). \end{aligned}$$

Applying Propositions 4.3 and 4.4, we have

$$\begin{aligned} & (\mu(B_t))^{-1-\frac{\beta}{2k+2}} \int_{B_t} M_{b,\beta} \chi_{B_t}(y) d\mu(y) \\ & \leq (\mu(B_t))^{-1-\frac{\beta}{2k+2}} \|M_{b,\beta} \chi_{B_t}\|_{\vec{q},\vec{p},\alpha^*} \|\chi_{B_t}\|_{\vec{q}',\vec{p}',\alpha^{*'}} \\ & \leq (\mu(B_t))^{-1-\frac{\beta}{2k+2}} \|\chi_{B_t}\|_{q,p,\alpha} \|\chi_{B_t}\|_{\vec{q}',\vec{p}',\alpha^{*'}} \\ & \leq (\mu(B_t))^{-1-\frac{\beta}{2k+2}} 4^{\frac{1}{q}} \|\chi_{B_t}\|_{\alpha} 4^{\frac{1}{q'}} \|\chi_{B_t}\|_{\alpha^{*'}} \\ & \leq 4^{\frac{1}{q}(2-\frac{\alpha\beta}{2k+2})} (\mu(B_t))^{-1-\frac{\beta}{2k+2}+\frac{1}{\alpha}+\frac{1}{\alpha^{*'}}} = 4^{\frac{1}{q}(2-\frac{\alpha\beta}{2k+2})}. \end{aligned}$$

Thus, there exists a constant $C > 0$ such that

$$(\mu(B_t))^{-1} \int_{B_t} |\tau_x b(y) - b_{B_t}| d\mu(y) \leq C.$$

By taking in the left-hand side the supremum over all $t > 0$ and $x \in \mathbb{R}$, we obtain $\|b\|_{BMO_k} < \infty$. This ends the proof. $\square$

Proof (Proof of Corollary 4.2) Let $1 < q \leq \alpha \leq p < \infty$, $0 < \beta < \frac{2k+2}{\alpha}$ and $f \in (L^q, L^p)^\alpha(\mu)$. Put

$$\frac{1}{\alpha^*} = \frac{1}{\alpha} - \frac{\beta}{2k+2}, \quad \frac{1}{\vec{p}} = \frac{1}{p} \left(1 - \frac{\alpha\beta}{2k+2}\right) \text{ and } \frac{1}{\vec{q}} = \frac{1}{q} \left(1 - \frac{\alpha\beta}{2k+2}\right).$$

Assume that $b \in BMO_k(\mathbb{R})$ and $b \geq 0$. Proceeding as in the proof of Corollary 4.1, we get

$$|[b, M_\beta]f| \leq M_{b,\beta} f. \quad (4.13)$$

We conclude the proof by taking the norm $\|\cdot\|_{\vec{q},\vec{p},\alpha^*}$ on the both sides of (4.13) and by applying Theorem 4.3. $\square$

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Chapter 5

The Ubiquitous Appearance of BUPUs



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Abstract While partitions of unity are a standard tool in differential geometry or the theory of PDEs, they seem to play a minor role in the context of Harmonic Analysis. In the present note we give a survey of the use of the so-called BUPUs, i.e., Bounded Uniform Partitions of Unity, demonstrating that they form a universal tool for many branches of Fourier Analysis, including engineering applications. Although all these tools can be established in the context of LCA (locally compact Abelian) groups, we present the results here in the context of the Euclidean spaces $\mathbb{R}^d$.

BUPUs are also in many ways a crucial tool for an alternative approach to Fourier Analysis developed by the author, avoiding the classical pathway via Lebesgue integration and Abstract Harmonic Analysis, usually starting with the classical theory of Fourier series and then going on to discuss the Fourier transform which appears to be naturally defined on the Banach convolution algebra $(L^1(\mathbb{R}^d), \|\cdot\|_1)$. The approach given here is based on the Segal algebra $(\mathcal{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathcal{S}_0})$ (known as Feichtinger's algebra), and the dual space $(\mathcal{S}'_0(\mathbb{R}^d), \|\cdot\|_{\mathcal{S}'_0})$, whose elements are called *mild distributions*. In this way it is possible to develop most of the applications of Fourier Analysis relevant for engineering applications, but also for abstract Harmonic Analysis from scratch, using only elementary functional analytic tools.

The goal of this chapter is to point out the crucial role played by BUPUs in this context. Details are given in a series of papers (many of which are cited in the current text).

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5.1 Fourier Standard Spaces

Looking at the content of various books on Fourier or Harmonic Analysis which are currently used, one finds a large variety of Banach spaces of functions or the so-called distributions (consisting of continuous linear functionals on spaces of test functions) playing important roles in the classical approach to Fourier Analysis. For this reason let us provide a list of such function spaces, with some indications concerning their relevance in the context of Fourier analysis, or in a wider sense in time–frequency analysis (or perhaps the theory of pseudo-differential operators, and so on). For the sake of simplicity we restrict our attention to the Euclidian case, i.e., to Banach spaces of functions or (tempered) distributions over $\mathbb{R}^d$. Here the Haar measure, the (up to a constant uniquely determined) nonnegative translation-invariant measure on $C_c(\mathbb{R}^d)$ (continuous complex-valued functions with compact support), can be realized via the usual Riemann integral. While most of these spaces have weighted variants, we focus on the unweighted cases. For those readers interested in the zoo of function spaces and the various construction principles which allow to create new spaces from old one, we can recommend the article [25]:

1. According to Reiter [56] (resp., Reiter/Stegeman [57]) the main object of Harmonic Analysis is $(L^1(\mathbb{R}^d), \|\cdot\|_1)$, the space of all Lebesgue integrable functions on $\mathbb{R}^d$ (with $C_c(\mathbb{R}^d)$ as a dense subspace); one has to study the structure of the closed ideals of $(L^1(\mathbb{R}^d), \|\cdot\|_1)$ in order to deal with the question of *spectral analysis* and *spectral synthesis*.
2. $(L^1(\mathbb{R}^d), \|\cdot\|_1)$ can be shown to form a commutative Banach algebra with respect to convolution (being well defined almost everywhere by the usual integral). The Gelfand transform is given by the Fourier transform, i.e., closed maximal ideals are those of the form $I_\xi = \{f \mid \widehat{f}(\xi) = 0\}$ for some $\xi \in \widehat{\mathbb{R}^d}$.
3. It also appears as a natural domain for the Fourier transform, viewed as an integral transform (defined pointwise); moreover, convolution makes sense pointwise almost everywhere, and the Convolution Theorem can be established: $\widehat{f * g} = \widehat{f} \widehat{g}$.
4. The Riemann–Lebesgue Lemma can thus be described as the observation that $f \mapsto \widehat{f}$ is a non-expansive, but injective algebra homomorphism from $(L^1(\mathbb{R}^d), \|\cdot\|_1)$ (with convolution) to $(C_0(\mathbb{R}^d), \|\cdot\|_\infty)$, the space of bounded, continuous functions vanishing at infinity, with the sup-norm.
5. The usual Fourier transform defined on $C_c(\mathbb{R}^d)$ (or $L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$) can be extended to $(L^2(\mathbb{R}^d), \|\cdot\|_2)$ due to the isometry due to the mapping $\mathcal{F} : f \mapsto \widehat{f}$. In this way the Fourier transform can be described as a *unitary* automorphism of $(L^2(\mathbb{R}^d), \|\cdot\|_2)$. Thus we have Plancherel's Theorem.
6. Since the $\mathcal{F}$ is an injective mapping, one can consider $\mathcal{FL}^1(\mathbb{R}^d) := \{\widehat{f} \mid f \in L^1(\mathbb{R}^d)\}$, the so-called *Fourier algebra*, which is a Banach algebra with pointwise multiplication, with the norm $\|\widehat{f}\|_{\mathcal{FL}^1} := \|f\|_{L^1}$, which is properly contained in $C_0(\mathbb{R}^d)$. Equivalently, the two norms are not equivalent on $\mathcal{FL}^1(\mathbb{R}^d)$.

7. Functional analysis suggests to consider dual spaces, such as $(\mathbf{M}_b(\mathbb{R}^d), \|\cdot\|_{\mathbf{M}_b})$ (the space of bounded measures), which can be introduced as the dual space of $(\mathbf{C}_0(\mathbb{R}^d), \|\cdot\|_\infty)$. It is also a commutative Banach algebra with respect to convolution and contains $(\mathbf{L}^1(\mathbb{R}^d), \|\cdot\|_1)$ as a closed ideal (the so-called *absolutely continuous elements*), but also the discrete measures as a closed subalgebra, the closed linear span of all the finite linear combinations of the Dirac measures $\delta_x : f \mapsto f(x)$, for some $x \in \mathbb{R}^d$ (see [26]).
8. The dual space of $(\mathbf{L}^1(\mathbb{R}^d), \|\cdot\|_1)$ can be identified with $(\mathbf{L}^\infty(\mathbb{R}^d), \|\cdot\|_\infty)$, the space of (essentially) bounded, measurable functions on $\mathbb{R}^d$, with the norm

$$\|h\|_\infty = \text{ess sup}(|h(x)|).$$

9. For certain applications also $(\mathbf{PM}(\mathbb{R}^d), \|\cdot\|_{\mathbf{PM}(\mathbb{R}^d)})$, the space of *pseudo-measures* can be introduced, as the dual to $(\mathcal{FL}^1(\mathbb{R}^d), \|\cdot\|_{\mathcal{FL}^1})$ (see [52], Chap. 4.2).
10. Traditionally, one views the spaces $(\mathbf{L}^p(\mathbb{R}^d), \|\cdot\|_p)$, with $1 \leq p \leq \infty$ as a *scale of Banach spaces*, closed under *complex interpolation* and duality (under the usual restrictions). In particular, one has *reflexivity* of these spaces (exactly) for $1 < p < \infty$.
11. There are the more complicated Herz–Figa–Talamanca spaces $(\mathbf{A}_p(\mathbb{R}^d), \|\cdot\|_{\mathbf{A}_p(\mathbb{R}^d)})$ which allow to characterize the multipliers of $(\mathbf{L}^p(\mathbb{R}^d), \|\cdot\|_p)$ via p -pseudo-measures, i.e., bounded linear functionals on these spaces (see [52], or the extensive work of A. Derighetti [12], or [31] for a recent contribution).
12. For the treatment of PDEs or stochastic process, these spaces appear to be less suited, so that one typically resorts to the use of the *Schwartz spaces* $\mathcal{S}(\mathbb{R}^d)$ of rapidly decreasing functions (as prototypes of nice functions), or the dual spaces, the space of *tempered distributions*, as a large reservoir of objects (engineers would say signals) which “have a Fourier transform” (going back to L. Schwartz, [61]).

With the exception of the dual space $(\mathbf{L}^\infty(\mathbb{R}^d), \|\cdot\|_\infty)$, $(\mathbf{M}_b(\mathbb{R}^d), \|\cdot\|_{\mathbf{M}_b})$ or $(\mathbf{PM}(\mathbb{R}^d), \|\cdot\|_{\mathbf{PM}(\mathbb{R}^d)})$ all the other spaces are the so-called *homogeneous Banach spaces* in the spirit of Y. Katznelson ([50]). This means that they are Banach spaces of locally integrable functions with the property that translations define a strongly continuous group representation on $(\mathbf{B}, \|\cdot\|_{\mathbf{B}})$, i.e., with

$$\|T_x f\|_{\mathbf{B}} = \|f\|_{\mathbf{B}}, \quad \text{and} \quad \lim_{x \rightarrow 0} \|T_x f - f\|_{\mathbf{B}} = 0, \quad \forall f \in \mathbf{B}. \quad (5.1)$$

A somehow more cumbersome but slightly more general description (avoiding local integrability) is given in [30].

Despite the richness (at first sight) of spaces, there are serious problems when it comes to describe the boundedness of operators as they arise in Fourier analysis, using only these spaces. Classical function spaces, such as Besov–Triebel–Lizorkin

spaces (as found in the books of H. Triebel and E. Stein), have emerged and found natural characterizations using wavelet theory.

Still another family of function spaces, the so-called modulation spaces, were needed and more and more discussed in the context of time–frequency analysis, meanwhile with many publications in mathematical analysis but also with engineering applications. Let us just mention the books [40] and [41] for early summaries of the subject, the book [44], which is by now the standard reference in the field (together with [42]). More recently, there are the following books covering this subject ([6, 71], and [11]). In all these constructions BUPUs play an important role, as we are going to point out in the course of this chapter. In fact, they are important, but in a different way if one looks into [62], Theorem 1.4.6.

5.2 Motivation or Introduction

While *partitions of unity* play a crucial role in differential geometry (in order to make use of local charts), in particular for Riemannian geometry (see [51, 67] or [8]) and the theory of Lie groups (see [46] for a standard reference), they are not so much used in the context of Abstract Harmonic Analysis (AHA) or applied Fourier Analysis, as it is relevant for engineering applications or physics. On the other hand (smooth) partitions of unity are employed in order to localize objects (such as distributions) and are needed in order to develop the concept of the *support* of a distribution ([61] or [47]).

It is the purpose of this note to demonstrate the high relevance of (suitable) partitions of unity, the so-called **BUPUs** (Bounded Uniform Partitions of Unity) over locally compact Abelian (LCA) groups. In fact, the term “partition of unity” is rarely found in the index of any of the classical books on Fourier Analysis or books on Signal Processing for engineers.

5.3 Defining General BUPUs

The idea of BUPUs is to make use of partitions of unity with some mild extra properties, expressed by the terms “uniform” (referring to the uniform size of the supports of the members of such a family) and “bounded” (referring to some form of boundedness, typically is measured by the norm in some Banach algebra of “smooth” functions). One more condition is required, which is used for certain applications, e.g., for the definition of *Wiener amalgam spaces*: We assume to have some control over the amount of overlap between neighboring elements, in the sense that there is a uniform control of their number. In fact, if one replaces a partition of unity arising by the indicator functions of a partition of $\mathbb{R}^d$, these sets (or supports of the indicator functions) are (almost) disjoint. In contrast, these supports have to overlap if one requires the partition to be constituted by continuous functions. As the

various papers using BUPUs show the mentioned “finite overlap condition” appears to be a good replacement for the disjoint support condition, we do not want to have a more detailed discussion here.

For our presentation here we restrict the description of BUPUs to the case of the Euclidean group $\mathbb{R}^d$, but the definitions are valid for general LCA groups.¹

Definition 5.1 A bounded family $\Psi = (\psi_i)_{i \in I}$ in some normed algebra $(A, \|\cdot\|_A)$ is called an **A-Uniform Partition of Unity of size $|\Psi| \leq \delta > 0$** if one has:

1. $\sup_{i \in I} \|\psi_i\|_A = C_A < \infty$.
2. There is a family of points $(x_i)_{i \in I}$ such that

$$\text{supp}(\psi_i) \subseteq B_\delta(x_i) \quad \forall i \in I.$$

3. There is uniform control over the number of overlapping supports:

$$\sup_{i \in I} \#\{j \mid B_\delta(x_j) \cap B_\delta(x_i) \neq \emptyset\} = C_X < \infty.$$

- 4.

$$\sum_{i \in I} \psi_i(x) \equiv 1 \quad \text{over } \mathbb{R}^d.$$

Remark 5.1

- (a) Note that we do not assume positivity of the members ψ_i constituting the partition of unity. Nevertheless, condition (3.) implies that the sum in (4.) is pointwise just a finite sum, so there is no issue concerning (e.g., unconditional) pointwise convergence.
- (b) We could also have restricted our attention to countable families, but such a viewpoint suggests to label the elements (ψ_n) with, say, indices $n \in \mathbb{N}$. However, thinking of well-spread points in the plane, one quickly observes that there is no natural order for the family $(x_n)_{n \in \mathbb{N}}$, and one might have to emphasize each time that the expressions used are independent from the specific order chosen.
- (c) Often it is enough to restrict the attention to nonnegative partitions of unity, but then $0 \leq \sum_{i \in I} \psi_i(x) = 1$ implies $\|\psi\|_\infty \leq 1$, i.e., boundedness in the pointwise Banach algebra $(C_0(\mathbb{R}^d), \|\cdot\|_\infty)$ of continuous functions vanishing at infinity, with the sup-norm.
- (d) By saying that a quantity X_Ψ tends to the limit κ for $|\Psi| \rightarrow 0$ just means that for every $\varepsilon > 0$ one can find some $\delta > 0$ such that one has: For any BUPUs with $|\Psi| \leq \delta$, one has $|X_\Psi - \kappa| < \varepsilon$. In other words, the family of BUPUs forms a

¹ Using suitable topological terminology, using neighborhoods of the identity instead of balls of radius $\delta > 0$.

net in the same spirit as Riemannian sums arising from partitions $(I_i)_{i \in I}$ of the interval $[a, b]$, and the points x_i correspond to the usual points $\xi_i \in I_i$ in the corresponding disjoint intervals.

(e) Illustrations showing typical BUPUs are given in Figs. 5.1 and 5.2.

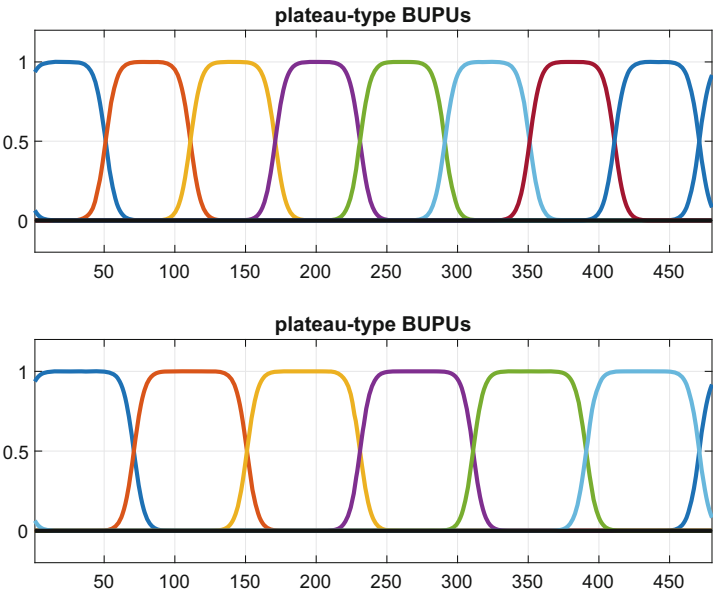


Fig. 5.1 Two different, typical regular BUPUs on the real line

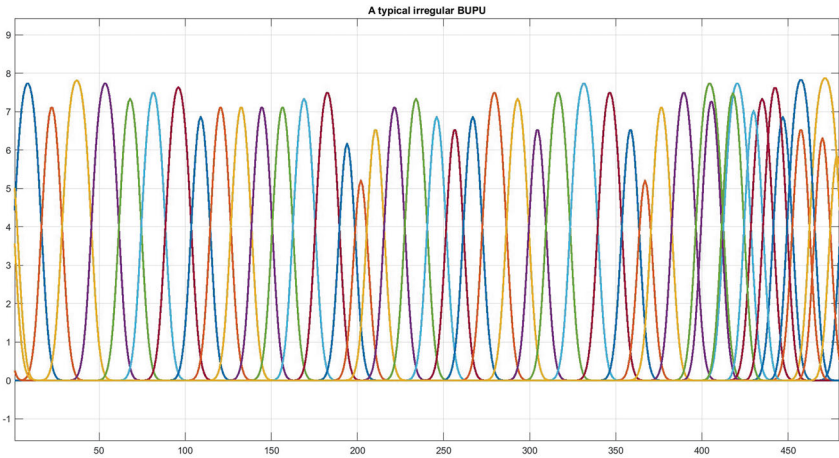


Fig. 5.2 A typical irregular BUPU on the real line

For many applications it suffices to restrict the attention to what we call *regular BUPUs*, where the set of “centers” X is just a *lattice* $\Lambda \triangleleft \mathbb{R}^d$, meaning a discrete (hence closed) and co-compact subgroup. In this case the finite overlap condition is automatically satisfied for any compactly supported function ψ . Let us give the definition in the general LCA setting:

Definition 5.2 A *coherent family* $\Psi = (\psi_\lambda)_{\lambda \in \Lambda} = (T_\lambda \psi)_{\lambda \in \Lambda}$ in a Banach algebra $(A, \|\cdot\|_A)$ forms a **regular A-Bounded Uniform Partition of Unity** generated by the pair (ψ, Λ) if ψ has compact support, if the family is bounded in $(A, \|\cdot\|_A)$ and

$$\sum_{\lambda \in \Lambda} T_\lambda \psi(x) = \sum_{\lambda \in \Lambda} \psi(x - \lambda) \equiv 1 \quad \text{for all } x \in \mathcal{G}.$$

Remark 5.2 Recall that for the case $\mathcal{G} = \mathbb{R}^d$ any lattice Λ can be described as a deformation of the standard lattice $\mathbb{Z}^d$, i.e., one can find some (in fact nonunique) non-singular matrix A such that $\Lambda = A * \mathbb{Z}^d \triangleleft \mathbb{R}^d$. In such cases it may be natural to choose Λ or $\mathbb{Z}^d$ as index set instead of the abstract symbol I . The meaning of closeness of indices is then referring typically to the Euclidian distance of the corresponding points in $\mathbb{R}^d$.

One of the advantages of discussing (regular) BUPUs in the context of $\mathcal{G} = \mathbb{R}^d$ is the possibility of applying contractions in order to create “arbitrary fine” partitions of unity, based on the following simple observation, making use of the standard dilation operator $D_\rho : f \mapsto D_\rho(f)$, given by

$$D_\rho f(x) = f(\rho x), \quad \rho > 0, x \in \mathbb{R}^d.$$

More generally, we can define dilations using any non-singular matrix instead of the diagonal matrix, describing the automorphism $x \mapsto \rho x, x \in \mathbb{R}^d$ of the group $(\mathbb{R}^d, +)$.

Lemma 5.1 *We assume that A is dilation invariant, i.e., that any D_ρ defines a bounded linear operator on $(A, \|\cdot\|_A)$. Then we have:*

- (i) *Given any δ -BUPU $(\psi_i)_{i \in I}$ and $\rho > 0$ the family $(D_\rho(\psi_i))_{i \in I}$ is a BUPU of size δ/ρ . Thus by letting $\rho \rightarrow \infty$ one obtains arbitrary fine BUPUs.*
- (ii) *A similar statement is of course valid for regular BUPUs, since (even anisotropic) dilations preserve the regularity of BUPUs.*

At a technical level we note that it is of course possible to obtain smooth BUPUs by regularization, and tensorization of BUPUs allows to create BUPUs in a multi-dimensional setting from lower-dimensional BUPUs. We restrict our attention to the case of simple tensor products.

In the classical setting BUPUs appear in a slightly hidden form, using fundamental domains. Just consider $Q = [-1/2, 1/2)^d$ as the fundamental domain of the standard lattice $\Lambda = \mathbb{Z}^d$, creating a partition of $\mathbb{R}^d$ into disjoint cubes of the form

$Q_k = k + Q$, with $k \in \mathbb{Z}^d$. Then we find, after applying the non-singular matrix A , that the family $A * Q_k$, $k \in \mathbb{Z}^{d2}$ defines a partition of $\mathbb{R}^d$ as well. We may then rewrite this condition as a partition of unity property for the corresponding indicator functions:

$$\mathbf{1} = \sum_{k \in \mathbb{Z}^d} \psi_k(x), x \in \mathbb{R}^d \quad \text{with} \quad \psi_k = \mathbf{1}_{A * Q_k}.$$

Obviously, these building blocks are uniformly bounded in $(L^1(\mathbb{R}^d), \|\cdot\|_1)$ (with the slight disadvantage of being discontinuous), since $\|\psi_k\|_1 = |A * Q|$, for any $k \in \mathbb{Z}^d$, which can be expressed by $\det(A)$. Thus one can use them as starting point for BUPUs with respect to any of the relevant Banach algebras $(A, \|\cdot\|_A)$:

Lemma 5.2 *Assume that $(A, \|\cdot\|_A)$ is a homogeneous Banach space on $\mathbb{R}^d$, i.e., a Banach space of locally integrable functions on $\mathbb{R}^d$ with the additional property that $L^1(\mathbb{R}^d) * A \subset A$ and the corresponding norm inequality*

$$\|g * f\|_A \leq \|g\|_{L^1} \|f\|_A, \quad g \in L^1, f \in A.$$

*Then for any nonnegative function $\varphi \in A$ with compact support the family generated by $\psi := \mathbf{1}_{A * Q} * \varphi$ defines a regular BUPU on $\mathbb{R}^d$ which is bounded in $(A, \|\cdot\|_A)$. Obviously one can create arbitrary fine BUPUs in this way if A contains such functions φ with arbitrary small support.*

5.3.1 General Λ -BUPUs

In a more general context we have the following definition for BUPUs over general LC groups $\mathcal{G}$ (regular A -BUPUs):

Definition 5.3 A coherent family $\Psi = (\psi_\lambda)_{\lambda \in \Lambda} = (T_\lambda \psi)_{\lambda \in \Lambda}$ in a Banach algebra $(A, \|\cdot\|_A)$ forms a regular **A-Bounded Uniform Partition of Unity** generated by the pair (ψ, Λ) if ψ has compact support, if the family of Λ -translates is bounded in $(A, \|\cdot\|_A)$ and

$$\sum_{\lambda \in \Lambda} T_\lambda \psi(x) = \sum_{\lambda \in \Lambda} \psi(x - \lambda) \equiv 1 \quad \text{for all } x \in \mathcal{G}.$$

For many examples Λ can be a lattice (discrete subgroup, e.g., $\Lambda = A * \mathbb{Z}^d$ in $\mathbb{R}^d$), but this is not a requirement.

² Here we use the symbol $*$ to describe matrix multiplication.

5.3.2 Generating BUPUs for a Given Algebra A

It is quite easy to generate BUPUs which are simply continuous or infinitely differentiable. In fact, the standard method in such cases is to add up a collection of localized, typically nonnegative bump functions and then divide through their (nonvanishing) some. Obviously inversion preserves pointwise inversion or smoothness (essentially thanks to the quotient rule). The case of the Fourier algebra $\mathcal{FL}^1(\mathbb{R}^d)$ (and in a similar way Fourier–Beurling algebras $\mathcal{FL}_w^1(\mathbb{R}^d)$) is quite different. Here one still has a local inversion theorem, but unlike the regular case there is no quantitative control of the norm of the inverse elements (over so many different configurations) known or even possible. This is why we want to point to a construction explained in [33] for an even more general context. In fact, the comment below allows to construct BAPUs, as they are used for the construction of *decomposition spaces*, but let us specialize the statement (following directly from Theorem of [33]) to the BUPU case.

Proposition 5.1 *Assume that $(A, \|\cdot\|_A) \hookrightarrow (C_0(\mathbb{R}^d), \|\cdot\|_\infty)$ is a homogeneous Banach space over some LCA group, which is also a Banach algebra with respect to pointwise multiplication (inherited from $C_0(\mathbb{R}^d)$) and has nonnegative functions with arbitrary small support. Then for any well-spread family $(x_i)_{i \in I}$ as it appears in the definition of a BUPU which is δ -dense, there exists a BUPU in $(A, \|\cdot\|_A)$ of size 2δ .*

Proof Using the fact that $L^1 * A \subset A$ we can create a plateau function in $(A, \|\cdot\|_A)$ which satisfies $\varphi(z) \equiv 1$ on $B_\delta(0)$ and with $\text{supp}(\varphi) \subset B_{2\delta}(0)$. Choosing a maximal family $(x_i)_{i \in I}$ in $\mathbb{R}^d$, we find that the balls of $B_{2\delta}(x_i)$ are a covering satisfying the required conditions of a BUPU. The sum of the functions $\varphi_i := T_{x_i}\varphi$ is nonzero (in fact ≥ 1 for any $x \in \mathbb{R}^d$). But instead of dividing φ_i by $H(x) = \sum_{i \in I} \varphi_i(x)$, we apply the algebraic method described in the proof of Theorem 4.2 of [22]. It shows how to construct from a *bounded control system* a partition of unity and that both of them can be used to characterize the membership of a function in a given decomposition space. $\square$

5.4 BUPUs and Bounded Measures

BUPUs are an important tool in the fresh approach to the convolution of bounded measures as given in [27] and continued in [30] later on. There are two key steps employing BUPUs in the verification of the fact that any TILS on $(C_0(\mathbb{R}^d), \|\cdot\|_\infty)$, i.e., any bounded and linear operator on $(C_0(\mathbb{R}^d), \|\cdot\|_\infty)$, which commutes with translation, is just a convolution operator by some bounded measure, given by $\mu * f(x) = \mu(T_x f^\vee)$, where $f^\vee(x) = f(-x)$. The norm of μ as a functional equals the norm of the corresponding convolution operator $C_\mu : f \mapsto \mu * f$ on $(C_0(\mathbb{R}^d), \|\cdot\|_\infty)$. This isomorphism can be used to transfer the composition law

for TILS to the generating measures, by enforcing the identity $C_{\mu_1} \circ C_{\mu_2} = C_{\mu_1 \star \mu_2}$, which is obviously associative (and commutative, via approximation by convolution of discrete measures, see below).

In this correspondence δ_x corresponds to the shift by x , i.e.,

$$\delta_x * f(z) = f(z - x) = [T_x f](z), \quad x, z \in \mathbb{R}^d.$$

Consequently, we have the expected relation $\delta_x \star \delta_y = \delta_{x+y} = \delta_y \star \delta_x$, and corresponding simple convolution laws for bounded, discrete measures (which are absolutely convergent sums of Dirac measures!).

Let us display the two arguments employing BUPUs explicitly here, which help to describe the convolution of measures in a (as we believe) rather natural way: First recall that we can view $(\mathbf{M}_b(\mathbb{R}^d), \|\cdot\|_{\mathbf{M}_b})$, the space of bounded (regular Borel) measures simply as the dual to $(C_0(\mathbb{R}^d), \|\cdot\|_\infty)$, i.e., all the functionals on μ on $(C_0(\mathbb{R}^d), \|\cdot\|_\infty)$ satisfying

$$\sup_{\|f\|_\infty \leq 1} |\mu(f)| = \|\mu\|_{\mathbf{M}_b(\mathbb{R}^d)} < \infty.$$

For each such $\mu \in \mathbf{M}_b(\mathbb{R}^d)$ and $h \in C_b(\mathbb{R}^d)$, one can define

$$(\mu \cdot h)(f) = \mu(h \cdot f), \quad f \in C_0(\mathbb{R}^d).$$

With the measures $\mu\psi_i$ being well defined the relevant results read as follows:

Lemma 5.3 *For any $\mu \in \mathbf{M}_b(\mathbb{R}^d)$ and any BUPU $\Psi = (\psi_i)_{i \in I}$, one has*

$$\sum_{i \in I} \|\mu\psi_i\|_{\mathbf{M}_b} = \|\mu\|_{\mathbf{M}_b}. \quad (5.2)$$

The net $D_\Psi \mu$, consisting of bounded, discrete measures given by the formula

$$D_\Psi \mu = \sum_{i \in I} \mu(\psi_i) \delta_{x_i},$$

that satisfy

$$\|D_\Psi \mu\|_{\mathbf{M}_b} \leq \|\mu\|_{\mathbf{M}_b},$$

is thus bounded and w^ -convergent to μ :*

$$\lim_{|\Psi| \rightarrow 0} D_\Psi \mu(f) = \mu(f), \quad f \in C_0(\mathbb{R}^d).$$

In the course of this approach, one can also show that one has

$$\mu_1 \star \mu_2(f) = \lim_{|\Psi| \rightarrow 0} (D_\Psi \mu_1 * D_\Psi \mu_2)(f). \quad (5.3)$$

Just to fix notations let us introduce the (standard) notation:

$$[f \otimes g](x, y) := f(x) \cdot g(y), \quad x \in \mathbb{R}^k, y \in \mathbb{R}^m$$

is a function defined on $\mathbb{R}^d$ (with $d = k + m$).

Some obvious construction principles follow. For simplicity let us specialize to $A = C_0$, respectively, $A = \mathcal{FL}^1$, the Fourier algebra.

Lemma 5.4 *Given two BUPUs $\Psi = (\psi_i)_{i \in I}$ on $\mathbb{R}^k$ and $\Phi = (\phi_j)_{j \in J}$ on $\mathbb{R}^m$, respectively, then their tensor product*

$$\Psi \otimes \Phi = (\psi_i \otimes \phi_j)_{(i,j) \in I \times J}$$

is a BUPU on $\mathbb{R}^d$.

The proof is left to the reader.

Even if one is mostly interested in regular BUPUs, one has to observe that pointwise products of two different BUPUs are not regular BUPUs anymore. This, together with the flexibility of irregular centers (or the lack of fine lattices in general LCA groups), is the reason for the specific approach to BUPUs as provided in Definition 5.1.

Lemma 5.5 *Given two partitions $\Psi = (\psi_i)_{i \in I}$ and $\Phi = (\phi_j)_{j \in J}$ on $\mathbb{R}^d$ of size δ_1 and δ_2 , respectively, then the joint refinement (or pointwise product BUPU) $\Psi \cdot \Phi$ is the collection of all products of the form $(\psi_i \cdot \phi_j)_{(i,j)}$. It is a BUPU as well, of size $\delta \leq \min(\delta_1, \delta_2)$.³*

These simple but rather important examples explain why it is important to allow for non-regular BUPUs.

5.5 Wiener Amalgam Spaces (WASPs)

One of the main applications where BUPUs play an important role is the theory of *Wiener Amalgam spaces* (WASPs), originally introduced by the author in [20] and [19], by then under the name of *Wiener-type spaces*. They have been designed in order to allow subsequently the definition of *modulation spaces* in [21], but published only in [23]. The idea was to modify the characterization of the classical function spaces, in particular *Besov spaces* or *Triebel–Lizorkin spaces*, which makes use of dyadic partitions of unity arising from sufficiently smooth functions, by BUPUs.

In order to allow the characterization of such spaces which belong locally to the Fourier algebra, it was necessary to allow for BUPUs which are bounded

³ Obviously, one can reduce the index set to be the set of all pairs $(i, j) \in I \times J$ with $B_{\delta_1}(x_i) \cap B_{\delta_2}(y_j) \neq \emptyset$.

in the Fourier algebra $(\mathcal{FL}^1(\mathbb{R}^d), \|\cdot\|_{\mathcal{FL}^1})$. Finally the idea was developed in a wider context, leading to the general Wiener amalgam spaces of the form $(W(\mathbf{B}, \mathbf{C}), \|\cdot\|_{W(\mathbf{B}, \mathbf{C})})$. They allow us to create function spaces in Triebel's sense (meaning Banach spaces of say tempered distributions on $\mathbb{R}^d$), by the specification of the *global behavior of some local property*. One starts from a *local component* $(\mathbf{B}, \|\cdot\|_{\mathbf{B}})$, i.e., function spaces describing local properties of a function (say membership in some Sobolev space or $\mathcal{FL}^1$) and a *global component*, the (solid and translation invariant) BF space $(\mathbf{C}, \|\cdot\|_{\mathbf{C}})$, which allows to impose decay and/or summability conditions on the local behavior of a given function, as measured by a *local control function*.

There are different, equivalent ways to introduce *Wiener amalgam spaces* over general locally compact groups, using either a *continuous* method (using the mentioned control function) or a discrete variant, making use of BUPUs, which we want to illustrate here, as it is easier to explain and to understand how the basic properties of such spaces arises. Again let us restrict our presentation to the Euclidian context and global components which are of the form $(\mathbf{C}, \|\cdot\|_{\mathbf{C}}) = (L_w^p(\mathbb{R}^d), \|\cdot\|_{p,w})$, for some *moderate* weight function w . Let us assume for simplicity that we have *polynomial moderateness*, i.e., satisfying for some $s \geq 0$:

$$w(x+y) \leq C(1+|y|^2)^s w(x), \quad x, y \in \mathbb{R}^d.$$

As pointed out in [20] in a general form and in more detail in [45], such spaces can be defined with the help of BUPUs as

$$W(\mathbf{B}, \ell_w^q) := \left\{ f \in B_{loc}, \left(\sum_{i \in I} \|f \psi_i\|_{\mathbf{B}}^q w^q(x_i) \right)^{1/q} < \infty \right\}. \quad (5.4)$$

The expression which has to be finite defines a norm, and of course, one obtains the same space and equivalent norms if one replaces the given BUPU by another one.

Wiener amalgam spaces are very useful for many applications and satisfy a number of (expected) properties concerning duality or interpolation. Whenever the test functions, say the Schwartz space $\mathcal{S}(\mathbb{R}^d)$, are dense in the local component $(\mathbf{B}, \|\cdot\|_{\mathbf{B}})$, and the finite sequences are dense in the corresponding sequence spaces (so typically $q < \infty$), the Schwartz space is also dense in $W(\mathbf{B}, \ell_w^q)$.

Many of the good properties of Wiener amalgam spaces rely on the fact that often operators can be considered “*componentwise*.” For example, one can apply the duality theorem (Theorem 2.8 in [33]), stating that in the above situation one has (for $q < \infty$) the following natural duality:

$$W(\mathbf{B}, \ell_w^q)' = W(\mathbf{B}', \ell_{1/w}^{q'}). \quad (5.5)$$

The fact that complex interpolation can also be applied coordinatewise is provided in [18], and the pointwise multiplier question is settled in [33], Theorem 2.11.

The perhaps most useful feature of WASps already established in [23] is the fact that *convolution results* can also be done coordinatewise. For the classical case of WASps (without smoothness), such results have been given already in [7]. Compared with the discrete description of Wiener amalgam spaces the continuous version of the norm allows to reduce the global estimate to a true convolution of *continuous control functions* (see [23] for details). In contrast, inclusion results (obtained also coordinatewise) are better discussed in the discrete setting, using BUPUs.

A good survey over the results on WASps (with some historical background) can be found in the survey article [43].

Let us just summarize the findings of the earlier section: We have $M_b(\mathbb{R}^d) = W(M_b, \ell^1)(\mathbb{R}^d)$ and $L^1(\mathbb{R}^d) = W(L^1, \ell^1)(\mathbb{R}^d)$, always with equivalence of norms, and of course $L^2(\mathbb{R}^d) = W(L^2, \ell^2)(\mathbb{R}^d)$. One of the interesting applications are summarized in the following lemma, making use of the space $W(L^2, \ell^1)$ which has already been used by N. Wiener in his early work.

Lemma 5.6 *As a consequence of the basic properties of WASPs, we have:*

1. $W(L^2, \ell^1) * W(L^2, \ell^1) \subseteq W(\mathcal{FL}^1, \ell^1)$
2. $\mathcal{F}(W(L^2, \ell^1)) \subset W(\mathcal{FL}^1, \ell^2) \hookrightarrow W(C_0, \ell^2) \hookrightarrow L^2 \cap C_0$

together with the corresponding norm estimates.

Remark 5.3 We would like to point out that the finite overlap condition is not required for spaces of the form $W(B, \ell^1)$, such as Wiener's algebra $W(C_0, \ell^1)(\mathbb{R}^d)$, Feichtinger's algebra $S_0(\mathbb{R}^d) = W(\mathcal{FL}^1, \ell^1)(\mathbb{R}^d)$, or similar constructions of minimal Banach algebras (see [16, 17, 19]).

In contrast, one has to use this condition for the case of spaces of the form $W(B, \ell^p)$ for $p > 1$ because there one might otherwise repeat $(1/n)\psi_n$ n times and still get a partition of unity, but with a different behavior of the ℓ^p -norm of the sequence of local norms, compared with $\|f\psi_n\|_B$ in $(\ell^p, \|\cdot\|_p)$.

5.6 Modulation Spaces

The construction of general Wiener amalgam spaces was really motivated by the idea of establishing over general LCA groups spaces similar to the well-known Besov spaces (see the books of E. Stein [62] or H. Triebel [68–70], and so on).

On the Fourier transform side, these spaces should be Fourier transforms of elements of $(L^p(\mathbb{R}^d), \|\cdot\|_p)$, or shortly members of $(\mathcal{FL}^p(\mathbb{R}^d), \|\cdot\|_{\mathcal{FL}^p})$, with the norm $\|\widehat{f}\|_{\mathcal{FL}^p} = \|f\|_{L^p}$. Although one might use sharp cutoff functions for $p > 1$, it is clear that localization in the Fourier algebra $\mathcal{FL}^1(\mathbb{R}^d)$ has to make use of functions that belong to $\mathcal{FL}^1(\mathbb{R}^d) \subset C_0(\mathbb{R}^d)$, and that consequently the members of a partition of unity have to have some overlap. One can say that the observation of a limited amount of overlap was the crucial fact allowing to establish most of the basic results for Wiener amalgams.

According to the original intention of this author *modulation spaces* are designed as function spaces (in Triebel’s sense) derived from a suitable space $(\mathbf{B}, \|\cdot\|_{\mathbf{B}})$ on $\mathbb{R}^d$, by decomposing their Fourier transforms into blocks of equal size (or in an engineering terminology into band-pass signals of equal bandwidth), and requiring a certain decay and summability behavior of the high-frequency bounds. Due to the established (in [23]) equivalence between continuous and discrete norms, the formal definition given in [21] makes use of the “more elegant” continuous description of such spaces. At that time it was slowly recognized that the continuous description of the spaces $\mathcal{F}(\mathbf{W}(\mathcal{FL}^p, \ell_w^q))$ with moderate weights of polynomial growth can be rewritten by a mixed-norm condition applied to the short-time Fourier transform. This is the way how modulation spaces have been introduced (hiding to some extent to role of Wiener amalgams) in [21] and then published 20 years later in [23].

Subsequently, several authors made the observations that by applying the opposite order in the definition of mixed-norm spaces (very much in the spirit of the transition from Besov spaces to Triebel–Lizorkin spaces, e.g., by describing them via the continuous wavelet transform), one obtains Wiener amalgam spaces, thus returning to the original motivation.

In order to make the approach better digestible to the readers it was then decided to reduce the level of generality and introduce (over $\mathbb{R}^d$) the by now classical modulation spaces $(M_{p,q}^s(\mathbb{R}^d), \|\cdot\|_{M_{p,q}^s})$ in the following form (using BUPUs in $(\mathcal{FL}^1(\mathbb{R}^d), \|\cdot\|_{\mathcal{FL}^1})$):

Definition 5.4

$$M_{p,q}^s(\mathbb{R}^d) := \{f \in \mathcal{S}'(\mathbb{R}^d), \sum_{k \in \mathbb{Z}^d} \|\hat{f} \cdot \psi_k\|_{\mathcal{FL}^p}^q (1 + |k|)^{sq} < \infty.\}$$

Using obvious results such as $\mathcal{FL}^1 \cdot \mathcal{FL}^p \subset \mathcal{FL}^p$ or $\mathcal{FL}^\infty * \mathcal{FL}^p \subset \mathcal{FL}^p$ and the general results for Wiener amalgam spaces, one can derive many basic properties of modulation spaces easily, e.g., with respect to convolution and pointwise multiplication or inclusion results.

Overall one can say that modulation spaces are by now well established, and the reader can learn a lot about their usefulness in the books [6, 44, 71] and [11]. They are not only a natural tool in the context of time–frequency analysis (because they allow to describe operators well by their effect in the phase space representation of the involved signals or distributions) but also for the treatment of various types of pseudo-differential operators. We avoid to give a long list of papers here.

5.7 The Segal Algebra $(S_0(\mathbb{R}^d), \|\cdot\|_{S_0})$

The prototypical example of a WAsp which is Fourier invariant, i.e., which is both a modulation space and a WASp, is the Segal algebra $(S_0(\mathbb{R}^d), \|\cdot\|_{S_0})$, which was introduced (in the context of LCA groups) in [19] as follows:

Definition 5.5

$$(\mathbf{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathbf{S}_0}) := (W(\mathcal{FL}^1, \ell^1)(\mathbb{R}^d), \|\cdot\|_{W(\mathcal{FL}^1, \ell^1)}).$$

The Fourier invariance of $\mathbf{S}_0(\mathbb{R}^d)$ is one of the important features of $\mathbf{S}_0(\mathbb{R}^d)$, making it a good substitute for $\mathcal{S}(\mathbb{R}^d)$ (or the corresponding Schwartz–Bruhat space, see [55], or [58] for the use of this space, later replaced by $\mathbf{S}_0(G)$, in [59]), which is actually contained as a dense subspace in $(\mathbf{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathbf{S}_0})$ (see [48]). Using as a (equivalent) norm the L^1 -norm of the Short-Time Fourier Transform (STFT) with respect to the Gaussian window, we have

$$\|f\|_{g_0, \mathbf{S}_0} := \int_{\mathbb{R}^d \times \widehat{\mathbb{R}^d}} |V_{g_0}(f)(t, s)| dt ds \quad (5.6)$$

as a norm, with $g_0(t) = e^{-\pi|t|^2}$ and

$$V_g f(t, s) = \int_{\mathbb{R}^d} f(y) g(y - t) e^{-2\pi i s y} dy.$$

With respect to this norm, which can be used to define $\mathbf{S}_0(\mathbb{R}^d)$ as a subspace of $L^2(\mathbb{R}^d)$, the Fourier transform is even isometric, due to the fact that one has as a consequence of Plancherel’s Theorem (see [44], or [48]):

$$V_g f(t, s) = e^{-2\pi i s t} V_{\widehat{g}} \widehat{f}(s, -t).$$

Thus, for the Gauss function $g_0 = \mathcal{F}(g_0)$, the Fourier transform looks like a rotation by 90° in the clockwise sense over phase space and is thus obviously isometric on $(\mathbf{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathbf{S}_0})$. Many other properties of $(\mathbf{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathbf{S}_0})$ have been developed in the last years, and a very good summary is provided by the article of M. Jakobsen [48], listing many properties and equivalent characterizations of $\mathbf{S}_0(G)$ for LCA groups G . Among others all functions in $\mathcal{FL}^1(\mathbb{R}^d)$ with compact support belong to $\mathbf{S}_0(\mathbb{R}^d)$, but also (via Fourier transform) all the band-limited functions $f \in L^1(\mathbb{R}^d)$ (because $\widehat{f}$ has compact support and belongs to $\mathcal{FL}^1(\mathbb{R}^d)$).

Among others one shows that $\mathcal{S}(\mathbb{R}^d) \hookrightarrow (\mathbf{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathbf{S}_0}) \hookrightarrow \mathcal{S}'(\mathbb{R}^d)$ forms a chain of dense embeddings and that Dirac sequence $(g_\rho)_{\rho>0}$ obtained by compression of a function $g \in \mathbf{S}_0(\mathbb{R}^d)$ with $\widehat{g}(0) = \int_{\mathbb{R}^d} g(y) = 1$ acts as expected on $(\mathbf{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathbf{S}_0})$, i.e.,

$$\lim_{\rho \rightarrow 0} \|g_\rho * f - f\|_{\mathbf{S}_0} = 0, \quad \forall f \in \mathbf{S}_0(\mathbb{R}^d). \quad (5.7)$$

Going to the Fourier transform, one observes that $\mathcal{F}(g_\rho)$ is just a typical summability kernel (in fact all the ones used in the literature belong to $\mathbf{S}_0(\mathbb{R}^d)$), and thus

also

$$\lim_{\rho \rightarrow 0} \|\mathcal{F}(g_\rho) \cdot h - h\|_{\mathcal{S}_0} = 0, \quad \forall h \in \mathcal{S}_0(\mathbb{R}^d). \quad (5.8)$$

By the general duality theory for WASps the dual space $\mathcal{S}'_0(\mathbb{R}^d)$ can be naturally identified with $\mathcal{W}(\mathcal{FL}^\infty, \ell^\infty)(\mathbb{R}^d)$. Its members are called *mild distributions*. Inside of $\mathcal{S}'(\mathbb{R}^d)$ mild distributions σ can be characterized by the fact that they have a bounded STFT $V_g(\sigma)$, e.g., with respect to the Gaussian window $g = g_0$. Moreover, norm convergence in $(\mathcal{S}'_0(\mathbb{R}^d), \|\cdot\|_{\mathcal{S}'_0})$ is equivalent to uniform convergence in the STFT domain.

As we can already see in the context of bounded measures, the functional norm on a dual space, in this case on $(\mathcal{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathcal{S}_0})$, given by

$$\|\sigma\|_{\mathcal{S}'_0} := \sup_{\|f\|_{\mathcal{S}_0} \leq 1} |\sigma(f)|$$

is too sensitive for many applications.

Just as a simple example, let us note that $\delta_x : f \mapsto f(x)$ is a member of $\mathcal{S}'_0(\mathbb{R})$, with $\|\delta_x\|_{\mathcal{S}'_0} = 1$, but for $x \neq y$ one has $\|\delta_x - \delta_y\|_{\mathcal{S}'_0} \geq 1$, as can easily be seen by noting that the triangular functions belong to $\mathcal{FL}^1(\mathbb{R})$ and have one (independent of their width). Thus the difference of two such triangular functions, located at x and y , respectively, belongs to $\mathcal{FL}^1(\mathbb{R})$, with compact support, hence to $(\mathcal{S}_0(\mathbb{R}), \|\cdot\|_{\mathcal{S}_0})$, with norm ≤ 2 . But the response to $\delta_x - \delta_y$ is 2.

Therefore it is important to allow for a more flexible alternative to norm convergence. The most natural one is of course w^* -convergence. Let us recall that a bounded net $(\alpha_\alpha)_{\alpha \in I}$ converges to σ_0 the w^* -sense if one has

$$\lim_{\alpha \rightarrow \infty} \alpha_\alpha(f) = \sigma_0(f), \quad f \in \mathcal{S}_0(\mathbb{R}^d). \quad (5.9)$$

Due to the boundedness it is sufficient to test convergence on a subset of $(\mathcal{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathcal{S}_0})$ whose closed linear span is all of $\mathcal{S}_0(\mathbb{R}^d)$.

The following characterization is not difficult to establish and shows some light on the concept of w^* -convergence:

Lemma 5.7 *A bounded net $(\alpha_\alpha)_{\alpha \in I}$ in $\mathcal{S}'_0(\mathbb{R}^d)$ is w^* -convergent to σ_0 if and only if one has uniform convergence over compact subsets of the corresponding STFTs over compact subsets of phase space. Spelled out more concretely, this means, if and only if one has: For every compact set $K \subset \mathbb{R}^d \times \widehat{\mathbb{R}}^d$ and $\varepsilon > 0$, one can find some index α_0 such that for $\alpha \succeq \alpha_0$ one has*

$$|V_g(\alpha_\alpha)(t, s) - V_g(\sigma_0)(t, s)| \leq \varepsilon, \quad \forall (t, s) \in K.$$

A further application of the basic principles of WASps allows to show the regularizing effect of product-convolution operators.

Proposition 5.2

$$(\mathcal{S}'_0(\mathbb{R}^d) * \mathcal{S}_0(\mathbb{R}^d)) \cdot \mathcal{S}_0(\mathbb{R}^d) \subseteq \mathcal{S}_0(\mathbb{R}^d) \quad \text{and} \quad (\mathcal{S}'_0(\mathbb{R}^d) \cdot \mathcal{S}_0(\mathbb{R}^d)) * \mathcal{S}_0(\mathbb{R}^d).$$

As a consequence $\mathcal{S}_0(\mathbb{R}^d)$ is w^* -dense in $\mathcal{S}'_0(\mathbb{R}^d)$.

Proof Convolution relations for WASPs imply

$$W(\mathcal{FL}^1, \ell^1)(\mathbb{R}^d) * W(\mathcal{FL}^\infty, \ell^\infty)(\mathbb{R}^d) \subseteq W(\mathcal{FL}^1, \ell^\infty)(\mathbb{R}^d),$$

shows that the convolution product belongs to the space of pointwise multipliers of $\mathcal{S}_0(\mathbb{R}^d)$, because the pointwise relation implies

$$W(\mathcal{FL}^1, \ell^\infty)(\mathbb{R}^d) \cdot W(\mathcal{FL}^1, \ell^1)(\mathbb{R}^d) \subseteq W(\mathcal{FL}^1, \ell^1)(\mathbb{R}^d) = \mathcal{S}_0(\mathbb{R}^d).$$

Combining these observations with the claims concerning approximate units in $(L^1(\mathbb{R}^d), \|\cdot\|_1)$ (Dirac sequences for convolution) and $(\mathcal{FL}^1(\mathbb{R}^d), \|\cdot\|_{\mathcal{FL}^1})$ (summability kernels for pointwise multiplication) as formulated above allows to conclude that the regularized versions of σ belong to $\mathcal{S}_0(\mathbb{R}^d)$ and converge to σ in the w^* -sense, as $\rho \rightarrow 0$, for any $\sigma \in \mathcal{S}'_0$. $\square$

These findings also allow to demonstrate that there is an equivalent *sequential approach* to *mild distributions* in the spirit of Lighthill's approach to tempered distribution ([53]), see [29] and the references given there. In fact, basic functional analytic arguments (such as the Banach–Steinhaus Theorem) allow to derive from Proposition 5.2 that mild distributions can be identified with ECmCs, i.e., *equivalence classes of mild Cauchy sequences*. So, in a rather natural sense $\mathcal{S}'_0(\mathbb{R}^d)$ can be viewed as a kind of completion of $\mathcal{S}_0(\mathbb{R}^d)$.

5.8 The Banach–Gelfand Triple $(\mathcal{S}_0, L^2, \mathcal{S}'_0)(\mathbb{R}^d)$

Together with the Hilbert space $(L^2(\mathbb{R}^d), \|\cdot\|_2)$ the three spaces form a so-called *Banach–Gelfand Triple*, based on the chain of continuous inclusions

$$(\mathcal{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathcal{S}_0}) \hookrightarrow (L^2(\mathbb{R}^d), \|\cdot\|_2) \hookrightarrow (\mathcal{S}'_0(\mathbb{R}^d), \|\cdot\|_{\mathcal{S}'_0}) \quad (5.10)$$

with density of embedding only in the w^* -sense for the second inclusion. There are a couple of papers meanwhile outlining the advantage of this setting, see [5, 10, 24, 28], or [32].

Recall that a *BGT homomorphism* is (by definition) continuous at each of the three layers, with the additional w^*w^* -continuity requirement at the outer layer. In this sense one can view the Fourier transform, which can be defined on $(\mathcal{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathcal{S}_0})$ just using Riemannian integrals, as a Banach–Gelfand Triple. By

norm approximation it extends to a unitary automorphism of $(L^2(\mathbb{R}^d), \|\cdot\|_2)$, and by duality (or alternatively via the sequential approach), it extends to a bounded operator on $S'_0(\mathbb{R}^d)$, which is not only norm to norm continuous, but also w^* - w^* -continuous.

It is generally a good recipe to view the inner space (i.e. $(S_0(\mathbb{R}^d), \|\cdot\|_{S_0})$) as a decent space, where integrals converge or transformations work well. Then one extends the operator to the Hilbert space, and finally by duality one reaches the continuous extension (with respect to both topologies on $S'_0(\mathbb{R}^d)$) by duality tricks. In fact, any bounded linear mapping between BGTs is the adjoint (in the sense of Hilbert spaces) of another BGT isomorphism, using the fact that the dual of $S'_0(\mathbb{R}^d)$, endowed with the w^* -topology, is just $S_0(\mathbb{R}^d)$. Thus $(S_0(\mathbb{R}^d), \|\cdot\|_{S_0})$ is not a reflexive Banach space but behaves often very much like a Hilbert space (some people call this situation a *rigged Hilbert space*).

The usual trick to extend the Fourier transform (from $S_0(\mathbb{R}^d)$ to $S'_0(\mathbb{R}^d)$) is of course not new, using the definition

$$\widehat{\sigma}(f) := \sigma(\widehat{f}), \quad f \in S_0(\mathbb{R}^d). \quad (5.11)$$

One has to verify that it is the unique, w^* - w^* -continuous extension of the automorphism of $S_0(\mathbb{R}^d)$ given by the Fourier transform (and restricted to $(L^2(\mathbb{R}^d), \|\cdot\|_2)$ one recovers Plancherel's Theorem). The fact that all kinds of "signals" of interest to engineers may be contained in $S'_0(\mathbb{R}^d)$ (but not in any of the spaces $(L^p(\mathbb{R}^d), \|\cdot\|_p)$, with $1 \leq p \leq \infty$), makes this definition a convenient tool, also to compare different settings via w^* -approximation, not just in a heuristic way.

Let us give a short list of examples. We will use the symbol $\Lambda \triangleleft \mathbb{R}^d$ here to describe lattices in $\mathbb{R}^d$, i.e., discrete and co-compact groups inside of $\mathbb{R}^d$. The standard lattice is just $\mathbb{Z}^d \triangleleft \mathbb{R}^d$ with fundamental domain $Q = Q_{\mathbb{Z}^d} := [-1/2, 1/2)^d$. Any such lattice has a fundamental domain Q_Λ , such that $(\lambda + Q_\lambda)_{\lambda \in \Lambda}$ provides a partition of $\mathbb{R}^d$. Since any such lattice is of the form $A(\mathbb{Z}^d)$ for some non-singular matrix A , we can take $Q_\Lambda = A(Q)$.

1. Given any bounded sequence $(c_\lambda) \in \ell^\infty(\Lambda)$ the weighted Dirac comb

$$\sigma_c := \sum_{\lambda \in \Lambda} c_\lambda \delta_\lambda$$

belongs to $S'_0(\mathbb{R}^d)$, and one finds for some constant $C'_\Lambda > 0$

$$\|\sigma_c\| \leq C'_\Lambda \|(c_\lambda)\|_{\ell^\infty(\Lambda)}.$$

2. For any given Λ any Λ -periodic and locally integrable function g defines a regular mild distribution, given by

$$\sigma_g(f) = \int_{\mathbb{R}^d} f(x)g(x)dx, \quad f \in S_0(\mathbb{R}^d).$$

In fact, there is a constant $C_\Lambda > 0$ such that for any such g one has:

$$\|\sigma_g\|_{S'_0(\mathbb{R}^d)} \leq C_\Lambda \|g \cdot \mathbf{1}_{Q_\Lambda}\|_1.$$

The (extended) Fourier transform then just provides the classical Fourier series expansion of g , with the interpretation that the *Fourier coefficients* are just the coefficients of a weighted Dirac comb, taken over the dual lattice $\Lambda^\perp$ (for the case $\Lambda = \mathbb{Z}^d$ we can identify this with $\mathbb{Z}^d$ itself).

3. The discrete Fourier transform defined on $\ell^1(\Lambda)$ can be seen as the restriction of the inverse Fourier transform back to the space of periodic functions. It extends of course also to the setting of bounded coefficients, creating periodic pseudo-measures.
4. Recall the definition of Wiener's algebra $(A(\mathbb{T}), \|\cdot\|_A)$ of absolutely convergent Fourier series. For the case $d = 1$ and $\Lambda = \Lambda^\perp = \mathbb{Z}$, they describe the range of the discrete Fourier transform. The generalized FT extends this isomorphism between $S_0(\mathbb{Z}) = \ell^1(\mathbb{Z})$ and $S_0(\mathbb{T}) = A(\mathbb{T})$ to a Banach–Gelfand Triple isomorphism between the *canonical Gelfand Triple* $(\ell^1, \ell^2, \ell^\infty)(\mathbb{Z})$ and the BGT $(A, L^2, PM)(\mathbb{T})$.
5. Finally, one can restrict the extended Fourier transform to periodic discrete signals and find that they are mapped onto discrete periodic signals. So in principle the Fourier transform in this setting can be viewed as the Fourier transform between two finite groups (of equal order, identifiable as a pair of dual finite Abelian groups). Consequently, (or by direct computation) one can show that this mapping restricts (in the corresponding natural bases) to the usual DFT, which can be implemented by the FFT.

The fact that the Segal algebra $S_0(\mathbb{R}^d)$ is invariant under automorphisms (in particular dilations) and under multiplication by the so-called *chirp* functions (also called characters of second degree by A. Weil in [72]) implies that $(S_0(\mathbb{R}^d), \|\cdot\|_{S_0})$ is invariant under the full *metaplectic group*. This fact deserves a separate detailed study of the subject, but specializing to the unitary transformations (there are many choices for the multi-dimensional case) called *Fractional Fourier Transforms* (FrFT), we can formulate immediately the following theorem:

Theorem 5.1 *For any (real multi-) parameter α , the fractional Fourier transform defines a unitary Banach–Gelfand Triple automorphism of $(S_0, L^2, S'_0)(\mathbb{R}^d)$.*

In particular, we observe that $\mathcal{F}_\alpha$ defines an automorphism of $(S_0(\mathbb{R}^d), \|\cdot\|_{S_0})$, and thus both $\mathcal{F}_\alpha$ and its inverse (with parameter $-\alpha$) are well defined on $(S_0(\mathbb{R}^d), \|\cdot\|_{S_0})$, for any α . The unitary version can be viewed as the extension of this integral transform to the Hilbert space $(L^2(\mathbb{R}^d), \|\cdot\|_2)$, while the extension (one might call it the generalized FrFT to $S'_0(\mathbb{R}^d)$) can be obtained by w^ -approximation (or by duality), as in the case of the ordinary Fourier transform.*

Proof The key arguments for this statement are the metaplectic invariance of $(S_0(\mathbb{R}^d), \|\cdot\|_{S_0})$ already established as Lemma 12 in the original paper [19] (see also Reiter's Lecture Notes [59] or the book of M. de Gosson, Proposition 82 in [13]

and [14]) combined with Theorem 7.3.3 of [39], describing the extension of unitary Gelfand Triple isomorphisms from $(S_0(\mathbb{R}^d), \|\cdot\|_{S_0})$ to the $(S_0, L^2, S'_0)(\mathbb{R}^d)$. $\square$

5.8.1 Cyclic Convolution of Periodic Functions

The purpose of this subsection is to demonstrate that (regular) BUPUs can be used in order to establish a clear connection between the extended FT of Λ -periodic functions on $\mathbb{R}^d$ and the classical theory of Fourier series.

We are used to describe a periodic function by its restriction to a fundamental domain. Integrability or square integrability of such a function is measured by the local L^1 or L^2 -norm over such a fundamental domain. Equivalently, one views such functions as functions on the quotient group $\mathcal{G}/\Lambda$.

In order to illustrate the general setting, it is enough to choose $\mathcal{G} = \mathbb{R}^d$ and $\Lambda = \mathbb{Z}^d$. By suitable reinterpretation of the periodicity lattice $\mathbb{Z}^d$ on the time side and the orthogonal lattice $\Lambda^\perp = \mathbb{Z}^d$ on the frequency side, the general case can be obtained by suitable substitution of terms in the following reasoning. In fact, it may be enough to first discuss the case $d = 1$. In that case the traditional way of looking at $\mathbb{Z}$ -periodic functions on $\mathbb{R}$ is to restrict them to $[0, 1)$, which is identified with the torus $\mathbb{T} = \{z \mid |z| = 1\}$, with the pure frequencies of the form $\chi_k(t) = \exp(2\pi ikt)$, $k \in \mathbb{Z}$.

Obviously, the characteristic function of the fundamental domain $[0, 1)$ generates a (discontinuous) partition of unity with respect to the lattice $\mathbb{Z}$. Alternatively, one might prefer to choose the symmetric version and take $\psi = \mathbf{I}_{[-1/2, 1/2)}$ (with the Fourier transform SINC) as such a generator, which obviously gives $f = \sqcup \sqcup * (f \cdot \psi)$. But this is true in general. We add a hopefully interesting comment concerning *cyclic convolution*.

Proposition 5.3 *Let Ψ be any BUPU (with respect to the lattice $\mathbb{Z}^d$). Then one has: Any $\mathbb{Z}^d$ -periodic, locally integrable function f on $\mathbb{R}^d$ is of the form*

$$f = \sqcup \sqcup * (f\psi), \quad (5.12)$$

with $f\psi \in L^1$, having compact support. Moreover, the cyclic convolution of two such functions f, g can be realized at the level of the localized versions, i.e., we have

$$f *_{\text{cycl}} g = \sqcup \sqcup * f\psi * g\psi, \quad (5.13)$$

where in fact convolution is associative and commutative.

Proof Identity (5.12), valid for any regular $\mathbb{Z}^d$ -BUPU, follows from

$$\sqcup \sqcup * f\psi = \sum_{k \in \mathbb{Z}^d} T_k(f\psi) = \sum_{k \in \mathbb{Z}^d} T_k(f)T_k(\psi) = f \cdot \sum_{k \in \mathbb{Z}^d} T_k(\psi) = f. \quad (5.14)$$

Taking the Fourier transform in the sense of $\mathcal{S}'_0(\mathbb{R}^d)$ gives

$$\mathcal{F}(\sqcup * f\psi) = \mathcal{F}(\sqcup) \cdot \mathcal{F}(f\psi) = \sqcup \cdot \mathcal{F}(f\psi) \quad (5.15)$$

Computing the samples of the Fourier transform of $f\psi$ over $\mathbb{Z}^d$ gives (writing Q for any fundamental domain of $\mathbb{R}^d$ w.r.t. $\Lambda = \mathbb{Z}^d$)

$$\mathcal{F}(f\psi)(n) = \int_{\mathbb{R}^d} f(x)\psi(x)e^{-2\pi i n x} dx = \sum_{k \in \mathbb{Z}^d} \int_{k+Q} f(x)\psi(x)e^{-2\pi i n x} dx. \quad (5.16)$$

Using the periodicity of both f and the pure frequency $\chi_n(x) = e^{-2\pi i n x}$, we can rewrite each term as $\int_Q \psi(x-k)f(x)e^{-2\pi i n x} dx$ and obtain thus, after changing the order of summation and integration (it is just a finite sum),

$$\begin{aligned} \mathcal{F}(f\psi)(n) &= \int_Q \left[\sum_{k \in \mathbb{Z}^d} \psi(x-k) \right] f(x) e^{-2\pi i n x} dx = \int_Q f(x) e^{-2\pi i n x} dx \\ &=: [\widehat{f}(n)]_{class} \end{aligned} \quad (5.17)$$

giving the classical Fourier coefficients and thus a w^* -convergent series representation:

$$f = \sum_{n \in \mathbb{Z}^d} [\widehat{f}(n)]_{class} \chi_n. \quad (5.18)$$

Let us write from now on $\widehat{f}(n)$ for $[\widehat{f}(n)]_{class}$.⁴

In other words, $\widehat{f}$ is a mild distribution, namely a weighted Dirac comb of the form $\sum_{k \in \mathbb{Z}^d} \widehat{f}(n)_{class} \delta_n$, and the identities above do not depend on the choice of the BUPU.

Now we note that the convolution of two periodic functions is not meaningful (in the same way as the pointwise product of two weighted Dirac combs is not meaningful, as a product of distributions). However, multiplication of the weights appearing makes perfect sense. It is equivalent to convolution of the corresponding integrable functions $f\psi$ and $g\psi$, respectively.

So $f *_{cycl} g$ is well defined as a mild distribution, in fact a locally integrable function, since $f\psi * g\psi$ is the convolution product of two (compactly supported) elements of $L^1(\mathbb{R}^d)$, hence belongs to $L^1(\mathbb{R}^d)$, and

$$\begin{aligned} [f\psi * g\psi] * \sqcup &\subseteq L^1(\mathbb{R}^d) * W(M, \ell^\infty) \\ &= W(L^1, \ell^1) * W(M, \ell^\infty) \subseteq W(L^1, \ell^\infty) \hookrightarrow \mathcal{S}'_0(\mathbb{R}^d). \end{aligned}$$

⁴ We think that there is no danger of confusion, since $\widehat{f}(n)$ could not be defined by the usual definition of the Fourier transform using Lebesgue integration, which makes sense for $f \in L^1(\mathbb{R}^d)$.

On the other hand it has a Fourier transform of the form

$$\sqcup \cdot \widehat{f\psi} \cdot \widehat{g\psi} = \sum_{n \in \mathbb{Z}^d} \widehat{f}(n) \widehat{g}(n) \delta_n.$$

□

Remark 5.4 These considerations show that Eq.(5.13) does not depend on Ψ , but also that cyclic convolution just corresponds to pointwise multiplication of the (classical) Fourier coefficients of f and g , respectively.

Remark 5.5 Note that we cannot expect to have (e.g., simply by reasoning about the size of the support of the two sides)

$$[f *_{\text{cycl}} g] \cdot \psi = (f\psi) * (g\psi).$$

Corollary 5.1 *For any BUPU Ψ in $\mathcal{FL}^1(\mathbb{R}^d)$ the statements concerning periodic elements extend to periodic elements $f \in \mathcal{S}'_0(\mathbb{R}^d)$. In fact, the (extended) Fourier transform allows to identify this subspace of $(\mathcal{S}'_0(\mathbb{R}^d), \|\cdot\|_{\mathcal{S}'_0})$ with the space of bounded sequences over the dual group $\mathbb{Z}^d$, i.e., with $(\ell^\infty(\mathbb{Z}^d), \|\cdot\|_{\ell^\infty})$, with norm equivalence. It is an algebra with respect to cyclic convolution.*

The identification of periodic distributions with distributions on the torus group $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ allows to identify this space with the space of pseudo-measures on the torus, the dual of the Wiener algebra $A(\mathbb{T}^d)$ of absolutely convergent Fourier series. Cyclic convolution corresponds then simply to pointwise multiplication in the pointwise algebra $\ell^\infty(\mathbb{Z}^d)$.

Remark 5.6 Using the terminology of Banach–Gelfand Triples, one can state the observation formulated in Corollary 5.1 as the fact that the extended Fourier transform establishes a unitary BGTr isomorphism between $(A, L^2, \mathbf{PM})(\mathbb{T}^d)$ and $(\ell^1, \ell^2, \ell^\infty)(\mathbb{Z}^d)$. Here the second BGTr can be identified with a triple of subspaces of mild distributions (corresponding to the viewpoint of considering them as periodic elements defined over all of $\mathbb{R}^d$) which are supported on the orthogonal lattice to the periodicity lattice, which is just $\mathbb{Z}^d$ in the simplified version. Alternatively, they can be viewed as subspaces of the corresponding coefficient sequences in $\ell^\infty(\mathbb{Z}^d)$, where pointwise multiplication makes sense. This pointwise multiplication taken back to the setting of periodic elements results in the (usual, and now reinterpreted) version of *cyclic convolution*.

5.9 Operators and Shift-Invariant Spaces

The setting of the Banach–Gelfand Triple and the fact that Dirac combs of the form $\sqcup_\Lambda := \sum_{\lambda \in \Lambda} \delta_\lambda$ for some discrete, co-compact group belong to $\mathcal{S}'_0(\mathbb{R}^d)$, with a Fourier transform which is (by Poisson’s formula valid for $\mathcal{S}'_0(\mathbb{R}^d)$) up to normal-

ization just $\sqcup_{\Lambda^\perp}$ (Dirac comb over the orthogonal lattice) allow to establish various fundamental principles which are also highly useful for engineering applications. In fact, regular sampling (i.e., sampling of a continuous function $h \in C_b(\mathbb{R}^d)$ along such a lattice) can be viewed as a multiplication operator, which corresponds to periodization on the Fourier transform side. Hence, if this does not produce overlaps (i.e., in the case of band-limited functions), perfect recovery is possible. Details are given in [32], but for illustration of the general principle let us show a demonstration for the case of $\Lambda = \mathbb{Z}^d \triangleleft \mathbb{R}^d$.

For this particular choice the Shannon sampling theorem can be formulated in the context of the Banach–Gelfand Triple setting as follows:

Theorem 5.2 *Let Ω be some compact subset of the open cube $(-1/2, 1/2)^d \subset \mathbb{R}^d$. Then we have for any $f \in (\mathcal{S}_0, L^2, \mathcal{S}'_0)$ with $\text{supp}(f) \subseteq \Omega$: For any function $g \in L^1(\mathbb{R}^d)$ with $\widehat{g}(y) \equiv 1$ on Ω and $\text{supp}(\widehat{g}) \cap (k + \Omega)$ for all $k \neq 0$ in $\mathbb{Z}^d$, we have*

$$f = \sum_{n \in \mathbb{Z}^d} f(n) T_n g. \quad (5.19)$$

Proof Note that $g \in L^1(\mathbb{R}^d)$ with $\widehat{g}$ compactly supported implies $g \in \mathcal{S}'_0(\mathbb{R}^d)$. Since $f \in \mathcal{S}'_0(\mathbb{R}^d)$, we have $\text{supp}(\widehat{f}) \in \mathcal{S}'_0(\mathbb{R}^d)$, and thus the support is perfectly well defined. Moreover, we can reproduce $\widehat{f}$ by pointwise multiplication with $\widehat{g}$ (i.e., $\widehat{f} \cdot \widehat{g} = \widehat{f}$) and consequently

$$f = f * g \in W(\mathcal{FL}^\infty, \ell^\infty) * W(\mathcal{FL}^1, \ell^1) \subset W(C_0, \ell^\infty) = C_b(\mathbb{R}^d).$$

For $f \in L^2(\mathbb{R}^d)$ the same argument gives $f \in W(C_0, \ell^2)$, and for $f \in L^1(\mathbb{R}^d)$ we have $f \in W(\mathcal{FL}^1, \ell^1) \hookrightarrow W(C_0, \ell^1)$.

The available information is of the form $\sqcup \cdot f = \sum_{n \in \mathbb{Z}^d} f(n) \delta_n$, a weighted Dirac comb with bounded coefficients $(f(n)) \in \ell^\infty(\mathbb{Z}^d)$, and thus belonging to $\mathcal{S}'_0(\mathbb{R}^d)$. Its FT in the sense of $\mathcal{S}'_0(\mathbb{R}^d)$ is thus (using the convolution theorem and the Fourier invariance of $\sqcup$):

$$\mathcal{F}(f \cdot \sqcup) = \mathcal{F}(f) * \mathcal{F}(\sqcup) = \widehat{f} * \sqcup = \sum_{n \in \mathbb{Z}^d} T_n \widehat{f}.$$

Clearly the support of this distribution is the disjoint union of the sets $n + \Omega$, $n \in \mathbb{Z}^d$, and thus recovery of $\widehat{f}$ from this periodized version can be obtained by pointwise multiplication with $\widehat{g}$. But this, taken back to the time side, means that the sampled version of f has to be convolved with g , so we get

$$f = (f \cdot \sqcup) * g = \sum_{n \in \mathbb{Z}^d} f(n) (\delta_n * g) = \sum_{n \in \mathbb{Z}^d} f(n) T_n g.$$

Here we observe that the sum is absolutely convergent in $(\mathcal{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathcal{S}_0})$ for $f \in \mathcal{S}_0(\mathbb{R}^d)$, while it is convergent in the $L^2(\mathbb{R}^d)$ (but also uniformly), because

$$(f \cdot \sqcup) * g \in [W(C_0, \ell^2) \cdot W(M, \ell^\infty)] * g \subset W(M, \ell^2) * W(C_0, \ell^1) \subset W(C_0, \ell^2).$$

Not that $\hookrightarrow L^2 \cap C_0$; hence convergence implies uniform convergence. For the $\mathcal{S}'_0(\mathbb{R}^d)$ -case, we have of course only w^* -convergence in $\mathcal{S}'_0(\mathbb{R}^d)$, but in fact a bounded series in $(C_b(\mathbb{R}^d), \|\cdot\|_\infty)$, with uniform convergence over compact subsets. $\square$

Remark 5.7 Note that we have used, as a part of the proof, the observation that the sampling procedure $f \mapsto (f(n))_{n \in \mathbb{Z}^d}$ is a Banach–Gelfand Triple homomorphism from $(\mathcal{S}_0, L^2, \mathcal{S}'_0)(\mathbb{R}^d)$ into $(\ell^1, \ell^2, \ell^\infty)(\mathbb{Z}^d)$, while the convolution operator (semi-discrete convolution of $f \cdot \sqcup$ with g) is another mapping back into $(\mathcal{S}_0, L^2, \mathcal{S}'_0)(\mathbb{R}^d)$.

We have seen that the setting of mild distributions allows to deal with discrete and continuous objects at equal footing inside of $\mathcal{S}'_0(\mathbb{R}^d)$. For band-limited functions recovery from the samples was possible by convolution with specific functions $g \in \mathcal{S}_0(\mathbb{R}^d)$. If we assume that $0 \in \Omega$ (typically one assumes that $\Omega = -\Omega$), then one has $\widehat{g}(n) = 1$ and $\widehat{g}(n) = 0$ for $0 \neq n \in \mathbb{Z}^d$. Thus we observe that the family $(T_n g)_{n \in \mathbb{Z}^d}$ almost forms a BUPU in $\mathcal{FL}^1(\mathbb{R}^d)$. The reservation comes from the fact that g itself will not have compact support because it is an analytic function. Therefore the term UCPU (uniformly concentrated partition of unit), as it is used in [15],⁵ is more appropriate.

Remark 5.8 Observe that the BUPU property can be viewed as a special case of the above theorem, since one has (obviously) $\mathcal{F}(\mathbf{1}) = \delta_0$ (e.g., by the observation that $\mathcal{F}(\mathcal{F}(f)) = f^\vee$, with $f^\vee(x) = f(-x)$ and the natural fact that $\mathcal{F}(\delta_0) = \mathbf{1}$).

A very similar argument can be used in order to verify the following characterization of BUPUs in $\mathcal{FL}^1(\mathbb{R}^d)$:

Lemma 5.8 *Given a compactly supported function in $\mathcal{FL}^1(\mathbb{R}^d)$, one has:*

The family $(T_n g)_{n \in \mathbb{Z}^d}$ forms a BUPU if and only if one has

$$g(n) = \delta_{0,n} \quad (\text{Kronecker Delta}).$$

Remark 5.9 This characterization reconfirms the fact that one can convolve any g creating a BUPU by any other compactly supported function $h \in L^1(\mathbb{R}^d)$ with $\int_{\mathbb{R}^d} h(y) dy = \widehat{h}(0) = 1$. In fact, $\mathcal{F}(g * h)(0) = \widehat{g}(0)\widehat{h}(0) = 1$ and is zero for any other $n \in \mathbb{Z}^d$.

Using the density of compactly supported functions in $(\mathcal{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathcal{S}_0})$ and the invariance under the Fourier transform, it is plausible that fine sampling allows to reconstruct the main, i.e., band-limited part of a function in $f \in \mathcal{S}_0(\mathbb{R}^d)$. This makes

⁵ Actually following a suggestion by this author.

plausible (but it is a highly nontrivial result) that one can approximately recover any function in $\mathbf{S}_0(\mathbb{R}^d)$ from sufficiently fine samples, using corresponding fine BUPUs. This result was formulated in detail in [49], which in turn is based on the results about *quasi-interpolation* developed in [38].

Basically the results says that one can factorize the (continuous, integral version of the) Fourier transform by applying the FFT (Fast Fourier transform), followed by quasi-interpolation.

Recall the following definition:

Definition 5.6 Given a BUPU Ψ we call the operator $\text{Sp}_\Psi : f \mapsto \text{Sp}_\Psi(f)$, given by

$$\text{Sp}_\Psi f(x) = \sum_{i \in I} f(x_i) \psi_i,$$

the *quasi-interpolation operator arising from Ψ* .

It is not difficult to check that the family of operators Sp_Ψ (with $|\Psi| \leq 1$) is uniformly bounded on $(\mathbf{C}_0(\mathbb{R}^d), \|\cdot\|_\infty)$ and $(\mathbf{W}(\mathbf{C}_0, \ell^1)(\mathbb{R}^d), \|\cdot\|_{\mathbf{W}})$ and that (due to the density of $\mathbf{C}_c(\mathbb{R}^d)$) one has

$$\lim_{\rho \rightarrow 0} \|\text{D}_{1/\rho} \Psi f - f\|_{\mathbf{W}(\mathbf{C}_0, \ell^1)} = 0, \quad f \in \mathbf{W}(\mathbf{C}_0, \ell^1)(\mathbb{R}^d).$$

In fact, the operator $\text{D}_\Psi : \mu \mapsto \text{D}_\Psi \mu$ is the adjoint operator to this operator (reconfirming w^* -convergence of these operators to the identity operator).

In order to illustrate the main statement concerning the realization of the FT via the FFT, we use the dilation operators D_ρ given by $\text{D}_\rho f(x) = f(\rho x)$, $\rho > 0$, $x \in \mathbb{R}^d$. Correspondingly, we define $\text{D}_{1/\rho} \Psi$ to be the family $(\text{D}_{1/\rho} \psi_i)_{i \in I}$. For a given BUPU of size δ $\text{D}_{1/\rho} \Psi$ is a BUPU of size $\rho\delta$.

A typical special case of the results in [49] now reads as follows:

Proposition 5.4 For any $f \in \mathbf{S}_0(\mathbb{R}^d)$ and any BUPU Ψ one has

$$\lim_{\rho \rightarrow 0} \|\text{D}_{1/\rho} \Psi f - f\|_{\mathbf{S}_0} = 0, \quad f \in \mathbf{S}_0(\mathbb{R}^d).$$

Moreover, applying the FFT to the finite discrete sequence of sampling values $(a_k)_{k=0}^{n-1}$ describing the basic period of the periodized version of f with period p and sampling rate p/n , one can recover a good approximation (in the norm of $(\mathbf{S}_0(\mathbb{R}^d), \|\cdot\|_{\mathbf{S}_0})$) by applying the quasi-interpolation operator to the sequence $(b_k)_{k=0}^{n-1}$, with $\mathbf{b} = \text{FFT}(\mathbf{a})$.

5.10 Sampling and Co-orbit Spaces

The finite overlap conditions play an important role for the description of WASps, but here the user can choose the admissible BUPUs allowing to characterize such spaces. In contrast, fine partitions of unity appeared quite early in the work on the reconstruction of band-limited functions f from irregular (meaning given) samples $(f(x_i))_{i \in I}$, where only the approximation error between the quasi-interpolation $Qf(x) = \sum_{i \in I} f(x_i)\psi_i(x)$ and the original function f (estimated uniformly for all the partitions with $|\Psi| \leq q$) is important for the possibility of perfect reconstruction (through iterations), with convergence in a variety of function spaces, such as weighted L^p -spaces.

This question gave rise to a whole direction of research. Somehow it is even the motivation for the SampTA Conference Series and was the motivation for the journal “Sampling Theory and Applications” which has been turned into “SaSiDa” and is published by Birkhäuser. Some of the early papers in this direction (since 1989) describing such iterative methods for families of function spaces are [36] and [37], for example. It was interesting to observe that various special cases (such as the method of nearest neighborhood approximation, using step functions, or piecewise linear methods) have been found useful by engineers, as one can see in the paper [60], without having a proof of convergence of the irregular case.

5.10.1 Iterative Reconstruction of Band-Limited Functions

The typical situation treated in these papers is the following: Assume you have a band-limited two-dimensional function f , with (up to a minor error, which can be controlled) $\text{supp}(\widehat{f}) \subset \Omega$, some compact set in $\mathbb{R}^2$. Furthermore assume that one has measurements $(f(x_i))_{i \in I}$, and the task is to (approximately) reconstruct f .

There are many different methods, including frame theoretic methods which work in the Hilbert space setting, i.e., if one can assume that $f \in L^2$. For more general spaces, such as weighted L^p -spaces the first generation of iterative algorithms can be described as follows, assuming that we want to reconstruct f iteratively from the data:

1. Use the information arising from the data in order to combine them with some form of quasi-interpolation using BUPUs which are centered at the points $(x_i)_{i \in I}$. Such BUPUs could be Voronoi domains (it is not problem to allow discontinuous BUPUs) or (for $d = 1$) some form of interpolation, such as piecewise linear interpolation. It is just important to know that the mapping $f \mapsto \sum_{i \in I} f(x_i)\psi_i$ is a bounded operator on the given space $\mathcal{B}$.
2. In the second step one has to project the function back to the space of band-limited functions. This can be done using the perfect filter (multiplying by the indicator function $\mathbf{1}_\Omega$ on the Fourier transform side), but it is better to use a well-localized version, typically a band-limited function g (as in Shannon’s Theorem)

from some Beurling algebra $(L_w^1(\mathbb{R}^d), \|\cdot\|_{1,w})$, in order to push back the first (rough) approximation to the space of band-limited functions with slightly larger bandwidth (corresponding to $\text{supp}(\widehat{g})$ instead of Ω).

3. The key step of this approach using BUPUs (they do NOT have to satisfy the finite overlap condition in this context, but rather have to be fine enough, typically strictly better than the Nyquist rate that one would expect) is to show that the approximation obtained in this form, i.e., by $A(f) = \sum_{i \in I} f(x_i)(\psi_i * g)$ (another quasi-interpolation operator), is getting somehow close to the identity, meaning that one has for some $\gamma < 1$;

$$\|A(f) - f\|_{\mathbf{B}} \leq \gamma \|f\|_{\mathbf{B}}, \quad \forall f \in \mathbf{B}, \text{supp}(\widehat{f}) \subseteq \Omega. \quad (5.20)$$

4. Thus one can start an iterative reconstruction procedure which allows to guarantee convergence in $(\mathbf{B}, \|\cdot\|_{\mathbf{B}})$, at the rate of γ^n . At each step a set of coefficients $(c_i^n)_{i \in I}$ is recovered, and the final reconstruction uses the coefficients $c_i := \sum_n c_i^n$.
5. In fact, there are some extra considerations required because obviously g cannot be a convolution idempotent, since $\widehat{g}(0) = 1$ and $\lim_{|s| \rightarrow \infty} \widehat{g}(s) = 0$ by the Riemann–Lebesgue Lemma, but we ignore such technical details here.

5.10.2 Generalizations: Averages and Spline-Type Spaces

The key arguments for the reconstruction are the following ones, which we illustrate for simplicity for the function spaces $(L^p(\mathbb{R}^d), \|\cdot\|_p)$, with $1 \leq p < \infty$:

1. There are reproducing functions (band-limited functions g as in the proof of Shannon’s Theorem 5.2) in $\mathcal{S}_0(\mathbb{R}^d)$, with $f = f * g$.
2. On the spaces of Ω -band-limited elements the norms of $L^p(\mathbb{R}^d)$ and of $W(C_0, \ell^p)(\mathbb{R}^d)$ are equivalent (uniformly with respect to p !).
3. For sufficiently fine partitions of unity, one can guarantee that the quasi-interpolator $Q_\Psi f$ provides a (sufficiently) good approximation in order to allow for an iterative process, which is convergent at the rate γ^n , as $n \rightarrow \infty$.
4. It can be shown that the approach is rather robust, i.e., the achievable estimate for $\gamma < 1$ will be better the smaller $|\Psi|$ is; moreover, there is robustness toward, e.g., *jitter error* imprecise information about the exact location of the points $(x_i)_{i \in I}$.

Later it turned out the same strategy can be applied for the so-called *spline-type* spaces, such as typically spaces of cubic spline functions over the real line, even if they are not band-limited. But still they have a particular structure (and knowledge of the generating family of perhaps multi-window systems is required). This direction was started in [1], and a good survey is [3]. In the approximation theory community such spaces are discussed by the name of *shift-invariant* spaces, see, e.g., [4, 9], or [64], but also the recent paper [54].

Another direction, where still the same principles are applied, is the reconstruction from local averages, which to our best knowledge started with [2]. The application of this approach to spline-type spaces was done in [65] or [66]. A further extension was provided in [63]. Since then, a large number of similar papers have been published by many authors, for different families of function spaces and other ingredients.

5.10.3 Co-orbit Spaces

Let us finally mention that very similar ideas are at the basis of *co-orbit theory*, which has been developed jointly by the author and K. Gröchenig (see, e.g., [34] for the main paper in this direction, which has started a whole new research area with many subsequent results in the mathematical literature based on this approach).

Here the starting point is the fact that the so-called *voice transform*, which in concrete case corresponds to the *continuous wavelet transform* or the Short-Time Fourier Transform (STFT) of a given distribution f with respect to a “decent” window g , has many properties comparable to a band-limited function on $\mathbb{R}^d$. In fact, $V_g(f)$ arises from a given irreducible and integrable representation π of some locally compact group $\mathcal{G}$ on some Hilbert space by the (extended) scalar product:

$$F(x) := V_g(f)(x) = \langle f, \pi(x)g \rangle, \quad x \in \mathcal{G}. \quad (5.21)$$

Writing $G := V_g(g)$, one finds that there is a remarkable reproduction property for F , namely $F = F * G$, for g suitably chosen, such that $G \in L^1(\mathcal{G})$. In particular, one has $G * G = G$ (where convolution is taken in the sense of the L^1 -algebra over $\mathcal{G}$ with respect to the Haar measure on this group).

Depending on the group (and perhaps the concrete version of the representation π under consideration), one obtains discrete representations of different types. The wavelet transform is related to the natural representation of the affine, the so-called $ax + b$ -group on $(L^2(\mathbb{R}^d), \|\cdot\|_2)$, which is non-unimodular and thus a bit more tricky compared to the STFT.

Time–Frequency Analysis is centered around the use of TF shifts and the STFT (see [35] for an early description of the links between the two areas, or [44]). From the point of view of co-orbit theory, it arises essentially from the Schrödinger representation of the Heisenberg groups via TF shifts. Here the discrete characterizations are the so-called *Gabor expansions of distributions*, which are well suited to characterize the membership of distributions in the modulation spaces introduced above.

At the technical level the reconstruction from the samples of $(F(x_i))_{i \in I}$ can be performed by a similar strategy, assuming that the family of sampling points in the group is dense enough in $\mathcal{G}$. In other words, given g (or the information about $\|T_x G - G\|_1$ for x near $0 \in \mathcal{G}$), one can find some neighborhood U of the neutral

element $0 \in \mathcal{G}$ such that reconstruction is possible, if the left translates of U , i.e., $(x_i U)$, form a covering of $\mathcal{G}$.

Again, the crucial first step for this theory is the approximation of F by a quasi-interpolation operator using a BUPU on $\mathcal{G}$, adapted to the (irregular) family $(x_i)_{i \in I}$. The corresponding arguments make use of WASPs over these groups.

As an alternative, and more prominent feature of the approach, one also obtains atomic representations of F as a sum of translates of the form $T_{x_i} G$, but this involves similar general ideas, but more refined estimates, using Wiener amalgams. Translating back to the original domain (say the Hilbert spaces which is used as a starting point) one obtains atomic representations of f as

$$f = \sum_{i \in I} c_i \pi(x_i) g. \quad (5.22)$$

Here the coefficients depend in a linear way on the distribution/signal f , and depending on the quality of the function f , one can guarantee decay or summability conditions for the coefficient family $(c_i)_{i \in I}$.

Because the Heisenberg group is an [IN]-group, meaning it has compact neighborhoods of the identity Q which are invariant under inner automorphism, i.e., satisfy $xQ = Qx$ for all $x \in \mathcal{G}$, one still has the expected convolution results for WASPs, which helps a lot in deriving the atomic (Gabor) characterizations of modulation spaces.

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Chapter 6

Estimate of the L^p -Fourier Transform Norm for Some Lie Groups



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Abstract This chapter aims to survey some recent results on the estimate of the norm of the L^p -Fourier transform ($1 < p \leq 2$) of some Lie groups. We focus on the setting of connected nilpotent Lie groups, the semidirect product $\mathbb{R}^n \rtimes K$ (and also some universal covering groups), where K stands for a compact subgroup of automorphisms of $\mathbb{R}^n$, and the case of compact extensions of locally compact groups. We highlight on the fact that the orbit method and the Plancherel theory play a crucial role in getting the results of the estimates, without any hope to get extremal functions, except in some particular cases.

6.1 Introduction

For $\mathbb{R}^n$ with a standard inner product $\langle \cdot, \cdot \rangle$, we identify $\mathbb{R}^{n*}$ with $\mathbb{R}^n$ by $\mathbb{R}^n \ni x \mapsto \ell_x \in \mathbb{R}^{n*}$ so that $\ell_x(y) = \langle x, y \rangle$, $y \in \mathbb{R}^n$. Let dx be the Lebesgue measure on $\mathbb{R}^n$ such that the unit cube has mass 1. Let $f \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$, and define the Fourier transform of the function f as

$$\widehat{f}(x) = \int_{\mathbb{R}^n} f(y) e^{i\langle x, y \rangle} dy. \quad (6.1)$$

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Then, by the Plancherel classical formula, we have

$$\int_{\mathbb{R}^n} |\widehat{f}(y)|^2 \frac{dy}{(2\pi)^n} = \int_{\mathbb{R}^n} |f(x)|^2 dx. \quad (6.2)$$

Let $1 < p \leq 2$, $q = \frac{p}{p-1}$, and $f \in L^1(\mathbb{R}^n) \cap L^p(\mathbb{R}^n)$. Then, by Beckner [8], we have

$$\left(\int_{\mathbb{R}^n} |\widehat{f}(y)|^q \frac{dy}{(2\pi)^n} \right)^{\frac{1}{q}} \leq A_p^n \left(\int_{\mathbb{R}^n} |f(x)|^p dx \right)^{\frac{1}{p}}, \quad (6.3)$$

where

$$A_p = \sqrt{\frac{p^{\frac{1}{p}}}{q^{\frac{1}{q}}}}.$$

Next, let $f(x) := e^{-\|x\|^2/2}$. Since $\widehat{f}(x) = (2\pi)^{\frac{n}{2}} f(x)$, with the normalized Lebesgue measures dx on $\mathbb{R}^n$ and $(2\pi)^{-n} dy$ on $\mathbb{R}^{n*}$, we get

$$\|\widehat{f}\|_q = A_p^n \|f\|_p, \quad (6.4)$$

which leads to equality in (6.3).

Let now G be a separable unimodular locally compact group of type I and $\widehat{G}$ its unitary dual, the set of equivalence classes of irreducible unitary representations, endowed with the Mackey–Borel structure. Let dg be the Haar measure on G . The operator-valued Fourier transform on G maps each $\varphi \in L^1(G)$ to the field $\mathcal{F}\varphi = (\pi(\varphi))_{\pi \in \widehat{G}}$ of bounded operators on $\widehat{G}$, where $\pi(\varphi)$ is defined by:

$$\pi(\varphi) = \int_G \varphi(g) \pi(g) dg,$$

which generalizes formula (6.1) in case $G = \mathbb{R}^n$. Let μ_G be the Plancherel measure on $\widehat{G}$, which is uniquely determined by the abstract Plancherel formula: For $\varphi \in L^1(G) \cap L^2(G)$,

$$\int_G |\varphi(g)|^2 dg = \int_{\widehat{G}} \text{tr}((\pi(\varphi)^* \pi(\varphi)) d\mu_G(\pi),$$

which is also the equivalent of formula (6.2) for $\mathbb{R}^n$. The Hausdorff–Young inequality, generalized by Kunze in [21] for separable locally compact unimodular groups, is the assertion

$$\left(\int_{\widehat{G}} \|\pi(\varphi)\|_{c_q}^q d\mu_G(\pi) \right)^{\frac{1}{q}} \leq \left(\int_G |\varphi(g)|^p dg \right)^{\frac{1}{p}}, \quad (6.5)$$

where $\varphi \in L^1(G) \cap L^p(G)$, $1 < p \leq 2$, q is the conjugate of p , i.e., $\frac{1}{p} + \frac{1}{q} = 1$, and $\|\pi(\varphi)\|_{c_q}$ is the q th Schatten norm of the operator $\pi(\varphi)$

$$\|\pi(\varphi)\|_{c_q}^q = \operatorname{tr} \left((\pi(\varphi)^* \pi(\varphi))^{\frac{q}{2}} \right).$$

Let $L^q(\widehat{G})$ be the Banach space of μ_G -measurable field of bounded operators F on $\widehat{G}$ with the norm $\|F\|_q < \infty$, where

$$\|F\|_q = \left(\int_{\widehat{G}} \|F(\pi)\|_{c_q}^q d\mu_G(\pi) \right)^{\frac{1}{q}}.$$

Then we can write (6.5) as

$$\|\mathcal{F}\varphi\|_q \leq \|\varphi\|_p,$$

and thus the map $\varphi \mapsto \mathcal{F}\varphi$ from $L^1(G) \cap L^p(G)$ to $L^q(\widehat{G})$ extends to a continuous operator $\mathcal{F}^p : L^p(G) \rightarrow L^q(\widehat{G})$ and its operator norm

$$\|\mathcal{F}^p(G)\| = \sup_{\|\varphi\|_p \leq 1} \|\mathcal{F}\varphi\|_q \leq 1.$$

For the abelian group $G = \mathbb{R}^n$, it is proved that

$$\|\mathcal{F}^p(\mathbb{R}^n)\| = A_p^n,$$

by Babenko [2] for $p = \frac{2k}{2k-1}$, where $k \geq 2$ is an integer, and by Beckner [8] for all p ($1 < p < 2$) and the Gaussian function turns out to be an extremal function as computed in Eq. (6.4).

On the other hand, it is shown by Fournier [14] that $\|\mathcal{F}^p(G)\| = 1$ if and only if G contains a compact open subgroup. For various classes of non-abelian, non-compact groups, further studies of Hausdorff–Young inequalities were made, and many results concerning better estimates of $\|\mathcal{F}^p(G)\|$ were obtained. See, e.g., [3–7, 10, 13, 16, 18, 19, 22, 25–29].

This chapter aims to record some results on the estimates of $\|\mathcal{F}^p(G)\|$ in the settings of connected nilpotent Lie groups and some compact extensions of locally compact groups.

6.2 The Settings of Nilpotent Connected Lie Groups

Throughout this section, $\mathfrak{g}$ will be an n -dimensional real nilpotent Lie algebra and $\widetilde{G}$ the associated Lie group. Then the exponential map $\exp : \mathfrak{g} \rightarrow \widetilde{G}$ is a global C^∞ -diffeomorphism from $\mathfrak{g}$ onto $\widetilde{G}$.

6.2.1 The Kirillov Theory

Let $\mathfrak{g}^*$ be the vector dual space of $\mathfrak{g}$. The Lie algebra $\mathfrak{g}$ acts on $\mathfrak{g}$ by the adjoint representation $\text{ad}_{\mathfrak{g}}$, that is, $\text{ad}_{\mathfrak{g}}(X)(Y) = \text{ad}(X)(Y) = [X, Y]$, $X, Y \in \mathfrak{g}$. The group $\tilde{G}$ acts on $\mathfrak{g}$ by the adjoint representation $\text{Ad}_{\tilde{G}}$, i.e.,

$$\text{Ad}_{\tilde{G}}(g)(Y) = \text{Ad}(g)(Y) = \sum_{n=0}^{\infty} \frac{(\text{ad}(X))^n}{n!} Y, \quad g = \exp X \in \tilde{G}, \quad \text{for } X \text{ and } Y \in \mathfrak{g}$$

and on $\mathfrak{g}^*$ by the coadjoint representation $\text{Ad}_{\tilde{G}}^*$, i.e.,

$$\langle \text{Ad}_{\tilde{G}}^*(g)\ell, X \rangle = \langle g \cdot \ell, X \rangle = \langle \ell, \text{Ad}(g^{-1})(X) \rangle, \quad g \in \tilde{G}, \quad \ell \in \mathfrak{g}^*, \quad X \in \mathfrak{g}.$$

The coadjoint orbit of ℓ is the set $\tilde{G} \cdot \ell = \{g \cdot \ell, g \in \tilde{G}\}$. The space of coadjoint orbits is denoted by $\mathfrak{g}^*/\tilde{G}$. Let $\mathfrak{g}(\ell) = \{X \in \mathfrak{g}; \ell([X, \mathfrak{g}]) = \{0\}\}$ be the stabilizer of $\ell \in \mathfrak{g}^*$ in $\mathfrak{g}$ which is actually the Lie algebra of the Lie subgroup $\tilde{G}(\ell) = \{g \in \tilde{G}, g \cdot \ell = \ell\}$. The unitary dual $\widehat{\tilde{G}}$ of $\tilde{G}$ is parameterized via the Kirillov orbit method by the space of coadjoint orbits. Let $\ell \in \mathfrak{g}^*$, and we take a real polarization $\mathfrak{h} = \mathfrak{h}[\ell]$ at ℓ . This means that $\mathfrak{h}[\ell]$ is a maximal dimensional isotropic subalgebra with respect to the skew-symmetric bilinear form B_{ℓ} defined by $B_{\ell}(X, Y) = \ell([X, Y])$, $X, Y \in \mathfrak{g}$. For such a polarization, we consider the unitary character χ_{ℓ} of $H[\ell] = \exp(\mathfrak{h}[\ell])$ defined by $\chi_{\ell}(\exp X) = e^{2\pi i \ell(X)}$, $X \in \mathfrak{h}[\ell]$, and the unitary representation

$$\pi_{\ell} = \pi_{\ell, \mathfrak{h}[\ell]} = \text{Ind}_{H[\ell]}^{\tilde{G}} \chi_{\ell}.$$

Then π_{ℓ} is an irreducible unitary representation of $\tilde{G}$, and its equivalence class $[\pi_{\ell, \mathfrak{h}[\ell]}]$ depends only on the coadjoint orbit of ℓ . Moreover, every irreducible unitary representation π is equivalent to an induced representation $\pi_{\ell, \mathfrak{h}}$ for some $\ell \in \mathfrak{g}^*$ and a polarization $\mathfrak{h}$ at ℓ . The following mapping, called the Kirillov mapping

$$\begin{aligned} K : \mathfrak{g}^*/\tilde{G} &\rightarrow \widehat{\tilde{G}} \\ \tilde{G} \cdot \ell &\mapsto [\pi_{\ell, \mathfrak{h}[\ell]}], \end{aligned}$$

is a homeomorphism (see [9]).

6.2.2 The Plancherel Formula

(See [9, section 4].) Let

$$\{0\} = \mathfrak{g}_0 \subset \mathfrak{g}_1 \subset \cdots \subset \mathfrak{g}_n = \mathfrak{g} \quad (6.6)$$

be a Jordan–Hölder sequence of the nilpotent Lie algebra $\mathfrak{g}$ (a flag of ideals of $\mathfrak{g}$ such that $\dim \mathfrak{g}_i = i$, $i = 0, \dots, n$). We extract from (6.6) a Jordan–Hölder basis $\mathcal{B} = \{X_1, \dots, X_n\}$ by taking $X_i \in \mathfrak{g}_i \setminus \mathfrak{g}_{i-1}$, $i = 1, \dots, n$. So, obviously $\mathfrak{g}_1 = \mathbb{R}X_1$ is a central ideal of $\mathfrak{g}$. Let $\mathcal{B}^* = \{X_1^*, \dots, X_n^*\}$ be the dual basis of $\mathfrak{g}^*$ dual to $\mathcal{B}$, which is a Jordan–Hölder basis for the coadjoint action of G on $\mathfrak{g}^*$. Let $\ell \in \mathfrak{g}^*$, an index $i \in \{1, \dots, n\}$ is said to be a jump index for ℓ if $\mathfrak{g}(\ell) + \mathfrak{g}_i \neq \mathfrak{g}(\ell) + \mathfrak{g}_{i-1}$. Let $e(\ell)$ be the set of jump indices for ℓ . This set contains exactly $\dim(\tilde{G} \cdot \ell)$ indices, which is necessarily an even number. Even more, there are two disjoint sets of indices S, T with $S \cup T = \{1, \dots, n\}$, and a $\tilde{G}$ -invariant non-empty Zariski open set $\mathcal{U}$ of $\mathfrak{g}^*$ such that $e(\ell) = S$ for all $\ell \in \mathcal{U}$. Let $\text{Pf}(\ell)$ denote the Pfaffian of the skew-symmetric matrix $M_S(\ell) = (\ell([X_i, X_j]))_{i,j \in S}$. Moreover, let $T = \{1, \dots, n\} \setminus S$ and $V_T = \mathbb{R}\text{-span}\{X_i^* : i \in T\}$. Then $\mathcal{W} = \mathcal{U} \cap V_T$ is a cross section for the generic orbits, and $\mathcal{W}$ supports the Plancherel measure on $\tilde{G}$. So, every $\tilde{G}$ -orbit in $\mathcal{U}$ related to a representation π meets $\mathcal{W}$ in a single unique element l_π which we call the *fundamental element* of π . Furthermore, if dl is the Lebesgue measure on $\mathcal{W}$, then $d\mu = |\text{Pf}(l)|dl$ is a Plancherel measure for $\hat{\tilde{G}}$ the unitary dual of $\tilde{G}$. The Plancherel formula reads

$$\|\varphi\|_2^2 = \int_{\tilde{G}} |\varphi(g)|^2 dg = \int_{\mathcal{W}} \|\pi_l(\varphi)\|_{c_2}^2 d\mu(l). \quad (6.7)$$

Then we have the following:

$$\|\mathcal{F}^p(\varphi)\|_q^q = \int_{\mathcal{W}} \|\pi_l(\varphi)\|_{c_q}^q d\mu(l).$$

6.2.3 Case of Simply Connected Nilpotent Lie Groups

We here provide an estimate of the L^p -Fourier transform norm for connected simply connected nilpotent Lie groups using the orbit method. The main result (Baklouti–Smaoui–Ludwig [7]) is the following:

Theorem 6.1 ([7, Theorem 2]) *Let G be a connected and simply connected nilpotent Lie group with Lie algebra $\mathfrak{g}$ and m be the dimension of the generic orbits. Then, for $1 < p < 2$,*

$$\|\mathcal{F}^p(G)\| \leq A_p^{\frac{2 \dim G - m}{2}}. \quad (6.8)$$

Remark 6.1 In [18], Inoue proved the inequality (6.8) in the setting of a restricted class of nilpotent Lie groups admitting an open dense subset $\mathcal{U}$ of $\mathfrak{g}^*$, such that the ideal generated by $\bigcup_{f \in \mathcal{U}} \mathfrak{g}(f)$ is abelian.

On the other hand, in the early work [26, Proposition 2], Russo obtained similar estimates based on the Plancherel formula of Dixmier [11, Théorème 4], [12,

Corollaire 1], and Pukanszky [24, p.108]. By Theorem 6.1 and [18], we found again the estimate of [26, Proposition 2] and specifically describe the upper bound in terms of coadjoint orbits.

Remark 6.2 Let us focus on the special case of even exponents $q = 2k$, where $k \geq 2$ is an integer. Let G be a separable unimodular locally compact group of type I. Then, for a continuous function φ on G with compact support and $\pi \in \widehat{G}$, we have

$$\begin{aligned} \|\pi(\varphi)\|_{c_q}^q &= \text{tr}((\pi(\varphi)^* \pi(\varphi))^{\frac{q}{2}}) = \text{tr}((\pi(\varphi)^* \pi(\varphi))^k) \\ &= \text{tr}(\pi(\Phi_k)^* \pi(\Phi_k)) = \|\pi(\Phi_k)\|_{c_2}^2, \end{aligned}$$

$$\text{where } \Phi_k = \varphi_1 * \varphi_2 * \cdots * \varphi_k, \quad \text{and } \varphi_j = \begin{cases} \varphi & (\text{if } j \text{ is odd}) \\ \varphi^* & (\text{if } j \text{ is even}), \end{cases} \quad 1 \leq j \leq k.$$

Thus we have

$$\int_{\widehat{G}} \|\pi(\varphi)\|_{c_q}^q d\mu_G(\pi) = \int_{\widehat{G}} \|\pi(\Phi_k)\|_{c_2}^2 d\mu_G(\pi) = \|\Phi_k\|_2^2$$

by the Plancherel formula, that is,

$$\|\mathcal{F}^p(\varphi)\|_q^q = \|\Phi_k\|_2^2. \quad (6.9)$$

In [19], Klein and Russo showed (6.9) and described the norm $\|\mathcal{F}^p(G)\|$ for $p = \frac{2k}{2k-1}$ by the best constant $Y_p(G)$ of the Young inequality for the convolutions of L^p functions $f_1, \dots, f_k$, that is,

$$\|\mathcal{F}^p(G)\| = Y_p(G)^{\frac{1}{k}}, \quad (6.10)$$

$$\text{where } Y_p(G) = \sup_{f_j \neq 0, j=1, \dots, k} \frac{\|f_1 * \cdots * f_k\|_2}{\prod_{j=1}^k \|f_j\|_p}$$

[19, Lemma 2.6]. (Note that they actually stated (6.9) and (6.10) under a general setting which is not necessarily type I.) Concerning Young's inequality, they obtained upper bounds for the best constants for general exponents greater than or equal to 1 on semidirect product groups of a general class including non-unimodular groups [19, Section 2].

By applying the results of Young's inequalities, in the setting of connected simply connected nilpotent Lie groups, they showed the inequality $\|\mathcal{F}^p(G)\| \leq A_p^{\dim G}$ for $p = \frac{2k}{2k-1}$ with $k \geq 2$ an integer [19, Corollary 2.8]. In particular, when G is the $2n+1$ -dimensional Heisenberg group, they obtained the equality $\|\mathcal{F}^p(G)\| = A_p^{2n+1}$ and showed that there are no extremal functions [19, Theorem 3] by the analysis of the Weyl operators developed in [19].

Recently, Cowling–Martini–Müller–Parcet [10, Corollary 2.7] proved that the equality $\|\mathcal{F}^p(G)\| = A_p^{\dim G}$ holds for $p = \frac{2k}{2k-1}$ with $k \geq 2$ an integer if G is an n -dimensional solvable Lie group admitting a chain of closed subgroups

$$\{e\} = G_0 < \cdots < G_n = G,$$

where G_j is normal in G_{j+1} and G_{j+1}/G_j is isomorphic to $\mathbb{R}$, namely the setting of completely solvable Lie groups which generalizes the case of nilpotent Lie groups discussed above.

The following result proves Theorem 6.1 as a direct consequence and makes use of a localized version of the Plancherel formula (6.7).

Theorem 6.2 ([7, Theorem 3]) *Let G be a connected and simply connected nilpotent Lie group and Z be the center of G . Let $A = \exp \mathfrak{a}$ be a closed connected subgroup of Z and $\chi = \chi_\psi$, $\psi \in \mathfrak{a}^*$, be a unitary character of A . Then, for $1 < p < 2$,*

$$\left(\int_{\widehat{G}_\chi} \|\pi(\varphi)\|_{c_q}^q d\mu_\chi(\pi) \right)^{\frac{1}{q}} \leq A_p^{\frac{2 \dim(G/A) - m_\psi}{2}} \|\varphi\|_{L^p(G/A)},$$

where $\varphi \in C_c(G/A, \psi)$ and $m_\psi = \sup \left\{ \dim(G \cdot \phi) ; \phi \in \mathfrak{g}_\psi^* \right\}$, $\mathfrak{g}_\psi^* = \psi + \mathfrak{a}^\perp$.

Here we briefly describe the details: Let $1 \leq p \leq 2$, $d_{G/A}$ be a Haar measure on G/A and $L^p(G/A, \chi)$ be the space of all measurable functions ϕ on G such that $\phi(gh) = \bar{\chi}_\lambda(h)\phi(g)$ for almost all $g \in G$ and $h \in A$ and

$$\|\phi\|_{L^p(G/A)}^p := \int_{G/A} |\phi(g)|^p d_{G/A}g < \infty.$$

Let $\widehat{G}_\chi = \{\tau \in \widehat{G}; \tau|_A = \chi\}$. For $\tau \in \widehat{G}_\chi$ and $\phi \in L^1(G/A, \chi)$, we define

$$\tau(\phi) := \int_{G/A} \tau(g)\phi(g) d_{G/A}g.$$

Let $d\mu_\chi$ be the Plancherel measure on $\widehat{G}_\chi$ associated with the Haar measure $d_{G/A}g$ on G/A , that is, there exists a Borel measure $d\mu_\chi$ on $\widehat{G}_\chi$ uniquely determined by

$$\int_{G/A} |\phi(g)|^2 d_{G/A}g = \int_{\widehat{G}_\chi} \|\tau(\phi)\|_{c_2}^2 d\mu_\chi(\tau), \quad \text{where} \quad \|\tau(\phi)\|_{c_2}^2 = \text{tr } \tau(\phi^* * \phi), \quad (6.11)$$

for all $\phi \in L^1(G/A, \chi) \cap L^2(G/A, \chi)$. We obtain therefore a continuous mapping $\mathcal{F}^p$ from $L^p(G/A, \chi)$ to $L^q(\widehat{G}_\chi)$ defined by

$$\mathcal{F}^p \phi(\tau) = \tau(\phi).$$

Based on the fact that the operators $\pi(\varphi)$, ($\pi \in \widehat{G}, \varphi \in C_c(G)$) are kernel operators, one of the main ingredients of the proof of Theorem 6.2 is the following Hausdorff–Young type’s inequality for integral operators.

6.2.4 Hausdorff–Young Type’s Inequality for Integral Operators

This inequality is due to Fournier and Russo [15]. Let $\mathcal{H}$ be a complex Hilbert space, $\mathcal{B}(\mathcal{H})$ be the space of bounded operators on $\mathcal{H}$, and $\mathcal{X}$ be a σ -finite measure space. Let $L^2(\mathcal{X}, \mathcal{H})$ be the Hilbert space of square integrable $\mathcal{H}$ -valued functions on $\mathcal{X}$ and K an integral operator on $L^2(\mathcal{X}, \mathcal{H})$ with an operator-valued kernel k defined as

$$K\eta(x) = \int_{\mathcal{X}} k(x, y)\eta(y)dy,$$

for all $\eta \in L^2(\mathcal{X}, \mathcal{H})$ and almost all $x \in \mathcal{X}$. Let $k^*(x, y) = \overline{k(y, x)}$, $x, y \in \mathcal{X}$.

For p, q in $[1, +\infty[$, let us define

$$\|k\|_{p,q} = \left(\int_{\mathcal{X}} \left(\int_{\mathcal{X}} \|k(x, y)\|_{\mathcal{C}_q}^p dx \right)^{\frac{q}{p}} dy \right)^{\frac{1}{q}}.$$

If $1 < p \leq 2$, q is the conjugate of p , and if $\|k\|_{p,q}$ and $\|k^*\|_{p,q}$ are finite, then the integral operator K belongs to $C_q(L^2(\mathcal{X}, \mathcal{H}))$ and

$$\|K\|_{C_q} \leq \|k\|_{p,q}^{\frac{1}{2}} \|k^*\|_{p,q}^{\frac{1}{2}}. \quad (6.12)$$

6.2.5 Removing the Simply Connectedness of G

Let G designate a connected nilpotent Lie group, $\widetilde{G} = \exp_{\widetilde{G}}(\mathfrak{g})$ its simply connected covering group, and Γ a discrete normal subgroup of $\widetilde{G}$ such that $G \simeq \widetilde{G}/\Gamma$. Following [9, section 4.4], we parametrize the unitary dual $\widehat{G}$: Let $P : \widetilde{G} \rightarrow G$ denote the natural projection, and $\Lambda = \log \Gamma$, which is an additive subgroup of the center $\mathfrak{z}$ of $\mathfrak{g}$. Set $\mathfrak{g}_{\Lambda}^* = \{\ell \in \mathfrak{g}^* : \ell(\Lambda) \subseteq \mathbb{Z}\}$ and $\mathfrak{h} = \mathbb{R}\text{-span}(\Lambda)$. It is clear that $\mathfrak{g}_{\Lambda}^*$ is an additive subgroup of $\mathfrak{g}^*$ containing $\mathfrak{h}^{\perp}$. Let $\{X_1, \dots, X_d\}$ be a basis for $(\Lambda, +)$, then we have $\mathfrak{g}_{\Lambda}^* = \mathbb{Z}\text{-span}\{X_1^*, \dots, X_d^*\} + \mathfrak{h}^{\perp}$, where $\mathfrak{h}^{\perp} = \{l \in \mathfrak{g}^* : l(\mathfrak{h}) = \{0\}\}$ and $X_j^* \in \mathfrak{g}^*$ satisfies $X_j^*(X_i) = \delta_{ij}$, the Kronecker symbol.

Let $\widehat{G}_{\Gamma} := \{\tilde{\pi} \in \widehat{\widetilde{G}}; \ker(\tilde{\pi}) \supset \Gamma\}$. We identify $\widehat{G}$ with $\widehat{G}_{\Gamma}$ through the correspondence

$$\widehat{G} \ni \pi, \leftrightarrow \tilde{\pi} = \pi \circ P \in \widehat{G}_{\Gamma}.$$

Noting that $\tilde{\pi}_\ell(\exp X) = e^{2\pi i \ell(X)} I$ for $X \in \mathfrak{z}$, we parameterize $\widehat{G}$ by $\mathfrak{g}_A^*/\widetilde{G}$. In [9], a Plancherel formula based on this description of $\widehat{G}$ is provided for the case of $\Gamma \simeq \mathbb{Z}$. We shall give an expression (6.14) for the general case of $\Gamma \simeq \mathbb{Z}^d$ by slightly modifying that in [9].

For $\lambda := (\lambda_1, \dots, \lambda_d) \in \mathbb{Z}^d$, define the following objects:

$$\lambda X^* := \lambda_1 X_1^* + \dots + \lambda_d X_d^* \text{ and } \widehat{G}_{\chi_\lambda} := \{\tilde{\pi}; \tilde{\pi}|_{\widetilde{H}} = \chi_\lambda := \chi_{\lambda X^*}\},$$

where $H = \exp \mathfrak{h}$, $\widetilde{H} = H/\Gamma$, and $\chi_\lambda(\exp X) = e^{2\pi i \lambda X^*(X)}$ for $X \in \mathfrak{h}$. Then we have the following parametrization:

$$\widehat{G} \simeq \widehat{G}_\Gamma = \coprod_{\lambda \in \mathbb{Z}^d} \widehat{G}_{\chi_\lambda} \simeq \mathfrak{g}_A^*/\widetilde{G} = \coprod_{\lambda \in \mathbb{Z}^d} (\lambda X^* + \mathfrak{h}^\perp)/\widetilde{G},$$

where $\lambda X^* + \mathfrak{h}^\perp = \{l \in \mathfrak{g}^*; l|_{\mathfrak{h}} = \lambda X^*\}$. We now apply Theorem 6.2 to our $\widetilde{G}$, $\widetilde{H}$, and λX^* . Taking a Haar measure $d_{\widetilde{G}/\widetilde{H}}g$ on $\widetilde{G}/\widetilde{H}$ and the Plancherel measure on $\widehat{G}_{\chi_\lambda}$ associated with $d_{\widetilde{G}/\widetilde{H}}g$, we have for $\phi \in L^1(\widetilde{G}/\widetilde{H}, \chi_\lambda) \cap L^p(\widetilde{G}/\widetilde{H}, \chi_\lambda)$

$$\left(\int_{\widehat{G}_{\chi_\lambda}} \|\tau(\phi)\|_{c_q}^q d\mu_{\chi_\lambda}(\tau) \right)^{\frac{1}{q}} \leq A_p^{(2 \dim(\widetilde{G}/\widetilde{H}) - m_\lambda)/2} \|\phi\|_{L^p(\widetilde{G}/\widetilde{H})}, \quad (6.13)$$

where

$$m_\lambda = \sup\{\dim(\text{Ad}^*(\widetilde{G})(\nu)); \nu \in \lambda X^* + \mathfrak{h}^\perp\}.$$

On the other hand, taking a Haar measure $d\dot{g}$ on $G = \widetilde{G}/\Gamma$ associated with the Haar measure on $\widetilde{G}$, we describe our Plancherel formula

$$\int_G |f(\dot{g})|^2 d\dot{g} = \sum_{\lambda \in \mathbb{Z}^d} \int_{\widehat{G}_{\chi_\lambda}} \|\pi(f)\|_{c_2}^2 d\mu_{\chi_\lambda}(\tilde{\pi}) \quad (6.14)$$

for $f \in L^1(G) \cap L^2(G)$ and the norm

$$\|\mathcal{F}^p f\|_q = \left(\sum_{\lambda \in \mathbb{Z}^d} \int_{\widehat{G}_{\chi_\lambda}} \|\pi(f)\|_{c_q}^q d\mu_{\chi_\lambda}(\tilde{\pi}) \right)^{\frac{1}{q}}.$$

Our main upshot in this section, which generalizes Theorem 6.1, is the following:

Theorem 6.3 ([3, Theorem 3]) *Let G be a connected nilpotent Lie group, $\widetilde{G}$ its universal covering group, and H the maximal compact subgroup of G . If m designates the maximal dimension of generic coadjoint orbits of $\widetilde{G}$, then for $1 < p \leq 2$, $\|\mathcal{F}^p(G)\| \leq A_p^{\dim(G/H) - \frac{m}{2}}$.*

We here mention that the integral inequality (6.13) also plays an important role in the proof.

6.3 The L^p -Fourier Transform for Compact Extensions of $\mathbb{R}^n$

Suppose G has a unimodular, closed normal subgroup N of type I such that the quotient group G/N is compact. This is, for instance, the case where $G = \mathbb{R}^n \rtimes K$, where K stands for a compact subgroup of the automorphism group $\text{Aut}(\mathbb{R}^n)$. In this case, $\mathbb{R}^n$ can be equipped with an inner product $\langle \cdot, \cdot \rangle$ which embeds the compact group K as a subgroup of orthogonal transformations. We show that $\|\mathcal{F}^p(G)\| = \|\mathcal{F}^p(\mathbb{R}^n)\|$ for all $1 < p < 2$. Using the Hausdorff–Young theorem for compact groups and the Hausdorff–Young inequality for integral operators due to Fournier and Russo [15], we proved that $\|\mathcal{F}^p(G)\|$ is dominated by $\|\mathcal{F}^p(\mathbb{R}^n)\|$, and we provide an extremal function by an extension of a Gaussian function of $\mathbb{R}^n$ associated with the inner product $\langle \cdot, \cdot \rangle$.

6.3.1 On Compact Extensions of $\mathbb{R}^n$

Suppose G is a semidirect product of a closed abelian normal subgroup $A = \mathbb{R}^n$ and a compact subgroup K . We have an isomorphism $A \rtimes K \ni (a, k) \mapsto ak \in G$, where $a = (a_1, \dots, a_n) \in A$, and $k \in K$. Let us remark that the continuous action of $k \in K$ on A defined by $A \ni a \mapsto kak^{-1}$ is a linear isomorphism on the vector space A , and we may assume that A is equipped with a K -invariant inner product $\langle \cdot, \cdot \rangle$. We take a Lebesgue measure $da := da_1 \cdots da_n$ on A as above and the normalized Haar measure $d\mu(k)$ on K . Then da is invariant by the action of K , and we have a Haar measure dg on G by $dg = dad\mu(k)$ along with the product $g = ak$.

We shall describe the irreducible representations of G by Mackey's theory. For $\gamma \in A^* \simeq A$, let $\chi_\gamma \in \widehat{A}$ be defined by $\chi_\gamma(x) := e^{i\langle \gamma, x \rangle}$, ($x \in \mathbb{R}^n$). We denote by $g \cdot \gamma$ the action of G on A defined by $\langle g \cdot \gamma, x \rangle := \langle \gamma, (g^{-1}xg) \rangle$ for $g \in G$, $x \in \mathbb{R}^n$. Then we have $\chi_{g \cdot \gamma} = g \cdot \chi_\gamma$, where G acts on $\widehat{A}$ by

$$g \cdot \chi_\gamma(x) := \chi_\gamma(g^{-1}xg),$$

and we identify $\widehat{A}$ with $\mathbb{R}^n$ by the mapping $\mathbb{R}^n \ni \gamma \mapsto \chi_\gamma \in \widehat{A}$.

We have that the stabilizer G_γ of χ_γ is $G_\gamma = AK_\gamma$, where $K_\gamma := \{k \in K; k \cdot \chi_\gamma = \chi_\gamma\}$. Let us take the normalized Haar measure $d\mu_\gamma$ on K_γ and a K -invariant measure $d\dot{\mu}$ on K/K_γ so that

$$\int_K \phi(k) d\mu(k) = \int_{K/K_\gamma} d\dot{\mu}(kK_\gamma) \int_{K_\gamma} \phi(kk') d\mu_\gamma(k').$$

We note that the measure $d\dot{\mu}$ is also normalized as $\int_{K/K_\gamma} d\dot{\mu} = 1$.

Concerning the action of K on $\widehat{A}$, let $d\bar{\gamma}$ be the image of the Lebesgue measure on $\mathbb{R}^n/K$ by the canonical projection $\mathbb{R}^n \ni \gamma \mapsto \bar{\gamma} := K \cdot \gamma \in \mathbb{R}^n/K$ so that

$$\int_{\mathbb{R}^n} \phi(\gamma) d\gamma = \int_{\mathbb{R}^n/K} d\bar{\gamma} \int_K \phi(k \cdot \gamma) d\mu(k).$$

Let $\sigma \in \widehat{K_\gamma}$ be an irreducible unitary representation of K_γ of degree d_σ on $\mathbb{C}^{d_\sigma}$ with the standard inner product $\langle \cdot, \cdot \rangle$ and $\pi_{\gamma, \sigma}$ be the representation of G induced from $\tau = \chi_\gamma \otimes \sigma \in \widehat{G_\gamma}$, that is, $\tau(ak') = \chi_\gamma(a)\sigma(k')$ for $a \in A$ and $k' \in K_\gamma$. Specifically, let $\mathcal{H}(K, \sigma)$ be the completion of the space of $\mathbb{C}^{d_\sigma}$ -valued measurable functions ϕ on K such that

$$\begin{aligned} \phi(kk') &= \sigma(k')^{-1} \phi(k) \quad \forall k \in K, \quad k' \in K_\gamma, \\ \int_K \|\phi(k)\|^2 d\mu(k) &< \infty \end{aligned}$$

with respect to the inner product

$$(\phi_1, \phi_2)_\sigma := d_\sigma \int_K \langle \phi_1(k), \phi_2(k) \rangle d\mu(k), \quad \phi_1, \phi_2 \in \mathcal{H}(K, \sigma).$$

We define a representation $\pi_{\gamma, \sigma}$ of G on $\mathcal{H}(K, \sigma)$ by

$$\pi_{\gamma, \sigma}(ak)\phi(h) := \chi_\gamma(h^{-1}ah)\phi(k^{-1}h) = h \cdot \chi_\gamma(a)\phi(k^{-1}h).$$

Then by Mackey's theory $(\pi_{\gamma, \sigma}, \mathcal{H}(K, \sigma))$ is an irreducible unitary representation of G , and we have the Plancherel formula for $f \in L^1(G) \cap L^2(G)$ by

$$\int_G |f(ak)|^2 dad\mu(k) = \int_{\mathbb{R}^n/K} \sum_{\sigma \in \widehat{K_\gamma}} d_\sigma \|\pi_{\gamma, \sigma}(f)\|_{c_2}^2 (2\pi)^{-n} d\bar{\gamma}, \quad (6.15)$$

where

$$\pi_{\gamma, \sigma}(f) = \int_G f(ak) \pi_{\gamma, \sigma}(ak) dad\mu(k).$$

(See [20, Theorem 4.4], [17, section 3].)

Note that in view of the Plancherel formula (6.15), we get for $f \in L^1(G) \cap L^p(G)$ the equation

$$\|\mathcal{F}^p f\|_q = \left(\int_{\mathbb{R}^n/K} \sum_{\sigma \in \widehat{K_\gamma}} d_\sigma \|\pi_{\gamma, \sigma}(f)\|_{c_q}^q (2\pi)^{-n} d\bar{\gamma} \right)^{\frac{1}{q}}.$$

Then we have the following:

Theorem 6.4 ([4, Theorem 3.1]) *Let $G = A \rtimes K$ be a semidirect product of a vector group $A = \mathbb{R}^n$ and a compact group K , and $1 < p \leq 2$. Then*

- (1) $\|\mathcal{F}^p f\|_q \leq A_p^n \|f\|_p$ for $f \in L^p(G)$.
- (2) A_p^n is the smallest constant in (1), that is, $\|\mathcal{F}^p(G)\| = A_p^n$.

Remark 6.3 In the case $G = \mathbb{R}^n \rtimes K$ above, define a function f on G by $f(ak) = e^{-\|a\|^2/2}$ for $a \in \mathbb{R}^n$ and $k \in K$. Then we have $\|\mathcal{F}^p f\|_q = \|f\|_p$. That is, f is an extremal function.

6.4 Case of Compact Extensions of Locally Compact Groups

6.4.1 Backgrounds and Notations

Let G be a separable unimodular locally compact group of type I, and let N be a closed normal subgroup. We assume that N is unimodular and type I and that G/N is compact.

We recall the Plancherel formula for group extensions obtained by Kleppner and Lipsman [20] and sketch it for our case of compact extensions. For the complete description, we refer the reader to [20].

The unitary dual of G is described by Mackey with the little group method. (See, e.g., [23].) For an irreducible representation γ of N , we denote by G_γ the stabilizer of γ under the action of G on the unitary dual $\widehat{N}$ of N , given by $(g \cdot \gamma)(n) = \gamma(g^{-1}ng)$ for $g \in G$ and $n \in N$. Then there exists a multiplier σ_γ on G_γ/N such that γ extends to a σ'_γ -representation γ' of G_γ , where σ'_γ is the lift of σ_γ to G_γ . Denote by $(\widehat{G_\gamma/N})^{\bar{\sigma}_\gamma}$ the set of irreducible $\bar{\sigma}_\gamma$ -representations of G_γ/N . Let $\rho \in (\widehat{G_\gamma/N})^{\bar{\sigma}_\gamma}$ and ρ' be its lift to G_γ . Then

$$\pi_{\gamma, \rho} := \text{Ind}_{G_\gamma}^G \gamma' \otimes \rho'$$

is irreducible, and every irreducible unitary representation of G is equivalent to some $\pi_{\gamma, \rho}$ by Mackey's theory. Let $\check{G}_\gamma$ be the set of $\tau \in \widehat{G}_\gamma$ such that $\tau|_N$ is a multiple of γ . Then every $\tau \in \check{G}_\gamma$ is described by $\tau = \gamma' \otimes \rho'$, and the multiplicity $n_\gamma(\tau) = \dim(\rho)$ of γ in $\tau|_N$ is finite.

Under the assumptions above, we have the following proposition:

Proposition 6.1 (Kleppner–Lipsman [20, Lemma 4.2]) *Let $\gamma \in \widehat{N}$. Then we have*

$$\text{Ind}_N^G \gamma \simeq \bigoplus_{\tau \in \check{G}_\gamma} n_\gamma(\tau) \text{Ind}_{G_\gamma}^G \tau.$$

Let dn and dv be Haar measures on N and G/N normalized by

$$\int_G f(g) dg = \int_{G/N} \int_N f(gn) dn dv(g) \text{ for } f \in L^1(G).$$

Let $d\mu_N$ be the Plancherel measure on $\widehat{N}$ normalized by

$$\int_N |\psi(n)|^2 dn = \int_{\widehat{N}} \text{tr}(\gamma(\psi)^* \gamma(\psi)) d\mu_N(\gamma) \text{ for } \psi \in L^1(N) \cap L^2(N).$$

Let $\bar{\mu}_N$ be the image of the Plancherel measure μ_N on $\widehat{N}$ by the canonical projection $\widehat{N} \ni \gamma \mapsto \bar{\gamma} := G \cdot \gamma \in \widehat{N}/G$ so that

$$\int_{\widehat{N}} \varphi(\gamma) d\mu_N(\gamma) = \int_{\widehat{N}/G} \int_{G/N} \varphi(g \cdot \gamma) dv(g) d\bar{\mu}_N(\bar{\gamma}) \text{ for } \varphi \in L^1(\widehat{N}). \quad (6.16)$$

For any $f \in L^1(G) \cap L^2(G)$, $\gamma \in \widehat{N}$, and $\rho \in (\widehat{G_\gamma/N})^{\bar{\sigma}_\gamma}$, the operator $\pi_{\gamma,\rho}(f)$ is a Hilbert–Schmidt operator. We denote by $\|\cdot\|_{\text{HS}}$ the Hilbert–Schmidt norm.

The Plancherel measure on $\widehat{G}$ is described by the following formula:

Theorem 6.5 (Kleppner–Lipsman [20, Theorem 4.4]) *For any $f \in L^1(G) \cap L^2(G)$,*

$$\int_G |f(g)|^2 dg = \int_{\widehat{N}/G} \sum_{\rho \in (\widehat{G_\gamma/N})^{\bar{\sigma}_\gamma}} \|\pi_{\gamma,\rho}(f)\|_{\text{HS}}^2 \dim(\rho) d\bar{\mu}_N(\bar{\gamma}). \quad (6.17)$$

Denoting $\pi_\gamma = \text{Ind}_N^G \gamma$ for $\gamma \in \widehat{N}$, let us remark that by Proposition 6.1 we have

$$\pi_\gamma \simeq \bigoplus_{\tau \in \check{G}_\gamma} n_\gamma(\tau) \text{Ind}_{G_\gamma}^G \tau = \bigoplus_{\rho \in (\widehat{G_\gamma/N})^{\bar{\sigma}_\gamma}} \dim(\rho) \pi_{\gamma,\rho}$$

and that the equality

$$\int_{\widehat{N}/G} \|\pi_\gamma(f)\|_{\text{HS}}^2 d\bar{\mu}_N(\bar{\gamma}) = \int_{\widehat{N}/G} \sum_{\rho \in (\widehat{G_\gamma/N})^{\bar{\sigma}_\gamma}} \|\pi_{\gamma,\rho}(f)\|_{\text{HS}}^2 \dim(\rho) d\bar{\mu}_N(\bar{\gamma}) \quad (6.18)$$

is obtained in the course of the proof of [20, Theorem 4.4].

6.4.1.1 L^p -Fourier Transforms

Let $1 < p \leq 2$ and $f \in L^1(G) \cap L^p(G)$. Our description of the L^p -Fourier transform is based on the Plancherel theorem given in Theorem 6.17.

$$\|\mathcal{F}f\|_q^q := \int_{\widehat{N}/G} \sum_{\rho \in (\widehat{G_\gamma/N})^{\bar{\sigma}_\gamma}} \|\pi_{\gamma,\rho}(f)\|_{c_q}^q \dim(\rho) d\bar{\mu}_N(\bar{\gamma}). \quad (6.19)$$

We have an analog of the computation of (6.18) above as follows: By Proposition 6.1 we have that the Schatten norm of $\pi_\gamma(f)$ for $f \in L^1(G) \cap L^p(G)$ is described by

$$\|\pi_\gamma(f)\|_{c_q}^q = \sum_{\rho \in (\widehat{G_\gamma/N})^{\bar{\sigma}_\gamma}} \|\pi_{\gamma,\rho}(f)\|_{c_q}^q \dim(\rho).$$

Thus we also obtain that the norm $\|\mathcal{F}f\|_q$ is computed by

$$\|\mathcal{F}f\|_q^q = \int_{\widehat{N}/G} \|\pi_\gamma(f)\|_{c_q}^q d\bar{\mu}_N(\bar{\gamma}). \quad (6.20)$$

6.4.2 Main Results

In this subsection, we assume that G is unimodular, and we shall provide a generalization of Russo's result [28, Theorem 2] of compact extensions for $p = \frac{2k}{2k-1}$ to that for general exponents p satisfying $1 < p < 2$. Our main result is the following:

Theorem 6.6 ([5, Theorem 1.1]) *Let G be a separable unimodular locally compact group of type I, and let N be a closed normal subgroup of G . Suppose that N is unimodular and type I and that G/N is compact. Then, for $1 < p \leq 2$, we have*

$$\|\mathcal{F}^p(G)\| \leq \|\mathcal{F}^p(N)\|. \quad (6.21)$$

Remark 6.4 We remark that the Hausdorff–Young theorem is generalized for non-unimodular groups by Terp [29], and there are several results treating some non-unimodular groups, e.g., in [6, 10, 13, 16, 18, 19, 25].

Concerning group extensions, Führ [16] obtained the following estimate in a setting of non-unimodular groups: Let a group extension $1 \rightarrow N \rightarrow G \rightarrow H \rightarrow 1$ be given with N, H unimodular, N a type I group, and G has a type I regular representation. Letting ν_N be the Plancherel measure on $\widehat{N}$, assume that there exists a Borel ν_N -conull subset $U \subset \widehat{N}$ with the following property: U is H -invariant with U/H standard. Moreover, for all $\gamma \in U$, the induced representation $\text{Ind}_N^G \gamma \in \widehat{G}$. Then the inequality $\|\mathcal{F}^p(G)\| \leq \|\mathcal{F}^p(N)\|$ holds [16, Theorem 2].

Let G be a compact extension of a closed normal subgroup N , and suppose N is unimodular and type I. Then Russo obtained the estimate $\|\mathcal{F}^p(G)\| \leq \|\mathcal{F}^p(N)\|$ for $p = \frac{2k}{2k-1}$, where $k \geq 2$ is an integer [28, Theorem 2]. In [19, Theorem 2], Klein and Russo proved that if $G = X \ltimes A$ is a semidirect product of locally compact unimodular groups X and A and if G is unimodular, then the inequality $\|\mathcal{F}^p(G)\| \leq \|\mathcal{F}^p(X)\| \|\mathcal{F}^p(A)\|$ holds for $p = \frac{2k}{2k-1}$ with $k \geq 2$ an integer.

6.4.3 Case of a Semidirect Product $G := K \ltimes N$

We next treat the case where G is a separable unimodular locally compact group of type I defined by $G := K \ltimes N$, where N is a separable unimodular locally compact group of type I and K stands for a compact subgroup of the automorphism group of N , equipped with the group law

$$(k_1, n_1) \cdot (k_2, n_2) = (k_1 k_2, n_1 k_1(n_2)), \quad ((k_1, n_1), (k_2, n_2)) \in G \times G.$$

We fix once for all a Haar measure $dg := dv \otimes dn$ of G , where dv and dn denote, respectively, the normalized Haar measure of K and N .

In this special case, the Mackey induction machinery resumes as follows: Let $(\gamma, \mathcal{H}_\gamma)$ be an irreducible unitary representation of N . We realize the induced representation $\pi_\gamma = \text{Ind}_N^G \gamma$ on the space $\mathcal{H}_{\pi_\gamma} \simeq L^2(K, \gamma)$ of the $\mathcal{H}_\gamma$ -valued L^2 -functions $\xi : K \rightarrow \mathcal{H}_\gamma$ with respect to

$$\|\xi\|^2 := \int_K \|\xi(k)\|_\gamma^2 dv(k) < \infty$$

and describe the induced representation π_γ by

$$\pi_\gamma(k, n)\xi(x) := \gamma(x^{-1}(n))\xi(k^{-1}x),$$

for $\xi \in \mathcal{H}_{\pi_\gamma}$, $k, x \in K$, $n \in N$.

We shall provide a sufficient condition, under which the equality holds in (6.21). It is a generalization of our previous result [4, Theorem 3.1]. We state the condition and some remarks from our results in [5].

Theorem 6.7 ([5, Theorem 1.2]) *Let $G = K \ltimes N$ be a semidirect product of a separable unimodular locally compact group N of type I and a compact subgroup K of the automorphism group of N . Then for $1 < p \leq 2$, we have the equality $\|\mathcal{F}^p(G)\| = \|\mathcal{F}^p(N)\|$ if in addition N has a K -invariant sequence of functions $(\varphi_j)_{j \in \mathbb{N}}$ in $L^1(N) \cap L^p(N)$ such that $\frac{\|\mathcal{F}\varphi_j\|_q}{\|\varphi_j\|_p}$ tends to $\|\mathcal{F}^p(N)\|$ when j goes to infinity.*

Remark 6.5 In general, we may have $\|\mathcal{F}^p(G)\| \neq \|\mathcal{F}^p(N)\|$: Let $G = \mathbb{R}$ and $N = \mathbb{Z}$. Then G/N is the torus group $\mathbb{T}$, and $\mathbb{R}$ is a compact extension of $\mathbb{Z}$. We have that $\|\mathcal{F}^p(\mathbb{Z})\| = 1$: In fact, let f be a function on $\mathbb{Z}$ defined by $f(0) = 1$, $f(m) = 0$ for $m \neq 0$. Then $\|f\|_p = 1$ and $\|\mathcal{F}f\|_q = 1$. Therefore, we have $\|\mathcal{F}^p(\mathbb{R})\| = A_p < \|\mathcal{F}^p(\mathbb{Z})\| = 1$ for $1 < p < 2$.

In this example, the group $G/N = \mathbb{R}/\mathbb{Z}$ trivially acts on $N = \mathbb{Z}$ by conjugation, and thus the extremal function f for $\mathcal{F}^p(N)$ is invariant under the action. Nevertheless, the equality does not hold for $1 < p < 2$.

Remark 6.6

1. Remark first that in [28, Lemma 4.1], Russo obtained the inequality $\|\mathcal{F}^p(G/K)\| \leq \|\mathcal{F}^p(G)\|$ for $1 < p < 2$ when G is a unimodular locally compact group of type I and K is a compact normal subgroup of G . Concerning the case of a direct product group $G = K \times N$, the inequality $\|\mathcal{F}^p(N)\| \leq \|\mathcal{F}^p(G)\|$ for $1 < p < 2$ follows from this result. Hence, for $p = \frac{2k}{2k-1}$ with $k \geq 2$ an integer, Russo's results [28, Theorem 2] and [28, Lemma 4.1] give the equality $\|\mathcal{F}^p(K \times N)\| = \|\mathcal{F}^p(N)\|$.
2. Now, we also remark that in the proof of [25, Theorem 2], Russo obtained the inequality $\|\mathcal{F}^p(\mathbb{R} \times H)\| \leq A_p \|\mathcal{F}^p(H)\|$ for a direct product of $\mathbb{R}$ and an arbitrary unimodular locally compact group H and $1 < p < 2$. For the case that $H := K$ is a compact group, this inequality and [28, Lemma 4.1] concerning a direct product $G = K \times \mathbb{R}$ give the equality $\|\mathcal{F}^p(K \times \mathbb{R})\| = A_p (\|\mathcal{F}^p(\mathbb{R})\|)$ for $1 < p < 2$.
3. For arbitrary $1 < p \leq 2$, let $G = K \times N$ be a direct product of a separable unimodular locally compact group N of type I and a compact group K . Then, since K trivially acts on N by conjugation, we can apply Theorem 6.7 and obtain the equality $\|\mathcal{F}^p(K \times N)\| = \|\mathcal{F}^p(N)\|$.

6.4.4 An Example of a Non-compact Extension

Here we discuss a concrete example of non-compact extension of $\mathbb{R}^n$ which is not nilpotent. Let $\widetilde{M(2)} = A \rtimes \mathbb{R}$, where $A = \mathbb{R}^2$, be the universal covering group of the Euclidean motion group of the plane $\mathbb{R}^2$. The action of $t \in \mathbb{R}$ on $a = (a_1, a_2) \in A$ is defined by

$$t \cdot a := (a_1 \cos t - a_2 \sin t, a_1 \sin t + a_2 \cos t),$$

and the group law is defined by

$$(a, t)(a', t') := (a + t \cdot a', t + t'), \quad (a, t), (a', t') \in A \rtimes \mathbb{R}.$$

We denote by $dadtdt = da_1 da_2 dt$ the Lebesgue measure on $A \rtimes \mathbb{R}$, which is a Haar measure on $\widetilde{M(2)}$, normalized so that the unit cube has mass 1. Concerning the standard basis $\{(1, 0), (0, 1)\}$ of $\mathbb{R}^2 = A$, we denote by $\{l_1, l_2\}$ the dual basis of $\mathbb{R}^{2*}$. For $r > 0$, let $\gamma := \gamma_r = rl_2 \in A^*$, that is, $\langle \gamma, a \rangle := ra_2$, and $\chi_\gamma \in \widehat{A}$ defined by $\chi_\gamma(a) = e^{i\langle \gamma, a \rangle} = e^{ira_2}$ for $a = (a_1, a_2)$. We note that the stabilizer of χ_γ in $\mathbb{R}$ for the action $t \cdot \chi_\gamma(a) = \chi_\gamma((-t) \cdot a)$ is $2\pi\mathbb{Z}$, and

$$\widehat{2\pi\mathbb{Z}} = \{\sigma_\nu; 0 \leq \nu < 1, \sigma_\nu(2\pi n) := e^{2\pi i \nu n} (n \in \mathbb{Z})\} \simeq \mathbb{T}.$$

Let $0 \leq \nu < 1$, and define a representation $\pi_{r,\nu}$ of $\widetilde{M(2)}$ inducing from the representation $\chi_\gamma \otimes \sigma_\nu$ of $A\Gamma$, where $\Gamma = 2\pi\mathbb{Z}$ is the stabilizer of χ_γ in $\mathbb{R}$. That is, let $\mathcal{H}(\mathbb{R}, \nu)$ be the completion of the space of continuous functions ϕ on $\mathbb{R}$ such that

$$\phi(t + 2\pi n) = e^{-2\pi i \nu n} \phi(t), \quad \forall n \in \mathbb{Z}, \forall t \in \mathbb{R}$$

with respect to the inner product

$$(\phi_1, \phi_2) := \int_{\mathbb{R}/2\pi\mathbb{Z}} \phi_1(t) \overline{\phi_2(t)} d\mu(t), \quad \phi_1, \phi_2 \in \mathcal{H}(\mathbb{R}, \nu),$$

where the invariant measure $d\mu$ is normalized so that

$$\int_{\mathbb{R}} \psi(t) dt = \int_{\mathbb{R}/2\pi\mathbb{Z}} \sum_{n \in \mathbb{Z}} \psi(t + 2\pi n) d\mu(t)$$

for any rapidly decreasing smooth functions ψ . On $\mathcal{H}(\mathbb{R}, \nu)$ we define

$$\begin{aligned} \pi_{r,\nu}(a, t)\phi(s) &:= \chi_\gamma((-s) \cdot a)\phi(s - t) = \chi_{s \cdot \gamma}(a)\phi(s - t), \quad \phi \in \mathcal{H}(\mathbb{R}, \nu), \\ s &\in \mathbb{R}, (a, t) \in A \rtimes \mathbb{R}. \end{aligned}$$

Then, by Mackey's theory, $\pi_{r,\nu}$ is an irreducible unitary representation, and the Plancherel measure is supported on the set $\{\pi_{r,\nu}; 0 < r, 0 \leq \nu < 1\}$. For $f \in L^1(A \rtimes \mathbb{R}) \cap L^2(A \rtimes \mathbb{R})$, the Plancherel formula is described by

$$\int_{A \rtimes \mathbb{R}} |f(a, t)|^2 da dt = \int_0^\infty r dr \int_0^1 \|\pi_{r,\nu}(f)\|_{\mathcal{C}_2}^2 (2\pi)^{-2} d\nu,$$

where

$$\pi_{r,\nu}(f) = \int_{A \rtimes \mathbb{R}} f(a, t) \pi_{r,\nu}(a, t) da dt.$$

(See [1, section 3].) In this setting, we have

$$\|\mathcal{F}^p(f)\|_q = \left(\int_0^\infty r dr \int_0^1 \|\pi_{r,\nu}(f)\|_{\mathcal{C}_q}^q (2\pi)^{-2} d\nu \right)^{\frac{1}{q}}.$$

By computing the Fourier transform similarly as in the case of compact extensions, we obtain an estimate of the norm $\|\mathcal{F}^p(\widetilde{M(2)})\|$.

Theorem 6.8 ([4, Theorem 5.1]) *Let $\widetilde{M(2)}$ be the universal covering group of the plane, and $1 < p \leq 2$. Then we have $\|\mathcal{F}^p(\widetilde{M(2)})\| \leq A_p^2$.*

Remark 6.7 When the exponent $p = \frac{2k}{2k-1}$ with $k \geq 2$ an integer, the estimate of Klein–Russo [19, Theorem 2] gives $\|\mathcal{F}^p(\widetilde{M(2)})\| \leq A_p^3$. Thus our result Theorem 6.8 is not sharp for those exponents.

Remark 6.8 If $n \geq 3$, the universal covering group $\widetilde{M(n)} = \mathbb{R}^n \rtimes \text{Spin}(n)$ of the Euclidean motion group $M(n) = \mathbb{R}^n \rtimes SO(n)$, where $\text{Spin}(n)$ is the universal covering group of $SO(n)$, is a compact extension of $\mathbb{R}^n$. Thus we have $\|\mathcal{F}^p(M(n))\| = A_p^n$ for $n \geq 3$ by Theorem 6.4.

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Chapter 7

Spherical Function of Type δ on Groupoids



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Abstract Let ϑ be a locally compact groupoid and K a compact subgroup of ϑ . In this chapter, we give a generalization of Fourier transform on the convolution algebra $I_{c,\delta}(\vartheta)$ of all continuous functions f with compact support on ϑ such that $\chi_\delta * f =_\delta f = f, \forall \delta \in \widehat{K}$; here, χ_δ denotes the character of a unitary irreducible representation of K times its dimension. We obtain an inversion formula for the spherical transform by using the Fourier inversion formula in ϑ .

7.1 Introduction

The study of harmonic analysis on locally compact spaces that are not groups has led some authors to define groupoids as a generalization of groups. In this chapter, we present an axiomatic approach that generalizes groups and has more advantages. We call locally compact groupoids a quadruplet $(\vartheta, s, r, \vartheta^{(0)})$, where ϑ is the total space, $\vartheta^{(0)}$ the basis, and s, r the source and goal maps. The idea of studying locally compact groupoids is suggested by the existence of locally compact spaces that are groupoids and are not groups. Groupoids preserve all the structures associated with groups. The law is explicit, which allows to define Gelfand pairs and spherical functions. In this chapter, we define the spherical functions of delta type and give some properties. Let ϑ be a topological locally compact Hausdorff groupoid of base $\vartheta^{(0)}$ and $C_c(\vartheta)$ the space of continuous complex functions on ϑ with compact support. A Radon system on ϑ is a measure set $\{\lambda^u, u \in \vartheta^{(0)}\}$ on ϑ such as: For

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all $u \in \vartheta^{(0)}$, the support

$$\sup p(\lambda^u) = r^{-1}(u)$$

and

$$f \in C_c(\vartheta), u \mapsto \lambda(f)(u) = \int f d\lambda^u$$

is a continuous application of $\vartheta^{(0)}$ in $\mathbb{C}$.

Let ϑ be a locally compact groupoid Hausdorff and K a compact subgroup of ϑ . In this chapter, we give a generalization of Fourier transform on the convolution algebra $I_{c,\delta}(\vartheta)$ of all continuous functions f with compact support on ϑ such that

$$\chi_\delta * f =_\delta f = f, \forall \delta \in \widehat{K};$$

here, χ_δ denotes the character of a unitary irreducible representation of K times its dimension. We obtain an inversion formula for the spherical transform by using the Fourier inversion formula in ϑ . ϕ (see [12]) is a continuous function on ϑ with values in $End_{\mathbb{C}}(E)$ such that

$$\int_K \chi_\delta(k^{-1}) \phi(kxk^{-1}y) d(k) = \phi(x)\phi(y), \forall (x, y) \in G^{(2)} \text{ and } x \in \vartheta^{r(k)},$$

where $(E, \vartheta^{(0)}, p)$ is a complex vector bundle space of finite dimension and $\chi_\delta = d(\delta)\xi_\delta$ such functions are the spherical function of type δ . In Sect. 7.3, we define positive-definite spherical functions and study their relation. The main goal of Sect. 7.4 is to obtain an inversion formula for the spherical transform for $f \in I_{c,\delta}(\vartheta)$. For any $f \in I_{c,\delta}(\vartheta)$, the spherical transform $\mathcal{F}(f)(\phi)$ (see [12]) is defined by

$$\mathcal{F}(f)(\phi) = \int_{\vartheta^u} f(x)\phi(x^{-1})d\lambda^u(x), \forall u \in \vartheta^{(0)},$$

where ϕ is assumed to be an irreducible spherical function of type δ .

7.2 Preliminaries

Definition 7.1 (See [11]) A groupoid is a space ϑ with multiplication xy , and the three applications define to $\vartheta \rightarrow \vartheta$ by $x \mapsto x^{-1}$, $x \mapsto s(x)$ et $x \mapsto r(x)$ such as:

- (i) The product xy of ϑ is defined if $s(x) = r(y)$.

- (ii) For all $x \in \vartheta$, the couple $(x, s(x))$ and $(r(x), x)$ is defined and $xs(x) = x$; $r(x)x = x$.
- (iii) If xy and yz are defined, then xyz and xyz exist $x(yz) = (xy)z$.
- (iv) For all $x \in \vartheta$, the product $x^{-1}x$ and x, x^{-1} exist and $x^{-1}x = s(x), x x^{-1} = r(x)$.

The subspace $\vartheta^{(0)} = \{s(x) : x \in \vartheta\} = \{r(x) : x \in \vartheta\}$ of ϑ is called the base of ϑ . We note by $\vartheta^{(2)} = \{(x, y) \in \vartheta \times \vartheta : s(x) = r(y)\}$ the subspace of $\vartheta \times \vartheta$. If $\vartheta^{(0)} = \{e\}$, then the groupoid is a group.

Definition 7.2 Let us consider two groupoids ϑ_1 and ϑ_2 . The application $c : \vartheta_1 \longrightarrow \vartheta_2$ is a homomorphism of groupoid if for all $(x, y) \in \vartheta_1^{(2)}$, $(c(x), c(y)) \in \vartheta_2^{(2)}$ and $c(x)c(y) = c(xy)$.

Example 7.1 (See [1, 2, 11])

- (1) Let ϑ be a groupoid, then $\vartheta^{(2)}$ is a groupoid such as the couple $((x, y), (x', z)) \in \vartheta^{(2)}$ if $x' = xy$. For all $x, y, z \in \vartheta$ such as (x, y) and $(xy, z) \in \vartheta^{(2)}$, we have $(x, y)(xy, z) = (x, yz)$. For all $(x, y) \in \vartheta^{(2)}$, we have $(x, y)^{-1} = (xy, y^{-1})$; $r(x, y) = (x, r(y))$, $s(x, y) = (xy, s(y))$, and the base $(\vartheta^{(2)})^{(0)} = \vartheta$.
- (2) Let ϑ be a groupoid, A a group, and $c : \vartheta \rightarrow A$ a homomorphism, then $\vartheta \times A$ is a groupoid such as the couple $((x, a), (y, b)) \in \vartheta \times A$ if $(x, y) \in \vartheta^{(2)}$ and $b = ac(x)$. For all $((x, y), a) \in \vartheta^{(2)} \times A$, we put $(x, a)(y, ac(x)) = (xy, a)$. For all $(x, a) \in \vartheta \times A$, we put $(x, a)^{-1} = (x^{-1}, ac(x))$; $s(x, a) = (s(x), ac(x))$ and $r(x, a) = (r(x), a)$. If $\vartheta \times A$ is a groupoid, it is noted $\vartheta(c)$ and the base $(\vartheta(c))^{(0)} = \vartheta^{(0)} \times A$.
- (3) Let ϑ be a groupoid and $Aut(\vartheta)$ the automorphism group of ϑ . Let A be a group, $\alpha : A \rightarrow Aut(\vartheta)$ a homomorphism, and for all $(x, a) \in \vartheta \times A$, $x \cdot a = [\alpha(a^{-1})](x)$. The space $\vartheta \times A$ is a groupoid such as the couple $((x, a), (z, b)) \in \vartheta \times A$ if $z = y \cdot a$ such as $(x, y) \in \vartheta^{(2)}$, for all $(x, y, a, b) \in \vartheta \times \vartheta \times A \times A$ such as $(x, y) \in \vartheta^{(2)}$, we put $(x, a)(y \cdot a, b) = (x, a) \cdot (y \cdot a, b) = (xy, ab)$, and for all $(x, a) \in \vartheta \times A$, we put $(x, a)^{-1} = (x^{-1} \cdot a, a^{-1})$; $s(x, a) = (s(x) \cdot a, e)$, $r(x, a) = (r(x), e)$, where $e \in A$ such as $ea = a$. The base $(\vartheta \times A)^{(0)} = \vartheta^{(0)}$.

The topological groupoid ϑ (see [5]) is a groupoid which is a topological locally compact Hausdorff space such as the topological separate subspace $\vartheta^{(0)}$ and the applications $m : \vartheta^{(2)} \longrightarrow \vartheta$, and $x \mapsto x^{-1}$ are continuous.

Let ϑ be a topological locally compact groupoid of base $\vartheta^{(0)}$ and $C_c(\vartheta)$ the space of continuous complex functions on ϑ with compact support. A Radon system on ϑ (see [3, 4]) is a measure set $\{\lambda^u, u \in \vartheta^{(0)}\}$ on ϑ such as: For all $u \in \vartheta^{(0)}$,

the support $\text{supp } p(\lambda^u) = r^{-1}(u)$ and $f \in C_c(\vartheta)$, $u \mapsto \lambda(f)(u) = \int f d\lambda^u$ is a continuous application of $\vartheta^{(0)}$ in $\mathbb{C}$. A left Haar system on ϑ is a Radon system on ϑ such as

$$x \in \vartheta, \forall f \in C_c(\vartheta), \int f(xy) d\lambda^{s(x)}(y) = \int f(y) d\lambda_{r(x)}(y)$$

with $y \in r^{-1}(s(x))$, and a right Haar system on ϑ is a Radon system $\{\lambda_u, u \in \vartheta^{(0)}\}$ on ϑ , such as

$$x \in \vartheta, \forall f \in C_c(\vartheta), \int f(yx) d\lambda_{r(x)}(y) = \int f(y) d\lambda^{s(x)}(y).$$

The convolution is defined on $C_c(\vartheta)$ (see [4, 5]) by

$$f * g(x) = \int f(y) g(y^{-1}x) d\lambda_{r(x)}(y) = \int f(xy^{-1}) g(y) d\lambda^{s(x)}(y).$$

Let $\{\lambda^u, u \in \vartheta^{(0)}\}$ be a left Haar system on ϑ , then $\{(\lambda^2)^x, x \in \vartheta\}$ is a left Haar system on $\vartheta^{(2)}$, such as

$$\int_{\vartheta^{(2)}} f d(\lambda^2)^x = \int f(x, z) d\lambda^{s(x)}(z); \forall f \in C_c(\vartheta^{(2)}) \text{ and } x \in \vartheta = [\vartheta^{(2)}]^{(0)}$$

For all $u \in \vartheta^{(0)}$, $\lambda_u = (\lambda^u)^{-1}$ is a right Haar system $\{\lambda_u, u \in \vartheta^{(0)}\}$ on ϑ . Let a Radon measure μ be on the base $\vartheta^{(0)}$, then

$$f \mapsto \int f dv = \int_{\vartheta^{(0)}} \int_{\vartheta^u} f(x) d\lambda^u(x) d\mu(u)$$

is a Radon measure on ϑ and

$$f \mapsto \int f dv^2 = \int_{\vartheta^{(0)}} \int_{\vartheta^{(2)}} f(x, y) d\lambda_u(x) d\lambda^u(y) d\mu(u)$$

is a Radon measure on $\vartheta^{(2)}$. A Radon measure μ on $\vartheta^{(0)}$ is quasi-invariant if the measure $\nu = \nu^{-1}$ on ϑ with ν^{-1} is inverse measure.

7.3 Spherical Function of Type δ on Groupoids

Let ϑ be a locally compact Hausdorff groupoid of base $\vartheta^{(0)}$, K a compact isotropy subgroup of ϑ , and dk a Haar measure on K . Let $\{\lambda^u, u \in \vartheta^{(0)}\}$ be a Haar system measure on ϑ , μ a measure on $\vartheta^{(0)}$, $(E, \vartheta^{(0)}, p)$ a vector bundle of finite dimension,

and $\text{End}_{\mathbb{C}}(E)$ the endomorphism groupoid of base $\vartheta^{(0)}$. We denote by $\widehat{K}$ the unitary dual of K . For all $\delta \in \widehat{K}$, let ξ_δ be a character of δ , $d(\delta)$ the degree of δ , and $\chi_\delta = d(\delta)\xi_\delta$. If $\check{\delta}$ is the class of contragredient representations of δ , we have $\bar{\chi}_\delta = \chi_{\check{\delta}}$ and we can easily verify, thanks to the Schur orthogonality relations, that

$$\chi_{\check{\delta}} * \chi_\delta = \chi_{\check{\delta}}.$$

Let $C_c(\vartheta)$ be a subspace of continuous complex functions on ϑ with compact support. Identifying χ_δ to a bounded measure on K , we put, for any function $f \in C_c(\vartheta)$,

$$_\delta f(x) = \chi_\delta * f(x) = \int_K \chi_\delta(k) f(kxk^{-1}) dk, \forall x \in \vartheta^u.$$

Let $J_c(\vartheta)$ be a subspace of $C_c(\vartheta)$ such that $f(kxk^{-1}) = f(x)$, $\forall k \in K, \forall x \in \vartheta^u$, and $I_\delta(\vartheta)$ the subspace of $C_c(\vartheta)$ such that

$$\chi_\delta * f =_\delta f = f, \forall \delta \in \widehat{K},$$

and we put

$$I_{c,\delta}(\vartheta) = J_c(\vartheta) \cap I_\delta(\vartheta).$$

The elements of $I_{c,\delta}(\vartheta)$ (see [12]) are said to be $K - \delta$ invariant.

Let $\delta \in \widehat{K}$ and

$$P(\delta) = \int_K \chi_\delta(k^{-1}) \delta(k) d(k),$$

then $P(\delta)$ is a continuous projection of E onto $P(\delta)E = E_\delta^u$, where $u \in \vartheta^{(0)}$.

Definition 7.3 (See [12])

- (1) A function ϕ defined on ϑ into $\text{End}_{\mathbb{C}}(E)$ is said to be positive-definite if the following condition

$$\sum_i^n \sum_j^n c_i \bar{c}_j \left(\phi(x_i x_j^{-1}) \zeta_{s(x_i)}, \zeta_{s(x_j)} \right) \geq 0$$

holds for every choice of $x_1, x_2, \dots, x_n$ in ϑ such that $s(x_i) = s(x_j)$ and for every choice of complex numbers $c_1, c_2, \dots, c_n$.

- (2) A spherical function of type δ is a quasi-bounded continuous function ϕ on ϑ into $\text{End}_{\mathbb{C}}(E)$ such that:

- (i) $\forall x \in \vartheta, \phi(x) \in B(E_{s(x)}, E_{r(x)})$.
- (ii) $\phi(kxk^{-1}) = \phi(x)$, $\forall k \in K, \forall x \in \vartheta^{r(k)}$ and $_\delta \phi(x) = \phi(x)$, $\forall \delta \in \widehat{K}$.

(iii) The map

$$f \mapsto X_\phi(f) = \phi(f) = \int_{\vartheta^u} f(x) \phi(x^{-1}) d\lambda^u(x), \forall u \in \vartheta^{(0)}$$

is an irreducible representation of the algebra $(I_{c,\delta}(\vartheta), s, \vartheta^{(0)})$.

Proposition 7.1 *Let ϕ be a positive-definite spherical function of type δ , then $\forall x \in \vartheta$, $\phi(x^{-1}) = (\phi(x))^*$, and $\|\phi(x)\| \leq 2$.*

Proof Let ϕ be a positive-definite spherical function of type δ . For any vector $\zeta \in E$, the function $\varrho : x \mapsto \langle \phi(x) \zeta_{s(x)}, \zeta_{r(x)} \rangle$ is positive-definite in $\mathbb{C}$. Therefore, if we take $n = 1$, $x_i = u \in \vartheta^{(0)}$, and $x_j = x$, then the matrix $\begin{pmatrix} \varrho(u) & \varrho(x) \\ \varrho(x^{-1}) & \varrho(u) \end{pmatrix}$ is positive semi-definite, and we have $\varrho(x) = \overline{\varrho(x^{-1})}$ and $\phi(x^{-1}) = (\phi(x))^*$.

Let $A \in \text{End}_{\mathbb{C}}(E)$ be a linear transformation such that $|\langle Av, v \rangle| \leq \langle v, v \rangle$, for all $v \in E$. If A is Hermitian or skew-Hermitian and $Av = \lambda v$, then $|\langle Av, v \rangle| = |\lambda| |\langle v, v \rangle| \leq \langle v, v \rangle \leq \langle v, v \rangle$. Therefore $|\lambda| \leq 1$. This says that $\|A\| \leq 1$. For any linear transformation A , we write $A = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*)$ and observe that the Hermitian and skew-Hermitian components of A also satisfy the same hypothesis that A . Then $\|A\| \leq \left\| \frac{1}{2}(A + A^*) \right\| + \left\| \frac{1}{2}(A - A^*) \right\|$. Now if we take $A = \phi(x)$,

$$\forall x \in \vartheta, \|\phi(x)\| \leq 2.$$

□

Theorem 7.1 *If $\phi : \vartheta \rightarrow \text{End}_{\mathbb{C}}(E)$ is a spherical function of type δ , then*

(i)

$$\int_K \chi_\delta(k^{-1}) \phi(kxk^{-1})y d(k) = \phi(x)\phi(y), \forall (x, y) \in G^{(2)} \text{ and}$$

$$\phi(u) = \text{Id}_{E_u}, \forall u \in \vartheta^{(0)}.$$

(ii) *For any function $f \in I_{c,\delta}(\vartheta)$, there exists an endomorphism $\phi(f)$ such that*

$$f * \phi(x) = \phi(f) \cdot \phi(x) \text{ and } \forall u \in \vartheta^{(0)}, f * \phi(u) = \phi(f).$$

Proof For all $f, g \in I_{c,\delta}(\vartheta)$, we have

$$\begin{aligned} \phi(f * g) &= \int_{\vartheta^u} \int_{\vartheta^u} f(y) g(y^{-1}x) \phi(x^{-1}) d\lambda^u(x) d\lambda^u(y) \\ &= \int_{\vartheta^u} \int_{\vartheta^u} f(y) g(x) \phi(x^{-1}y^{-1}) d\lambda^u(x) d\lambda^u(y) \end{aligned}$$

and

$$\phi(f)\phi(g) = \int_{\vartheta^u} \int_{\vartheta^u} f(y)g(x)\phi(x^{-1})\phi(y^{-1})d\lambda^u(x)d\lambda^u(y).$$

$$\begin{aligned} \phi(f * g) - \phi(f)\phi(g) &= \int \int (\phi(x^{-1}y^{-1}) - \phi(x^{-1})\phi(y^{-1})) \\ &\quad \times f(y)g(x)d\lambda(x)d\lambda(y) \\ &= \int \int \int_K (\phi((kx^{-1}k^{-1})(k'y^{-1}k'^{-1})) - \phi(x^{-1})\phi(y^{-1})) \\ &\quad \times f(y)g(x)d(k)d(k')d\lambda^{s(y)}(x)d\lambda^{s(x)}(y) \\ &= 0, \end{aligned}$$

then

$$\int_K \phi((kxk^{-1})(k'yk'^{-1}))d\beta(k)d\beta(k') = \phi(x)\phi(y), \forall (x, y) \in G^{(2)}.$$

For $x = u \in \vartheta^{(0)}$, we have

$$\phi(y) = \phi(u)\phi(y), \forall y \in \vartheta$$

and

$$\phi(u) = id_{E_u}, \forall u \in \vartheta^{(0)}.$$

If $f \in I_{c,\delta}(\vartheta)$, we have

$$\begin{aligned} f * \phi(x) &= \int_{\vartheta^{(0)}} \int_{\vartheta^u} f(y)\phi(y^{-1}x)d\lambda^u(y)d\mu(u) \\ &= \int_{\vartheta^{(0)}} \int_{\vartheta^u} \int_K f((k(y)))\phi(k(y^{-1})k(x))dkd\lambda^u(y)d\mu(u) \\ &= \int_{\vartheta^{(0)}} \int_{\vartheta^u} f((y))\phi(y^{-1})\phi(x)d\lambda^u(y)d\mu(u) \\ &= \phi(f) \cdot \phi(x) \end{aligned}$$

and

$$\phi(f * g) = \int_{\vartheta^u} \int_{\vartheta^u} f(xy^{-1})g(y)\phi(x^{-1})d\lambda^u(y)d\lambda^u(x)$$

$$\begin{aligned}
&= \int_{\vartheta^u} \int_{\vartheta^u} f(x)g(y)\phi(x^{-1}y)d\lambda^u(y)d\lambda^u(x) \\
&= \int_{\vartheta^u} \int_{\vartheta^u} \int_K f(x)g(y)\phi((kx^{-1}k^{-1})k'yk'^{-1}))dkdk'd\lambda^u(y)d\lambda^u(x) \\
&= \phi(f) \cdot \phi(g).
\end{aligned}$$

□

Theorem 7.2 *Let U be a unitary representation of (G, E) and E_δ^u , where $u \in \vartheta^{(0)}$ is finite dimensional and nonzero, then the spherical function $\phi^U : x \mapsto P(\delta)U(x)P(\delta)$ is positive-definite and unitary.*

Proof Let $x_1, x_2, \dots, x_n$ be in ϑ such that $s(x_i) = s(x_j)$ and for every choice of complex numbers $c_1, c_2, \dots, c_n$. Then

$$\begin{aligned}
&\sum_i^n \sum_j^n c_i \bar{c}_j \left(\phi(x_i x_j^{-1}) \zeta_{s(x_i)}, \zeta_{s(x_j)} \right) \\
&= \sum_i^n \sum_j^n c_i \bar{c}_j \left(P(\delta)U(x_i x_j^{-1})P(\delta) \zeta_{s(x_i)}, \zeta_{s(x_j)} \right) \\
&= \sum_i^n \sum_j^n c_i \bar{c}_j \left(U(x_i x_j^{-1})P(\delta) \zeta_{s(x_i)}, P(\delta) \zeta_{s(x_j)} \right) \\
&= \sum_i^n \sum_j^n c_i \bar{c}_j \left(U(x_i)P(\delta) \zeta_{s(x_i)}, U(x_j)P(\delta) \zeta_{s(x_j)} \right) \geq 0.
\end{aligned}$$

□

7.4 The Spherical Transform of Type δ on a Groupoid

Let $X(I_{c,\delta}(\vartheta))$ be a space of continuous representations of $I_{c,\delta}(\vartheta)$ on E_δ , $S_{c,\delta}(\vartheta)$ be a space of spherical functions of type δ , and Δ be a subspace of $S_{c,\delta}(\vartheta)$ of functions that are of positive-type.

Definition 7.4

- (1) For any function $f \in I_{c,\delta}(\vartheta)$, the Gelfand transform of f is a map $\varrho(f)$ from $X(I_{c,\delta}(\vartheta))$ into $\text{End}_{\mathbb{C}}(E)$ definite by

$$\varrho(f)(\phi) = \int_{\vartheta^u} f(y)\phi(y^{-1})d\lambda^u(y), \forall u \in \vartheta^{(0)}.$$

If $\varrho(f)$ is Gelfand transform of f , then there exists a map:

$$\mathcal{F}(f) : S_{c,\delta}(\vartheta) \rightarrow \text{End}_{\mathbb{C}}(E)$$

such that

$$\begin{aligned} \varrho(f) \circ F(\phi) &= \mathcal{F}(f)(\phi) \\ &= \int_{\vartheta^u} f(x) \phi(x^{-1}) d\lambda^u(x), \forall u \in \vartheta^{(0)}. \end{aligned}$$

The map $\mathcal{F}$ is called spherical Fourier transformation of type δ .

(2) Let $M^1(\Delta)$ be a space of Radon measure on Δ and $\lambda \in (M^1(\Delta), s, \vartheta^u)$. The function $\hat{\lambda}$ defined on Δ by

$$\hat{\lambda}(\phi) = \int_{\vartheta^u} \phi(x^{-1}) d\lambda^u(x), \forall u \in \vartheta^{(0)}$$

is called spherical Fourier transformation of λ of type δ .

Theorem 7.3 *If ϕ is a spherical function of type δ , then for any functions $f, g \in I_{c,\delta}(\vartheta)$, we have*

$$f * \phi = \mathcal{F}(f)(\phi) \cdot \phi$$

and

$$\mathcal{F}(f * g)(\phi) = \mathcal{F}(f)(\phi) \cdot \mathcal{F}(g)(\phi).$$

Proof Let $f \in I_{c,\delta}(\vartheta)$ and ϕ be a spherical function of type δ . We have

$$\begin{aligned} f * \phi(x) &= \int_{\vartheta^u} f(y) \phi(y^{-1}x) d\lambda^u(y) \\ &= \int_{\vartheta^u} f(y) \int_K \phi((ky^{-1}k^{-1}k'xk'^{-1}) dk dk' d\lambda^u(y) \\ &= \int_{\vartheta^u} f(y) \phi(y^{-1}) \phi(x) d\lambda^u(y) \\ &= \mathcal{F}(f)(\phi) \cdot \phi(x). \end{aligned}$$

□

Remark 7.1 If $\vartheta^{(0)} = \{e\}$, then ϑ is a group, and we have the classical Gelfand transform.

Theorem 7.4

- (i) Let $\{\gamma^u, u \in \vartheta^{(0)}\}$ be a Radon positive system on Δ , then the function defined by

$$\phi(x) = \int_{\Delta^u} \text{tr}(\omega(x)) d\gamma^u(\omega)$$

is of positive-type.

- (ii) If $\phi : \vartheta \rightarrow \text{End}_{\mathbb{C}}(E)$ is a spherical function of type δ and positive-type, then there exists a Radon system $\{\gamma^u, u \in \vartheta^{(0)}\}$ on Δ , such that

$$\phi(x) = \int_{\Delta^{s(x)}} \text{tr}(\omega(x)) d\gamma^{s(x)}(\omega).$$

Proof

- (i) Let $\phi : \vartheta \rightarrow \text{End}_{\mathbb{C}}(E)$ be a spherical function of type δ and positive-type; we have

$$\begin{aligned} {}_{\delta}\phi(x) &= \int_K \chi_{\delta}(k) \phi(kxk^{-1}) dk \\ &= \int_{\Delta^u} \int_K \chi_{\delta}(k) \text{tr}(\omega(kxk^{-1})) dk d\gamma^u(\omega) \\ &= \int_{\Delta^u} \text{tr}(\omega(x)) d\gamma^u(\omega) \end{aligned}$$

and

$$\begin{aligned} \sum_{i,j}^n c_i \bar{c}_j \left(\phi(x_i x_j^{-1}) \zeta_{s(x_i)}, \zeta_{r(x_j)} \right) &= \int_{\vartheta^{(0)}} \int_{\Delta^u} \sum_{i,j}^n c_i \bar{c}_j \left(\omega(x_i x_j^{-1}) \zeta_{s(x_i)}, \zeta_{r(x_j)} \right) \\ &\quad \times d\gamma^u(\omega) d\mu(u) \geq 0. \end{aligned}$$

- (ii) Let $P_{0,\delta}(\vartheta)$ be a space of $K - \delta$ invariant function ϕ such that $\|\phi\|_{\rho} \leq 1$. Using Banach–Alaoglu–Bourbaki theorem, $P_{0,\delta}(\vartheta)$ is a compact space. Let $M_0(\Delta)$ be the bundle space $\{\gamma^u, u \in \vartheta^{(0)}\}$ on Δ such that $\|\gamma^u\| \leq 1, \forall u \in \vartheta^{(0)}$.

Let $F : M_0(\Delta) \rightarrow P_{0,\delta}(\vartheta)$, the map defined by

$$F(\gamma)(x) = \int_{\Delta^{s(x)}} \phi(x) d\gamma^{s(x)}(\phi);$$

we have

$$\begin{aligned} f * F(\gamma)(x) &= \int_{\vartheta^u} f(x)F(\gamma)(x^{-1})d\lambda^u(x) = \int_{\vartheta^u} \int_{\Delta^u} \omega(x^{-1})f(x)d\gamma^u(\omega)d\lambda^u(x) \\ &= \int_{\Delta^u} \mathcal{F}(f)(\omega)d\gamma^u(\omega) = \hat{\gamma}(\mathcal{F}(f)) \end{aligned}$$

and

$$\begin{aligned} f * F(\gamma)(u) &= \mathcal{F}(f)(F(\gamma)), \forall u \in \vartheta^{(0)} \\ &= \hat{\gamma}(\mathcal{F}(f)) \end{aligned}$$

and then

$$\lim_{\gamma \mapsto 0} f * F(\gamma)(u) = 0, \forall u \in \vartheta^{(0)}.$$

F is continuous and $P_{0,\delta}(\vartheta)$ is convex. Using Milman–Krein theorem, we have

$$F(M_0(\Delta)) = P_{0,\delta}(\vartheta),$$

and F is surjective.

Let us denote by $I_{c,\delta}^2(\vartheta)$ the convolution algebra of square integrable $K - \delta$ invariant functions. $\square$

Theorem 7.5 *For all $f \in I_{c,\delta}(\vartheta)$, there exists a Radon system $\{\sigma^u, u \in \vartheta^{(0)}\}$ on Δ such that*

$$f(x) = \int_{\Delta^u} \text{tr}(\mathcal{F}(f)(\phi) \cdot \phi(x)) d\sigma^u(\phi).$$

If $f \in I_{c,\delta}^2(\vartheta)$, then

$$\int_{\vartheta^{(0)}} \int_{\vartheta^u} |f(x)|^2 d\lambda^u(x) d\mu(u) = \int_{\vartheta^{(0)}} \int_{\Delta^u} \|\mathcal{F}(f)(\phi)\|^2 d\sigma^u(\phi) d\mu(u).$$

Proof Let f, g be positive-type functions of $I_{c,\delta}(\vartheta)$, then

$$g(x) = \int_{\Delta^u} \text{tr}(\omega(x)) d\gamma^u(\omega)$$

and

$$\begin{aligned} f * g(x) &= \int_{\Delta^u} f * \text{tr}(\omega(x)) d\gamma^u(\omega) \\ &= \int_{\Delta^u} \text{tr}(\mathcal{F}(f)(\omega) \omega(x)) d\gamma^u(\omega). \end{aligned}$$

Let

$$\mathcal{F}(f)d\sigma^u = d\gamma^u.$$

Then we have

$$f(x) = \int_{\Delta^u} \text{tr}(\omega(x)\mathcal{F}(f)(\omega))d\sigma^u(\omega),$$

and if $f \in I_{c,\delta}^2(\vartheta)$, $h = f * f^*$ is positive-type, and

$$\begin{aligned} h(x) &= \int_{\vartheta(0)} \int_{\vartheta^u} f(xy) f^*(y^{-1}) d\lambda^u(y) d\mu(u) \\ &= \int_{\vartheta(0)} \int_{\Delta^u} \omega(x) \mathcal{F}(h)(\omega) d\sigma^u(\omega) d\mu(u). \end{aligned}$$

If $x = v \in \vartheta^{(0)}$, we have $\int_{\vartheta(0)} \int_{\vartheta^u} \|f(x)\|^2 d\lambda^u(x) d\mu(u) = \int_{\vartheta(0)} \int_{\Delta^u} \|\mathcal{F}(f)(\phi)\|^2 d\sigma^u(\phi) d\mu(u)$. $\square$

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Chapter 8

Convolution in Orlicz Spaces on Hypergroups



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Abstract Let G be a locally compact hypergroup, and let K be a compact sub-hypergroup of G . (G, K) is a Gelfand pair if $M_c(G//K)$, the algebra of measures with compact support on the double coset $G//K$, is commutative for the convolution.

In this chapter, we assume that (G, K) is a Gelfand pair, and we consider a pair (Φ_1, Φ_2) of Young functions satisfying the sequence condition. We prove a necessary condition in terms of aperiodic elements of the center of G , for the existence always everywhere of convolution $f * g$, where f and g are arbitrary elements of Orlicz spaces $L^{\Phi_1}(G)$ and $L^{\Phi_2}(G)$.

8.1 Introduction

The hypergroups generalize the locally compact groups. They are used when there is a convolution on the Banach space of all bounded Radon measures on a locally compact space which has the properties of a group convolution, except the fact that the convolution of two Dirac measures is a probability measure with compact support and not necessarily a Dirac measure. The intention was to unify harmonic analysis on duals of compact groups, double coset spaces $G//H$ (H a compact subgroup of a locally compact group G), and commutative convolution algebras associated with product linearization formulas of special functions. The notion of hypergroup has been sufficiently studied (see, e.g., [3, 5, 8, 10]). Harmonic analysis and probability theory on commutative hypergroups are well developed meanwhile where many results from group theory remain valid (see [2]). When G is

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a commutative hypergroup, the convolution algebra $M_c(G)$ consisting of measures with compact support on G is commutative. A typical example of commutative hypergroup is the double coset $G//K$ when G is a locally compact group, and K is a compact subgroup of G such that (G, K) is a Gelfand pair.

The well-known L^p -conjecture stated that if $1 < p < \infty$ and G is a locally compact group, the Lebesgue space $L^p(G)$ is a Banach algebra under the convolution product if and only if G is compact. In [1], the authors mentioned that for each $p > 2$, if $f * g$ exists a.e. for all $f, g \in L^p(G)$, then G is compact, and so automatically $f * g \in L^p(G)$. The case of hypergroups was studied in [12] under some conditions. Assuming that G is a commutative hypergroup, the authors in [9] give a necessary condition in terms of aperiodic elements of the center of G , for the convolution $f * g$ to exist a.e., where f and g are arbitrary elements of Orlicz spaces $L^{\Phi_1}(G)$ and $L^{\Phi_2}(G)$, respectively. When the hypergroup G is not commutative, it is possible to introduce a compact sub-hypergroup K of G leading to a commutative subalgebra of $M_c(G)$. In fact, if K is a compact sub-hypergroup of a hypergroup G , the pair (G, K) is said to be a Gelfand pair if $M_c(G//K)$, the convolution algebra of measures with compact support on $G//K$, is commutative. The notion of Gelfand pairs for hypergroups is well known (see [4, 11, 13]). The goal of this chapter is to study the condition for the existence of the convolution of two functions from Orlicz spaces on a hypergroup which is not necessarily commutative, but such that (G, K) is a Gelfand pair. We characterize the center $Z(G)$ of the hypergroup G and its action on G , and finally we establish our main result, namely the existence of convolution in Orlicz spaces in terms of aperiodic elements of $Z(G)$.

8.2 Notations and Preliminaries

Let G be a topological locally compact space. We denote by:

- $C(G)$ (respectively, $M(G)$) the space of continuous complex-valued functions (respectively, the space of Radon measures) on G
- $C_b(G)$ (respectively, $M_b(G)$) the space of bounded continuous functions (respectively, the space of bounded Radon measures) on G
- $\mathcal{K}(G)$ (respectively, $M_c(G)$) the space of continuous functions (respectively, the space of Radon measures) with compact support on G
- $\mathfrak{C}(G)$ the space of compact subspace of G
- δ_x the Dirac measure at $x \in G$
- $\text{supp}(\mu)$ the support of the measure μ
- $cl(A)$ the closure of the subset A in G

Let us note that the topology on $M(G)$ is the cône topology [5] and the topology on $\mathfrak{C}(G)$ is the topology of Michael [6].

Definition 8.1 G is said to be a *hypergroup* if the following assumptions are Satisfied:

(H1) There is a binary operator $*$ called convolution on $M_b(G)$ under which $M_b(G)$ is an associative algebra such that:

- (i) The mapping $(\mu, \nu) \mapsto \mu * \nu$ is continuous from $M_b(G) \times M_b(G)$ in $M_b(G)$.
- (ii) $\forall x, y \in G$, $\delta_x * \delta_y$ is a measure of probability with compact support.
- (iii) The mapping $(x, y) \mapsto \text{supp}(\delta_x * \delta_y)$ is continuous from $G \times G$ in $\mathfrak{C}(G)$.

(H2) There is a unique element e (called neutral element) in G such that $\delta_x * \delta_e = \delta_e * \delta_x = \delta_x, \forall x \in G$.

(H3) There is an involutive homeomorphism: $x \mapsto \bar{x}$ from G in G , named involution, such that:

- (i) $(\delta_x * \delta_y)^- = \delta_{\bar{y}} * \delta_{\bar{x}}, \forall x, y \in G$ with $\mu^-(f) = \mu(f^-)$, where $f^-(x) = f(\bar{x}), \forall f \in C(G)$ and $\mu \in M(G)$.
- (ii) $\forall x, y, z \in G, z \in \text{supp}(\delta_x * \delta_y)$ if and only if $x \in \text{supp}(\delta_z * \delta_{\bar{y}})$.

The assumption (H3)–(ii) is equivalent to the following: For $x, y \in G, e \in \text{supp}(\delta_x * \delta_y)$ if and only if $x = \bar{y}$.

For two subset A and B of $G, A * B = \bigcup_{x \in A, y \in B} \text{supp}(\delta_x * \delta_y)$. An element x of G is identified with $\{x\}$ and that $x * y = \{x\} * \{y\} = \text{supp}(\delta_x * \delta_y)$ for $x, y \in G$. Thus $x * y$ is a compact subset of G that is not necessarily be reduced to a single point. We have $\bar{x} * \bar{y} = \bar{y} * \bar{x}$.

The hypergroup G is commutative if $\delta_x * \delta_y = \delta_y * \delta_x, \forall x, y \in G$.

For $x, y \in G$ and for $f \in C(G)$, $(\delta_x * \delta_y)(f)$ is noted by $f(x * y)$. Thus,

$$f(x * y) = (\delta_x * \delta_y)(f) = \int_G f(z) d(\delta_x * \delta_y)(z).$$

The convolution of two measures μ, ν in $M_b(G)$ is defined by

$$\begin{aligned} (\mu * \nu)(f) &= \int_G \int_G (\delta_x * \delta_y)(f) d\mu(x) d\nu(y) \\ &= \int_G \int_G f(x * y) d\mu(x) d\nu(y), \forall f \in C(G). \end{aligned}$$

For μ in $M_b(G)$, $\mu^* = (\bar{\mu})^-$. So $M_b(G)$ is a $*$ -Banach algebra.

Example 8.1

- (i) All the locally compact groups are hypergroups.
- (ii) Let H be a compact group with the normalized Haar measure σ and G be a locally compact group. Let

$$\begin{aligned} G \times H &\longrightarrow G \\ (x, h) &\longmapsto x^h \end{aligned}$$

be a continuous action of H on G such that for $h \in H$, the mapping $x \mapsto x^h$ is an automorphism of G . Let the quotient space $G^H = \{x^H, x \in G\}$ have the quotient topology. Then the operation

$$\delta_{x^H} * \delta_{y^H} = \int_H \delta_{(x^h y)^H} d\sigma(h) = \int_H \delta_{(xy^t)^H} d\sigma(t)$$

defines a hypergroup structure on G^H with identity $1^H = \{1\}$. If $x \in G$, then $\overline{x^H} = (x^{-1})^H$.

- (iii) $\mathbb{R}^d / SO(d-1) \sim \mathbb{R}^+$ is a hypergroup in which δ_r means the surface measure of the sphere centered at 0 and of radius r ($\delta_{rS^{d-1}}$), and the convolution $r * s$ is the support of $\delta_{rS^{d-1}} * \delta_{sS^{d-1}}$ the usual convolution of measures in $\mathbb{R}^d$ so that $r * s$ is the interval $[a_+ - a_-, a_+ + a_-]$ with $a_+ = \max(r, s)$, $a_- = \min(r, s)$.
- (iv) Let S_4 denote the group of all permutations of the set $\{1, 2, 3, 4\}$. The elements of S_4 will be written in cyclic notation, and multiplication is computed from left to right. For example, $(12)(13) = (123)$. The subset $H = \{1, (12)\}$ is a compact subgroup of S_4 . H acts on the subgroup G of even permutation by inner automorphisms in S_4 . Applying (ii), the resulting hypergroup is $G^H = \{e, a, b, c, d, u, v\}$, where $e = \{1, (12)\}$, $a = \{(12)(34), (34)\}$, $b = \{(13)(24), (14)(23), (1324), (1423)\}$, $c = \{(123), (132), (13), (23)\}$, $d = \{(124), (142), (14), (24)\}$, $u = \{(134), (234), (1234), (1342)\}$ and $v = \{(143), (243), (1243), (1432)\}$. G^H is not commutative. In fact, $\delta_b * \delta_c = \frac{1}{2}\delta_d + \frac{1}{2}\delta_v$, while $\delta_c * \delta_b = \frac{1}{2}\delta_d + \frac{1}{2}\delta_u$.

Note that many of the basic constructions of hypergroups are to be found in [3, 5], and [2].

Definition 8.2 $H \subset G$ is a sub-hypergroup of G if the following conditions are Satisfied:

1. H is non-empty and closed in G .
2. $\forall x \in H, \bar{x} \in H$.
3. $\forall x, y \in H, \text{supp}(\delta_x * \delta_y) \subset H$.

Let us now consider a hypergroup G provided with a left Haar measure μ_G and K a compact sub-hypergroup of G with a normalized Haar measure ω_K . For $x \in G$, the double coset of x with respect to K is $K * \{x\} * K = \{k_1 * x * k_2; k_1, k_2 \in K\}$. We write simply KxK for a double coset and recall that $KxK = \bigcup_{k_1, k_2 \in K} \text{supp}(\delta_{k_1} * \delta_x * \delta_{k_2})$. All double cosets form a partition of G and the quotient topology with respect to the corresponding equivalence relation equip the double cosets space $G//K$ with a locally topology ([2], page 53). The natural mapping $p_K : G \longrightarrow G//K$ defined by

$$p_K(x) = KxK, x \in G$$

is an open surjective continuous mapping. A function $f \in C(G)$ is said to be invariant by K or K -invariant if $f(k_1 * x * k_2) = f(x)$ for all $x \in G$ and for all $k_1, k_2 \in K$. We denote by $C^\natural(G)$ (respectively, $\mathcal{K}^\natural(G)$) the space of continuous functions (respectively, continuous functions with compact support) which are K -invariant. For $f \in C^\natural(G)$, one defines the function $\tilde{f}$ on $G//K$ by

$$\tilde{f}(KxK) = f(x) \text{ for } x \in G.$$

$\tilde{f}$ is well defined, and it is continuous on $G//K$. Conversely, for all continuous function φ on $G//K$, the function $f = \varphi \circ p_K \in C^\natural(G)$. One has the obvious consequence that the mapping $f \mapsto \tilde{f}$ sets up a topological isomorphism between the topological vector spaces $C^\natural(G)$ and $C(G//K)$ (see [11, 13]). So, for any f in $C^\natural(G)$, $f = \tilde{f} \circ p_K$. Otherwise, we consider the K -projection $f \mapsto f^\natural$ (by identifying $f^\natural$ and $f^\natural$) from $C(G)$ into $C(G//K)$ where for $x \in G$, $f^\natural(x) = \int_K \int_K f(k_1 * x * k_2) d\omega_K(k_1) d\omega_K(k_2)$. If $f \in \mathcal{K}(G)$, then $f^\natural \in \mathcal{K}(G//K)$. For a measure $\mu \in M(G)$, one defines $\mu^\natural$ by

$$\mu^\natural(f) = \mu(f^\natural) \text{ for } f \in \mathcal{K}(G).$$

μ is said to be K -invariant if $\mu^\natural = \mu$, and we denote by $M^\natural(G)$ the set of all those measures. Considering these properties, one defines a hypergroup operation on $G//K$ by

$$\delta_{KxK} * \delta_{KyK}(\tilde{f}) = \int_K f(x * k * y) d\omega_K(k) \text{ (see [11] and [2]).}$$

This defines uniquely the convolution $(KxK) * (KyK)$ on $G//K$. The involution is defined by $\overline{KxK} = K\bar{x}K$, and the neutral element is K . Let us put $m = \int_G \delta_{KxK} d\mu_G(x)$, m is a left Haar measure on $G//K$. We say that (G, K) is a Gelfand pair if the convolution algebra $M_c(G//K)$ is commutative. $M_c(G//K)$ is topologically isomorphic to $M_c^\natural(G)$. Considering the convolution product on $\mathcal{K}(G)$ defined by

$$f * g(x) = \int_G f(x * \bar{y}) g(y) d\mu_G(y),$$

$\mathcal{K}(G)$ is a convolution algebra and $\mathcal{K}^\natural(G)$ is a subalgebra. Thus (G, K) is a Gelfand pair if and only if $\mathcal{K}^\natural(G)$ is commutative ([4], theorem 3.2.2).

Let us now recall some definitions and properties about Orlicz spaces.

Definition 8.3 (Young Function) A mapping $\Phi : \mathbb{R} \longrightarrow [0, +\infty] = \overline{\mathbb{R}^+}$ is called *Young function* if the following properties are verified:

- (i) Φ is even and convex.
- (ii) $\Phi(0) = 0$ and $\lim_{t \rightarrow +\infty} \Phi(t) = +\infty$.

The *complementary* of a Young function Φ is a function $\Psi : \mathbb{R} \longrightarrow [0, +\infty]$ defined by

$$\Psi(t) = \sup_{s \geq 0} \{s|t| - \Phi(s)\}, t \in \mathbb{R}.$$

It follows from the definitions that:

- (1) $\Psi(0) = 0$.
- (2) $\Psi(-t) = \Psi(t) \forall t \in \mathbb{R}$.
- (3) Ψ is a convex increasing function satisfying $\lim_{t \rightarrow +\infty} \Psi(t) = +\infty$.
- (4) The pair (Φ, Ψ) satisfies *Young's inequality*:

$$xy \leq \Phi(x) + \Psi(y).$$

Example 8.2

- (i) Let $\Phi(x) = \frac{|x|^p}{p}$ (respectively, $|x|^p$), $p \geq 1$. Φ is a continuous Young function such that $\Phi(0) = 0$; $\lim_{x \rightarrow +\infty} \Phi(x) = +\infty$ while $\Phi(x) < +\infty$ for $x \in \mathbb{R}$. The function Ψ defined by $\Psi(x) = \frac{|x|^q}{q}$ (respectively, $\frac{|x|^q}{qp^{\frac{1}{p-1}}}$) for $p > 1$ where $\frac{1}{p} + \frac{1}{q} = 1$ or by $\Psi(x) = \begin{cases} 1 & \text{if } 0 \leq x < 1 \\ +\infty & \text{if } x \geq 1 \end{cases}$ for $p = 1$ is the complementary of Φ .
- (ii) Let $\Phi(x) = \begin{cases} 0 & \text{if } 0 \leq x \leq a < +\infty \\ \Phi_1(x) & \text{if } a < x < b, \text{ where } \Phi_1 \text{ is a continuous increasing} \\ +\infty & \text{if } x \geq b \end{cases}$ convex function on $]a; b[$. For instance, $\Phi_1(x) = |x - a|^p$, $p \geq 1$ for $a \leq x < b$. Thus Φ is a young function which is continuous on $]0; b[$ and jump to ∞ at $b > 0$, and $\Phi(-x) = \Phi(x)$.
- (iii) Let Ψ be the function such that $\Psi(-x) = \Psi(x)$ and

$$\Psi(x) = \begin{cases} 0 & \text{if } 0 \leq x < a \\ +\infty & \text{if } x \geq a. \end{cases}$$

Ψ is complementary to Φ defined by $\Phi(x) = |x|$ with $a = 1$.

For more details on the Young functions, see [7].

For a Young function Φ , let $L^\Phi(G)$ be the set of all Borel measurable functions $f : G \longrightarrow \mathbb{C}$ such that for some $\alpha > 0$, $\int_G \Phi(\alpha |f(x)|) d\mu_G(x) < \infty$. $L^\Phi(G)$ is a vector space and is called an *Orlicz space*. Assuming that two elements of $L^\Phi(G)$ are the same if they are equal μ_G -almost everywhere, we endow $L^\Phi(G)$ with the

norm

$$\|f\|_{\Phi} = \sup \left\{ \int_G |f(x)v(x)| d\mu_G(x) : \int_G \Psi(|v(x)|) d\mu_G(x) < 1 \right\},$$

and $(L^{\Phi}(G), \|\cdot\|_{\Phi})$ is a Banach space (see [7], chapter III, proposition 11).

In Example 8.2, we obtain the classical L^p spaces.

8.3 Convolution in Orlicz Spaces

In this part, we first define an action of the center of a Gelfand pair (G, K) on G and finally establish our result.

8.3.1 An Action of the Center of the Hypergroup

Let us recall the following two definitions concerning locally compact groups.

Let H be a locally compact group, X be a topological locally compact space, and ν a nonnegative Radon measure on X . An *action* of H on the measure space (X, ν) (written $H \curvearrowright (X, \nu)$) is a continuous action of H on X such that ν is H -invariant, i.e., for $h \in H$ and any Borel subset E of X , $hE = \{hx : x \in E\}$ is also a Borel subset of X and $\nu(hE) = \nu(E)$.

Definition 8.4 An element $h \in H$ is called *aperiodic* with respect to the given action $H \curvearrowright (X, \nu)$ if for each compact subset $E \subset X$ there exists a nonnegative integer N such that for each $n \geq N$, $E \cap h^n E = \emptyset$.

For the following, we consider the hypergroup G such that (G, K) is a Gelfand pair as described in the preliminaries.

Let us recall that for a hypergroup G , the space $G_e = \{x \in G : x * \bar{x} = \bar{x} * x = e\}$ is a locally compact group which is called the *maximal group* of G (see [5]).

Definition 8.5 Let (G, K) be a Gelfand pair. The *center* of G is the set $Z(G) = \{x \in G : \forall y \in G, \exists \alpha_x(y) \in G \text{ with } x * y = y * x = \delta_{\alpha_x(y)}\}$.

Seen the definition, we can say that $x \in Z(G)$ if x commutes with every element $y \in G$ and the product is reduced to a single point.

$Z(G)$ is a closed subspace of G and $Z(G) \subset G_e$.

When G is a commutative hypergroup, $Z(G) = G_e$. In fact, let $x \in G_e$, $y \in G$, and choose $s \in x * y$. Then $\bar{x} * s \subset \bar{x} * x * y = y$. Thus $\bar{x} * s = y$, and $x * y = x * \bar{x} * s = s$. Since G is commutative, then $y * x = x * y = s$. So $x \in Z(G)$.

Let us give the following property that characterizes the center of a Gelfand pair.

Proposition 8.1 *Let (G, K) be a Gelfand pair. Then the function*

$$\begin{aligned} A : Z(G) \times G &\longrightarrow G \\ (x, y) &\longmapsto x * y = \alpha_x(y) \end{aligned}$$

defines an action of $Z(G)$ on (G, μ_G) .

Proof The center $Z(G)$ is a locally compact group. In fact, let $x, y \in Z(G)$. For any z belonging to G , $\alpha_x(\alpha_y(z)) \in G$. We have

$$\alpha_x(\alpha_y(z)) = x * (y * z) = (x * y) * z$$

and

$$\begin{aligned} \alpha_x(\alpha_y(z)) &= x * (y * z) \\ &= x * (z * y) \\ &= (x * z) * y \\ &= (z * x) * y \\ &= z * (x * y). \end{aligned}$$

Thus, $(x*y)*z = z*(x*y) = \alpha_x(\alpha_y(z))$. So $x*y \in Z(G)$ and $\alpha_{x*y}(z) = \alpha_x(\alpha_y(z))$. If $x \in Z(G)$ and $y \in G$, we have $\alpha_x(\overline{y}) = \overline{(x*y)} = y * \overline{x}$ and also $\alpha_x(\overline{y}) = \overline{(x*y)} = \overline{(y*x)} = \overline{y} * \overline{x}$; that is, $\overline{x} \in Z(G)$. Since $e \in Z(G)$, $Z(G)$ is a subgroup of the maximal group.

For any x belonging to $Z(G)$, the function

$$\begin{aligned} G &\longrightarrow G \\ y &\longmapsto x * y = \alpha_x(y) \end{aligned}$$

is a homeomorphism (see [5] 10.4B). Then A defines a continuous action on G , and for any Borel subset $E \subset G$, $x * E$ is a Borel subset of G . Furthermore,

$$\begin{aligned} \mu_G(x * E) &= \int_G \chi_{x*E}(y) d\mu_G(y), \text{ where } \chi_E \text{ is the characteristic function of } E \\ &= \int_G \chi_E(\overline{x} * y) d\mu_G(y) \\ &= \int_G \chi_E(y) d\mu_G(y), \text{ since } \mu_G \text{ is invariant} \\ &= \mu_G(E). \end{aligned}$$

That is, $Z(G) \curvearrowright (G, \mu_G)$. □

For $n \in \mathbb{N}$, let us put $x^n = \underbrace{x * x * \dots * x}_{n \text{ times}}$ and $x^{-n} = (\overline{x})^n$ for $x \in Z(G)$.

8.3.2 Convolution in Orlicz Spaces

Our study will focus on a certain class of Young functions introduced by the authors in [9].

Definition 8.6 Let Φ_1 and Φ_2 be two Young functions. The pair (Φ_1, Φ_2) is said to satisfy the *sequence condition* if there exist two sequences $(a_n)_n$ and $(b_n)_n$ of nonnegative real numbers such that

$$\sum_{n \geq 1} \Phi_1(a_n) < \infty, \sum_{n \geq 1} \Phi_2(b_n) < \infty \text{ and } \sum_{n \geq 1} a_n b_n = \infty.$$

Example 8.3 Let $\infty > p_1, p_2 > 2$ such that $\frac{1}{p_1} + \frac{1}{p_2} < 1$. We consider Φ_1, Φ_2 the Young functions defined by $\Phi_1(x) = |x|^{p_1}, \Phi_2(x) = |x|^{p_2} \forall x \in \mathbb{R}$. Let us take α, β such that $\alpha > \frac{1}{p_1}, \beta > \frac{1}{p_2}$ and $\alpha + \beta = 1$. For any integer $n \geq 1$, let put $a_n = \sum_{n \geq 1} n^{-\alpha}$ and $b_n = \sum_{n \geq 1} n^{-\beta}$. $\sum_{n \geq 1} \Phi_1(a_n) = \sum_{n \geq 1} \frac{1}{n^{\alpha p_1}}$ and $\sum_{n \geq 1} \Phi_2(b_n) = \sum_{n \geq 1} \frac{1}{n^{\beta p_2}}$ are two convergent Riemann series, while $\sum_{n \geq 1} a_n b_n = +\infty$. Thus the pair (Φ_1, Φ_2) satisfies the sequence condition.

After presenting the pair of Young functions satisfying the sequence condition, we have the following result which is our main theorem.

Theorem 8.1 Let (G, K) be a Gelfand pair and (Φ_1, Φ_2) be a pair of Young functions which satisfies the sequence condition. If for each $f \in L^{\Phi_1}(G)$ and $g \in L^{\Phi_2}(G)$, $(f * g)(x)$ exists for almost every $x \in G$, then the set of aperiodic elements of $Z(G)$ with respect to the action $Z(G) \curvearrowright (G, \mu_G)$ is empty.

Proof Let (Φ_1, Φ_2) be a pair of Young functions which satisfies the sequence condition and such that $f * g$ exists almost everywhere for any $f \in L^{\Phi_1}(G)$ and $g \in L^{\Phi_2}(G)$. Let us consider U , a compact symmetric neighborhood of e .

Let us suppose that there is an aperiodic element h of $Z(G)$ with respect to the action $Z(G) \curvearrowright (G, \mu_G)$. Then there exists a nonnegative integer N such that

$$U \cap (h^n * U) = \emptyset \text{ for } n \geq N. \text{ This implies } U \cap (h^{-n} * U) = \emptyset \text{ for } n \geq N. \quad (8.1)$$

Let $V \subset U$ be an open neighborhood of e . $\{e\}$ is a compact subset of G and $\{e\} * \{e\} = \{e\} \subset V$. By [5], Lemma 3.2D, there exist two open subsets V_1 and V_2 containing $\{e\}$ such that $V_1 * V_2 \subset V \subset U$. Set $I = V_1 \cap V_2$. I is an open subset containing $\{e\}$ and $I * I \subset U$. Let us put $W = cl(I)$; W is a compact neighborhood

of e , and $W * W \subset cl(I * I) \subset U$. Since U is symmetric, we can assume that W is symmetric. This being, for $n, m \geq N$, we have

$$(h^{-mN} * W) \cap (h^{-nN} * W) = \emptyset \text{ and } (h^{-mN} * W * W) \cap (h^{-nN} * W * W) = \emptyset. \quad (8.2)$$

Indeed, let us suppose that $(h^{-mN} * W) \cap (h^{-nN} * W) \neq \emptyset$. Since $h \in Z(h)$, then there exist $x, y \in W$ such that $h^{-mN} * x = h^{-nN} * y$. This implies $x = h^{(m-n)N} * y$, that is, $(h^{(m-n)N} * U) \cap U \neq \emptyset$, contradicting (8.1). Similar one can prove that $(h^{-mN} * W * W) \cap (h^{-nN} * W * W) = \emptyset$.

Let us now consider $(a_n)_n$ and $(b_n)_n$, two sequences of nonnegative real numbers such that

$$\sum_{n \geq 1} \Phi_1(a_n) < \infty, \sum_{n \geq 1} \Phi_2(b_n) < \infty \text{ and } \sum_{n \geq 1} a_n b_n = \infty.$$

We have

$$\sum_{n \geq N} \Phi_1(a_n) < \infty, \sum_{n \geq N} \Phi_2(b_n) < \infty \text{ and } \sum_{n \geq N} a_n b_n = \infty.$$

Let us define the following two functions:

$$f = \sum_{n \geq N} a_n \chi_{h^{-nN} * W} \text{ and } g = \sum_{n \geq N} b_n \chi_{h^{nN} * W * W}.$$

We have on the one hand

$$\begin{aligned} \int_G \Phi_1(f(x)) d\mu_G(x) &= \int_{\bigcup_{n \geq N} h^{-nN} * W} \Phi_1(f(x)) d\mu_G(x) \\ &= \sum_{n \geq N} \int_{h^{-nN} * W} \Phi_1(f(x)) d\mu_G(x) \text{ (see (8.2))} \\ &= \sum_{n \geq N} \Phi_1(a_n) \int_{h^{-nN} * W} d\mu_G(x) \\ &= \mu_G(W) \sum_{n \geq N} \Phi_1(a_n) < \infty \end{aligned}$$

and on the other hand

$$\begin{aligned}
 \int_G \Phi_2(g(x)) d\mu_G(x) &= \int_{\bigcup_{n \geq N} h^{nN} * W * W} \Phi_2(g(x)) d\mu_G(x) \\
 &= \sum_{n \geq N} \int_{h^{nN} * W * W} \Phi_2(g(x)) d\mu_G(x) \\
 &= \sum_{n \geq N} \Phi_2(b_n) \int_{h^{nN} * W * W} d\mu_G(x) \\
 &= \mu_G(W * W) \sum_{n \geq N} \Phi_2(b_n) < \infty.
 \end{aligned}$$

So $f \in L^{\Phi_1}(G)$ and $g \in L^{\Phi_2}(G)$. Let $x \in G$, we have

$$\begin{aligned}
 f * g(x) &= \int_G f(y) g(\bar{y} * x) d\mu_G(y) \\
 &= \int_{\bigcup_{n \geq N} h^{-nN} * W} f(y) g(\bar{y} * x) d\mu_G(y) \\
 &= \sum_{n \geq N} \int_{h^{-nN} * W} a_n g(\bar{y} * x) d\mu_G(y) \\
 &= \mu_G(W) \sum_{n \geq N} a_n b_n = \infty \text{ if } x \in W.
 \end{aligned}$$

This contradicts the fact that $f * g(x)$ exists for almost every x in G , and the proof is complete. $\square$

Corollary 8.1 *Let (G, K) be a Gelfand pair. If $Z(G)$ has an aperiodic element with respect to the action $Z(G) \curvearrowright (G, \mu_G)$, then for $\infty > p_1, p_2 > 2$, there are $f \in L^{p_1}(G)$ and $g \in L^{p_2}(G)$ such that $\mu_G(\{x \in G : f * g(x) \text{ does not exist}\})$ is not zero.*

Proof It follows from the fact that the functions $\Phi_1 : x \mapsto |x|^{p_1}$ and $\Phi_2 : x \mapsto |x|^{p_2}$ are two Orlicz functions such that the pair (Φ_1, Φ_2) satisfies the sequence condition (see Example 8.3). $\square$

Remark Since any commutative locally compact hypergroup G is such that $(G, \{e\})$ is a Gelfand pair, the results above are generalizations of those established on commutative locally compact hypergroups, specifically on commutative locally compact groups.

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Chapter 9

A Variational Approach to the Cauchy Problem for Nonlinear Elliptic Systems



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Abstract We discuss the Cauchy problem for a class of nonlinear elliptic equations with data on a piece $\mathcal{S}$ of the boundary surface by means of a variational approach known in the optimal control literature as “equation error method.” To this end, we use purely nonlinear methods such as successive iterations of a Zarembo-type mixed boundary value problem. In the case of real analytic data on a non-characteristic hypersurface $\mathcal{S}$, the existence of a local solution for the Cauchy problem near $\mathcal{S}$ is guaranteed by the Cauchy–Kowalewskaya theorem.

9.1 Introduction

We study the Cauchy problem for a class of nonlinear elliptic equations with data on a piece $\mathcal{S}$ of the boundary surface. By the Cauchy problem is meant any boundary value problem for an unknown function y in a domain X with the property that the data on $\mathcal{S}$, if combined with the differential equations in X , allow one to determine all derivatives of y on $\mathcal{S}$ by means of functional equations.

If $\mathcal{S}$ is an open piece of the boundary surface different from the whole boundary, then the Cauchy problem fails to be solvable for all Cauchy data of compact support in $\mathcal{S}$. Moreover, the problem is not stable.

As first observed by J. Hadamard in 1917, this problem is ill posed for the Laplace operator if the Cauchy data are posed on an open part $\mathcal{S}$ of the boundary surface.

Our motivation is the fact that the Cauchy problem the physicists were interested in was for nonlinear elliptic differential equations or systems with variable coefficients.

Let X be a bounded domain with smooth boundary denoted by ∂X in $\mathbb{R}^n$ and $L(x, y, p)$ a smooth function on the set $X \times \mathbb{R}^k \times \mathbb{R}^{k \times n}$ with values in $\mathbb{R}^\ell$.

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y is a smooth function on the set X with values in $\mathbb{R}^k$, i.e., $y = (y_1, \dots, y_k)$, and we interpret p as a column matrix $(p_{\cdot,1}, \dots, p_{\cdot,n})$, where $p_{\cdot,j}$ is a column with entries $p_{1,j}, \dots, p_{k,j}$. We will substitute a function $y(x)$ on X with values in $\mathbb{R}^k$ for y and its Jacobi matrix $y'(x) = (\partial_j y_i(x))$ for p .

Suppose S is a nonempty open piece of the surface ∂X , and let $1 < q < \infty$. In this chapter, we consider the Cauchy problem (9.1) for a function $y \in W^{1,q}(X)$, namely

$$\left\{ \begin{array}{l} L'_{y_i}(x, y, y') - \sum_{\ell=1}^n \partial_\ell L'_{p_{i,\ell}}(x, y, y') = 0 \quad \text{in } X, \\ y = y_0 \quad \text{on } S, \\ \sum_{\ell=1}^n L'_{p_{i,\ell}}(x, y, y') v_\ell = y_{1,i} \quad \text{on } S \end{array} \right. \quad (9.1)$$

for all $i = 1, \dots, k$.

$v(x) = (v_1(x), \dots, v_n(x))$ stands for the unit outward normal vector for ∂X , and y_0, y_1 are given functions on S .

The Cauchy problem for nonlinear partial differential equations can be handled within nonlinear mixed boundary value problems [1–3]. The classical mixed boundary value problem for the Laplace operator consists of finding a harmonic function in a domain X , if its values on a part of the boundary S and the normal derivative on $\partial X \setminus S$ are given. Zaremba was the first who studied this problem, cf. [4]. For this reason, we call a mixed problem of this type a Zaremba problem.

The connection between the Cauchy problem for linear elliptic equations and Zaremba problem goes back at least as far as [5].

The book [6] presents a modern treatment of the Cauchy problem for elliptic linear partial differential equations and systems.

Analytical methods developed in [6] allow one to describe those Cauchy data on S for which the Cauchy problem is solvable.

This chapter is aimed at elaborating an iterative approach to the Cauchy problem (9.1). We generalize the results of [7]. We develop successive iterations for this problem and show that they converge to the solution of the Cauchy problem, if there is anyone, see [8]. Our contribution consists of studying a class of mixed boundary problems with nonzero data in Sobolev spaces with negative smoothness.

9.2 Mixed Problems

The Cauchy problem (9.1) can be handled within nonlinear mixed boundary value problems which arise as Euler equations for suitable extremal problems. To this end, for any given function y_1 on $\partial X \setminus S$ with values in $\mathbb{R}^k$, we consider the

functional

$$f(y) = \int_{\mathcal{X}} L(x, y(x), y'(x)) dx - \int_{\partial\mathcal{X} \setminus \mathcal{S}} \langle y_1(x), y(x) \rangle ds, \quad (9.2)$$

ds being the surface measure on $\partial\mathcal{X}$. We are looking for extreme points of the functional over the set $\mathcal{A}_R$ (the set of admissible functions) which consists of all functions $y \in W^{1,q}(\mathcal{X}, \mathbb{R}^k)$ satisfying $y = y_0$ on $\mathcal{S}$ and $\|y\|_{W^{\frac{1}{q'},q}(\partial\mathcal{X} \setminus \mathcal{S})} \leq R$, such that $1/q + 1/q' = 1$.

y_0 is a given function on $\mathcal{S}$ with values in $\mathbb{R}^k$ and R a fixed positive constant.

It is clear that the functional f is well defined on $\mathcal{A}_R$.

Lemma 9.1 *As defined above, the extremals of the functional (9.2) satisfy the Euler equations*

$$\left\{ \begin{array}{l} L'_{y_i}(x, y, y') - \sum_{\ell=1}^n \partial_{\ell} L'_{p_{i,\ell}}(x, y, y') = 0 \quad \text{in } \mathcal{X}, \\ y = y_0 \quad \text{on } \mathcal{S}, \\ \sum_{\ell=1}^n L'_{p_{i,\ell}}(x, y, y') \nu_{\ell} = y_{1,i} \quad \text{on } \partial\mathcal{X} \setminus \mathcal{S} \end{array} \right. \quad (9.3)$$

for all $i = 1, \dots, k$.

Proof Assume that f has a local minimum at a function $y \in \mathcal{A}_R$. A familiar continuity argument shows that there is an $R' > R$, such that y is actually an extreme function of the functional over the slightly larger set $\mathcal{A}'$ consisting of all $y \in W^{1,q}(\mathcal{X}, \mathbb{R}^k)$ which satisfy $y = y_0$ on $\mathcal{S}$ and $\|y\|_{W^{\frac{1}{q'},q}(\partial\mathcal{X} \setminus \mathcal{S})} \leq R'$.

We now evaluate f at a small variation $(y_1 + \varepsilon_1 \delta_1, \dots, y_k + \varepsilon_k \delta_k)$ of the function y , where $\delta = (\delta_1, \dots, \delta_k)$ is an arbitrary smooth function on $\bar{\mathcal{X}}$ with values in $\mathbb{R}^k$ vanishing on $\mathcal{S}$. If $|\varepsilon| \ll 1$, such a variation still remains in the set $\mathcal{A}'$; hence the resulting function $\varepsilon \mapsto f(y_1 + \varepsilon_1 \delta_1, \dots, y_k + \varepsilon_k \delta_k)$ has an extreme value at the point $\varepsilon = 0$ if y is an extreme point of f . On differentiating in ε_i and setting $\varepsilon = 0$, we get

$$\int_{\mathcal{X}} \left(L'_{y_i}(x, y, y') \delta_i + \sum_{\ell=1}^n L'_{p_{i,\ell}}(x, y, y') \partial_{\ell} \delta_i \right) dx - \int_{\partial\mathcal{X} \setminus \mathcal{S}} y_{1,i} \delta_i ds = 0$$

for each $i = 1, \dots, k$.

Choosing δ of compact support in $\mathcal{X}$, we readily derive the system of differential equations in $\mathcal{X}$. Moving the derivatives ∂_{ℓ} from δ_i to $L'_{p_{i,\ell}}(x, y, y')$ by partial integration now yields

$$\int_{\partial X \setminus \mathcal{S}} \left(\sum_{\ell=1}^n L'_{p_{i,\ell}}(x, y, y') v_\ell - y_{1,i} \right) \delta_i ds = 0$$

for arbitrary functions $\delta \in C^\infty(\overline{X}, \mathbb{R}^k)$ which vanish on $\mathcal{S}$. This is a weak form of the boundary condition on $\partial X \setminus \mathcal{S}$, as desired. $\square$

Let $k = 1$ and $L(x, y, p) = 1/2 |p|^2$, where $p = (p_1, \dots, p_n)$. Then (9.3) becomes

$$\begin{cases} -\Delta y = 0 & \text{in } X, \\ y = y_0 & \text{on } \mathcal{S}, \\ \frac{\partial y}{\partial \nu} = y_1 & \text{on } \partial X \setminus \mathcal{S}, \end{cases}$$

which is precisely the problem of [9]. $\frac{\partial}{\partial \nu}$ stands for the unit outward normal vector derivative for ∂X .

Let $k = 1$ and $L(x, y, y') = 1/p |y'|^p$, where $p \geq 1$. Then (9.3) becomes

$$\begin{cases} -\Delta_p y = 0 & \text{in } X, \\ y = y_0 & \text{on } \mathcal{S}, \\ |\nabla y|^{p-2} \frac{\partial y}{\partial \nu} = y_1 & \text{on } \partial X \setminus \mathcal{S}, \end{cases}$$

where $\Delta_p y =: \operatorname{div}(|\nabla y|^{p-2} \nabla y)$ is the so-called p -Laplace operator, see, for instance, [10].

From now on, we make some assumptions on the Lagrangian $L(x, y, y'(x))$; namely, we suppose that it fulfills the properties

$$|L(x, y, p)| \leq C(1 + |y|^q + |p|^q) \quad (9.4)$$

$$|D_p L(x, y, p)| \leq C(1 + |y|^{q-1} + |p|^{q-1}) \quad (9.5)$$

$$|D_y L(x, y, p)| \leq C(1 + |y|^{q-1} + |p|^{q-1}) \quad (9.6)$$

$$L(x, y, p) \geq \alpha(|p|^q - \beta), \quad (9.7)$$

where $D_p L(x, y, p)$ is the $(k \times n)$ -matrix given by

$$D_p L(x, y, p) = \begin{pmatrix} L_{p_1^1} & \cdots & L_{p_n^1} \\ \vdots & \ddots & \vdots \\ L_{p_k^1} & \cdots & L_{p_n^k} \end{pmatrix}$$

while $D_y L(x, y, p) = (L_{y_1}, \dots, L_{y_k})$.

α, β, C are nonnegative real numbers.

We state a main lemma that will be of great importance for one of our main results below.

Lemma 9.2 *If $y \in W^{1,q}(X)$, then $\sum_{\ell=1}^n L'_{p_{i,\ell}}(x, y, y')$ and $L'_{y_i}(x, y, y')$ are elements of $L^{q'}(X)$ for all $i = 1, \dots, k$, where q' is the dual exponent of q .*

We just give the proof for the first term since the proof for the second term is analogous.

Proof Using (9.6), we have

$$\begin{aligned} \int_X |D_{y'} L(x, y, y')|^{q'} dx &\leq \int_X C^{q'} (1 + |y|^{q-1} + |y'|^{q-1})^{q'} dx \\ &\leq \int_X C^{q'} (1 + |y|^{\frac{q}{q'}} + |y'|^{\frac{q}{q'}})^{q'} dx \\ &\leq C^{q'} (\text{mes}(X) + \|y\|_{W^{1,q}(X)}^q + \|y'\|_{W^{1,q}(X)}^q) \\ &< \infty, \end{aligned}$$

where $\text{mes}(X)$ stands for the measure of the domain X , and the lemma is proved. $\square$

9.3 Existence of a Variational Solution

In this section, we show the solvability of the problem (9.3) which is one of our main results.

Theorem 9.1 *Assume that the Lagrangian function $L(x, y(x), y'(x))$ satisfies the conditions (9.4)–(9.7) and be convex in $y'(x)$ for each (x, y) . Then, the mixed boundary value problem (9.3) has at least one solution in $W^{1,q}(X)$.*

Proof Consider the functional

$$f(y) = \int_X L(x, y, y') dx - \int_{\partial X \setminus S} y_1 y ds,$$

where $y_1 = (y_{1,1}, \dots, y_{1,k})$. The functional will be minimized on the domain $\mathcal{A}_R$.

We first estimate the boundary integral by using the Hölder inequality and the trace theorem, cf. [6, p. 77]. We then get

$$\left| \int_{\partial\mathcal{X}\setminus\mathcal{S}} y_1 y ds \right| \leq \|y_1\|_{W^{\frac{1}{q'}, q'}(\partial\mathcal{X}\setminus\mathcal{S})} \|y\|_{W^{\frac{1}{q}, q}(\partial\mathcal{X}\setminus\mathcal{S})} \quad (9.8)$$

$$\leq C_0 \|y_1\|_{W^{\frac{1}{q'}, q'}(\partial\mathcal{X}\setminus\mathcal{S})} \cdot \|y\|_{W^{1,q}(\mathcal{X})}. \quad (9.9)$$

By Lemma 1.9.2 of [11], there exists a constant C depending only on $\mathcal{S}$, $\mathcal{X}$, q , and n such that

$$\|y\|_{W^{1,q}(\mathcal{X})} \leq C(\|y'\|_{L^q(\mathcal{X}, \mathbb{R}^n)} + \|y\|_{L^1(\mathcal{S})}), \quad (9.10)$$

for all $y \in W^{1,q}(\mathcal{X})$.

Combining the inequalities (3.3) and (3.4), we get

$$\left| \int_{\partial\mathcal{X}\setminus\mathcal{S}} y_1 y ds \right| \leq C_0 \|y_1\|_{W^{\frac{1}{q'}, q'}(\partial\mathcal{X}\setminus\mathcal{S})} \cdot C(\|y'\|_{L^q(\mathcal{X}, \mathbb{R}^n)} + \|y\|_{L^1(\mathcal{S})}) \quad (9.11)$$

$$\leq C_0 C \|y_1\|_{W^{\frac{1}{q'}, q'}(\partial\mathcal{X}\setminus\mathcal{S})} \cdot (\|y'\|_{L^q(\mathcal{X}, \mathbb{R}^n)} + \|y\|_{L^1(\mathcal{S})}) \quad (9.12)$$

$$\leq C_1 (\|y'\|_{L^q(\mathcal{X}, \mathbb{R}^n)} + \|y\|_{L^1(\mathcal{S})}), \quad (9.13)$$

where C_1 stands for the constant $C_0 C \|y_1\|_{W^{\frac{1}{q'}, q'}(\partial\mathcal{X}\setminus\mathcal{S})}$.

Using the properties of the Lagrangian function and the inequality (9.9), we are able to estimate our functional f from below. We then have

$$\begin{aligned} f(y) &\geq \alpha \|y'\|_{L^q(\mathcal{X}, \mathbb{R}^n)}^q - C' - C_1 (\|y'\|_{L^q(\mathcal{X}, \mathbb{R}^n)} + \|y\|_{L^1(\mathcal{S})}) \\ &\geq A_1 \|y'\|_{L^q(\mathcal{X}, \mathbb{R}^n)}^q - A_2 \|y'\|_{L^q(\mathcal{X}, \mathbb{R}^n)} - A_3, \end{aligned}$$

where A_1 , A_2 , and A_3 are nonnegative real numbers.

As we assume that $q > 1$, we then get that $A_1 \|y'\|_{L^q(\mathcal{X}, \mathbb{R}^n)}^q - A_2 \|y'\|_{L^q(\mathcal{X}, \mathbb{R}^n)} - A_3$ tends to ∞ as $\|y'\|_{L^q(\mathcal{X}, \mathbb{R}^n)}^q \rightarrow \infty$. Consequently, the functional f is coercive.

Let $y \in \mathcal{A}_R$. Then f has an infimum m on $\mathcal{A}_R$. Let $\{y_N\}$ be a minimizing sequence in $\mathcal{A}_R$, i.e., $f(y_N) \rightarrow m$. Therefore

$$\int_{\mathcal{X}} L(x, y_N, y'_N) dx - \int_{\partial\mathcal{X}\setminus\mathcal{S}} y_1 y_N ds \geq m. \quad (9.14)$$

Our next concern will be the boundedness of $\{y_N\}$ in $W^{1,q}(\mathcal{X})$. To prove this, we observe that

$$\|y_N\|_{W^{1,q}(\mathcal{X})}^q = \int_{\mathcal{X}} |y_N|^q dx + \int_{\mathcal{X}} |y'_N|^q dx; \quad \forall N \in \mathbb{N}. \quad (9.15)$$

Once again using the properties of the Lagrangian function L , we have

$$\int_{\mathcal{X}} \alpha(|y'_N|^q - \beta) dx \leq \int_{\mathcal{X}} L(x, y_N, y'_N) dx.$$

Moreover, an easy computation shows that

$$\|y'_N\|_{W^{1,q}(\mathcal{X})}^q \leq \mathcal{Q} \left(f(y_N) + \beta \text{mes}(\mathcal{X}) + \int_{\partial \mathcal{X} \setminus \mathcal{S}} y_1 y_N ds \right).$$

We also have

$$\|y_N\|_{W^{1,q}(\mathcal{X})}^q \leq \int_{\mathcal{X}} |y_N|^q dx + \mathcal{Q} \left(f(y_N) + \beta \text{mes}(\mathcal{X}) + \int_{\partial \mathcal{X} \setminus \mathcal{S}} y_1 y_N ds \right),$$

for all $N \in \mathbb{N}$.

Since $\{f(y_N)\}$ converges, this sequence is bounded, i.e., $|f(y_N)| < C_1$, where C_1 is a constant independent of N . By the Hölder inequality and the definition of admissible functions, we also get

$$\begin{aligned} \left| \int_{\partial \mathcal{X} \setminus \mathcal{S}} y_1 y_N ds \right| &\leq \|y_1\|_{W^{\frac{1}{q'}, q'}(\partial \mathcal{X} \setminus \mathcal{S})} \|y_N\|_{W^{\frac{1}{q}, q}(\partial \mathcal{X} \setminus \mathcal{S})} \\ &\leq R \|y_1\|_{W^{\frac{1}{q'}, q'}(\partial \mathcal{X} \setminus \mathcal{S})} \\ &:= C_2, \end{aligned}$$

C_2 being a constant.

It remains to estimate the third and the fourth terms. To this end, we use a well-known estimate (Korn's inequality)

$$\begin{aligned} \int_{\mathcal{X}} |y_N|^q dx &\leq C_3 \left(\left(\int_{\mathcal{X}} |y'_N|^q dx \right)^{\frac{1}{q}} + \left(\int_{\mathcal{S}} |y_N|^q dx \right)^{\frac{1}{q}} \right) \\ &= C_3 \mathcal{Q} \left(f(y_N) + \beta \text{mes}(\mathcal{X}) + \int_{\partial \mathcal{X} \setminus \mathcal{S}} y_1 y_N ds \right)^{\frac{1}{q}} \\ &\quad + C_3 \left(\int_{\mathcal{S}} |y_0|^q dx \right)^{\frac{1}{q}} \end{aligned}$$

$$\begin{aligned} &\leq C_4 \left(((C_1 + \beta \text{mes}(\mathcal{X}) + C_2))^{\frac{1}{q}} + \|y_0\|_{L^q(\mathcal{S})} \right) \\ &:= C_5, \end{aligned}$$

where C_5 is a constant independent of N .

Summarizing, we deduce that $\|y_N\|_{W^{1,q}(\mathcal{X})} < \infty$, showing the boundedness of $\{y_N\}$ in $W^{1,q}(\mathcal{X})$.

We now use the fact that $W^{1,q}(\mathcal{X})$ is a reflexive Banach space for all nonextreme q , and so each bounded set in this space is relatively compact. It follows that there is a subsequence of $\{y_N\}$ in $W^{1,q}(\mathcal{X})$, which converges to $y \in W^{1,q}(\mathcal{X})$ in the weak topology of this space. It will cause no confusion if we still write $\{y_N\}$ for this subsequence. If we prove that $y \in \mathcal{A}_R$ and $\liminf f(y_N) \geq f(y)$, the assertion follows, i.e., $f(y) = m$.

To prove that $y \in \mathcal{A}_R$, it suffices to show by a familiar Mazur theorem (cf. [12], p. 120) that $\mathcal{A}_R$ is convex and closed. We first prove the convexity. If $y, v \in \mathcal{A}_R$, then $ty + (1-t)v = y_0$ on $\mathcal{S}$ and

$$\begin{aligned} \|ty + (1-t)v\| &\leq t\|y\|_{W^{\frac{1}{q'},q}(\partial\mathcal{X}\setminus\mathcal{S})} + (1-t)\|v\|_{W^{\frac{1}{q'},q}(\partial\mathcal{X}\setminus\mathcal{S})} \\ &\leq R \end{aligned}$$

for all $t \in [0; 1]$, as desired.

We now show the closedness of $\mathcal{A}_R$. To this end, we pick a sequence $\{y_N\}$ in $\mathcal{A}_R$, which converges to $y \in W^{1,q}(\mathcal{X})$ in the norm of this space. We have to show that $y = y_0$ on $\mathcal{S}$. By definition of $\mathcal{A}_R$, each y_N coincides with y_0 on $\mathcal{S}$. Since $\{y_N\}$ converges to y in the $W^{1,q}(\mathcal{X})$ -norm, the Sobolev embedding theorem (see [6, p. 77]) readily yields $\|y - y_N\|_{W^{\frac{1}{q'},q}(\mathcal{X})} \rightarrow 0$ as $N \rightarrow \infty$. Hence it follows that

$$\begin{aligned} \|y_N - y\|_{W^{\frac{1}{q'},q}(\mathcal{S})} &\leq \|y_N - y\|_{W^{\frac{1}{q'},q}(\mathcal{X})} \\ &\rightarrow 0, \end{aligned}$$

as $N \rightarrow \infty$, and so $y = y_0$ on $\mathcal{S}$ as desired. Furthermore,

$$\begin{aligned} \|y\|_{W^{\frac{1}{q'},q}(\partial\mathcal{X}\setminus\mathcal{S})} &\leq \|y_N\|_{W^{\frac{1}{q'},q}(\partial\mathcal{X})} + \|y - y_N\|_{W^{\frac{1}{q'},q}(\partial\mathcal{X})} \\ &\leq R + \|y - y_N\|_{W^{\frac{1}{q'},q}(\partial\mathcal{X})} \end{aligned}$$

holds for all N . Hence we conclude, by $\|y - y_N\|_{W^{\frac{1}{q'},q}(\partial\mathcal{X})} \rightarrow 0$ as $N \rightarrow \infty$, that

$\|y\|_{W^{\frac{1}{q'},q}(\partial\mathcal{X}\setminus\mathcal{S})} \leq R$, implying $y \in \mathcal{A}_R$. The closedness of $\mathcal{A}_R$ is thus proved.

We now turn to the weak lower semicontinuity of f . Suppose that $\{y_N\}$ converges weakly in $W^{1,q}(\partial\mathcal{X})$ to y . Our purpose is to show that $f(y) \leq \liminf f(y_N)$ as $N \rightarrow \infty$, i.e.,

$$\begin{aligned} & \int_{\mathcal{X}} L(x, y, y') dx - \int_{\partial\mathcal{X} \setminus \mathcal{S}} y_1 y_N ds \\ & \leq \liminf_{N \rightarrow \infty} \left(\int_{\mathcal{X}} L(x, y_N, y'_N) dx - \int_{\partial\mathcal{X} \setminus \mathcal{S}} y_1 y_N ds \right). \end{aligned}$$

□

Any functional that fulfills the above inequality is said to be weak lower semicontinuous. Analysis similar to that in the proof of theorem 2 in [13, P456] shows that

$$\int_{\mathcal{X}} L(x, y, y') dx \leq \liminf_{N \rightarrow \infty} \int_{\mathcal{X}} L(x, y_N, y'_N) dx.$$

We are thus left with the task of proving

$$\int_{\partial\mathcal{X} \setminus \mathcal{S}} y_1 y_N ds \rightarrow \int_{\partial\mathcal{X} \setminus \mathcal{S}} y_1 y ds \quad (9.16)$$

as $N \rightarrow \infty$.

By the definition of $\mathcal{A}_R$, the sequence $\{y_N\}$ is bounded in $W^{\frac{1}{q'},q}(\partial\mathcal{X} \setminus \mathcal{S})$, if specified as a sequence in this space.

Indeed, we have $\|y_N\|_{W^{\frac{1}{q'},q}(\partial\mathcal{X} \setminus \mathcal{S})} \leq R$ for all N . Since $W^{\frac{1}{q'},q}(\partial\mathcal{X} \setminus \mathcal{S})$ is a reflexive Banach space, it follows that it contains a subsequence that converges weakly in $W^{\frac{1}{q'},q}(\partial\mathcal{X} \setminus \mathcal{S})$.

Thus, we can assume without restriction of generality that $\{y_N\}$ converges weakly in $W^{\frac{1}{q'},q}(\partial\mathcal{X} \setminus \mathcal{S})$ to a function $u \in W^{\frac{1}{q'},q}(\partial\mathcal{X} \setminus \mathcal{S})$, for if not, we pass to a subsequence.

We know, cf. [7], that if $\{y_N\} \subset W^{1,q}(\mathcal{X})$ converges weakly to a function y in $W^{1,q}(\mathcal{X})$ and the restriction of $\{y_N\}$ to $\partial\mathcal{X} \setminus \mathcal{S}$ converges weakly in $W^{\frac{1}{q'},q}(\partial\mathcal{X} \setminus \mathcal{S})$ to a function u , then the restriction of y to $\partial\mathcal{X} \setminus \mathcal{S}$ actually coincides with u . From this, equality (9.16) readily follows.

For some regularity and uniqueness results for solutions of (9.3) we refer the reader to [14, 15], and the references therein. From now on, we consider the class of (9.3) which possesses a unique solution and turn back to our original Cauchy problem (9.1).

9.4 Iterations

To solve the Cauchy problem,

$$\left\{ \begin{array}{l} L'_{y_i}(x, y, y') - \sum_{\ell=1}^n \partial_{\ell} L'_{p_{i,\ell}}(x, y, y') = 0 \quad \text{in } \mathcal{X}, \\ y = y_0 \quad \text{on } \mathcal{S}, \\ \sum_{\ell=1}^n L'_{p_{i,\ell}}(x, y, y') v_{\ell} = y_{1,i} \quad \text{on } \mathcal{S} \end{array} \right. \quad (9.17)$$

for all $i = 1, \dots, k$, and for some given y_0 and y_1 functions on $\mathcal{S}$, we describe an iterative procedure. As above, the natural Sobolev space for solving the Cauchy problem is $W^{1,q}(\mathcal{X})$, where the Cauchy data are chosen in $W^{\frac{1}{q'},q}(\mathcal{S})$ and $W^{\frac{-1}{q'},q'}(\mathcal{S})$, respectively (see [16, 17]).

We now use Theorem 9.1 to construct a solution to the Cauchy problem by iteratively solving the mixed boundary problem. The latter problem has the advantage of being of order 2.

Let $\tilde{y}_1$ be an arbitrary element of $(W^{\frac{1}{q'},q}(\partial\mathcal{X}))'$. We seek the initial approximation y^0 as a solution of the mixed boundary.

$$\left\{ \begin{array}{l} L'_{y_i}(x, y^0, y'^0) = \sum_{\ell=1}^n \partial_{\ell} L'_{p_{i,\ell}}(x, y^0, y'^0) \quad \text{in } \mathcal{X}, \\ y^0 = y_0 \quad \text{on } \mathcal{S}, \\ \sum_{\ell=1}^n L'_{p_{i,\ell}}(x, y^0, y'^0) v_{\ell} = \tilde{y}_{1,i} \quad \text{on } \partial\mathcal{X} \setminus \mathcal{S} \end{array} \right.$$

for all $i = 1, \dots, k$, which exists by Theorem 9.1. The next iterations for odd indices solve the problem

$$\left\{ \begin{array}{l} L'_{y_i}(x, y^{2t+1}, y'^{2t+1}) = \sum_{\ell=1}^n \partial_{\ell} L'_{p_{i,\ell}}(x, y^{2t+1}, y'^{2t+1}) \quad \text{in } \mathcal{X}, \\ y^{2t+1} = y_0 \quad \text{on } \partial\mathcal{X} \setminus \mathcal{S}, \\ \sum_{\ell=1}^n L'_{p_{i,\ell}}(x, y^{2t+1}, y'^{2t+1}) v_{\ell} = y_{1,i} \quad \text{on } \mathcal{S} \end{array} \right.$$

for all $i = 1, \dots, k$, and for even indices the problem

$$\left\{ \begin{array}{ll} L'_{y_i}(x, y^{2t+2}, y'^{2t+2}) = \sum_{\ell=1}^n \partial_{\ell} L'_{p_{i,\ell}}(x, y^{2t+2}, y'^{2t+2}) & \text{in } \mathcal{X}, \\ y^{2t+2} = y_0 & \text{on } \mathcal{S}, \\ \sum_{\ell=1}^n L'_{p_{i,\ell}}(x, y^{2t+2}, y'^{2t+2}) v_{\ell} = \sum_{\ell=1}^n L'_{p_{i,\ell}}(x, y^{2t+1}, y'^{2t+1}) v_{\ell} & \text{on } \partial \mathcal{X} \setminus \mathcal{S} \end{array} \right.$$

for all $i = 1, \dots, k$. The Zaremba problem is uniquely solvable on each step, and this is guaranteed by Theorem 9.1. Therefore, the iterative sequence $\{y^t\}$ is well defined. Obviously, if $y \in W^{1,q}(\mathcal{X})$ is a solution of the Cauchy problem (9.1) and we take

$$y_1 = \sum_{\ell=1}^n L'_{p_{i,\ell}}(x, y, y') v_{\ell} \text{ for all } i = 1, \dots, k,$$

then $y^t = y$ for $t = 0, 1, 2, 3, 4, \dots$

Corollary 9.1 *If the sequence $\{y^t\}_{t=0}^{\infty}$ converges in $W^{1,q}(\mathcal{X})$, then the limit function is the solution to the Cauchy problem (9.1).*

Proof Assume that $y^t \rightarrow y$ in the $W^{1,q}(\mathcal{X})$ -norm. Then

$$L'_{y_i}(x, y^t, (y')^t) - \sum_{\ell=1}^n \partial_{\ell} L'_{p_{i,\ell}}(x, y^t, (y')^t)$$

converges weakly to

$$L'_{y_i}(x, y, y') - \sum_{\ell=1}^n \partial_{\ell} L'_{p_{i,\ell}}(x, y, y')$$

in the $W^{-1,q'}(\mathcal{X})$ -norm by using the properties of the Lagrangian function. Hence, it follows that

$$L'_{y_i}(x, y, y') - \sum_{\ell=1}^n \partial_{\ell} L'_{p_{i,\ell}}(x, y, y') = 0$$

in $\mathcal{X}$.

The convergence in the space $W^{1,q}(\mathcal{X})$ leads to the convergence of the sequence $\{y^t|_{\mathcal{S}}\}_{t=0}^{\infty}$ to $y|_{\mathcal{S}} \in W^{\frac{1}{q'},q}(\mathcal{S})$. But the subsequence $\{y_{\mathcal{S}}^{2t+2}\}_{t=0}^{\infty}$ converges to y_0 ; therefore, $y = y_0$ on $\mathcal{S}$. This convergence implies also that the Neumann condition in the Cauchy problem is satisfied. $\square$

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Chapter 10

Large Deviations and Berry–Esseen Inequalities in the Stochastic Diffusion Driven by a Volterra Type Process



Armél Fabrice Evrard Yodé and Kouacou Tanoh

Abstract We derive large deviations and Berry–Esseen inequalities for the maximum likelihood estimator of the unknown parameter appearing linearly in the drift of stochastic differential equation driven by a Volterra process.

10.1 Introduction

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a complete probability space, i.e., for any $B \in \mathcal{A}$ with $\mathbb{P}(B) = 0$ and all $A \subset B$, we have $A \in \mathcal{A}$. Let $W = \{W_t, t \in [0, T]\}$, where $T > 0$, be the standard Brownian motion. A Volterra process $(B_t)_{t \geq 0}$ is a stochastic integral defined by

$$B_t = \int_0^t K(t, s) dW_s,$$

where the function K called kernel verifies the following assumptions:

$(\mathcal{H}_1)$ $K(t, s) = 0$ for all $s, t \in [0, T]$ such that $s > t$.

$(\mathcal{H}_2)$ There are constants $C, \beta > 0$ such that for all $s, t \in [0, T]$,

$$\int_0^T \left(K(t, r) - K(s, r) \right)^2 dr \leq C|t - s|^\beta.$$

$(\mathcal{H}_3)$ K is injective as a transformation of functions in $L^2([0, T])$:

$$(Kg)(t) = \int_0^T K(t, s)g(s)ds \quad g \in L^2([0, T]).$$

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Brownian motion and more generally fractional Brownian motion are typical examples of Volterra processes. Processes of this type have applications in various fields such as telecommunication, finance, and economics.

We consider the process $X_0^T = (X_t)_{0 \leq t \leq T}$ solution of the stochastic differential equation

$$dX_t = \theta f(t, X_t)dt + dB_t, \quad (10.1)$$

where $(B_t)_{t \geq 0}$ is a Volterra process, $f(\cdot, \cdot)$ is a known real-valued continuous function defined on $\mathbb{R}_+ \times \mathbb{R}$, and θ is an unknown parameter belonging to a closed interval Θ in $\mathbb{R}$. In this chapter, we consider the maximum likelihood estimator (MLE) $\hat{\theta}_T$ of θ . According to Hult [1], this estimator is strongly convergent and asymptotically Gaussian. We aim to establish a large deviation result and a Berry–Esseen bound for MLE $\hat{\theta}_T$. In the context of large deviations, the question here is to evaluate the rate of convergence to zero of $\mathbb{P}(|\hat{\theta}_T - \theta| > \varepsilon)$ for any ε strictly positive. We will show that $\mathbb{P}(|\hat{\theta}_T - \theta| > \varepsilon)$ goes to zero exponentially when $T \rightarrow +\infty$. In the case of the Berry–Esseen inequality, we say something about the convergence of the distribution function of MLE $\hat{\theta}_T$ to that of the normal distribution.

Our present work is the expanded version of Kouacou, N’zi, and Yodé [2]. The organization of this chapter is as follows. In Sect. 10.2, we give some preliminaries on fractional calculus, Hult’s work on the reproducing kernel Hilbert space associated with Gaussian process, stochastic integral with respect to Volterra Process, and maximum likelihood estimation. We also give some useful lemmas to prove our main results. The results of large deviations and Berry–Esseen type bounds are presented in Sect. 10.3.

10.2 Preliminaries

We consider $\Omega = C([0, T])$ the space of continuous functions, equipped with the Borel σ -field $\mathcal{A}$. The canonical process $B = \{B_t, t \in [0, T]\}$ is defined by $B_t(\omega) = \omega(t)$, and the probability measure $\mathbb{P}$ on $(\Omega, \mathcal{A})$ is the unique measure such that B is centered Gaussian process with covariance

$$\rho(t, s) = \int_0^T K(t, r)K(s, r)dr,$$

where $K : [0, T] \times [0, T] \rightarrow [0, \infty)$ is a function satisfying hypothesis $(\mathcal{H}_1)$, $(\mathcal{H}_2)$, and $(\mathcal{H}_3)$. Let $\mathbb{P}_\theta^T$ be the probability measure generated by the process X_0^T on the space $(\Omega, \mathcal{A})$. We denote by $\mathbb{E}_\theta^T$ the expectation with respect to $\mathbb{P}_\theta^T$. Let us put

$$Q_t = K^{-1} \left(\int_0^T f(r, X_r)dr \right) (t). \quad (10.2)$$

We assume the following conditions:

($\mathcal{H}_4$) $\mathbb{P}_{\theta_1}^T \neq \mathbb{P}_{\theta_2}^T$ for $\theta_1 \in \Theta$ and $\theta_2 \in \Theta$ such as $\theta_1 \neq \theta_2$.

($\mathcal{H}_5$) It is assumed that there are constants $0 < \alpha \leq 1$ and $C > 0$ such that

$$|f(t, x)| \leq C(1 + |x|^\alpha) \quad \forall (t, x) \in [0, T] \times \mathbb{R}.$$

($\mathcal{H}_6$) (i) $n_T = \mathbb{E}_\theta^T(I_T) < \infty$ with $n_T \rightarrow \infty$ as $T \rightarrow \infty$, where

$$I_T = \int_0^T Q_t^2 dt.$$

(ii) $\frac{I_T}{n_T} \xrightarrow{\mathbb{P}_\theta^T} 1$ as $T \rightarrow \infty$.

($\mathcal{H}_7$) Suppose that there exist $\gamma \geq 2$ and $C > 0$ such that for any $a \in \mathbb{R}$, we have

$$\left(\mathbb{E}_\theta^T \left[\exp \left[- \frac{a^2}{3} \frac{I_T}{n_T} \right] \right] \right)^{\frac{1}{4}} \leq \exp(-C|a|^\gamma).$$

The existence and uniqueness of the strong solution for Eq. (10.1) is proved under assumption ($\mathcal{H}_5$) in Nualart and Ouknine [3]. We need the stronger regularity condition ($\mathcal{H}_7$) in order to prove large deviations for maximum likelihood.

In the rest of the section, we first recall the work of Hult [1] on the stochastic integral with respect to the Volterra process and the construction of MLE $\hat{\theta}_T$ of parameter θ . We give some useful lemmas in the proofs of our main results.

10.2.1 Malliavin Calculus

We start with some concepts of Malliavin calculus. For further details, we refer to Nualart [4] and Üstünel [5]. Let $\mathcal{H}$ be a real separable Hilbert space that consists of real-valued functions on $[0, T]$ and equipped with scalar product denoted by $\langle \cdot, \cdot \rangle_{\mathcal{H}}$. The norm of an element $h \in \mathcal{H}$ will be denoted by $\|h\|_{\mathcal{H}}$.

Definition 10.1 We say that stochastic process $G = \{G(h), h \in \mathcal{H}\}$ defined in $(\Omega, \mathcal{A}, \mathbb{P})$ is an isonormal Gaussian process if G is a centered Gaussian family of random variables such that

$$\mathbb{E}(G(h)G(g)) = \langle h, g \rangle_{\mathcal{H}} \quad \text{for all } h, g \in \mathcal{H}.$$

A real-valued smooth cylindrical functional F is a real-valued functional $F : \Omega \rightarrow \mathbb{R}$ defined by

$$F = f(G(h_1), G(h_2), \dots, G(h_n))$$

where $h_1, \dots, h_n \in \mathcal{H}$ and f belongs to the space $C_b^\infty(\mathbb{R}^n)$ of bounded infinitely differentiable functions from $\mathbb{R}^n$ to $\mathbb{R}$. We write $\mathcal{S}$ for the space of all real-valued smooth cylindrical functionals. For $F \in \mathcal{S}$, the derivative of F is

$$\nabla F = \sum_{j=1}^n \partial_j f(G(h_1), \dots, G(h_n)) h_j,$$

where $\partial_j f = \frac{\partial f}{\partial x_j}$. By Nualart [4] or Üstünel [5], the derivative operator ∇ is a closable operator from $L^p(\Omega)$ to $L^p(\Omega, \mathcal{H})$ with $p > 1$. Thus, we can define the L^p -domain of ∇ .

Definition 10.2 We say that F is in $\text{Dom}_p(\nabla)$ if there exists a sequence $\{F_n, n \geq 1\} \subset \mathcal{S}$ such that $F_n \rightarrow F$ in $L^p(\Omega)$ and $\{\nabla F_n : n \geq 1\}$ is a Cauchy sequence in $L^p(\Omega, \mathcal{H})$. Then, we define

$$\nabla F = \lim_{n \rightarrow +\infty} \nabla F_n.$$

This extended operator ∇ is called the Gross–Sobolev derivative.

We denote by $\mathbb{D}_{p,1}$ the linear space $\text{Dom}_p(\nabla)$ equipped with the norm

$$\|F\|_{p,1}^p = \|F\|_{L^p(\Omega)}^p + \|\nabla F\|_{L^p(\Omega, \mathcal{H})}^p.$$

We introduce the divergence operator δ which is the formal adjoint of ∇ .

Definition 10.3 Let $U : \Omega \rightarrow \mathcal{H}$. We say that $U \in \text{Dom}_p(\delta)$, if for any $F \in \mathbb{D}_{q,1}$ with $\frac{1}{p} + \frac{1}{q} = 1$, there exists a constant $c_{p,q}(U)$ such that

$$\mathbb{E}[\langle \nabla F, U \rangle_{\mathcal{H}}] \leq c_{p,q}(U) \|F\|_q,$$

and in this case we define $\delta(U)$ by

$$\mathbb{E}[F \delta(U)] = \mathbb{E}[\langle \nabla F, U \rangle_{\mathcal{H}}] \quad \text{for each } F \in \mathbb{D}_{q,1}.$$

The element $\delta(U)$ is called the Skorohod integral of U .

10.2.2 A Stochastic Integral with Respect to a Volterra Type Process

Let H be the closure in $L^2(\Omega)$ of the space spanned by $\{B_t, t \in [0, T]\}$ equipped with the inner product $\langle \xi, \zeta \rangle_H = \mathbb{E}(\xi \zeta)$ for $\xi, \zeta \in H$. We denote by $\{\mathcal{F}_t^B, t \in$

$[0, T]$ the natural filtration generated by B . To the stochastic process $B = \{B_t, t \in [0, T]\}$, we can associate the isonormal Gaussian process $\{B(h) : h \in \mathcal{H}\}$ with $B(h) = R^{-1}(h)$ where for $\xi \in H$, $R(\xi)$ is a function defined on $[0, T]$ by

$$R(\xi)(t) = \mathbb{E}(\xi B_t).$$

and $\mathcal{H} = K(L^2([0, T]))$. Since $W = \{W_t : t \in [0, T]\}$ is a Brownian motion, we can also consider the process $\{W(f) : f \in L^2([0, T])\}$, where $W(f) = B(Kf)$. We denote by $\nabla^B, \delta^B, \nabla^W$, and δ^W the Gross–Sobolev derivative and the divergence associated with the Gaussian processes $\{B(h) : h \in \mathcal{H}\}$ and $\{W(f) : f \in L^2([0, T])\}$, respectively.

Definition 10.4 (Hult[1]) For a process $\{u(t, \omega) : t \in [0, T], \omega \in \Omega\} \in L^2([0, T] \times \Omega)$ such that $Ku \in \text{Dom}_2(\delta^B)$, the stochastic integral with respect to B is defined as

$$\int_0^t u_s \delta B_s = \delta^B(Ku).$$

The following proposition is a consequence of the above definition of the stochastic integral with respect to B .

Proposition 10.1 (Hult[1]) Assume that $\{u(t, \omega) : t \in [0, T], \omega \in \Omega\}$ is a process in $L^2([0, T] \times \Omega)$ such that Ku is in $\text{Dom}_2(\delta^X)$ and u is adapted to $\{\mathcal{F}_t^B : t \in [0, T]\}$. Then $M = \{M_t : t \in [0, T]\}$ defined by $M_t = \delta^B(K(u1_{[0,t]}))$ is a square integrable continuous martingale with respect to $\{\mathcal{F}_t^B : t \in [0, T]\}$ with associated Doob–Meyer process $\langle M \rangle_t = \int_0^t u^2(s) ds$.

10.2.3 Preliminary Lemmas

Lemma 10.1 It is assumed that hypothesis $(\mathcal{H}_5)$ is verified. We have

- (a) $\sup_{0 \leq t \leq T} |X_t|^\alpha \leq (C|\theta|T + \sup_{t \in [0, T]} |B_t|) \exp(C|\theta|T).$
- (b) $|X_t - X_s| \leq C(\theta, T)(1 + \sup_{u \in [0, T]} |X_u|^\alpha)|t - s| + |B_t - B_s| \quad 0 < s < t,$

where $C(\theta, T)$ is a constant depending on θ and T .

Proof

- (a) Using hypothesis $(\mathcal{H}_5)$, we obtain

$$|X_t| \leq C|\theta|t + \sup_{t \in [0, T]} |B_t| + \int_0^t C|\theta||X_s|^\alpha ds.$$

As $|X_t|^\alpha \leq 1 + |X_t|^\alpha 1_{|X_t| \geq 1}$, we obtain

$$\begin{aligned} |X_t| &\leq 2C|\theta| + \sup_{t \in [0, T]} |B_t| + \int_0^t C|\theta| |X_s|^\alpha 1_{|X_s| \geq 1} ds \\ &\leq C_1 \left(1 + \sup_{t \in [0, T]} |B_t|\right) + \int_0^t C|\theta| |X_t|^\alpha 1_{|X_t| \geq 1} ds. \end{aligned}$$

As $|X_t| 1_{|X_t| < 1} + |X_t|^\alpha 1_{|X_t| \geq 1} \leq |X_t|$, we have

$$|X_t|^\alpha 1_{|X_t| \geq 1} \leq C_1 \left(1 + \sup_{t \in [0, T]} |B_t|\right) + \int_0^t C|\theta| |X_t|^\alpha 1_{|X_t| \geq 1} ds.$$

According to Gronwall's lemma, we deduce

$$|X_t|^\alpha 1_{|X_t| \geq 1} \leq C_1 \left(1 + \sup_{t \in [0, T]} |B_t|\right) e^{C|\theta|T}.$$

Therefore, we deduce that

$$\begin{aligned} |X_t|^\alpha &\leq |X_t|^\alpha 1_{|X_t| < 1} + |X_t|^\alpha 1_{|X_t| \geq 1} \\ &\leq 2|X_t|^\alpha 1_{|X_t| \geq 1} \\ &\leq C_2 \left(1 + \sup_{t \in [0, T]} |B_t|\right) e^{C|\theta|T}. \end{aligned}$$

(b) For $0 < s < t$, we have

$$X_t - X_s = \theta \int_s^t f(u, X_u) du + B_t - B_s.$$

Using hypothesis $(\mathcal{H}_5)$, we obtain

$$\begin{aligned} |X_t - X_s| &\leq |\theta| \int_s^t |f(u, X_u)| du + |B_t - B_s| \\ &\leq C|\theta| \int_s^t (1 + |X_u|^\alpha) du + |B_t - B_s| \\ &\leq C|\theta|(t - s) \sup_{0 \leq t \leq T} (1 + |X_t|^\alpha) + |B_t - B_s|. \end{aligned}$$

Consider the canonical process defined in Sect. 10.2.

Lemma 10.2 (Fernique [6]) *For all $\lambda < \frac{1}{2}$, we have $\mathbb{E} \left(\exp \left(\lambda \left(\sup_{t \in [0, T]} |B_t| \right)^2 \right) \right) < \infty$.*

Lemma 10.3 (Ibragimov and Khasminskii [7]) *Let $\zeta(t)$ be a real valued random function defined on a closed subset F of the Euclidean space $\mathbb{R}^k$. We shall assume that the random process $\zeta(t)$ is measurable and separable. Assume that the following condition is fulfilled: There exist numbers $m \geq r > k$ and a function $H : \mathbb{R}^k \rightarrow \mathbb{R}$ bounded on compact set such that for all $x, p \in F, x + p \in F$,*

$$\mathbb{E}|\zeta(x)|^m \leq H(x), \quad \mathbb{E}|\zeta(x+p) - \zeta(x)|^m \leq H(x)\|p\|^r.$$

Then with probability one the realizations of ζ are continuous functions of F .

Let $w(\delta, \zeta, L) = \sup_{\substack{x, y \in F \\ \|x\|, \|y\| \leq L \\ \|x-y\| \leq \delta}} |\zeta(x) - \zeta(y)|$ be the random variable.

Then

$$\mathbb{E}(w(\delta, \zeta, L)) \leq B_0 \left(\sup_{\|x\| < L} H(x) \right)^{\frac{1}{m}} L^k h^{\frac{r-k}{m}} \log(h^{-1}),$$

where $B_0 = 64^k(1 - 2^{-(r-k)m})^{-1} + (2^{-(m-r)m} - 1)^{-1}$ et $\|p\| = h$.

10.2.4 Maximum Likelihood Estimator

Now we give Girsanov formula for the Volterra type process B .

Proposition 10.2 (Hult[1]) *Let $B = \{B_t, t \in [0, T]\}$ be a Volterra type process on $(\Omega, \mathcal{A}, \mathbb{P})$ with kernel K , and let $u = \{u(t, \omega) : t \in [0, T], \omega \in \Omega\}$ be a process in $L^2([0, T] \times \Omega)$ with $Ku \in \text{Dom}_2(\delta^B)$ and $u(t, \cdot) \mathcal{F}_t^B$ -measurable. Define the probability measures $\mathbb{P}_\theta, \theta \in \mathbb{R}$ by*

$$\frac{d\mathbb{P}_\theta}{d\mathbb{P}} \Big|_{\mathcal{F}_t^B} = \exp \left(\theta \delta^B \left(K(u1_{[0,t]}) \right) - \frac{\theta^2}{2} \int_0^t u_s^2 ds \right).$$

Then the process $(X_t, t \in [0, T])$ defined by

$$X_t = B_t - \theta \int_0^t K(t, s) u_s ds$$

is a Volterra type process with kernel K under $\mathbb{P}_\theta$.

The proposition below ensures that $\mathbb{P}_\theta$ is absolutely continuous with respect to $\mathbb{P}$ for all θ .

Proposition 10.3 *Assume that the process $(f(t, X_t))_{t \in [0, T]}$ is adapted with integrable paths, and consider the process Q defined by (10.2). Then the process $M = \{M_t, t \in [0, T]\}$ defined by $M_t = \delta^B(K(Q1_{[0,t]}))$ is a square integrable continuous martingale w.r.t. $\{\mathcal{F}_t^B, t \in [0, T]\}$.*

Proof To prove the proposition, we show that:

- (a) $Q \in L^2([0, T] \times \Omega)$.
 - (b) $Q(t, \cdot)$ is $\mathcal{F}_t^B$ -measurable.
 - (c) $K(Q1_{[0, T]}) \in \text{Dom}_2(\delta)$.
- (a) Let us put $h_t = \int_0^t f(s, X_s)ds$. Using the Cauchy–Schwarz inequality, the condition $(\mathcal{H}_5)$, and Lemma 10.1, we obtain

$$\int_0^T h_t^2 dt \leq 2C^2 T^3 (1 + 2(C^2 \theta^2 T^2 + (\sup_{t \in [0, T]} |B_t|)^2 \exp(2|\theta|T))).$$

Observing that $x \leq \exp(\frac{1}{e}x)$ for any positive real x , we have

$$(\sup_{t \in [0, T]} |B_t|)^2 \leq \exp(\frac{1}{e}(\sup_{t \in [0, T]} |B_t|)^2)$$

Using Lemma 10.2, we obtain $\mathbb{E}(\int_0^T h_t^2 dt) < \infty$, and we deduce

$$\int_0^T h_t^2 dt < \infty \quad a.s.$$

The application K is a continuous linear bijective of $L^2([0, T])$ and $\mathcal{H}$, which are Banach spaces. It is therefore a homeomorphism. Consequently, K^{-1} is a continuous linear application. We deduce that

$$\|Q\|_{L^2([0, T] \times \Omega)} = \|K^{-1}h\|_{L^2([0, T] \times \Omega)} \leq \|K^{-1}\| \|h\|_{L^2([0, T])}. \quad (10.3)$$

According to (10.3), we deduce the process $Q \in L^2([0, T] \times \Omega)$.

- (b) Q_t is measurable as a composite of measurable functions.
- (c) We have

$$\begin{aligned} \mathbb{E}(\langle \nabla F, K(Q1_{[0, T]}) \rangle_{L^2([0, T])}) &= \mathbb{E}(\langle \nabla F, h \rangle_{L^2([0, T])}) \\ &= \mathbb{E}(F\delta(h)) \\ &\leq (\mathbb{E}(F^2))^{\frac{1}{2}} (\mathbb{E}(\delta(h)^2))^{\frac{1}{2}} \\ &\leq C_{2,2}(h) \|F\|_2. \end{aligned}$$

We conclude that the process $K(Q1_{[0, t]}) \in \text{Dom}_2(\delta)$. □

Suppose now that $\mathbb{P}_\theta^T$ is absolutely continuous with respect to $\mathbb{P}$, and let $L_T(\theta)$ be the Radon–Nikodym derivative of $\mathbb{P}_\theta^T$ with respect to $\mathbb{P}$. According to Proposition 10.2, we have

$$L_T(\theta) = \exp \left(\theta \delta^B \left(K(Q1_{[0,T]}) \right) - \frac{\theta^2}{2} \int_0^T Q_t^2 dt \right). \quad (10.4)$$

The MLE $\hat{\theta}_T$ of θ on the basis of the observations of X_0^T is given by

$$\hat{\theta}_T = \arg \max_{\theta \in \Theta} L_T(\theta).$$

Therefore, we obtain

$$\hat{\theta}_T = \frac{\delta^B \left(K(Q1_{[0,T]}) \right)}{\int_0^T Q_t^2 dt}.$$

Recall that the process W defined by $W_t = \delta^B \left(K(1_{[0,T]}) \right)$ is a Brownian motion under $\mathbb{P}$ and in particular

$$\delta^B \left(K(Q1_{[0,T]}) \right) = \delta^W \left(Q1_{[0,T]} \right) = \int_0^T Q_t dW_t.$$

Therefore, we have

$$\hat{\theta}_T = \frac{\int_0^T Q_s dW_s}{\int_0^T Q_t^2 dt}.$$

Proposition 10.4 (Hult[1]) *Suppose that B , Q , M , and $\hat{\theta}_T$ are as above and $\langle M \rangle_t \rightarrow +\infty$ $\mathbb{P}$ -p.s. as $T \rightarrow +\infty$. Then, we have the following statement:*

1. $\hat{\theta}_T \rightarrow \theta$, $\mathbb{P}_\theta$ -p.s. as $T \rightarrow +\infty$.
2. $\langle M \rangle_t^{1/2} \left(\hat{\theta}_T - \theta \right) \rightarrow \mathcal{N}(0, 1)$ as $T \rightarrow +\infty$.

For any $\theta \in \Theta$, let us consider the likelihood ratio process

$$\Psi_T(a) = \frac{d\mathbb{P}^T}{d\mathbb{P}_\theta^T}.$$

Lemma 10.4 *Under the assumptions $(\mathcal{H}_4)$, $(\mathcal{H}_5)$, $(\mathcal{H}_6)$, $(\mathcal{H}_7)$, for any a and b , we have*

$$(a) \sup_{\theta \in \Theta} \mathbb{E}_\theta^T \left[\Psi_T^{\frac{1}{2}}(a) - \Psi_T^{\frac{1}{2}}(b) \right]^2 \leq \frac{1}{2}(a - b)^2.$$

$$(b) \sup_{\theta \in \Theta} \mathbb{E}_{\theta}^T \left[\Psi_T^{\frac{1}{2}}(a) \right] \leq \exp(-C|a|^{\gamma}).$$

Proof

(a) Let us put $\theta_1 = \theta + an_T^{-\frac{1}{2}}$, $\theta_2 = \theta + bn_T^{-\frac{1}{2}}$, and $R_T = \left(\frac{d\mathbb{P}_{\theta_2}^T}{d\mathbb{P}_{\theta_1}^T} \right)^{\frac{1}{2}}$. Therefore, using Jensen's inequality, we have

$$\begin{aligned} & \mathbb{E}_{\theta}^T [\Psi_T^{\frac{1}{2}}(a) - \Psi_T^{\frac{1}{2}}(b)]^2 \\ &= 2(1 - \mathbb{E}_{\theta_1}^T(R_T)) \\ &= 2(1 - \mathbb{E}_{\theta_1}^T[\exp[-\frac{1}{2}(\frac{b-a}{\sqrt{n_T}}) \int_0^T Q_t \delta B_t - \frac{1}{4} \frac{(b-a)^2}{n_T} \int_0^T Q_t^2 dt]]) \\ &\leq 2(1 - [\exp[\mathbb{E}_{\theta_1}^T(-\frac{1}{2}(\frac{b-a}{\sqrt{n_T}}) \int_0^T Q_t \delta B_t - \frac{1}{4} \frac{(b-a)^2}{n_T} \int_0^T Q_t^2 dt)])]. \end{aligned}$$

Using the fact that $\mathbb{E}_{\theta}^T(\int_0^T Q_t \delta B_t) = 0$ by Proposition 10.3, we have

$$\mathbb{E}_{\theta_1}^T [\Psi_T^{\frac{1}{2}}(a) - \Psi_T^{\frac{1}{2}}(b)]^2 \leq 2(1 - \exp[-\frac{1}{4} \frac{(b-a)^2}{n_T} \mathbb{E}_{\theta_1}^T \int_0^T Q_t^2 dt]).$$

Thus, we deduce

$$\begin{aligned} \mathbb{E}_{\theta_1}^T [\Psi_T^{\frac{1}{2}}(a) - \Psi_T^{\frac{1}{2}}(b)]^2 &\leq \frac{1}{2} \frac{(b-a)^2}{n_T} \mathbb{E}_{\theta_1}^T \int_0^T Q_t^2 dt \\ &\leq \frac{1}{2} (b-a)^2 \mathbb{E}_{\theta_1}^T \left(\frac{I_T}{n_T} \right) \leq \frac{1}{2} (a-b)^2. \end{aligned}$$

(b) Using Hölder's inequality, we obtain

$$\begin{aligned} \mathbb{E}_{\theta_1}^T (\Psi_T^{1/2}(a)) &= \mathbb{E}_{\theta_1}^T [\exp \frac{1}{2} \frac{-a}{\sqrt{n_T}} \int_0^T Q_t \delta B_t - \frac{1}{6} \frac{a^2}{n_T} \\ &\quad \times \int_0^T Q_t^2 dt] \exp[-\frac{1}{12} \frac{a^2}{n_T} \int_0^T Q_t^2 dt] \\ &\leq (\mathbb{E}_{\theta}^T [\exp \frac{1}{2} \frac{-a}{\sqrt{n_T}} \int_0^T Q_t \delta B_t - \frac{1}{6} \frac{a^2}{n_T} \end{aligned}$$

$$\begin{aligned}
& \times \int_0^T Q_t^2 dt]^{4/3})^{3/4} (\mathbb{E}_\theta^T [\exp - \frac{1}{12} \frac{a^2}{n_T} \int_0^T Q_t^2 dt]^{4/3})^{1/4} \\
& \leq (\mathbb{E}_\theta^T [\exp(-\frac{2}{3} \frac{a}{\sqrt{n_T}} \int_0^T Q_t \delta B_t - \frac{2}{9} \frac{a^2}{n_T} \\
& \quad \times \int_0^T Q_t^2 dt)]^{4/3})^{3/4} (\mathbb{E}_\theta^T [\exp[-\frac{1}{3} \frac{a^2}{n_T} \int_0^T Q_t^2 dt]^{4/3})^{1/4} \\
& \leq (\mathbb{E}_\theta^T [\exp[-\frac{1}{3} \frac{a^2}{n_T} \int_0^T Q_t^2 dt]^{4/3})^{1/4}.
\end{aligned}$$

From the assumption $(\mathcal{H}_7)$, it follows that

$$\sup_{\theta \in \Theta} E_Q^T \left[\Psi_T^{1/2}(a) \right] \leq \exp(-C|a|^\gamma).$$

10.3 Main Results

10.3.1 Large Deviations

Theorem 10.1 *Under the assumptions $(\mathcal{H}_1)$ – $(\mathcal{H}_7)$, for any $s > 0$, we have*

$$\sup_{\theta \in \Theta} \mathbb{P}_\theta^T \left(n_T^{1/2} |\widehat{\theta}_T - \theta| \geq s \right) \leq B \exp(-\beta(s\sqrt{n_T})^\gamma)$$

for some positive constants B and β independent of T .

Proof Let us put

$$S_T = \{a : \theta + an_T^{-1/2} \in \Theta\}.$$

According to Bishwal [8], we can write that

$$\mathbb{P}_\theta^T (n_T^{1/2} |\widehat{\theta}_T - \theta| \geq s) \leq \sum_{r=0}^{\infty} \mathbb{P}_\theta^T \left\{ \sup_{a \in \Gamma_r} \Psi_T(a) \geq 1 \right\},$$

where $\Gamma_r = \{a : s+r \leq |a| \leq s+r+1\}$. From Lemma 10.3 with $\zeta(a) = \Psi_T^{1/2}(a)$, there exists a constant $D > 0$ such as

$$\sup_{\theta \in \Theta} E_\theta^T \left\{ \sup_{\substack{|a-b| \leq h \\ |a|, |b| \leq l}} \left| \Psi_T^{1/2}(a) - \Psi_T^{1/2}(b) \right| \right\} \leq Dlh^{1/2} \log h^{-1}. \quad (10.5)$$

Divide Γ_r into subintervals of length at most h . The number n of subintervals is clearly less than or equal to $[\frac{1}{h}] + 1$. Let $\Gamma_r^{(j)}$, $1 \leq j \leq n$, be the subintervals chosen. Let $b_j \in \Gamma_r^{(j)}$. We have

$$\mathbb{P}_\theta^T[\sup_{a \in \Gamma_r} \Psi_T(a) \geq 1] \leq \sum_{i=1}^n \mathbb{P}_\theta^T[\Psi_T^{\frac{1}{2}}(b_i) \geq \frac{1}{2}] + \mathbb{P}_\theta^T\left\{\sup_{\substack{|a-b| \leq h \\ |a|, |b| \leq s+r+1}} |\Psi_T^{\frac{1}{2}}(a) - \Psi_T^{\frac{1}{2}}(b)| \geq \frac{1}{2}\right\}.$$

Using Markov inequality and Lemma 10.3, we have

$$\begin{aligned} \mathbb{P}_\theta^T[\sup_{a \in \Gamma_r} \Psi_T(a) \geq 1] &\leq 2 \sum_{i=1}^n E_\theta^T \left[\Psi_T^{\frac{1}{2}}(b_i) \right] + 2D(s+r+1)h^{\frac{1}{2}} \log(h^{-1}), \\ &\leq 2 \sum_{i=1}^n \exp(-C|b_i|^\gamma) + 2D(s+r+1)h^{\frac{1}{2}} \log(h^{-1}) \\ &\leq 2 \left(\left[\frac{1}{h} \right] + 1 \right) \exp(-C(s+r)^\gamma) + 2D(s+r+1)h^{\frac{1}{2}} \log(h^{-1}). \end{aligned}$$

Choosing $h = \exp\left(\frac{-C(s+r)^\gamma}{2}\right)$ and the relation that is the generalized binomial formula $(s+r)^\gamma \geq s^\gamma + \gamma r s^{\gamma-1}$, we obtain

$$\begin{aligned} \sum_{r=0}^{\infty} \mathbb{P}_\theta^T[\sup_{a \in \Gamma_r} \Psi_T(a) \geq 1] &\leq 4 \exp\left[-\frac{1}{2}Cs^\gamma\right] \sum_{r=0}^{\infty} \exp\left(\frac{-C}{2}\gamma r s^{\gamma-1}\right) \\ &\quad + DC \exp\left[-\frac{1}{4}Cs^\gamma\right] \sum_{r=0}^{\infty} (s+r+1)^{\gamma+1} \exp\left(\frac{-C}{4}\gamma r s^{\gamma-1}\right). \end{aligned}$$

Then, we conclude that

$$\sup_{\theta \in \Theta} \mathbb{P}_\theta^T\left(n_T^{\frac{1}{2}} |\widehat{\theta}_T - \theta| \geq s\right) \leq B \exp\left(\frac{-C}{2}(s\sqrt{n_T})^\gamma\right).$$

10.3.2 Berry–Esseen Inequality

The following two lemmas are recalled for the proof of Berry–Esseen inequalities.

Lemma 10.5 (Michel and Pfanzagl[9]) *Let Y and Z be two random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{Q})$ with $\mathbb{Q}(Z > 0) = 1$. Then, for all $\varepsilon > 0$, we have*

$$\sup_{x \in \mathbb{R}} \left| \mathbb{Q} \left\{ \frac{Y}{Z} \leq x \right\} - \Phi(x) \right| \leq \sup_{x \in \mathbb{R}} |\mathbb{Q}\{Y \leq x\} - \Phi(x)| + \mathbb{Q}\{|Z - 1| > \varepsilon\} + \varepsilon,$$

where $\Phi(\cdot)$ is the distribution function of the standard normal distribution.

Lemma 10.6 (Hall and Heyde[10]) Let $(\widetilde{W}_t, t \geq 0)$ be a standard Brownian process and Z be a nonnegative random variable. Then, for every $x \in \mathbb{R}$ and $\varepsilon > 0$, we have

$$|\mathbb{Q}(\widetilde{W}_Z \leq x) - \Phi(x)| \leq (2\varepsilon)^{\frac{1}{2}} + \mathbb{Q}(|Z - 1| > \varepsilon),$$

where $\Phi(\cdot)$ is the distribution function of the standard normal distribution.

Thus, we deduce our result.

Theorem 10.2 Under the assumptions $(\mathcal{H}_1) - (\mathcal{H}_7)$,

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}_\theta^T \left(\sqrt{n_T}(\widehat{\theta}_T - \theta) \leq x \right) - \Phi(x) \right| \leq 2\mathbb{P}_\theta^T \left(\left| \frac{I_T}{n_T} - 1 \right| > \varepsilon_T \right) + 3\sqrt{\varepsilon_T}$$

for any $\varepsilon_T \downarrow 0$ as $T \rightarrow \infty$ and where $\Phi(\cdot)$ is the distribution function of the standard normal distribution.

Proof Let $(M_t)_{0 \leq t \leq T}$ be the process such that

$$M_t = \int_0^t Q_s \delta B_s,$$

where Q_t is defined by (10.2). Then, we can write

$$\sqrt{n_T}(\widehat{\theta}_T - \theta) = -\frac{n_T^{-\frac{1}{2}} M_T}{n_T^{-1} I_T}.$$

Therefore, we deduce that

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}_\theta^T \left(\sqrt{n_T}(\widehat{\theta}_T - \theta) \leq x \right) - \Phi(x) \right| = \sup_{x \in \mathbb{R}} \left| \mathbb{P}_\theta^T \left(-\frac{n_T^{-\frac{1}{2}} M_T}{n_T^{-1} I_T} \leq x \right) - \Phi(x) \right|.$$

Using Lemma 10.5, we have

$$\begin{aligned} \sup_{x \in \mathbb{R}} \left| \mathbb{P}_\theta^T \left(\sqrt{n_T}(\widehat{\theta}_T - \theta) \leq x \right) - \Phi(x) \right| &\leq \sup_{x \in \mathbb{R}} \left| \mathbb{P}_\theta^T (-n_T^{-\frac{1}{2}} M_T \leq x) - \Phi(x) \right| \\ &\quad + \mathbb{P}_\theta^T \left(\left| \langle -n_T^{-\frac{1}{2}} M \rangle_T - 1 \right| > \varepsilon_T \right) + \varepsilon_T. \end{aligned}$$

As $(M_t)_{t \in [0; T]}$ is a square integrable continuous martingale, from the results on the representation of locally square integrable continuous martingales (Ikeda and Watanabe [11]), it follows that there exists a standard Wiener $(\tilde{W}_t)_{t \in [0; T]}$ adapted to $(\mathcal{F}_t)_{t \in [0; T]}$ such that process M_t is equal to $\tilde{W}_{\langle M \rangle_t}$. Then, using Lemma 10.6 with $Z = \langle n_T^{-\frac{1}{2}} M \rangle_T$, we have

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}_\theta \left(-n_T^{-\frac{1}{2}} M_T \leq x \right) - \Phi(x) \right| \leq \sup_{x \in \mathbb{R}} \left| \mathbb{P}_\theta \left(\tilde{W}_Z \leq x \right) - \Phi(x) \right|.$$

According to Lemma 10.6, we have

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}_\theta \left(\tilde{W}_Z \leq x \right) - \Phi(x) \right| \leq (2\varepsilon_T)^{\frac{1}{2}} + \mathbb{P}_\theta \left(\left| \frac{I_T}{n_T} - 1 \right| > \varepsilon_T \right).$$

Consequently, we obtain

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P}_\theta^T \left(\sqrt{n_T} (\hat{\theta}_T - \theta) \leq x \right) - \Phi(x) \right| \leq 2\mathbb{P}_\theta^T \left(\left| \frac{I_T}{n_T} - 1 \right| > \varepsilon_T \right) + 3\sqrt{\varepsilon_T}.$$

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Chapter 11

Entropy Stable Godunov Numerical Schemes for the Euler Equations



Ludovic Martaud and Christophe Berthon

Abstract This chapter concerns the design of entropy stable numerical schemes for the Euler equations. The fully discrete entropy inequality is reached involving a local condition incorporated in the scheme design. The method yields explicit schemes. Numerical experiments are carried out to assess the performances of the proposed schemes.

11.1 Introduction

The present work is devoted to the numerical approximation of the weak solutions of the Euler equations in one space dimension given by

$$\begin{aligned}\partial_t \rho + \partial_x(\rho u) &= 0, \\ \partial_t \rho u + \partial_x(\rho u^2 + p) &= 0, \\ \partial_t \rho E + \partial_x((\rho E + p)u) &= 0.\end{aligned}$$

This nonlinear model governs the density $\rho > 0$, the velocity $u \in \mathbb{R}$, and the total energy $\rho E > 0$ of a compressible fluid. The unknown state vector $w = (\rho, \rho u, \rho E)^T$ is assumed to be in the set $\Omega = \{(\rho, \rho u, \rho E) \in \mathbb{R}^3 \mid \rho > 0, \rho u \in \mathbb{R}, E - u^2/2 > 0\}$. The function $p : \Omega \rightarrow \mathbb{R}_*^+$ denotes the fluid pressure. In this chapter, we consider the perfect gas model in which

$$p(w) = (\gamma - 1) \left(\rho E - \frac{\rho u^2}{2} \right), \quad \forall w \in \Omega, \quad (11.1)$$

where $\gamma \in (1, 3]$ stands for the adiabatic constant.

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We consider $w_0 : \mathbb{R} \rightarrow \Omega$ a given measurable function of $L^1_{\text{loc}}(\mathbb{R})$ as initial condition of w at time $t = 0$, and we study the following Cauchy problem:

$$\begin{cases} \partial_t w + \partial_x f(w) = 0, & x \in \mathbb{R}, t > 0, \\ w(x, t = 0) = w_0(x), & x \in \mathbb{R}, \end{cases} \quad (11.2)$$

where $f(w) = (\rho u, \rho u^2 + p, (\rho E + p)u)^T$.

The Euler system (11.2) is a nonlinear hyperbolic system of conservation laws [9], and it is well known that its solutions may develop discontinuities in a finite time. In this case, the uniqueness of the solution is lost [13–15, 18], and the selection of a solution is done according to entropy inequalities [13, 14, 18] that write

$$\partial_t \eta(w) + \partial_x G(w) \leq 0, \quad (11.3)$$

where the entropy $\eta : \Omega \rightarrow \mathbb{R}$ denotes a convex function and the entropy flux $G : \Omega \rightarrow \mathbb{R}$ is such that $\nabla G(w)^T = \nabla \eta(w)^T \nabla f(w)$ for all $w \in \Omega$. According to [2, 6, 13] and for a function $\mathcal{K} : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\mathcal{K}'(y) > 0, \quad \mathcal{K}'(y) - \gamma \mathcal{K}''(y) > 0, \quad \forall y \in \mathbb{R},$$

the couple convex entropy–entropy flux (η, G) for the Euler system (11.2) is given by

$$\eta(w) = -\rho \mathcal{K}(\ln(p/\rho^\gamma)), \quad G(w) = -(\rho u) \mathcal{K}(\ln(p/\rho^\gamma)), \quad \forall w \in \Omega.$$

In the sequel, a single entropy inequality (11.3) is considered, and it is defined by $\mathcal{K}(y) = y$ in the above formulations. As a consequence, the considered couple (η, G) is given by

$$\eta(w) = -\rho \ln(p/\rho^\gamma), \quad G(w) = -(\rho u) \ln(p/\rho^\gamma), \quad \forall w \in \Omega. \quad (11.4)$$

From a numerical point of view, we consider uniform meshes in space $(x_{i+\frac{1}{2}})_{i \in \mathbb{Z}} \subset \mathbb{R}$ of constant size $\Delta x > 0$. Thus, we have $x_{i+\frac{1}{2}} = x_{i-\frac{1}{2}} + \Delta x$ for all $i \in \mathbb{Z}$. We also consider uniform meshes in time $(t^n)_{n \in \mathbb{N}} \subset [0, +\infty)$ of constant size $\Delta t > 0$, and we have $t^{n+1} = t^n + \Delta t$ for all n in $\mathbb{N}$. At time t^n , the quantity $w(\cdot, t^n)$ is approximated by a piecewise constant function defined by a sequence $(w_i^n)_{i \in \mathbb{Z}}$ such that

$$w_i^n \approx \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} w(x, t^n) dx, \quad \forall i \in \mathbb{Z}.$$

Since the sequence $(w_i^0)_{i \in \mathbb{Z}}$ is given by $w_i^0 = \frac{1}{\Delta x} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} w_0(x) dx$, a numerical approximation of $w(\cdot, t^{n+1})$ is entirely defined by a numerical scheme that gives

the updated sequence $(w_i^{n+1})_{i \in \mathbb{Z}}$ from the sequence $(w_i^n)_{i \in \mathbb{Z}}$. Such schemes write

$$w_i^{n+1} = w_i^n - \frac{\Delta t}{\Delta x} (\mathcal{F}_{i+\frac{1}{2}} - \mathcal{F}_{i-\frac{1}{2}}). \quad (11.5)$$

However, a suitable updated sequence $(w_i^{n+1})_{i \in \mathbb{Z}}$ has to satisfy a discrete version of the inequality (11.3). Denoting $\mathcal{G}_{i+1/2}$ a consistent approximation of G defined in (11.4), these discrete inequalities read

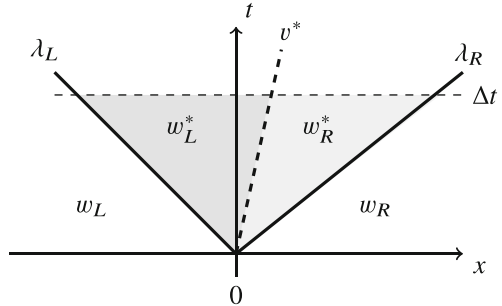
$$\frac{\eta(w_i^{n+1}) - \eta(w_i^n)}{\Delta t} + \frac{\mathcal{G}_{i+\frac{1}{2}} - \mathcal{G}_{i-\frac{1}{2}}}{\Delta x} \leq 0, \quad \forall i \in \mathbb{Z}, \quad (11.6)$$

and they are of main importance. On the one hand, they avoid the scheme convergence toward a nonentropic solution [4, 12, 16, 21], and on the other hand, they drastically reduce spurious oscillations [7].

Several schemes verifying the discrete entropy inequality (11.6) or a semidiscrete version have already been introduced [8, 10, 11, 20]. However, Riemann solvers defined by two intermediate states (see Fig. 11.1) and satisfying a fully discrete inequality are rare. They are composed of the HLLC scheme [22], relaxation schemes such as Suliciu relaxation approaches [1, 3, 5], and a scheme introduced in [10] (Section b.ii). However, the proof of the discrete entropy inequality (11.6) requires technical computations [3] or depends on implicit parameters [10, 22], which can make their implementation complex.

In this chapter, we propose to design numerical schemes from sufficient conditions that ensure the weak consistency and the discrete entropy inequality (11.6). The chapter is organized as follows. In Sect. 11.2, we define consistent, entropic approximate Riemann solvers for the Riemann problem associated with the Euler system (11.2). In Sect. 11.3, we propose to define an approximate Riemann solver with quadratic equations to enforce the sufficient conditions derived in Sect. 11.2. These quadratic equations are always well-posed, and they give explicit approximate Riemann solvers that are used in Sect. 11.4 to define consistent, entropic numerical schemes (11.5). Finally, in Sect. 11.5, numerical tests are carried out to illustrate the performance of the numerical schemes.

Fig. 11.1 Approximate Riemann solver made of two intermediate constant states w_L^* and w_R^* in the plane (x, t)



11.2 Two Constant States' Approximate Riemann Solver

In this section, we introduce two constant states approximate Riemann solver (ARS). Let us consider two constant states $(w_L, w_R) \in \Omega^2$ such that the initial condition w_0 reads

$$w_0(x) = \begin{cases} w_L & \text{if } x < 0, \\ w_R & \text{otherwise.} \end{cases} \quad (11.7)$$

In this case, the Cauchy problem (11.2), (11.7) recasts into Riemann problem. An approximate Riemann solver is a given function $\tilde{w} : \mathbb{R} \times \Omega^2 \rightarrow \Omega$ that defines an (L^2) approximation of the exact solution of the Riemann problem (11.2), (11.7). The choice of an approximate Riemann solver has to satisfy some criteria that are now defined.

Definition 11.1 (Approximate Riemann Solver [10]) Assume $\Delta t > 0$ such that the following CFL condition holds:

$$\frac{\Delta t}{\Delta x} \max_{w \in \{w_L, w_R\}} \mu_{\nabla f(w)}(w) \leq \frac{1}{2}, \quad (11.8)$$

where $\mu_{\nabla f(w)}(w)$ denotes the spectral radius of the Jacobian matrix $\nabla f(w)$. A given function $\tilde{w} : \mathbb{R} \times \Omega^2 \rightarrow \Omega$ is an ARS if the following statements hold:

- (i) $\xi \mapsto \tilde{w}(\xi, w_L, w_R) \in L^1_{\text{Loc}}(\mathbb{R})$ for all $(w_L, w_R) \in \Omega^2$.
- (ii) The function $\tilde{w}$ is consistent that writes $\tilde{w}(\xi, w, w) = w$ for all $(\xi, w) \in \mathbb{R} \times \Omega$.
- (iii) Under the CFL condition (11.8), the function $\tilde{w}$ satisfies a consistency integral relation that writes

$$\frac{1}{\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \tilde{w}(x/\Delta t, w_L, w_R) dx = \frac{w_L + w_R}{2} - \frac{\Delta t}{\Delta x} (f(w_R) - f(w_L)). \quad (11.9)$$

- (iv) Under the CFL condition (11.8), the function $\tilde{w}$ satisfies the following entropy condition:

$$\frac{1}{\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \eta(\tilde{w}(x/\Delta t, w_L, w_R)) dx \leq \frac{\eta(w_L) + \eta(w_R)}{2} - \frac{\Delta t}{\Delta x} (G(w_R) - G(w_L)). \quad (11.10)$$

The consistency given in (ii) means that an approximate Riemann solver preserves the constants of Ω that are obvious solutions of the Euler system (11.2).

Now, considering three given reals $\lambda_L, v^*, \lambda_R$ such that $\lambda_L < v^* < \lambda_R$, an approximate Riemann solver $\tilde{w}(\cdot, w_L, w_R) : \mathbb{R} \rightarrow \Omega$ made of two intermediate constant states is displayed in Fig. 11.1. Denoting $w_L^* := w_L^*(w_L, w_R)$ and $w_R^* :=$

$w_R^*(w_L, w_R)$ the two intermediate states, such an approximate Riemann solver writes

$$\tilde{w}(x/\Delta t, w_L, w_R) = \begin{cases} w_L & \text{if } \frac{x}{\Delta t} \leq \lambda_L, \\ w_L^* & \text{if } \lambda_L < \frac{x}{\Delta t} \leq v^*, \\ w_R^* & \text{if } v^* < \frac{x}{\Delta t} \leq \lambda_R, \\ w_R & \text{if } \lambda_R < \frac{x}{\Delta t}. \end{cases} \quad (11.11)$$

According to [10], the fan of the exact solution must be included within the (x, t) -plane cone defined by λ_L and λ_R (see Fig. 11.1). This restriction, once again called CFL condition, is given by

$$\begin{aligned} \lambda_L &< \min_{\alpha \in \{L, R\}} (u_\alpha \pm \sqrt{\gamma p(w_\alpha)/\rho_\alpha}, u_\alpha) \leq \max_{\alpha \in \{L, R\}} (u_\alpha \pm \sqrt{\gamma p(w_\alpha)/\rho_\alpha}, u_\alpha) < \lambda_R, \\ 0 &< \max_{\alpha \in \{L, R\}} |\lambda_\alpha| \frac{\Delta t}{\Delta x} \leq \frac{1}{2}. \end{aligned} \quad (11.12)$$

In addition, a two constant states ARS (11.11) obviously satisfies Definition 11.1-(i). The consistency condition (ii) will be immediate according to the definition which will be given to w_L^* and w_R^* .

As a consequence, we now have to determine the two intermediate states w_L^* and w_R^* according to the consistency integral relation (11.9) and to the entropy condition (11.10). The following lemma presents the satisfied constraints by entropic, consistent, intermediate states w_L^* and w_R^* .

Lemma 11.1 *Consider an approximate Riemann solver $\tilde{w}(\cdot, w_L, w_R)$ (11.11) that approximates the solution of the Riemann problem for the Euler equations given by (11.2), (11.7). Assume the CFL condition (11.12) holds, and let us set*

$$\delta_L = \frac{v^* - \lambda_L}{\lambda_R - \lambda_L} > 0, \quad \delta_R = \frac{\lambda_R - v^*}{\lambda_R - \lambda_L} > 0, \quad (11.13a)$$

$$\begin{aligned} w^{\text{HLL}} &= (\rho^{\text{HLL}}, (\rho u)^{\text{HLL}}, (\rho E)^{\text{HLL}})^T = \\ &\quad \frac{\lambda_R w_R - \lambda_L w_L}{\lambda_R - \lambda_L} - \frac{f(w_R) - f(w_L)}{\lambda_R - \lambda_L} \in \mathbb{R}^3, \end{aligned} \quad (11.13b)$$

$$\eta^{\text{HLL}} = \frac{\lambda_R \eta(w_R) - \lambda_L \eta(w_L)}{\lambda_R - \lambda_L} - \frac{G(w_R) - G(w_L)}{\lambda_R - \lambda_L} \in \mathbb{R}. \quad (11.13c)$$

There is an equivalence between the relations (11.9) and (11.10) and the relations

$$\delta_L w_L^* + \delta_R w_R^* = w^{\text{HLL}}, \quad (11.14a)$$

$$\delta_L \eta(w_L^*) + \delta_R \eta(w_R^*) \leq \eta^{\text{HLL}}. \quad (11.14b)$$

The relations (11.14) give four constraints on the intermediate states w_L^* and w_R^* . The three equations (11.14a) ensure the weak consistency, while the inequality (11.14b) enforces the entropy condition. However, the states w_L^* and w_R^* are made of six unknowns that write

$$w_\alpha^* = (\rho_\alpha^*, \rho_\alpha^* u_\alpha^*, \rho_\alpha^* E_\alpha^*)^\top, \quad \forall \alpha \in \{L, R\}. \quad (11.15)$$

As a consequence, the relations (11.14) underdetermine w_L^* and w_R^* , and several additional relations will be given in the next section. The proof of Lemma 11.1 is given below.

Proof of Lemma 11.1 For an approximate Riemann solver (11.11), if the CFL condition (11.12) holds, then the left-hand side of the equality (11.9) writes

$$\begin{aligned} & \frac{1}{\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \tilde{w}(x/\Delta t, w_L, w_R) dx \\ &= \frac{1}{\Delta x} \int_{-\frac{\Delta x}{2}}^{\lambda_L \Delta t} w_L dx + \frac{1}{\Delta x} \int_{\lambda_L \Delta t}^{v^* \Delta t} w_L^* dx + \frac{1}{\Delta x} \int_{v^* \Delta t}^{\lambda_R \Delta t} w_R^* dx + \frac{1}{\Delta x} \int_{\lambda_R \Delta t}^{\frac{\Delta x}{2}} w_R dx, \\ &= \left(\frac{\lambda_L \Delta t}{\Delta x} + \frac{1}{2} \right) w_L + \frac{\Delta t}{\Delta x} (v^* - \lambda_L) w_L^* + \frac{\Delta t}{\Delta x} (\lambda_R - v^*) w_R^* + \left(\frac{1}{2} - \frac{\lambda_R \Delta t}{\Delta x} \right) w_R. \end{aligned}$$

Therefore, the equality (11.9) is equivalent to

$$\begin{aligned} \left(\frac{\lambda_L \Delta t}{\Delta x} + \frac{1}{2} \right) w_L + \frac{\Delta t}{\Delta x} (v^* - \lambda_L) w_L^* + \frac{\Delta t}{\Delta x} (\lambda_R - v^*) w_R^* + \left(\frac{1}{2} - \frac{\lambda_R \Delta t}{\Delta x} \right) w_R = \\ \frac{w_L + w_R}{2} - \frac{\Delta t}{\Delta x} (f(w_R) - f(w_L)). \end{aligned}$$

Using the quantities δ_L , δ_R , and w^{HLL} defined in (11.13) in the above equality, we deduce the equivalent formulation (11.14a).

Concerning the inequality (11.14b), according to the approximate Riemann solver structure (11.11), and since the CFL condition (11.12) holds, the left-hand side of the inequality (11.10) writes

$$\frac{1}{\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \eta(\tilde{w}(x/\Delta t, w_L, w_R)) dx$$

$$\begin{aligned}
&= \frac{1}{\Delta x} \int_{-\frac{\Delta x}{2}}^{\lambda_L \Delta t} \eta(w_L) dx + \frac{1}{\Delta x} \int_{\lambda_L \Delta t}^{v^* \Delta t} \eta(w_L^*) dx \\
&\quad + \frac{1}{\Delta x} \int_{v^* \Delta t}^{\lambda_R \Delta t} \eta(w_R^*) dx + \frac{1}{\Delta x} \int_{\lambda_R \Delta t}^{\frac{\Delta x}{2}} \eta(w_R) dx, \\
&= \left(\frac{\lambda_L \Delta t}{\Delta x} + \frac{1}{2} \right) \eta(w_L) + \frac{\Delta t}{\Delta x} (v^* - \lambda_L) \eta(w_L^*) \\
&\quad + \frac{\Delta t}{\Delta x} (\lambda_R - v^*) \eta(w_R^*) + \left(\frac{1}{2} - \frac{\lambda_R \Delta t}{\Delta x} \right) \eta(w_R).
\end{aligned}$$

As a consequence, an equivalent form of the inequality (11.10) is given by

$$\begin{aligned}
&\left(\frac{\lambda_L \Delta t}{\Delta x} + \frac{1}{2} \right) \eta(w_L) + \frac{\Delta t}{\Delta x} (v^* - \lambda_L) \eta(w_L^*) + \frac{\Delta t}{\Delta x} (\lambda_R - v^*) \eta(w_R^*) \\
&\quad + \left(\frac{1}{2} - \frac{\lambda_R \Delta t}{\Delta x} \right) \eta(w_R) \leq \\
&\quad \frac{\eta(w_L) + \eta(w_R)}{2} - \frac{\Delta t}{\Delta x} (G(w_R) - G(w_L)).
\end{aligned}$$

Using the quantities δ_L , δ_R , and η^{HLL} given by (11.13) in the above inequality, we deduce the equivalent formulation (11.14b) that concludes the proof.

The state vector w^{HLL} defined in (11.13) refers to the standard HLL approximate Riemann solver detailed in [10]. According to the robustness properties of the HLL approximate Riemann solver, the following inequalities hold (see Appendix for the proof):

$$\rho^{\text{HLL}} > 0, \quad (\rho E)^{\text{HLL}} > 0, \quad \eta(w^{\text{HLL}}) \leq \eta^{\text{HLL}}. \quad (11.16)$$

As a consequence, we can define

$$u^{\text{HLL}} = \frac{(\rho u)^{\text{HLL}}}{\rho^{\text{HLL}}}, \quad E^{\text{HLL}} = \frac{(\rho E)^{\text{HLL}}}{\rho^{\text{HLL}}} > 0, \quad (11.17)$$

and w^{HLL} given by (11.13) also reads

$$w^{\text{HLL}} = (\rho^{\text{HLL}}, \rho^{\text{HLL}} u^{\text{HLL}}, \rho^{\text{HLL}} E^{\text{HLL}})^T.$$

The next section derives additional equations to entirely define the intermediate states w_L^* , w_R^* and to satisfy the relations (11.14).

11.3 Consistent Entropy Stable Approximate Riemann Solver

In this section, we consider the conditions (11.14) to design an ARS (11.11) for the Euler equations (11.2). As the conditions (11.14) are underdetermined, below the first lemma gives two closure equations that impose the velocity u and the pressure p continuity in the intermediate states of the solver. Such continuities have already been introduced for the HLLC scheme [22], for instance.

Lemma 11.2 *Consider an approximate Riemann solver $\tilde{w}(\cdot, w_L, w_R)$ (11.11) that approximates the solution of the Riemann problem for the Euler equations (11.2), (11.7) endowed with the pressure function $w \mapsto p(w)$ given by (11.1). Assume the CFL condition (11.12) holds, and let us also consider the quantities $\delta_L > 0$, $\delta_R > 0$, w^{HLL} , and u^{HLL} , respectively, defined in (11.13), (11.17). If the intermediate states denoted w_L^* and w_R^* are defined by the equations*

$$w_L^* \delta_L + w_R^* \delta_R = w^{\text{HLL}}, \quad (11.18a)$$

$$u_L^* = u_R^* \quad (11.18b)$$

$$p(w_L^*) = p(w_R^*), \quad (11.18c)$$

then,

$$u_L^* = u_R^* = u^{\text{HLL}}, \quad (11.19a)$$

$$p(w_L^*) = p(w_R^*) = p(w^{\text{HLL}}). \quad (11.19b)$$

Proof Writing component by component the consistency integral relation (11.18a) verified by the states w_L^* , w_R^* , we have

$$\delta_L \rho_L^* + \delta_R \rho_R^* = \rho^{\text{HLL}}, \quad (11.20a)$$

$$\delta_L \rho_L^* u_L^* + \delta_R \rho_R^* u_R^* = \rho^{\text{HLL}} u^{\text{HLL}}, \quad (11.20b)$$

$$\delta_L \rho_L^* E_L^* + \delta_R \rho_R^* E_R^* = \rho^{\text{HLL}} E^{\text{HLL}}, \quad (11.20c)$$

where u^{HLL} and E^{HLL} are given in (11.17). As $u_L^* = u_R^*$, let us denote $u^* = u_L^* = u_R^*$. Equation (11.20b) associated with (11.20a) gives

$$\begin{aligned} \rho^{\text{HLL}} u^{\text{HLL}} &= \delta_L \rho_L^* u_L^* + \delta_R \rho_R^* u_R^*, \\ &= (\delta_L \rho_L^* + \delta_R \rho_R^*) u^*, \\ &= \rho^{\text{HLL}} u^*. \end{aligned} \quad (11.21)$$

Since $(w_L, w_R) \in \Omega^2$, the robustness of the standard HLL solver [10] enforces the strict inequality $\rho^{\text{HLL}} > 0$ (see Lemma 11.5-(ii) for the proof). Therefore, multiplying equation (11.21) by $1/\rho^{\text{HLL}}$, we deduce the result related to u_L^*, u_R^* .

Using the equalities (11.19a), we also deduce that $p(w_L^*)$ and $p(w_R^*)$ write

$$p(w_\alpha^*) = (\gamma - 1) \left(\rho_\alpha^* E_\alpha^* - \frac{\rho_\alpha^* (u^{\text{HLL}})^2}{2} \right), \quad \forall \alpha \in \{L, R\}. \quad (11.22)$$

With the above result, we can compute the quantity $\delta_L p(w_L^*) + \delta_R p(w_R^*)$, and using Eqs. (11.20a)–(11.20c), we obtain

$$\begin{aligned} \delta_L p(w_L^*) + \delta_R p(w_R^*) &= (\gamma - 1) \left(\delta_L \rho_L^* E_L^* + \delta_R \rho_R^* E_R^* - \frac{\delta_L \rho_L^* + \delta_R \rho_R^*}{2} (u^{\text{HLL}})^2 \right), \\ &= (\gamma - 1) \left(\rho^{\text{HLL}} E^{\text{HLL}} - \frac{\rho^{\text{HLL}}}{2} (u^{\text{HLL}})^2 \right), \\ &= p(w^{\text{HLL}}). \end{aligned}$$

As $\delta_L + \delta_R = 1$ and since the equality $p(w_L^*) = p(w_R^*)$ holds, we deduce the result. $\square$

The closure equations of the above lemma give the formulation of the quantities u_L^* , u_R^* , $p(w_L^*)$, and $p(w_R^*)$. In addition, if the quantities ρ_L^* and ρ_R^* are known, then the computation of E_L^* , E_R^* is direct with Eq. (11.22). As a consequence, we only have to determine ρ_L^* , ρ_R^* according to the constrained equation

$$\rho_L^* \delta_L + \rho_R^* \delta_R = \rho^{\text{HLL}}, \quad (11.23a)$$

$$\delta_L \eta(w_L^*) + \delta_R \eta(w_R^*) \leq \eta^{\text{HLL}}. \quad (11.23b)$$

Since the entropy nonlinearity complexifies the solving of the constrained equation (11.23), the constrain (11.23b) will be substituted by a simpler quadratic equation. This new equation is defined from an upper bound of $\delta_L \eta(w_L^*) + \delta_R \eta(w_R^*)$ that is now detailed.

Lemma 11.3 (Quadratic Upper Bound of $\delta_L \eta(w_L^*) + \delta_R \eta(w_R^*)$) *Consider an approximate Riemann solver $\tilde{w}(\cdot, w_L, w_R)$ (11.11) defined by the system (11.18). Let us also consider the entropy $w \mapsto \eta(w)$ given by (11.4) and the quantities $\delta_L > 0$, $\delta_R > 0$, w^{HLL} defined in (11.13). If $\rho_L^* > 0$ and $\rho_R^* > 0$, then*

$$\eta(w_L^*) \delta_L + \eta(w_R^*) \delta_R \leq \frac{\gamma \delta_L}{\rho^{\text{HLL}}} (\rho_L^* - \rho^{\text{HLL}})^2 + \frac{\gamma \delta_R}{\rho^{\text{HLL}}} (\rho_R^* - \rho^{\text{HLL}})^2 + \eta(w^{\text{HLL}}). \quad (11.24)$$

Proof At first, if $\rho_L^* = \rho_R^*$, then Eq. (11.20a) imposes $\rho_L^* = \rho_R^* = \rho^{\text{HLL}}$. In this case, the inequality (11.24) is obviously verified in the sense of the equality. Let us

now consider the case $\rho_L^* \neq \rho_R^*$. Using the definition of the entropy $\eta(w)$ given by (11.4) and Eqs. (11.19b)–(11.20a), we have

$$\delta_L \eta(w_L^*) + \delta_R \eta(w_R^*) = -\delta_L \rho_L^* \ln \left(\frac{p(w^{\text{HLL}})}{(\rho_L^*)^\gamma} \right) - \delta_R \rho_R^* \ln \left(\frac{p(w^{\text{HLL}})}{(\rho_R^*)^\gamma} \right). \quad (11.25)$$

Considering equation (11.20a) in the quantity $\eta(w^{\text{HLL}})$, we also have

$$\eta(w^{\text{HLL}}) = -\delta_L \rho_L^* \ln \left(\frac{p(w^{\text{HLL}})}{(\rho^{\text{HLL}})^\gamma} \right) - \delta_R \rho_R^* \ln \left(\frac{p(w^{\text{HLL}})}{(\rho^{\text{HLL}})^\gamma} \right).$$

Therefore, subtracting the above equation to Eq. (11.25), we obtain

$$\frac{\delta_L \eta(w_L^*) + \delta_R \eta(w_R^*) - \eta(w^{\text{HLL}})}{\gamma \rho^{\text{HLL}}} = \delta_L \frac{\rho_L^*}{\rho^{\text{HLL}}} \ln \left(\frac{\rho_L^*}{\rho^{\text{HLL}}} \right) + \delta_R \frac{\rho_R^*}{\rho^{\text{HLL}}} \ln \left(\frac{\rho_R^*}{\rho^{\text{HLL}}} \right). \quad (11.26)$$

Up to a permutation, we can assume $\rho_L^* < \rho_R^*$. In this case, a direct proof by contradiction associated with Eq. (11.20a) gives $\rho_L^* < \rho^{\text{HLL}}$. Since the inequalities $\rho_L^* > 0$ and $\rho_R^* > 0$ are assumed satisfied, we deduce the existence of reals r_L^*, r_R^* such that

$$\frac{\rho_L^*}{\rho^{\text{HLL}}} = 1 - r_L^*, \quad \frac{\rho_R^*}{\rho^{\text{HLL}}} = 1 + r_R^*, \quad \text{with, } r_R^* \geq 0, r_L^* \in [0, 1). \quad (11.27)$$

Reformulating equation (11.26) with r_L^* and r_R^* , we have

$$\begin{aligned} \frac{\delta_L \eta(w_L^*) + \delta_R \eta(w_R^*) - \eta(w^{\text{HLL}})}{\gamma \rho^{\text{HLL}}} &= \delta_L (1 - r_L^*) \ln (1 - r_L^*) \\ &\quad + \delta_R (1 + r_R^*) \ln (1 + r_R^*). \end{aligned}$$

But, using a Taylor expansion with an integral form of the reminder, we obtain

$$\begin{aligned} \ln(1 - r_L^*) &= -r_L^* - \int_0^{r_L^*} \frac{(r_L^* - s)}{(1 - s)^2} ds, \\ (1 + r_R^*) \ln(1 + r_R^*) &= r_R^* + \frac{(r_R^*)^2}{2} - \frac{1}{2} \int_0^{r_R^*} \left(\frac{r_R^* - s}{1 + s} \right)^2 ds. \end{aligned}$$

Since $0 \leq r_L^* < 1$ and $0 \leq r_R^*$, both residues of the above Taylor expansions are signed, and consequently, we deduce

$$\delta_L (1 - r_L^*) \ln (1 - r_L^*) + \delta_R (1 + r_R^*) \ln (1 + r_R^*)$$

$$\begin{aligned}
&\leq -\delta_L r_L^* (1 - r_L^*) + \delta_R \left(r_R^* + \frac{(r_R^*)^2}{2} \right), \\
&\leq \delta_R r_R^* - \delta_L r_L^* + \delta_L (r_L^*)^2 + \delta_R (r_R^*)^2.
\end{aligned}$$

Rewriting equation (11.20a) with the quantities r_L^* , r_R^* defined in (11.27), we obtain $\delta_L r_L^* = \delta_R r_R^*$. Using this result in the above inequality, we have

$$\begin{aligned}
\frac{\delta_L \eta(w_L^*) + \delta_R \eta(w_R^*) - \eta(w^{\text{HLL}})}{\gamma \rho^{\text{HLL}}} &= \delta_L (1 - r_L^*) \ln(1 - r_L^*) \\
&\quad + \delta_R (1 + r_R^*) \ln(1 + r_R^*), \\
&\leq \delta_L (r_L^*)^2 + \delta_R (r_R^*)^2.
\end{aligned}$$

Using once again the definition (11.27) to rewrite the above inequality with ρ_L^* , ρ_R^* , we eventually deduce the inequality (11.24) that concludes the proof. $\square$

The upper bound of the inequality (11.24) is not the sharpest, but it is proposed in this work for the sake of simplicity. Imposing this upper bound equals η^{HLL} , we obtain the last equation that preserves the inequality $\delta_L \eta(w_L^*) + \delta_R \eta(w_R^*) \leq \eta^{\text{HLL}}$. Therefore, the approximate Riemann solver (11.11) proposed for the Euler system (11.2) is defined by the equations

$$w_L^* \delta_L + w_R^* \delta_R = w^{\text{HLL}}, \quad (11.28a)$$

$$u_L^* = u_R^*, \quad (11.28b)$$

$$p(w_L^*) = p(w_R^*), \quad (11.28c)$$

$$\frac{\gamma \delta_L}{\rho^{\text{HLL}}} (\rho_L^* - \rho^{\text{HLL}})^2 + \frac{\gamma \delta_R}{\rho^{\text{HLL}}} (\rho_R^* - \rho^{\text{HLL}})^2 + \eta(w^{\text{HLL}}) = \eta^{\text{HLL}}. \quad (11.28d)$$

Considering the above system and the results of Lemma 11.2, we now achieve the approximate Riemann solver design.

Theorem 11.1 (Entropic Preserving Two Intermediate States' Approximate Riemann Solver for the Euler Equations) *Consider an approximate Riemann solver $\tilde{w}(\cdot, w_L, w_R)$ (11.11) that approximates the solution of the Riemann problem for the Euler equations (11.2), (11.7) endowed with the pressure function $w \mapsto p(w)$ given by (11.1) and supplemented with the convex entropy–entropy flux (η, G) given by (11.4). Assume the CFL condition (11.12) holds, and let us also consider the quantities $\delta_L > 0$, $\delta_R > 0$, w^{HLL} , η^{HLL} , and u^{HLL} , respectively, defined in (11.13), (11.17). If the intermediate states denoted w_L^* and w_R^* write*

$$u_L^* = u_R^* = u^{\text{HLL}}, \quad (11.29a)$$

$$p(w_L^*) = p(w_R^*) = p(w^{\text{HLL}}), \quad (11.29b)$$

$$\rho_L^* = \rho^{\text{HLL}} \mp \sqrt{\frac{\rho^{\text{HLL}}}{\gamma} \frac{\delta_R}{\delta_L} (\eta^{\text{HLL}} - \eta(w^{\text{HLL}}))}, \quad (11.29c)$$

$$\rho_R^* = \rho^{\text{HLL}} \pm \sqrt{\frac{\rho^{\text{HLL}}}{\gamma} \frac{\delta_L}{\delta_R} (\eta^{\text{HLL}} - \eta(w^{\text{HLL}}))}, \quad (11.29d)$$

then the following statements are verified:

- (i) The approximate Riemann solver (11.11) is consistent with the Euler equations (11.2).
- (ii) If $\rho_L^* > 0$ and $\rho_R^* > 0$, then approximate Riemann solver (11.11) verifies the entropy condition (11.10).

In the definitions (11.29), both symbols $\pm$ and $\mp$ mean $+$ or $-$, but both are self-dependent. If $\pm$ is positive (*respectively*, negative), then $\mp$ is negative (*respectively*, positive). As a consequence, the above theorem yields two distinct approximate Riemann solvers, and the selection of one of them is done with an arbitrary choice for $\pm$. The proof of Theorem 11.1 is given below.

Proof of Theorem 11.1 At first, we show that the formulations (11.29) is the solutions of the system (11.28). The results (11.29a)–(11.29b) that, respectively, concern u_L^* , u_R^* , and $p(w_L^*)$, $p(w_R^*)$ have already proved in Lemma 11.2. Now, we have to prove the formulations of ρ_L^* , ρ_R^* from Eqs. (11.28a) and (11.28d) that read

$$\delta_L \rho_L^* + \delta_R \rho_R^* = \rho^{\text{HLL}}, \quad (11.30a)$$

$$\frac{\gamma \delta_L}{\rho^{\text{HLL}}} (\rho_L^* - \rho^{\text{HLL}})^2 + \frac{\gamma \delta_R}{\rho^{\text{HLL}}} (\rho_R^* - \rho^{\text{HLL}})^2 + \eta(w^{\text{HLL}}) = \eta^{\text{HLL}}. \quad (11.30b)$$

Writing equation (11.30a) in Eq. (11.30b), we obtain

$$\frac{\gamma \delta_R \delta_L}{\rho^{\text{HLL}}} (\rho_R^* - \rho_L^*)^2 = \eta^{\text{HLL}} - \eta(w^{\text{HLL}}). \quad (11.31)$$

As the CFL condition (11.12) holds, the inequality $\eta^{\text{HLL}} - \eta(w^{\text{HLL}}) \geq 0$ is ensured thanks to the robustness the standard HLL approximate Riemann solver [10] (see Lemma 11.5-(iii) for the proof). In addition, since $\delta_L > 0$ and $\delta_R > 0$, we can multiply equation (11.31) by $\rho^{\text{HLL}}/\gamma\delta_L\delta_R$, then using the square root on both sides of the result, we deduce

$$\rho_R^* - \rho_L^* = \pm \sqrt{\rho^{\text{HLL}} (\eta^{\text{HLL}} - \eta(w^{\text{HLL}})) / \gamma \delta_L \delta_R}.$$

Since $\delta_L + \delta_R = 1$, the above equation associated with Eq. (11.30a) leads to the formulations of ρ_L^* , ρ_R^* . This last remark achieves to show that (11.29) are the solutions of the system (11.28).

For the consistency (i), according to the intermediate states definition given by (11.29), the equality $\tilde{w}(\cdot, w, w) = w$ is satisfied. In addition, the approximate Riemann solver is designed from Eq. (11.14a) which is equivalent to the consistency of integral relation (11.9) (see Lemma 11.1). As a consequence, the approximate Riemann solver is consistent with the Euler system (11.2).

Concerning statement (ii), according to Lemma 11.3, Eq. (11.30b) ensures the inequality $\delta_L \eta(w_L^*) + \delta_R \eta(w_R^*) \leq \eta^{\text{HLL}}$. Hence, using Lemma 11.1, the approximate Riemann solver satisfies the entropy condition (11.10) that achieves the proof.

Both approximate Riemann solvers of the above theorem are entropic, but their definitions require the strict inequalities $\rho_L^* > 0$ and $\rho_R^* > 0$ that may not be satisfied. However, these strict inequalities can be enforced with a local correction that uses the standard HLL approximate Riemann solver [10]. Indeed, denoting $\tilde{w}_R^*$ and $\tilde{w}_L^*$ two states defined by the formulations (11.29), a robust version of Theorem 11.1 approximate Riemann solver is given by

$$(w_L^*, w_R^*) = \begin{cases} (\tilde{w}_L^*, \tilde{w}_R^*), & \text{if } (\tilde{w}_L^*, \tilde{w}_R^*) \in \Omega^2, \\ (w^{\text{HLL}}, w^{\text{HLL}}), & \text{otherwise.} \end{cases} \quad (11.32)$$

According to Lemma 11.1, the above procedure ensures $\rho_L^*, \rho_R^* > 0$ and preserves the consistency and the entropy condition, and it also guarantees the preservation of the set Ω .

Theorem 11.1 achieves to define the approximate Riemann solver that is now used to define a numerical scheme (11.5).

11.4 Consistent Entropy Stable Godunov Type Scheme

At time t^n , the solution $w(\cdot, t^n)$ of the Euler system (11.2) is approximated by a piecewise constant function defined by a sequence $(w_i^n)_{i \in \mathbb{Z}}$ in Ω . As a consequence, the sequence $(w_i^n)_{i \in \mathbb{Z}}$ can be reinterpreted as a juxtaposition of Riemann problems each of them defined on one interface of the mesh.

As displayed in Fig. 11.2, a restriction of the time step $\Delta t > 0$ imposes a non-interaction between each Riemann problem. This restriction refers to the CFL condition enunciated in Definition 11.1-(i). As a consequence and by definition of the approximate Riemann solver detailed in Sects. 11.2 and 11.3, an approximation of $w(\cdot, t^{n+1})$ is given by the L^2 -projection of the sequence $(\tilde{w}(\cdot, w_i^n, w_{i+1}^n))_{i \in \mathbb{Z}}$ on piecewise constant functions that reads

$$w_i^{n+1} = \frac{1}{\Delta x} \int_0^{\frac{\Delta x}{2}} \tilde{w}(x/\Delta t, w_{i-1}^n, w_i^n) dx + \frac{1}{\Delta x} \int_{-\frac{\Delta x}{2}}^0 \tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n) dx. \quad (11.33)$$

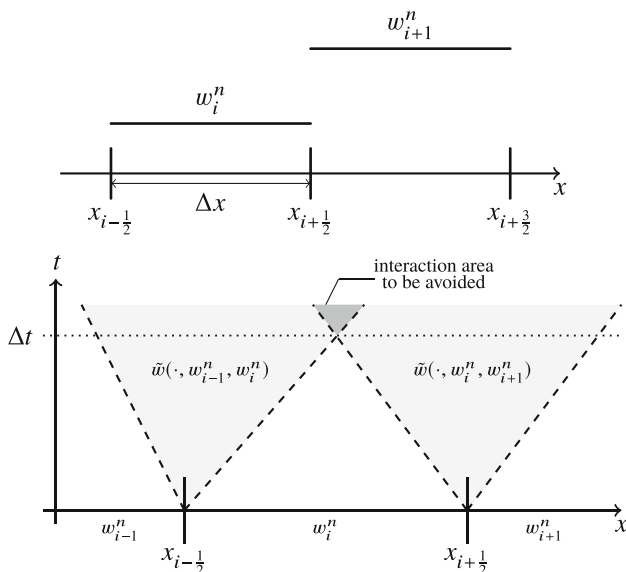


Fig. 11.2 On the top, piecewise constant approximation of $w(\cdot, t^n)$ on the mesh defined by interfaces $(x_{i+\frac{1}{2}})_{i \in \mathbb{Z}}$. On the bottom, Riemann problems juxtaposition in the plane (x, t)

The above numerical scheme is a Godunov type scheme [10] which is entirely defined by an approximate Riemann solver $\tilde{w}$. The next lemma establishes a more usual formulation of the Godunov type scheme (11.33) and also shows the discrete entropy inequality from the entropy condition (11.10).

Lemma 11.4 (Godunov Type Scheme Properties) *Consider a Godunov type scheme (11.33) defined by an approximate Riemann solver $\tilde{w}$ satisfying Definition 11.1 for the Euler equations (11.2) endowed with the convex entropy–entropy flux (η, G) given by (11.4). Let us also consider the functions $\mathcal{F} : (\mathbb{R}^3)^2 \rightarrow \mathbb{R}^3$ and $\mathcal{G} : (\mathbb{R}^3)^2 \rightarrow \mathbb{R}$ such that*

$$\begin{aligned} \mathcal{F}(w_i^n, w_{i+1}^n) &= \frac{f(w_{i+1}^n) + f(w_i^n)}{2} - \frac{\Delta x}{4\Delta t} (w_{i+1}^n - w_i^n) \\ &\quad + \frac{1}{2\Delta t} \int_0^{\frac{\Delta x}{2}} \tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n) dx \\ &\quad - \frac{1}{2\Delta t} \int_{-\frac{\Delta x}{2}}^0 \tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n) dx, \end{aligned} \quad (11.34)$$

$$\begin{aligned}
\mathcal{G}(w_i^n, w_{i+1}^n) &= \frac{G(w_{i+1}^n) + G(w_i^n)}{2} - \frac{\Delta x}{4\Delta t} (\eta(w_{i+1}^n) - \eta(w_i^n)) \\
&\quad + \frac{1}{2\Delta t} \int_0^{\frac{\Delta x}{2}} \eta(\tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n)) \, dx \\
&\quad - \frac{1}{2\Delta t} \int_{-\frac{\Delta x}{2}}^0 \eta(\tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n)) \, dx.
\end{aligned} \tag{11.35}$$

We have the following statements:

(i) Both functions $\mathcal{F}$ and $\mathcal{G}$ are, respectively, consistent with f and G that write

$$\mathcal{F}(w, w) = f(w), \quad \mathcal{G}(w, w) = G(w), \quad \forall w \in \Omega. \tag{11.36}$$

(ii) The Godunov type scheme (11.33) is equivalent to

$$w_i^{n+1} = w_i^n - \frac{\Delta t}{\Delta x} (\mathcal{F}(w_i^n, w_{i+1}^n) - \mathcal{F}(w_{i-1}^n, w_i^n)), \quad \forall i \in \mathbb{Z}. \tag{11.37}$$

(iii) The updated sequence $(w_i^{n+1})_{i \in \mathbb{Z}}$ given by the Godunov type scheme (11.33) satisfies the following discrete entropy inequality:

$$\frac{\eta(w_i^{n+1}) - \eta(w_i^n)}{\Delta t} + \frac{\mathcal{G}(w_i^n, w_{i+1}^n) - \mathcal{G}(w_{i-1}^n, w_i^n)}{\Delta x} \leq 0, \quad \forall i \in \mathbb{Z}. \tag{11.38}$$

Proof For (i), using the consistency of the approximate Riemann solver given by Definition 11.1-(iii), we have

$$\begin{aligned}
\mathcal{F}(w, w) &= \frac{f(w) + f(w)}{2} - \frac{\Delta x}{4\Delta t} (w - w) \\
&\quad + \frac{1}{2\Delta t} \int_0^{\frac{\Delta x}{2}} \tilde{w}(x/\Delta t, w, w) \, dx - \frac{1}{2\Delta t} \int_{-\frac{\Delta x}{2}}^0 \tilde{w}(x/\Delta t, w, w) \, dx, \\
&= f(w) + \frac{1}{2\Delta t} \int_0^{\frac{\Delta x}{2}} w \, dx - \frac{1}{2\Delta t} \int_{-\frac{\Delta x}{2}}^0 w \, dx, \\
&= f(w).
\end{aligned}$$

Since similar computations lead to the consistency for the numerical entropy flux, namely $\mathcal{G}(w, w) = G(w)$, that shows the statement (i).

For the reformulation (ii), since the CFL condition (11.12) holds, we can split the integrals in (11.33) to rewrite a Godunov type scheme (11.33) under the form

$$\begin{aligned} w_i^{n+1} = & \frac{1}{2\Delta x} \int_{-\frac{\Delta x}{2}}^0 \tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n) dx + \frac{1}{2\Delta x} \int_0^{\frac{\Delta x}{2}} \tilde{w}(x/\Delta t, w_{i-1}^n, w_i^n) dx \\ & + \frac{1}{2\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \tilde{w}(x/\Delta t, w_{i-1}^n, w_i^n) dx + \frac{1}{2\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n) dx \\ & - \frac{1}{2\Delta x} \int_{-\frac{\Delta x}{2}}^0 \tilde{w}(x/\Delta t, w_{i-1}^n, w_i^n) dx - \frac{1}{2\Delta x} \int_0^{\frac{\Delta x}{2}} \tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n) dx. \end{aligned}$$

Using the consistency integral relation (11.9) in the above equality, we deduce

$$\begin{aligned} w_i^{n+1} = & \frac{1}{2\Delta x} \int_{-\frac{\Delta x}{2}}^0 \tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n) dx + \frac{1}{2\Delta x} \int_0^{\frac{\Delta x}{2}} \tilde{w}(x/\Delta t, w_{i-1}^n, w_i^n) dx \\ & + \frac{1}{2} \left(\frac{w_{i-1}^n + w_i^n}{2} - \frac{\Delta t}{\Delta x} (f(w_i^n) - f(w_{i-1}^n)) \right) \\ & + \frac{1}{2} \left(\frac{w_i^n + w_{i+1}^n}{2} - \frac{\Delta t}{\Delta x} (f(w_{i+1}^n) - f(w_i^n)) \right) \\ & - \frac{1}{2\Delta x} \int_{-\frac{\Delta x}{2}}^0 \tilde{w}(x/\Delta t, w_{i-1}^n, w_i^n) dx - \frac{1}{2\Delta x} \int_0^{\frac{\Delta x}{2}} \tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n) dx. \end{aligned}$$

Considering the definition of the function $\mathcal{F}$ given by (11.34) in the above formulation, we deduce the form (11.37). The converse is direct.

Concerning the discrete entropy inequality (iii), as the CFL condition (11.12) holds, we can split the integrals in (11.33), and since the entropy $w \mapsto \eta(w)$ is a convex function, the Jensen inequality [17] gives

$$\begin{aligned} \eta(w_i^{n+1}) \leq & \frac{1}{2\Delta x} \int_{-\frac{\Delta x}{2}}^0 \eta(\tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n)) dx + \frac{1}{2\Delta x} \int_0^{\frac{\Delta x}{2}} \eta(\tilde{w}(x/\Delta t, w_{i-1}^n, w_i^n)) dx \\ & + \frac{1}{2\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \eta(\tilde{w}(x/\Delta t, w_{i-1}^n, w_i^n)) dx + \frac{1}{2\Delta x} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} \eta(\tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n)) dx \\ & - \frac{1}{2\Delta x} \int_{-\frac{\Delta x}{2}}^0 \eta(\tilde{w}(x/\Delta t, w_{i-1}^n, w_i^n)) dx - \frac{1}{2\Delta x} \int_0^{\frac{\Delta x}{2}} \eta(\tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n)) dx. \end{aligned}$$

Using the entropy condition (11.10) in the above inequality, we obtain

$$\begin{aligned}
 \eta(w_i^{n+1}) &\leq \\
 &\frac{1}{2\Delta x} \int_{-\frac{\Delta x}{2}}^0 \eta(\tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n)) dx + \frac{1}{2\Delta x} \int_0^{\frac{\Delta x}{2}} \eta(\tilde{w}(x/\Delta t, w_{i-1}^n, w_i^n)) dx \\
 &+ \frac{1}{2} \left(\frac{\eta(w_{i-1}^n) + \eta(w_i^n)}{2} - \frac{\Delta t}{\Delta x} (G(w_i^n) - G(w_{i-1}^n)) \right) \\
 &+ \frac{1}{2} \left(\frac{\eta(w_i^n) + \eta(w_{i+1}^n)}{2} - \frac{\Delta t}{\Delta x} (G(w_{i+1}^n) - G(w_i^n)) \right) \\
 &- \frac{1}{2\Delta x} \int_{-\frac{\Delta x}{2}}^0 \eta(\tilde{w}(x/\Delta t, w_{i-1}^n, w_i^n)) dx - \frac{1}{2\Delta x} \int_0^{\frac{\Delta x}{2}} \eta(\tilde{w}(x/\Delta t, w_i^n, w_{i+1}^n)) dx.
 \end{aligned}$$

Reorganized the above terms with the definition of the function $\mathcal{G}$ given by (11.35), we deduce the discrete entropy inequality (11.38) that achieves the proof. $\square$

Associating the above lemma and the results of Sects. 11.2 and 11.3, we finally define a consistent entropic Godunov type scheme for the Euler equations (11.2).

Theorem 11.2 (Consistent, Entropic Godunov Type Scheme for the Euler Equations) *Consider a Godunov type scheme (11.33) that approximates the solution of the Euler equations (11.2) endowed with the pressure function $w \mapsto p(w)$ given by (11.1) and completed with the convex entropy–entropy flux (η, G) given by (11.4). Assume this numerical scheme is defined by an approximate Riemann solver made of two intermediate states (11.11). Let us also assume the CFL condition (11.12) holds, and the intermediate states of the approximate Riemann solver are given by (11.32). The following statements are satisfied:*

- (i) *The Godunov type scheme (11.33) is consistent with the Euler equations (11.2).*
- (ii) *The updated sequence $(w_i^{n+1})_{i \in \mathbb{Z}}$ given by the Godunov type scheme (11.33) satisfies a discrete entropy inequality given by (11.38).*
- (iii) *If the sequence $(w_i^n)_{i \in \mathbb{Z}}$ defines a stationary contact state characterized by $(u_i^n)_{i \in \mathbb{Z}} = 0$ and $(p(w_i^n))_{i \in \mathbb{Z}} = \text{cste}$, then the Godunov type scheme (11.33) exactly preserves the variables u and p . This property writes*

$$\text{if } \left\{ \begin{array}{l} u_i^n = 0, \\ p(w_i^n) = \text{cste}, \end{array} \right. \quad \forall i \in \mathbb{Z}, \quad \text{then } \left\{ \begin{array}{l} u_i^{n+1} = 0, \\ p(w_i^{n+1}) = \text{cste}, \end{array} \right. \quad \forall i \in \mathbb{Z}.$$

Proof For statements (i) and (ii), according to Theorem 11.1, the approximate Riemann solver design includes the weak consistency and the entropy condition. As a consequence, the proof of (i) and (ii) is direct with Lemma 11.4.

Concerning statement (iii), for the sake of clarity, the local notations $w_i^n = w_L$ and $w_{i+1}^n = w_R$ are adopted. It is now sufficient to show that if

$$u_L = u_R = 0, \quad (11.39a)$$

$$p(w_L) = p(w_R) = p_0, \quad (11.39b)$$

then $u_L^* = u_R^* = 0$ and $\rho_\alpha E_\alpha = \rho_\alpha^* E_\alpha^*$ for all α in $\{L, R\}$. At first, the equalities (11.29a) read $u_L^* = u_R^* = u^{\text{HLL}}$, but according to the definition of u^{HLL} given by (11.17), we have

$$\begin{aligned} u^{\text{HLL}} &= \frac{(\rho u)^{\text{HLL}}}{\rho^{\text{HLL}}}, \\ &= \frac{1}{\rho^{\text{HLL}}(\lambda_R - \lambda_L)} \left(\lambda_R \rho_R u_R - \lambda_L \rho_L u_L - \rho_R u_R^2 + \rho_L u_L^2 \right. \\ &\quad \left. - (p(w_R) - p(w_L)) \right), \\ &= 0. \end{aligned} \quad (11.40)$$

As a consequence, using the above equation, we deduce that if the equalities (11.39) are satisfied, then $u_L^* = u_R^* = 0$. In addition and according to the pressure function $p(w)$ defined in (11.1), if the conditions (11.39) hold, then a straightforward computation shows

$$\rho_\alpha E_\alpha = \frac{p_0}{\gamma - 1}, \quad \forall \alpha \in \{L, R\}. \quad (11.41)$$

Combining the above result with the equalities (11.40)–(11.39a) in the definition of w^{HLL} given by (11.13), we obtain

$$\begin{aligned} \frac{p(w^{\text{HLL}})}{\gamma - 1} &= (\rho E)^{\text{HLL}} - \frac{\rho^{\text{HLL}}(u^{\text{HLL}})^2}{2}, \\ &= \frac{\lambda_R \rho_R E_R - \lambda_L \rho_L E_L}{\lambda_R - \lambda_L} - \frac{(\rho_R E_R + p(w_R))u_R - (\rho_L E_L + p(w_L))u_L}{\lambda_R - \lambda_L}, \\ &= \frac{\lambda_R \rho_R E_R - \lambda_L \rho_L E_L}{\lambda_R - \lambda_L}, \\ &= \frac{p_0}{\gamma - 1}. \end{aligned}$$

Since the equalities $u_L^* = u_R^* = 0$ have been proved, the association of the above equality to the results (11.29b)–(11.41) infers that if the conditions (11.39)

hold then

$$\rho_\alpha^* E_\alpha^* = \frac{p(w_\alpha^*)}{\gamma - 1} = \frac{p(w^{\text{HLL}})}{\gamma - 1} = \frac{p_0}{\gamma - 1} = \rho_\alpha E_\alpha, \quad \forall \alpha \in \{L, R\}.$$

These last equalities conclude the proof of statement (iii) and the proof of Theorem 11.2. $\square$

The next section concerns the numerical experiments.

11.5 Numerical Results

This section presents some numerical experiments to assess the performances of the numerical schemes derived in Theorem 11.2. The CFL condition is given by (11.12), and for all the interfaces of the mesh, we fix $v^* = u^{\text{HLL}}$, where u^{HLL} is defined in (11.17). For each test case, one of the two approximate Riemann solvers given by (11.29) is randomly selected. The results are compared to the HLLC scheme [22] and to the scheme detailed in [10] (Section b.ii). This last scheme is symmetrized fixing $\lambda_R = -\lambda_L$, and its two implicit constants are both fixed to 10^{-7} .

The first test case is the Sod shock tube problem for a diatomic gas ($\gamma = 1.4$) [19]. The spatial domain $[0, 1]$ is discretized with 400 cells. The initial condition is

$$\rho_0(x) = \begin{cases} 1 & \text{if } x < 0.5, \\ 0.125 & \text{otherwise,} \end{cases} \quad u_0(x) = 0, \quad p_0(x) = \begin{cases} 1 & \text{if } x < 0.5, \\ 0.1 & \text{otherwise.} \end{cases} \quad (11.42)$$

Homogeneous Neumann conditions are imposed on both boundaries of the spatial domain. The final time is $t = 0.2$, and Fig. 11.3 displays the results.

The results of the schemes defined in Theorem 11.2 are very satisfying and conform to the literature standards.

The second test case concerns the simulation of a stationary contact wave defined by the following initial condition:

$$\rho_0(x) = \begin{cases} 1 & \text{if } x < 0.5, \\ 0.125 & \text{otherwise,} \end{cases} \quad u_0(x) = 0, \quad p_0(x) = 1. \quad (11.43)$$

Homogeneous Neumann conditions are imposed on both boundaries of the spatial domain. The final time is $t = 1.0$, and Fig. 11.4 displays the results and the numerical errors for several norms.

The HLLC scheme [22] exactly preserves the contact discontinuity defined by the initial condition (11.43). The scheme derived in [10] (Section b.ii) commits a slight error on the variable ρ that is clearly inferior to the first-order error committed by the

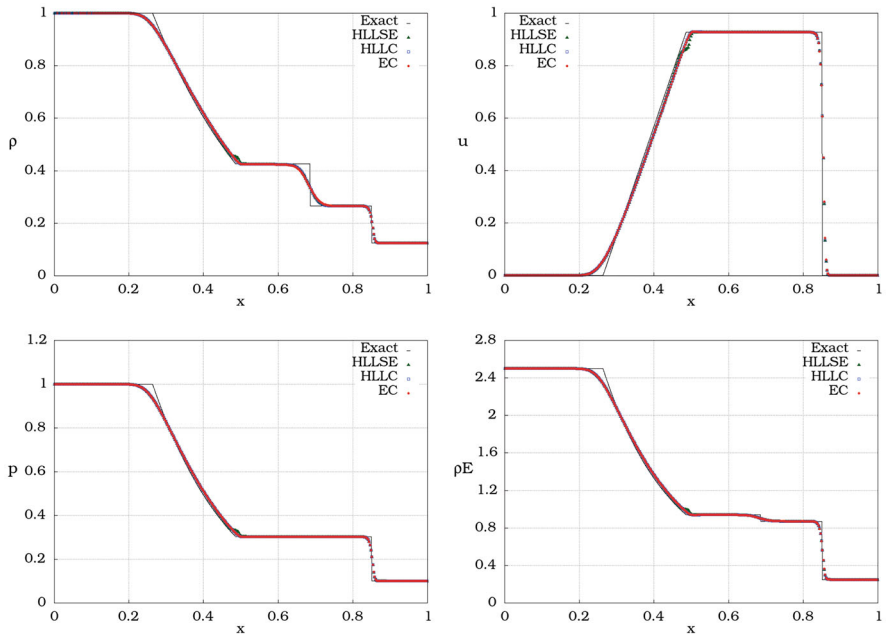


Fig. 11.3 Numerical results for the Sod test case (11.42), at time $t = 0.2$, with the legend EC: schemes derived in Theorem 11.2, and HLLC and HLLSE: schemes, respectively, detailed in [10–20, 22] (Section b.ii)

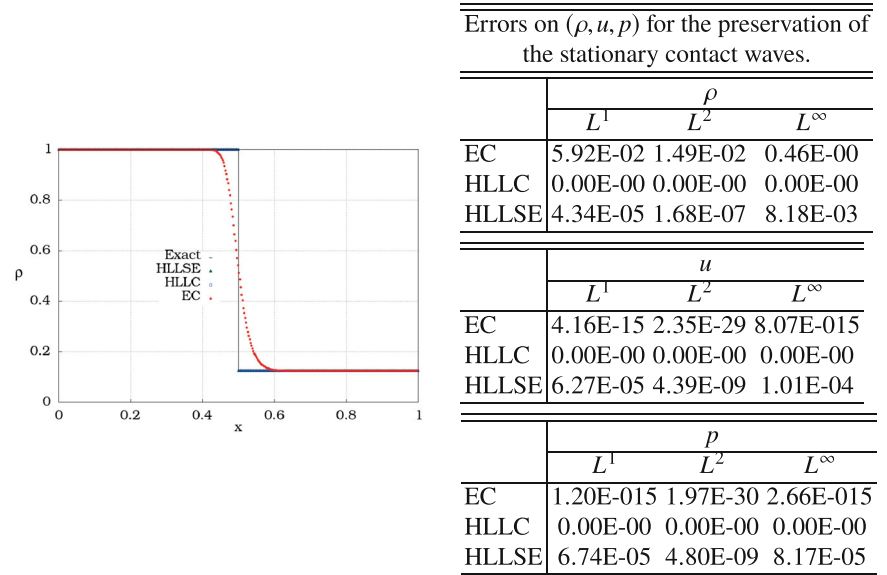


Fig. 11.4 On the left, numerical results at time $t = 1.0$ for the stationary contact problem (11.43) with the legend EC: schemes derived in Theorem 11.2, HLLC, HLLSE: schemes respectively detailed in [10–20, 22] (Section b.ii). On the right, numerical errors for the variables ρ , u and p at time $t = 1.0$

schemes defined in Theorem 11.2. For this variable, the schemes of Theorem 11.2 are more diffusive than the existing numerical schemes. Nevertheless, the schemes proposed in Theorem 11.2 exactly preserve the quantities u and p that is not the case of the scheme detailed in [10] (Section b.ii).

11.6 Conclusion

We have presented entropic Godunov type scheme for the Euler equations. The schemes are defined with ARSs made of two constant intermediate states and include sufficient conditions for the consistency and for the discrete entropy stability. The approximate Riemann solver is defined by quadratic equations that always admit explicit real solutions. As a consequence, the entropic schemes are simple. Their numerical results conform to the literature standards devoted to the first-order schemes.

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Appendix

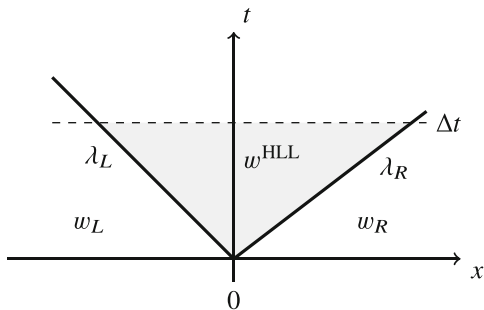
This appendix concerns the standard HLL approximate Riemann solver [10]. Considering the formalism of Sect. 11.2, this approximate Riemann solver is illustrated in Fig. 11.5, it is denoted $\tilde{w}^{\text{HLL}}(x/\Delta t, w_L, w_R)$, and it writes

$$\tilde{w}^{\text{HLL}}(x/\Delta t, w_L, w_R) = \begin{cases} w_L & \text{if } \frac{x}{\Delta t} \leq \lambda_L, \\ w^{\text{HLL}} & \text{if } \lambda_L < \frac{x}{\Delta t} < \lambda_R, \\ w_R & \text{if } \lambda_R < \frac{x}{\Delta t}. \end{cases} \quad (11.44)$$

The CFL condition for the HLL approximate Riemann solver (11.44) is also given by (11.12). Using the consistency integral relation (11.9) in the definition of the HLL approximate Riemann solver given by (11.44), a straightforward computation shows

$$w^{\text{HLL}} = (\rho^{\text{HLL}}, (\rho u)^{\text{HLL}}, (\rho E)^{\text{HLL}})^T = \frac{\lambda_R w_R - \lambda_L w_L}{\lambda_R - \lambda_L} - \frac{f(w_R) - f(w_L)}{\lambda_R - \lambda_L}. \quad (11.45)$$

Fig. 11.5 HLL approximate Riemann solver in the plane (x, t)



As a consequence, the HLL approximate Riemann solver (11.44) satisfies the consistency condition $\tilde{w}^{\text{HLL}}(\cdot, w, w) = w$, and the following lemma gives additional robustness properties of the state vector w^{HLL} .

Lemma 11.5 (Robustness Properties of the HLL Approximate Riemann Solver) *Consider the Riemann problem for the Euler equations given by (11.2), (11.7). Assume the CFL condition (11.12) holds, and let us also assume the solution of the Riemann problem $(\rho, \rho u, \rho E)^T(x, \Delta t)$ is in the set Ω for all $x \in [\lambda_L \Delta t, \lambda_R \Delta t]$. The following statements are satisfied:*

(i) *An equivalent form of w^{HLL} is given by*

$$w^{\text{HLL}} = \frac{1}{(\lambda_R - \lambda_L) \Delta t} \int_{\lambda_L \Delta t}^{\lambda_R \Delta t} w(x, \Delta t) dx.$$

(ii) $\rho^{\text{HLL}} > 0$, $(\rho E)^{\text{HLL}} > 0$.

(iii) $\eta(w^{\text{HLL}}) \leq \eta^{\text{HLL}}$.

Proof Integrating equation (11.2) over the domain $[0, \Delta t] \times [-\Delta x/2, \Delta x/2]$ and using the initial condition (11.7), we have

$$\begin{aligned} 0 &= \int_0^{\Delta t} \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} (\partial_t w + \partial_x f(w)) dx dt, \\ &= \int_{-\frac{\Delta x}{2}}^{\frac{\Delta x}{2}} (w(x, \Delta t) - w_0(x)) dx + \int_0^{\Delta t} (f(w(\Delta x/2, t)) - f(w(-\Delta x/2, t))) dt, \\ &= \int_{-\frac{\Delta x}{2}}^{\lambda_L \Delta t} w_L dx + \int_{\lambda_L \Delta t}^{\lambda_R \Delta t} w(x, \Delta t) dx + \int_{\lambda_R \Delta t}^{\frac{\Delta x}{2}} w_R dx \\ &\quad - \left(\int_{-\frac{\Delta x}{2}}^0 w_L dx + \int_0^{\frac{\Delta x}{2}} w_R dx \right) \end{aligned}$$

$$\begin{aligned}
& + \Delta t (f(w_R) - f(w_L)), \\
& = \int_{\lambda_L \Delta t}^{\lambda_R \Delta t} w(x, \Delta t) dx + \lambda_L \Delta t w_L - \lambda_R \Delta t w_R + \Delta t (f(w_R) - f(w_L)).
\end{aligned} \tag{11.46}$$

Using the above result in the definition of w^{HLL} given by (11.45), we deduce the formulation given by (i).

For the inequalities (ii), since the solution of the Riemann problem (11.2), (11.7) is assumed to belong to Ω , writing the first and the last components of w^{HLL} given by (i), we have

$$\begin{aligned}
\rho^{\text{HLL}} &= \frac{1}{(\lambda_R - \lambda_L) \Delta t} \int_{\lambda_L \Delta t}^{\lambda_R \Delta t} \rho(x, \Delta t) dx > 0, \\
(\rho E)^{\text{HLL}} &= \frac{1}{(\lambda_R - \lambda_L) \Delta t} \int_{\lambda_L \Delta t}^{\lambda_R \Delta t} (\rho E)(x, \Delta t) dx > 0.
\end{aligned}$$

Concerning the inequality (iii), as $\lambda_L < \lambda_R$, the convexity of the entropy $w \mapsto \eta(w)$ associated with the Jensen inequality [17] gives

$$\eta(w^{\text{HLL}}) \leq \frac{1}{(\lambda_R - \lambda_L) \Delta t} \int_{\lambda_L \Delta t}^{\lambda_R \Delta t} \eta(w(x, \Delta t)) dx. \tag{11.47}$$

Since $w(x, \Delta t)$ is the exact solution of the Riemann problem (11.2), (11.7), it verifies the entropy inequality (11.3). As a consequence, integrating this inequality (11.3) over the domain $[0, \Delta t] \times [-\Delta x/2, \Delta x/2]$, an analogous computation to (11.46) yields the following inequality:

$$\int_{\lambda_L \Delta t}^{\lambda_R \Delta t} \eta(w(x, \Delta t)) dx + \lambda_L \Delta t \eta(w_L) - \lambda_R \Delta t \eta(w_R) + \Delta t (G(w_R) - G(w_L)) \leq 0.$$

Finally, associating the above inequality with (11.47), we deduce

$$\begin{aligned}
\eta(w^{\text{HLL}}) &= \eta\left(\frac{1}{(\lambda_R - \lambda_L) \Delta t} \int_{\lambda_L \Delta t}^{\lambda_R \Delta t} w(x, \Delta t) dx\right), \\
&\leq \frac{1}{(\lambda_R - \lambda_L) \Delta t} \int_{\lambda_L \Delta t}^{\lambda_R \Delta t} \eta(w(x, \Delta t)) dx, \\
&\leq \frac{\lambda_R \eta(w_R) - \lambda_L \eta(w_L)}{\lambda_R - \lambda_L} - \frac{G(w_R) - G(w_L)}{\lambda_R - \lambda_L}.
\end{aligned}$$

Since the right-hand side of the above inequality coincides with η^{HLL} given by (11.13) that achieves the proof.

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Chapter 12

Time Convergence of Bloch-Type Model with Pauli's Master Equation and Wave–Matter Interaction



Kolé Keita and Mahamar Dicko

Abstract In this chapter, we show the time convergence towards equilibrium for Bloch-type model which depends on the terms such that Pauli's master equation and wave interaction terms. After this convergence, we discretize the Bloch-type model with a version of Strang's scheme whose evolution operators are computed by decoupling matrices. Numerical density matrix obtained with Strang scheme conserves the qualitative proprieties.

12.1 Introduction

Bloch-type model describes the time evolution of particle presence probabilities in quantum systems such as atoms and quantum dots. This model is obtained from Schrodinger equation or Heisenberg equation (see, e.g., the reviews [2, 7, 8]). In the literature, different Bloch-type models take into account the relaxation terms (see, e.g., [1, 3]), nonlinear terms such as Coulomb interaction (see, e.g., [7, 9, 10]) and electron–phonon interaction (see, e.g., [6]). In these papers, analytical, mathematical, and physical results of quantum systems (atoms and quantum dots) are studied. The study of the Bloch-type model explains the response of medium after this interaction with laser sources (electromagnetic wave).

In the case of two-level quantum model and if we take into account the Pauli relaxation terms, there exist the works that are about the comparison of numerical methods such as the Crank–Nicolson scheme [3] and splitting schemes [13].

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Splitting scheme is used to approach the nonlinear Bloch-type models coupled with Maxwell equations (see, e.g., [6, 7]). In these papers, they showed that the numerical solution obtained with splitting scheme conserves all the physical properties (trace and hermicity), and they realized self-induced transparency phenomenon (given in [7, 11, 15]) in the case of two-level and three-level quantum models.

The model with relaxation terms studied in [1, 3, 13] is about two-level quantum models only. In this chapter, we study three-level quantum models in considering the interaction with electromagnetic wave (wave–matter interaction). In these studies, we show that Bloch-type model can be described by a real nonautonomous dynamical system. Nonautonomous dynamical system has a unique solution and bounded on time interval. Next, we discretize the Bloch-type model with two versions of splitting scheme (Strang scheme [12, 14]) whose evolution operators are, respectively, computed with second-order Taylor series of matrix exponential and with reduction matrices. The qualitative properties (trace and hermiticity) are verified with the numerical solutions obtained by Strang method. In the following, we compare the two numerical methods, and we valid those obtained with reduction matrices in terms of stability and CPU times. This method is used to simulate the time convergence of populations and to realize self-induced transparency phenomenon.

This chapter is organized as follows: In Sect. 12.2, we present a Bloch-type model for three-level quantum model, and we describe this model as a nonautonomous dynamical system. Section 12.3 is about the results of existence and uniqueness of Cauchy problem. We determine equilibrium states and some proprieties of dynamical system in Sect. 12.4. Section 12.5 provides the construction step of numerical methods 1 and 2. Finally, the numerical simulations are realized in Sect. 12.6.

12.2 Bloch-Type Model for Three Energy Levels

The variable of Bloch-type model is a matrix called density matrix and denoted ρ . Their diagonal elements of density matrix ρ_{jj} called populations correspond to the occupation probabilities and their off-diagonal elements $\rho_{jk} (j \neq k)$ called coherences whose absolute values correspond to transition probabilities.

In this chapter, we consider a three-level quantum model. The general form of Bloch-type model with the relaxation and wave–matter interaction terms is given by

$$\begin{cases} \partial_t \rho = W(\rho) - \frac{i}{\hbar} ([E_0, \rho] + e(x_0, t)[\mathbf{m}, \rho]), \forall t > 0, \\ \rho(0) = \rho^0, \end{cases} \quad (12.1)$$

where ρ_0 is the 3×3 matrix, E_0 is the diagonal matrix (free energies) with $E_{0,j} = e_j$ for all $j \in \{1, 2, 3\}$, $\mathbf{M}(x_0, t) = e(x_0, t) \cdot \mathbf{m}$ with $\mathbf{m}$ is the dipolar moment matrix ($\mathbf{M}^*(x_0, t) = \mathbf{M}(x_0, t)$), $e(x_0, t)$ is the electromagnetic wave function, x_0

is the location in space of quantum system, and $W(\rho)$ is Pauli's master equation (relaxation terms). These relaxation terms take into account physical phenomena such as spontaneous emission of light, collisions vibration in crystal lattices, and thermal perturbations in fluids.

Pauli's master equation is defined by

$$\begin{aligned} \forall j, k \in \{1, 2, 3\}, \quad W(\rho)_j &= \sum_{k=1, k \neq j}^3 (w_{jk} \rho_k - w_{kj} \rho_j), \\ W(\rho)_{jk} &= \gamma_{jk} \rho_{jk}, \end{aligned}$$

where $\forall j, k \in \{1, 2, 3\}, w_{jk}, \gamma_{jk} \in \mathbb{R}$.

Bloch-type model (12.1) verifies some proprieties such as:

- **Hermicity:** We consider the equation of system (12.1) and apply adjoint operator, and then we obtain

$$\forall t \geq 0, \quad \partial_t \rho^* = W(\rho^*) + \frac{i}{\hbar} (-[E_0, \rho^*] - [\mathbf{M}(x_0, t), \rho^*]).$$

Thanks to uniqueness of the solution then we have $\rho^* = \rho$. This implies that

$$\begin{aligned} \forall j, \quad \Im(\rho_{jj}) &= 0, \\ \forall j \neq k, \quad \Re(\rho_{jk}) &= \Re(\rho_{kj}) \text{ and } \Im(\rho_{jk}) = -\Im(\rho_{kj}), \end{aligned}$$

where $\Re$ and $\Im$ are, respectively, real and imaginary parts of complex number.

- **Trace conservation:** We have

$$\forall t \geq 0, \quad \text{tr}(\partial_t \rho) = 0 \implies \forall t \geq 0, \quad \text{tr}(\rho(t)) = \text{tr}(\rho^0).$$

Dynamical System From the elements of matrix ρ , we define the following vectors $U_p = (\rho_{11}, \rho_{22}, \rho_{33})$, $U_c = (\rho_{12}, \rho_{13}, \rho_{23})$, $U = (U_p, \Re(U_c), \Im(U_c))$ and two operators denoted diag , and Diag :

- Diag creates a diagonal matrix from a vector, and the elements of the vector are on the diagonal of created matrix.
- Diag creates a vector with diagonal elements of a matrix.

We ask the following block matrix:

$$A(x_0, t) = \begin{pmatrix} W_1 & \Im(A_1(x_0, t)) & \Re(A_1(x_0, t)) \\ -\frac{1}{2} \Im(A_1^*(x_0, t)) & \Im(A_2(x_0, t)) + D_\gamma & \Re(A_3(x_0, t)) \\ -\frac{1}{2} \Re(A_1^*(x_0, t)) & -\Re(A_2(x_0, t)) & \Im(A_3(x_0, t)) + D_\gamma + \mathcal{E}_0 \end{pmatrix},$$

where

$$W_1 = \begin{pmatrix} -(w_{21} + w_{31}) & w_{21} \frac{a_1}{a_2} & w_{31} \frac{a_1}{a_3} \\ w_{21} & -\left(w_{21} \frac{a_1}{a_2} + w_{32}\right) & w_{32} \frac{a_2}{a_3} \\ w_{31} & w_{32} & -\left(w_{31} \frac{a_1}{a_3} + w_{32} \frac{a_2}{a_3}\right) \end{pmatrix},$$

with $\forall j, a_j = \exp\left(\frac{e_j}{k_B T}\right)$, k_B is Boltzmann's constant, and T is the temperature,

$$D_\gamma = \text{diag}(\gamma_{12}, \gamma_{13}, \gamma_{23}), \quad \mathcal{E}_0 = -\frac{1}{\hbar} \text{diag}(e_1 - e_2, e_1 - e_3, e_2 - e_3),$$

$$A_1(x_0, t) = \frac{2e(x_0, t)}{\hbar} \begin{pmatrix} -\mathbf{m}_{21} & -\mathbf{m}_{31} & 0 \\ \mathbf{m}_{21} & 0 & -\mathbf{m}_{32} \\ 0 & \mathbf{m}_{31} & \mathbf{m}_{32} \end{pmatrix},$$

$$A_2 = \frac{e(x_0, t)}{\hbar} \begin{pmatrix} \mathbf{m}_{11} - \mathbf{m}_{22} & -\mathbf{m}_{32} & \mathbf{m}_{13} \\ -\mathbf{m}_{23} & \mathbf{m}_{11} - \mathbf{m}_{33} & \mathbf{m}_{12} \\ -\mathbf{m}_{13} & \mathbf{m}_{21} & \mathbf{m}_{22} - \mathbf{m}_{33} \end{pmatrix}$$

$$A_3 = \frac{e(x_0, t)}{\hbar} \begin{pmatrix} \mathbf{m}_{11} - \mathbf{m}_{22} & -\mathbf{m}_{32} & -\mathbf{m}_{13} \\ -\mathbf{m}_{23} & \mathbf{m}_{11} - \mathbf{m}_{33} & \mathbf{m}_{12} \\ \mathbf{m}_{13} & \mathbf{m}_{21} & \mathbf{m}_{22} - \mathbf{m}_{33} \end{pmatrix}.$$

The vector U and its first derivative respect the following Cauchy problem:

$$\begin{cases} \partial_t U(t) = A(x_0, t) U(t), & t > 0 \\ U(0) = U^0. \end{cases} \quad (12.2)$$

The entries of matrix are reals, and then we have $t \mapsto A(x_0, t) \in \mathcal{M}_9(\mathbb{R})$ and $t \mapsto U(t) \in \mathbb{R}^9$. The new system (12.2) verifies the trace conservation:

$$\partial_t \sum_{j=1}^3 U_j = \sum_{j=1}^3 (W_1 U_{1:3} + \Im(A_1(x_0, t)) U_{4:6} + \Re(A_1(x_0, t)) U_{7:9})_j = 0_{\mathbb{R}^3}.$$

Then

$$\forall t \geq 0, \quad \sum_{j=1}^3 U_j(t) = \sum_{j=1}^3 U_j^0.$$

12.3 Existence and Uniqueness of Solution

The Cauchy problem (12.2) can be present so

$$(S) : \begin{cases} \partial_t U(t) = f(t, U(x_0, t)), & t > 0 \\ U(0) = U^0, \end{cases}$$

where $f(t, U(x_0, t)) = A(x_0, t) U(x_0, t)$ and the entries of matrix $A(x_0, t)$ are denoted $A_{jk}(x_0, t) : \mathbb{R}_+ \longrightarrow \mathbb{R}$ for all $j, k \in \{1, 2, \dots, 9\}$. The studies of their entries permit us to establish the following lemma.

Lemma 12.1 *Let x_0 be given and $t \in \mathbb{R}_+$. If the function $t \mapsto e(x_0, t)$ is holomorphic, then their entries of the matrix $A(x_0, t)$ belong to $C^1(\mathbb{R}_+, \mathbb{R})$.*

Definition 12.1 A function $g : \mathbb{R}_+ \times C^1(\mathbb{R}^p) \longrightarrow C^1(\mathbb{R}^n)$ ($p, n \geq 2$) is called locally Lipschitz with respect to the last variables belonging to $\mathbb{R}^p$ if for all $t \in \mathbb{R}_+$, $U \in C^1(\mathbb{R}^p)$, there exist the reals k_U and $r > 0$ such that

$$\|g(t, U_1) - g(t, U_2)\| \leq k_U \|U_1 - U_2\|$$

for all $U_1, U_2 \in B(U, r)$ (closed ball of radius r centered at a point U). If $k_U \in [0, 1[$, then g is called a contraction.

Let g be a function. We denote by J_g the Jacobian matrix or the matrix of all its first-order partial derivatives of g .

Theorem 12.1 *If $\forall t \in \mathbb{R}_+$, the function $t \mapsto e(x_0, t)$ is holomorphic and locally Lipschitz with respect to U , then Cauchy problem (S) has a unique solution defined on a time interval given by (12.3).*

Proof The integral solution of system (S) gives us

$$\forall t \geq 0, \quad U(x_0, t) = U^0(x_0) + \int_0^t f(s, U(x_0, s)) ds,$$

where $U(x_0, 0) = U^0(x_0) \in \mathbb{R}^9$.

We define the application $T : \mathbb{R}^9 \longrightarrow \mathbb{R}^9$ by

$$T(U(t)) = U^0(x_0) + \int_0^t f(s, U(x_0, s)) ds.$$

We have the matrix $s \mapsto A(x_0, s) \in \mathcal{M}_9(C^1(\mathbb{R}_+))$ because the function $s \mapsto e(x_0, s)$ is holomorphic.

Now we consider a closed ball of radius R centered at an initial point $U^0(x_0)$ for the norm $\|\cdot\|_{C^0}$ denoted B_R .

- We suppose that $U_1, U_2 \in B_R$, and we compute the norm of $T(U_1) - T(U_2)$.

$$\begin{aligned} \|T(U_1) - T(U_2)\| &\leq \int_0^t \|f(s, U_1(x_0, s)) - f(s, U_2(x_0, s))\| ds \\ &\leq \|J_f\|_\infty \int_0^t \|U_1(x_0, s) - U_2(x_0, s)\|_\infty ds. \end{aligned}$$

We have

$$\|U_1 - U_2\|_\infty = \sup_{s \in [0, t]} |U_1(x_0, s) - U_2(x_0, s)|,$$

then

$$\|T(U_1) - T(U_2)\| \leq t \|J_f\|_\infty \|U_1 - U_2\|_\infty.$$

The application T is a δ -contraction ($\delta \in [0, 1[$) if $t \in [0, \delta / \|J_f\|_\infty[$.

- f is bounded because for all $U \in B_R \implies \exists \mu \in \mathbb{R}, \|f(\cdot, U)\| \leq \mu \|U\| \leq \mu R$.

We can show that $\forall U \in B_R \implies T(U) \in B_R$ if $t \in [0, 1/\mu]$.

$$\|T(U) - U^0\| \leq \int_0^t \|f(s, U)\| ds \leq \mu R t.$$

Then the Cauchy problem (S) has a unique solution defined on

$$[0, \min\{1/\mu, \delta / \|J_f\|_\infty\}]. \quad (12.3)$$

□

12.4 Study of Convergence and Stability

It is difficult to study the stability of equilibrium solutions and the convergence toward equilibrium solutions (system (S)) because of the complexity of the matrix $A(x_0, t)$. However, under certain assumptions, we can obtain important properties on the solution of the system (S). For this, we decompose the system (S) as follows:

$$\begin{cases} \partial_t U(x_0, t) = A_L U(x_0, t) + A_N(x_0, t) U(x_0, t), & t > 0 \\ U(0) = U^0, \end{cases}$$

where

$$A_L = \begin{pmatrix} W_1 & 0 & 0 \\ 0 & D_\gamma & 0 \\ 0 & 0 & D_\gamma + \mathcal{E}_0 \end{pmatrix} \quad \text{and} \quad A_N(x_0, t) = e(x_0, t) \begin{pmatrix} 0 & \Im(\mathbf{m}_1) & \Re(\mathbf{m}_1) \\ -\frac{1}{2}\Im(\mathbf{m}_1^\dagger) & \Im(\mathbf{m}_2) & \Re(\mathbf{m}_3) \\ -\frac{1}{2}\Re(\mathbf{m}_1^\dagger) & -\Re(\mathbf{m}_2) & \Im(\mathbf{m}_3) \end{pmatrix}.$$

This means that without the electromagnetic wave, the system (S) can be reduced to

$$\begin{cases} \partial_t U_L(x_0, t) = A_L U_L(x_0, t), & t > 0 \\ U_L(0) = U_L^0. \end{cases} \quad (12.4)$$

This last system is autonomous, and we obtain the following result.

Theorem 12.2 *For the large times, the population of autonomous system (12.4) converges towards equilibrium, and the coherences are null.*

Proof The spectrum of a matrix A_L is given by

$$\text{sp}(A_L) = \{0, \lambda_1, \lambda_2, -\gamma_{12}, -\gamma_{13}, -\gamma_{23}, -\gamma_{12} + e_1 - e_2, -\gamma_{13} + e_1 - e_3, -\gamma_{23} + e_2 - e_3\},$$

with $\Re(\lambda_i) < 0, \forall i \in \{1, 2\}$.

The change-of-basis matrix and its inverse are thus given as follows:

$$P = \begin{pmatrix} P_1 & 0 \\ 0 & I_6 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} P_1^{-1} & 0 \\ 0 & I_6 \end{pmatrix} \quad \text{with}$$

$$P_1 = \begin{pmatrix} a_1 & \alpha_1 & \alpha_2 \\ a_2 & \beta_1 & \beta_2 \\ a_3 & \delta_1 & \delta_2 \end{pmatrix}, \quad a_j = \exp\left(\frac{e_j}{k_B T}\right), \quad (12.5)$$

and the eigenvectors are denoted as $E_{\lambda_i} = \text{vect}\left\{\begin{pmatrix} \alpha_i \\ \beta_i \\ \delta_i \end{pmatrix}\right\}, \forall i \in \{1, 2\}$.

We denote $U_L = (U_{L,1}, U_{L,2}, U_{L,3}) \in \mathbb{R}^9$, the solution of the autonomous system. After some basic operations, we obtain

$$U_{L,1}(t) = P_1 \begin{pmatrix} 1 & 0 & 0 \\ 0 \exp(\lambda_1 t) & 0 \\ 0 & 0 & \exp(\lambda_2 t) \end{pmatrix} P_1^{-1} U_{L,1}^0,$$

$$U_{L,2}(t) = \begin{pmatrix} \exp(-\gamma_{12} t) & 0 & 0 \\ 0 & \exp(-\gamma_{13} t) & 0 \\ 0 & 0 & \exp(-\gamma_{23} t) \end{pmatrix} U_{L,2}^0,$$

and

$$U_{L,3}(t) = \begin{pmatrix} \begin{bmatrix} \exp((- \gamma_{12} \\ + e_1 - e_2) t) \end{bmatrix} & 0 & 0 \\ 0 & \begin{bmatrix} \exp((- \gamma_{13} \\ + e_1 - e_3) t) \end{bmatrix} & 0 \\ 0 & 0 & \begin{bmatrix} \exp((- \gamma_{23} \\ + e_2 - e_3) t) \end{bmatrix} \end{pmatrix} U_{L,3}^0.$$

When $t \rightarrow +\infty$, we show easily that the solution $U_L \rightarrow$
 $\left(\begin{pmatrix} a_1 & 0 & 0 \\ a_2 & 0 & 0 \\ a_3 & 0 & 0 \end{pmatrix} P_1^{-1} U_{L,1}^0, 0_{\mathbb{R}^3}, 0_{\mathbb{R}^3} \right).$ $\square$

Theorem 12.3 *Let $T \in \mathbb{R}_+$. If the function $t \mapsto e(x_0, t) \in L_{loc}^1(\mathbb{R})$, then for any $t \in [0, T]$,*

$$\|U(x_0, t)\|_\infty \leq C \|U^0\|_1,$$

where $C \in \mathbb{R}_+$ is a constant independent of t .

Proof Let $T \in \mathbb{R}_+$. We denote

$$\mathcal{D}_1 = \text{diag}(0, \lambda_1, \lambda_2), \quad (12.6)$$

$\mathcal{D}_2 = \text{diag}(-\gamma_{12}, -\gamma_{13}, -\gamma_{23}, -\gamma_{12} + e_1 - e_2, -\gamma_{13} + e_1 - e_3, -\gamma_{23} + e_2 - e_3)$,
 and $\mathcal{D} = \begin{pmatrix} \mathcal{D}_1 & 0 \\ 0 & \mathcal{D}_2 \end{pmatrix}$.

The integral solution of system (S) becomes

$$\begin{aligned} U(x_0, t) = & \begin{pmatrix} P_1 \exp(\mathcal{D}t) P_1^{-1} & 0 \\ 0 & \exp(\mathcal{D}t) \end{pmatrix} U^0 \\ & + \int_0^t \begin{pmatrix} P_1 \exp(\mathcal{D}(t-s)) P_1^{-1} & 0 \\ 0 & \exp(\mathcal{D}(t-s)) \end{pmatrix} A_N(x_0, s) U(x_0, s) ds. \end{aligned}$$

We introduce the notation

$$\mathcal{R}(t) = \begin{pmatrix} P_1 \exp(\mathcal{D}t) P_1^{-1} & 0 \\ 0 & \exp(\mathcal{D}t) \end{pmatrix}$$

The integral solution of (S) can be summarized so

$$U(x_0, t) = \mathcal{R}(t)U^0 + \int_0^t \mathcal{R}(t-s)A_N(x_0, s)U(x_0, s)ds.$$

We calculate the uniform norm of this solution, and we obtain

$$\begin{aligned} & \|U(x_0, t)\|_\infty \\ & \leq \|\mathcal{R}(t)\|_\infty \|U^0\|_1 + \int_0^t \|\mathcal{R}(t-s)\|_\infty \|A_N(x_0, s)\|_\infty \|U(x_0, s)\|_1 ds. \end{aligned}$$

We know that

$$A_N(x_0, s) = \mathbf{m} e(x_0, s) \implies \|A_N(x_0, s)\|_\infty \leq \|\mathbf{m}\|_\infty |e(x_0, s)|,$$

then

$$\begin{aligned} & \|U(x_0, t)\|_\infty \\ & \leq \|\mathcal{R}(t)\|_\infty \|U^0\|_1 + \|\mathbf{m}\|_\infty \int_0^t \|\mathcal{R}(t-s)\|_\infty |e(x_0, s)| \|U(x_0, s)\|_1 ds. \end{aligned}$$

For any time $t \geq 0$, we have

$$\|\mathcal{R}(t)\|_\infty \leq \|P_1\|_\infty \|P_1^{-1}\|_\infty = M_0,$$

and thus

$$\|U(x_0, t)\|_\infty \leq M_0 \|U^0\|_1 + \|\mathbf{m}\|_\infty M_0 \int_0^t |e(x_0, s)| \|U(x_0, s)\|_1 ds.$$

Using the Gronwall lemma, we obtain

$$\begin{aligned} \|U(x_0, t)\|_\infty & \leq M_0 \|U^0\|_1 \exp\left(\|\mathbf{m}\|_\infty M_0 \int_0^t |e(x_0, s)| ds\right) \\ & \leq M_0 \|U^0\|_1 \exp\left(\|\mathbf{m}\|_\infty M_0 \int_0^T |e(x_0, s)| ds\right). \end{aligned}$$

Using the hypothesis $t \mapsto e(x_0, t) \in L^1_{loc}(\mathbb{R})$, then there is a constant $M \in \mathbb{R}_+$ such as

$$\int_0^T |e(x_0, s)| ds < M.$$

Hence,

$$\|U(x_0, t)\|_\infty \leq C \|U^0\|_1 \quad \text{with } C = M_0 \exp(\|\mathbf{m}\|_\infty M_0 M).$$

□

12.5 Discretization

12.5.1 Motivation

The exact solution of Bloch-type model without relaxation terms is given by

$$\begin{aligned} \rho(t) = & \exp \left(-\frac{i}{\hbar} \left(E_0 t + \mathbf{m} \cdot \int_0^t e(x_0, s) \, ds \right) \right) \cdot \\ & \rho^0 \exp \left(\frac{i}{\hbar} \left(E_0 t + \mathbf{m} \cdot \int_0^t e(x_0, s) \, ds \right) \right). \end{aligned} \quad (12.7)$$

We choose the step time δt and define $t_n = n\delta t$ ($n \in \mathbb{N}$). We can be stated symbolically $\rho(n\delta t) = \rho^n$.

If the electromagnetic wave function $t \mapsto e(x_0, t)$ admits an explicit primitive integral $t \mapsto E(x_0, t)$, then the solution (12.7) becomes

$$\begin{aligned} \rho^{n+1} = & \rho^0 + t_{n+1} \left(-\frac{i}{\hbar} [E_0 + \Delta_{n+1}, \rho^0] \right) \\ & + t_{n+1}^2 \left(-\frac{1}{\hbar^2} [E_0 + \Delta_{n+1}, [E_0 + \Delta_{n+1}, \rho^0]] \right) + O(t_{n+1}^3), \end{aligned}$$

where $\Delta_{n+1} = \frac{E(x_0, t_{n+1}) - E(x_0, 0)}{t_{n+1}}$. For small time t_{n+1} ($\lim_{t_{n+1} \rightarrow 0} \Delta_{n+1} = e(x_0, 0)$), then we have

$$\rho^{n+1} = \rho^0 + t_{n+1} \partial_t \rho(0) + t_{n+1}^2 \partial_t^2 \rho(0) + O(t_{n+1}^3).$$

If the step time is large, the difference between approximate and exact analytical solutions is greater ($O(t_{n+1}^3)$). For this, we must calculate the solution ρ^{n+1} in terms of ρ^n to increase the approximation order.

12.5.2 Strang Scheme

The application of splitting scheme (see, e.g., [12, 14]) needs to calculate the solutions of subequations (exactly or numerically) and to apply the formulae of schemes (Strang method). From Bloch-type model (12.1), the subequations are defined by

$$\partial_t Z_1 = W(Z_1), \partial_t Z_2 = -\frac{i}{\hbar} [E_0, Z_2] \text{ and } \partial_t Z_3 = -\frac{i}{\hbar} [\mathbf{M}(x_0, t), Z_3], \quad (12.8)$$

where Z_1 , Z_2 , and Z_3 are the matrices belonging in $\mathcal{M}_3(\mathbb{C})$.

For all $t \in \mathbb{R}$, exact solutions of previous subequations (12.8) are given by

$$\begin{aligned} \text{Diag}(Z_1(t + \delta t)) &= \exp(W_1 \delta t) \text{Diag}(Z_1(t)) \\ Z_{1,jk}(t + \delta t) &= \exp(\gamma_{jk} \delta t) Z_{1,jk}(t + \delta t), \quad \forall j \neq k \\ Z_2(t + \delta t) &= \exp\left(-\frac{i}{\hbar} E_0 \delta t\right) Z_2(t) \exp\left(\frac{i}{\hbar} E_0 \delta t\right) \\ Z_3(t + \delta t) &= \exp\left(-\frac{i}{\hbar} \mathbf{m} \int_t^{t+\delta t} e(x_0, s) ds\right) \\ &\quad \times Z_3(t) \exp\left(\frac{i}{\hbar} \mathbf{m} \int_t^{t+\delta t} e(x_0, s) ds\right). \end{aligned}$$

In the exact solutions given by the previous equations, we have the matrix exponential. The approximations of exact solutions consist to approximate the matrix exponential. For this, we use two different methods.

Method 1 We define an operator $\mathcal{L}$ that uses the Taylor series of the matrix exponential. Let A be a square matrix, and the operator $\mathcal{L}$ is given by

$$\mathcal{L}(A) = I + A + \frac{1}{2} A^2.$$

The steps to determine the numerical solution are:

1. Calculate $\rho^{n+1} = \mathcal{L}\left(-\frac{i}{\hbar} E_0 \frac{\delta t}{2}\right) \rho^n \mathcal{L}\left(\frac{i}{\hbar} E_0 \frac{\delta t}{2}\right)$.
2. Calculate $\rho^{n+1} = \mathcal{L}\left(-\frac{i}{\hbar} \mathbf{m} \int_t^{t+\frac{\delta t}{2}} e(x_0, s) ds\right) \rho^{n+1} \mathcal{L}\left(\frac{i}{\hbar} \mathbf{m} \int_t^{t+\frac{\delta t}{2}} e(x_0, s) ds\right)$.
3. Calculate $\text{Diag}(\rho^{n+1}) = \mathcal{L}(W_1 \delta t) \text{Diag}(\rho^{n+1})$.
4. Calculate $\rho_{jk}^{n+1} = \exp(\gamma_{jk} \delta t) \rho_{jk}^{n+1}$.
5. Calculate $\rho^{n+1} = \mathcal{L}\left(-\frac{i}{\hbar} E_0 \frac{\delta t}{2}\right) \rho^{n+1} \mathcal{L}\left(\frac{i}{\hbar} E_0 \frac{\delta t}{2}\right)$.
6. Calculate $\rho^{n+1} = \mathcal{L}\left(-\frac{i}{\hbar} \mathbf{m} \int_t^{t+\frac{\delta t}{2}} e(x_0, s) ds\right) \rho^{n+1} \mathcal{L}\left(\frac{i}{\hbar} \mathbf{m} \int_t^{t+\frac{\delta t}{2}} e(x_0, s) ds\right)$.

Method 2 The matrix W_1 can be described as $W_1 = P_1 \mathcal{D}_1 P_1^{-1}$. The matrices $\mathcal{D}_1$ and P_1 are given by formulas (12.6) and (12.5).

The dipolar matrix $\mathbf{m}$ admits a simple decomposition denoted R . We have $\mathbf{m} = P_2 R P_2^{-1}$. We obtain

$$\begin{aligned} \text{Diag}(Z_1(t + \delta t)) &= P_1 \exp(\mathcal{D}_1 t) P_1^{-1} \text{Diag}(Z_1(t)) \\ Z_{1,jk}(t + \delta t) &= \exp(\gamma_{jk} \delta t) Z_{1,jk}(t + \delta t), \quad \forall j \neq k \\ Z_2(t + \delta t) &\equiv \mathcal{L}_1\left(-\frac{i}{\hbar} E_0 \delta t; Z_2(t)\right), \end{aligned}$$

where $\mathcal{L}_1(A, B) \in \mathcal{M}_3(\mathbb{C})$, $\forall j, k \in \{1, 2, 3\}$, $\forall A, B \in \mathcal{M}_3(\mathbb{C})$, and $\mathcal{L}_1(A, B)_{jk} = \exp(\lambda(A)_j - \lambda(A)_k) B_{jk}$ with $\lambda(A)$ the spectrum of A .

For the last subequation, if R is a diagonal matrix D_2 , then we have

$$Z_3(t + \delta t) \equiv P_2 \mathcal{L}_1 \left(-\frac{i}{\hbar} D_2 \left(\int_t^{t+\delta t} e(x_0, s) ds \right)^2; \tilde{Z}_3(t) \right) P_2^{-1},$$

with $\tilde{Z}_3(t) = P_2^{-1} Z_3(t) P_2$.

We notice

$$\int_t^{t+\delta t} \mathbf{M}(x_0, s) ds = \mathbf{m} \int_t^{t+\delta t} e(x_0, s) ds = P_2 D_2 P_2^{-1} \int_t^{t+\delta t} e(x_0, s) ds.$$

The computation of numerical solution obtained with method 2 can be summarized as follows:

1. Determine the eigenvalues of the matrix W_1 and a non-singular square matrix P_1 and its inverse P_1^{-1} .
2. Determine the eigenvalues of the dipolar matrix $\mathbf{m}$ and a non-singular square matrix P_2 and its inverse P_2^{-1} .
3. For all $n \in \mathbb{N}$, the computation of solution ρ^{n+1} in terms of ρ^n is given by:
 - a. Calculate $\rho^{n+1} = \mathcal{L}_1 \left(-\frac{i}{\hbar} E_0 \frac{\delta t}{2}; \rho^n \right)$.
 - b. Determine $\rho^{n+1} = P_2^{-1} \rho^{n+1} P_2$.
 - c. If R is a diagonal matrix, calculate $\rho^{n+1} = P_2 \mathcal{L}_1 \left(-\frac{i}{\hbar} D_2 \int_t^{t+\frac{\delta t}{2}} e(x_0, s) ds; \rho^{n+1} \right) P_2^{-1}$.
 - d. Calculate $\text{Diag}(\rho^{n+1}) = P_1 \exp(\mathcal{D}_1 \delta t) P_1^{-1} \text{Diag}(\rho^{n+1})$.
 - e. Calculate $\rho_{jk}^{n+1} = \exp(\gamma_{jk} \delta t) \rho_{jk}^{n+1}$, $\forall j \neq k$.
 - f. Calculate $\rho^{n+1} = \mathcal{L}_1 \left(-\frac{i}{\hbar} E_0 \frac{\delta t}{2}; \rho^{n+1} \right)$.
 - g. Determine $\rho^{n+1} = P_2^{-1} \rho^{n+1} P_2$.
 - h. If R is a diagonal matrix, calculate $\rho^{n+1} = P_2 \mathcal{L}_1 \left(-\frac{i}{\hbar} D_2 \int_{t+\frac{\delta t}{2}}^{t+\delta t} e(x_0, s) ds; \rho^{n+1} \right) P_2^{-1}$.

Theorem 12.4 *The results of method 2 verify the following proprieties:*

- *The numerical solutions are second order.*
- *Hermicity and the trace are conserved.*

Proof

- The order of method 2 is deduced from the order of Strang scheme.
- If $\text{sp}(A) \subset \mathbb{R}$, B is Hermitian, and knowing that $\overline{\mathcal{L}_1(iA, B)_{jk}} = \mathcal{L}_1(iA, B)_{kj}$, then, $\forall n \geq 0$, $(\rho^{n+1})^* = \rho^{n+1}$.

For all real matrices A and B , we have $\forall j, \mathcal{L}_1(iA, B)_{jj} = B_{jj}$, and for all vector $v \in \mathbb{R}^3$,

$$\begin{aligned} \sum_{j=1}^3 \left(P_1 \exp(\mathcal{D}_1 \delta t) P_1^{-1} v \right)_j &= \sum_{j=1}^3 v_j + \sum_{j=1}^3 \left(W_1 \sum_{k=0}^{\infty} P_1 \mathcal{D}_1^k P_1^{-1} v (\delta t)^{k+1} \right)_j \\ &= \sum_{j=1}^3 v_j, \end{aligned}$$

then

$$\forall n \geq 0, \quad \text{tr}(\rho^n) = \text{tr}(\rho^0).$$

□

12.6 Simulation

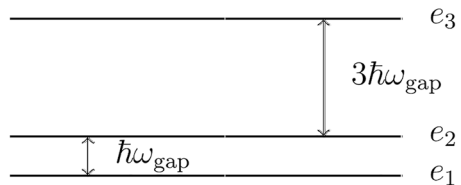
In this section, we present the numerical simulation of Bloch-type model. First, we compare two numerical methods developed in Sect. 12.5.2. After that, the best numerical method obtained is used to study the effects of the electromagnetic wave (wave–matter interaction term) about time evolution of the populations.

In the simulations, we choose three-level quantum model (represented in Fig. 12.1) with the energies $e_1 = -\hbar\omega_{\text{gap}}$, $e_2 = 0$, and $e_3 = 3\hbar\omega_{\text{gap}}$ (ω_{gap} is gap frequency). Initially, the first level is fully occupied ($\rho_{11}(0) = 1$), and the other levels are empty ($\rho_{22}(0) = 0$ and $\rho_{33}(0) = 0$).

12.6.1 Model Without Electromagnetic Wave

In this part, we compare the two numerical solutions used in Strang scheme (method 1 and method 2). This comparison is established in the case of Bloch-type model without wave–matter interaction term.

Fig. 12.1 Electronic configuration of quantum model (three-level) with $e_2 - e_1 = \hbar\omega_{\text{gap}}$ and $e_3 - e_2 = 3\hbar\omega_{\text{gap}}$



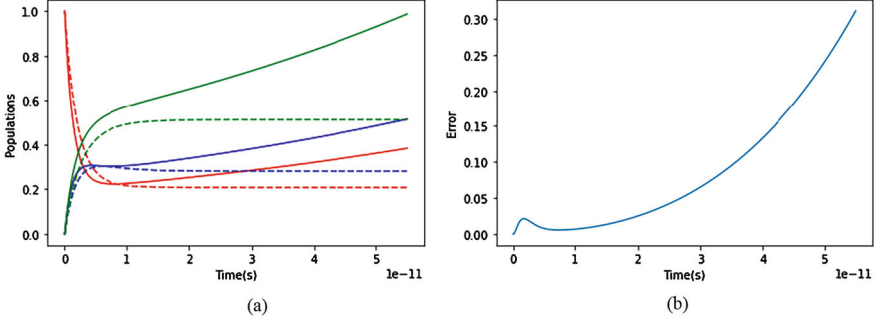


Fig. 12.2 (a) Time evolution of numerical solutions given by method 1 and method 2. (b) Square difference between the two numerical solutions

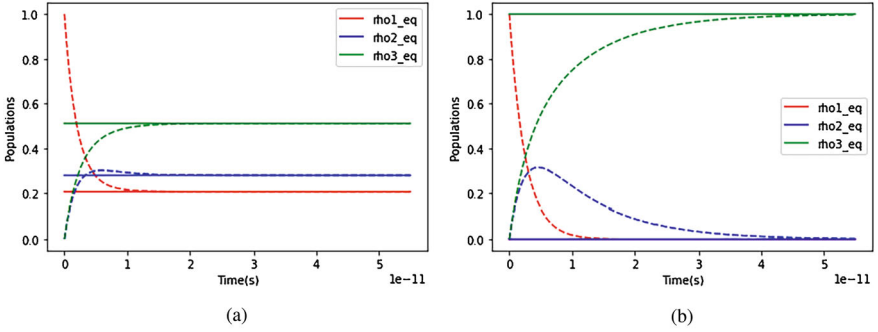


Fig. 12.3 Time evolution of the populations for the autonomous model. (a) $\omega_{\text{gap}} = 1.18 \cdot 10^{13}$. (b) $\omega_{\text{gap}} = 1.18 \cdot 10^{15}$

In Fig. 12.2a, numerical solutions obtained with method 1 are represented in solid lines, and those obtained with method 2 are represented in dotted lines. The populations ρ_{11} , ρ_{22} , and ρ_{33} are, respectively, represented in red, blue, and green curves. We remark a large absolute deviation between the two types of numerical solutions represented in Fig. 12.2b, and this deviation increases in terms of time. Time evolution of this absolute deviation proves that method 1 destroys the natural stability of splitting scheme unlike method 2 that conserves this stability. In addition, method 1 gives an explosive solution with a large step time.

The solutions obtained with method 1 do not converge to equilibrium, but those provided by method 2 converge to equilibrium. The illustration of this convergence is established in Fig. 12.3a and b. Also, we remark a rapid convergence of the three populations to equilibrium in the absence of the electromagnetic wave and time convergence depends to gap frequency value (these results are proved in Sect. 12.4 in the case of autonomous model). With the gap frequency $\omega_{\text{gap}} \approx 1.18 \cdot 10^{15}$, we assist a full occupation of an initially empty energy level (see Fig. 12.3b).

Table 12.1 CPU times of numerical methods (numerical Strang scheme)

Mesher number (times)		2000	4000	10000	50000	100000
CPU times (second)	Method 1	0.29	0.57	1.40	6.56	14.07
	Method 2	0.17	0.34	0.79	3.70	8.35

To approximate the time where the populations converge to equilibrium, we calculated the first time T_c such as the function $\sum_{j=1}^3 |\rho_{jj}(T_c) - \rho_{jj}^{eq}|$ is less than ϵ , where ρ_{jj}^{eq} is a population value obtained at equilibrium.

In Table 12.1, we present the CPU times in terms of meshes number. These results prove that method 2 is better in terms of CPU time.

In the autonomous model (without wave–matter interaction term), the results given in Figs. 12.2 and 12.3 and the natural stability conservation show that the numerical solutions calculated with method 2 (Strang scheme whose evolution operators are computed by reduction matrices) are near to exact solutions. This confirms the convergence of the method. Also, we remark that the trace conservation and positiveness of numerical solutions are verified with method 2.

12.6.2 Population Inversion

In this section, the simulations are realized with the numerical method 2 because it conserves the best proprieties (hermicity and the trace are conserved, also the convergence to the equilibrium).

12.6.2.1 Sinusoidal Plane Wave

In this section, we add the wave–matter interaction term in the previous model simulated in order to study this effect. Let a sinusoidal plane wave defined by

$$e(x_0, t) = e_0 \sin \left(\omega_0 \left(t - \frac{x_0}{v} \right) \right),$$

where ω_0 is the frequency, e_0 the maximum amplitude of the field depends to $\mathcal{T} = \hbar \omega_{\text{gap}} / \|\mathbf{m}\|_{\infty}$. The exact solution of integral is

$$\int_{t_n}^{t_n + \delta t} e(x_0, s) \, ds = \frac{e_0}{\omega_0} \left(-\cos \left(\omega_0 \left(t_n + \delta t - \frac{x_0}{v} \right) \right) + \cos \left(\omega_0 \left(t_n - \frac{x_0}{v} \right) \right) \right). \quad (12.9)$$

In the simulations, we choose $x_0 = 0$ and $\omega_0 = \omega_{\text{gap}}$. We consider a Bloch-type model without taking into account the relaxation terms ($W_1 = 0$ and $\forall j, k, \gamma_{jk} = 0$). The quantum model used is represented in Fig. 12.1.

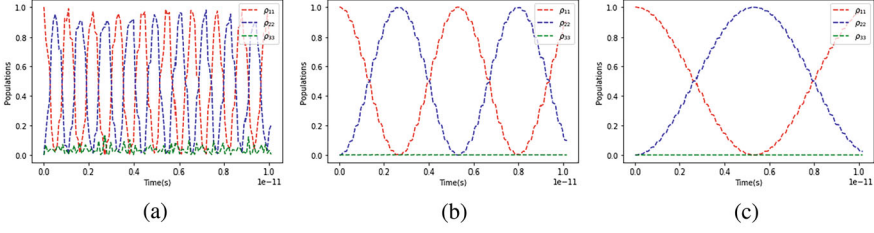


Fig. 12.4 Time evolution of populations without the relaxation terms (considering free energy and wave–matter interaction term). (a) $\delta t = \frac{1}{\omega_0}$. (b) $\delta t = \frac{1}{5\omega_0}$. (c) $\delta t = \frac{1}{10\omega_0}$

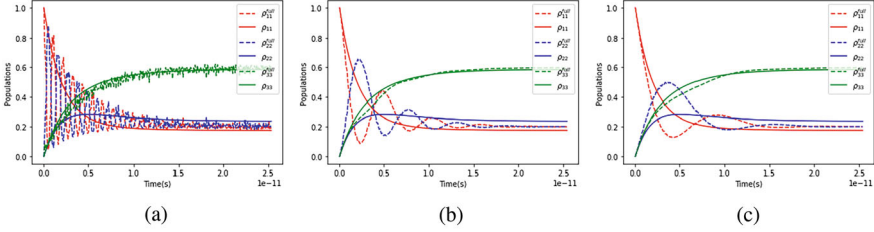


Fig. 12.5 Time evolution of populations in the case of autonomous model and full model. (a) $\delta t = \frac{1}{\omega_0}$. (b) $\delta t = \frac{1}{5\omega_0}$. (c) $\delta t = \frac{1}{10\omega_0}$

We choose a different time step and represent time evolution of populations in Fig. 12.4. In these experiments, electromagnetic wave inverts the levels whose energy gaps are proportional to its frequency and small oscillations appear during these inversions. With the small step time, we remark that population inversion is slowly realized in time and small amplitude oscillations are difficult to identify. These results can be explain by integral value (12.9) that is equivalent to $-\frac{\epsilon_0 \delta t}{\omega_0} \sin(\omega_0 t_n)$; then, inversion time depends to step time. In the case of small time step, we have a very slow inversion of populations in considering Bloch-type model without the relaxation terms.

Now, we add the relaxation terms to the model studied above. The simulation results are illustrated in Fig. 12.5. Time evolution of populations in the case of model without wave–matter interaction (autonomous model) is represented in solid lines, and time evolution of full model populations is represented in dotted lines. The results show us that the populations tend to the values which depend on the physical parameters.

Taking into account the relaxation and wave–matter interaction terms, electromagnetic wave having a frequency proportional to an energy gap of quantum model cannot to invert the populations, but we assist to time evolution of coherences (see Fig. 12.6). From Fig. 12.6, we can deduce that density matrix remains Hermitian, and absolute values of coherences are less than or equal to 1. These results are obtained with step time $\delta t = \frac{1}{10\omega_0}$ and $\omega_{\text{gap}} = 1.18 \cdot 10^{13}$.

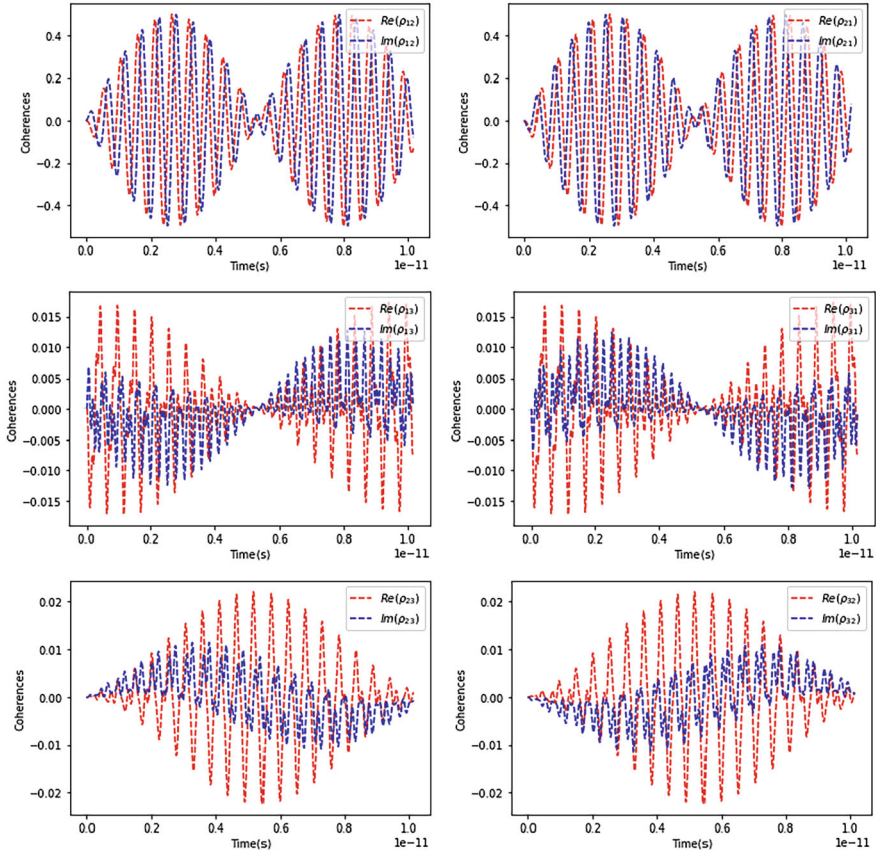


Fig. 12.6 Time evolution of coherences

In addition to these results, we remark that electromagnetic wave and the populations vary at a different scale time. This property is assumed in the study provided by [4, 5].

12.6.2.2 Self-induced Transparency

Self-induced transparency (SIT) is a phenomenon that realizes an exact population inversion using a particular resonant light pulse. The light pulse transmits energy to the medium, but the matter unchanged after the wave passage. This phenomenon is firstly realized in [6, 13, 15]. In [6] and [13], SIT phenomenon is studied to validate the numerical schemes. In the papers [15] and [13], the authors are not used the relaxation terms. Here, we test the numerical simulation of SIT with the relaxation terms.

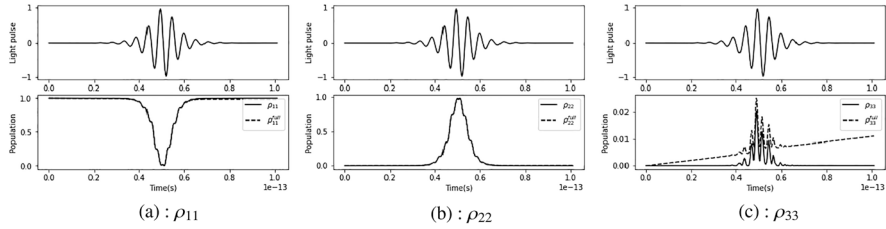


Fig. 12.7 The SIT 2π pulse in the case of autonomous model and full model. (a) ρ_{11} . (b) ρ_{22} . (c) ρ_{33}

Consider the light pulse of SIT phenomenon given by

$$e(x_0, t) = e_0 \operatorname{sech} \left(\frac{t - \frac{x_0}{v}}{\tau_p} \right) \sin \left(\omega_0 \left(t - \frac{x_0}{v} \right) \right).$$

The numerical solution of integral obtained trapezoidal rule:

$$\int_{t_n}^{t_n + \delta t} e(x_0, s) \, ds \approx \delta t \frac{e(x_0, t_n + \delta t) + e(x_0, t_n)}{2}.$$

The results of experiments SIT are represented in Fig. 12.7. The populations obtained with the autonomous model are represented in solid lines, and those obtained with full Bloch-type model are represented in dotted lines.

We remark population inversion in the case without the relaxation terms, and these inversions are slowly destroyed with full model. In the case of full model, the third population ρ_{33} is excited without electromagnetic wave.

12.7 Conclusion

In this chapter, we describe classical Bloch-type model as dynamical system that admits a unique solution and locally time bound. We show that dynamical system conserves qualitative proprieties verified by classical model. In the absence of wave–matter interaction, we demonstrate the time convergence to equilibrium of Bloch-type solution.

The splitting schemes are known as method having numerical stability, but in the application about the Bloch-type model, we obtain evolution operators whose computer programming requires an approximation of this operator. In this work, we show that a bad approximation of evolution operators can destroy the stability of splitting schemes. For this, we develop a numerical method that includes the reduction matrices. This numerical method is better in terms of CPU time, preserves the numerical stability, and conserves the qualitative proprieties verified

by continuous model. Without wave–matter interaction term, we show that the numerical solution converge to equilibrium value calculated with dynamical system.

After the time evolution of the Bloch-type model, we will solve numerically Maxwell–Bloch model with finite element method taking into account the time scales of two models and realize some physical experiments in higher spatial dimension.

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